Looked Up or Copied? Later Mentions Decide Whether a Model Reads an In-Context Value Through Its Key or Its Value

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Abstract

A token that states a value in context, such as *shelf* in "the candle is moved to the shelf", can reach a later answer in two ways: a later token can match its cached key and look the value up, or the answer can copy its cached value. Clamping this writing token's key or value separately, we find that later mentions decide the route. In stories, when the candidates are named again after the token, its key carries the value's identity (most of it after a list, part after a neutral sentence, none after a sentence naming no candidate); without a later mention, or with the mention before the token, the identity is copied. On fresh stories in seven model families (7–14B), scored on the forms the models emit, the route decides generated answers: clamping only the key changes them when the candidates are listed after the story, and clamping only the value does the reverse. On counterfactual reading-comprehension passages, an options list added after the passage likewise makes every evaluable model read the answer entity through its keys, and without it through its values; a sentence naming candidates after the passage opens only a small key read. In two 7B models, heads enriched for duplicate-token heads write at each later mention, besides the word itself, a shared component (a flag) that names no candidate; moved to an absent option's row, it moves the answer there and is read by key, but it is neither specific nor the only carrier. A preregistered law that edits of the written value keep the model's own key-to-value ratio failed, even for near-natural steering vectors; a remap trained at the writing token is read far less through its key than that ratio predicts. Of the predictions we judged risky before the run, 10 of 27 held (expected 12.05), as did 6 of 6 replications and 16 of 19 extrapolations; we report every failure.

1. Introduction

Mechanistic accounts often credit one cache channel: keys act as routers (Ma et al., 2025), Llama models carry binding information mainly in key vectors (Oh & Demberg, 2026), and refusal steering acts mainly through the OV circuit (Cheng et al., 2026). Later tokens see a token only through its cached key and value, so a written value can reach the answer in two ways (Figure 1a): it is *looked up* when a later token that already holds a candidate matches the writing token's key, so that the key carries which value was written, and *copied* when the identity travels in the value. What decides which route a model takes, and what follows for interventions credited to one channel?

In the belief stories of Anonymous (2026), one token, which they call the critical token and we call the *writing token*, states where an object was moved. We clamp its cached key and value separately and vary whether and where the candidates are named again: in 10 discovery-sample models from three families; in a preregistered confirmation on 150 newly drawn stories in eight models from seven families, four never examined before; and on counterfactual reading-comprehension passages. Before the final round we recorded a probability of holding for each of its 69 predictions and, for each primary one, the text to print for each outcome; we name every failure. From Anonymous (2026) we take only the story template and, in Section 5, one released remap and its training code.

Findings. Later mentions decide the route (Section 3): a key-only clamp changes 76–100 % of generated answers in all eight confirmation models when the candidates are listed after the story, a value-only clamp most of them without a later mention (Figure 1b,c); on passages a later options list opens the key read. In two 7B models the lookup is a duplicate-token match writing a flag that the answer reads by key, sufficient but neither specific nor the only carrier (Section 4). A preregistered law predicting an edit's apparent channel from the model's own channel ratio failed

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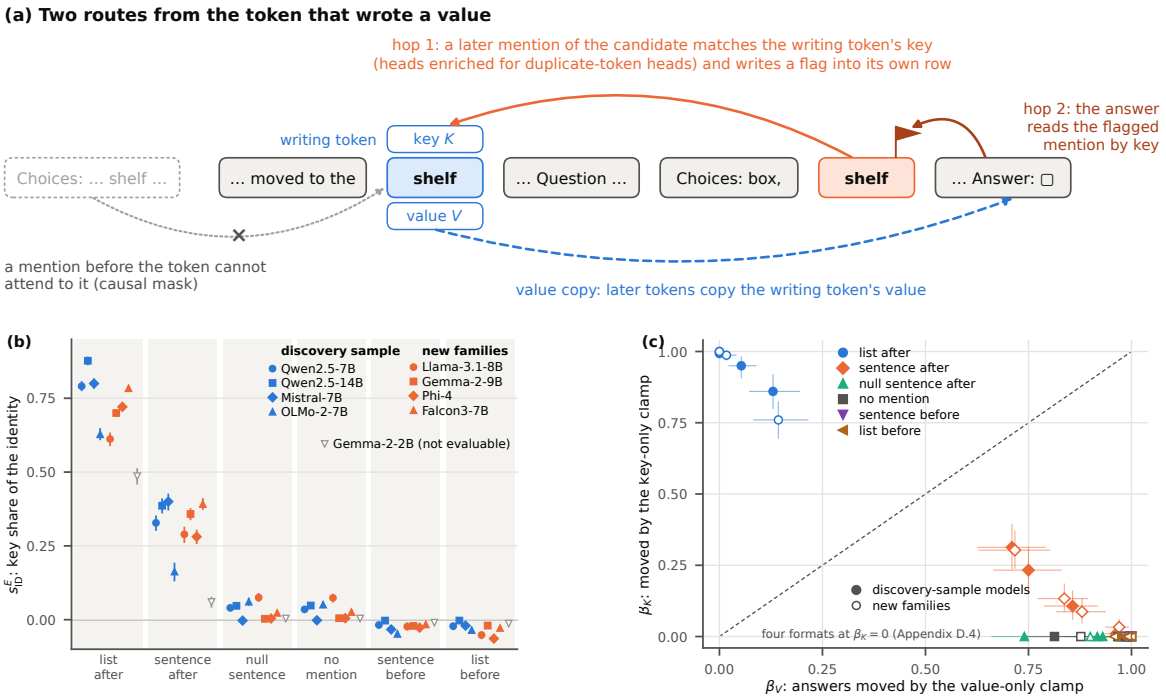


Figure 1. **Later mentions decide whether a written value is looked up through its key or copied from its value.** (a) The writing token *shelf* leaves a key K and a value V in the cache at every layer; we clamp them separately. A later mention matches K and writes a flag into its own row (hop 1), which the answer reads by key (hop 2); without one, the identity reaches the answer in V. (b) Fresh stories, eight models: key share of the identity, s^E_{ID} (E: emitted-form score), by where the candidates are named; 95 % two-stage cluster intervals (98.75 % for the new families); grey: Gemma-2-2B, not evaluable. (c) Share of generated answers moved to the clamped value by the key-only clamp (β_K) against the value-only clamp (β_V); intervals as in (b); formats at zero key flips: Appendix D.4.

(Section 5).

What is new. Every attention head factors into a QK circuit, which decides where it attends, and an OV circuit, which decides what it moves (Elhage et al., 2021); known circuits attend to earlier copies, select among options and retrieve bound entities (Table 1; Tulchinskii et al., 2024; Sun et al., 2026; Feng & Steinhardt, 2023; Dai et al., 2024; Prakash et al., 2024; 2026; Gur-Arieh et al., 2025). None of this says which route a model takes, because the copy route is always open. We claim no new head type. We add, each by a preregistered test, that where the candidates are mentioned decides which route carries the identity, in every family we tested and in generated answers; and that the duplicate-token match writes into the mention's row a component that is the same whichever value was written and suffices to move the answer (though structured controls move it too), which the answer selects by key: the duplicate signal is consumed through keys, and which value was written is read at the later mention, not looked back for at the token that states it. The same reader heads carry the read whether the question asks for a mentioned or an unmentioned option, the sign being applied after them (exploratory); a key read of opposite sign to the value read, as with IOI's re-mentions, is consistent with S-inhibition (Wang et al., 2022).

2. Setup

Stories, runs and formats. We use the belief stories of Anonymous (2026) ("[...] {agent} watches as the {o} is moved to the {LOC}. [...] Where does {agent} believe the {o} is?"; Appendix B). Base and source stories B and S differ only at the writing token {LOC}, at position p; a third value X is absent from the story. OPTIONS-AFTER lists the candidates after the question ("Answer with exactly one choice"), LETTERS-AFTER labels them A–F; under "Answer with one word", LIST-AFTER lists them there, SENTENCE-AFTER names them in a neutral sentence after the story, NULL-SENTENCE replaces it by one of the same length that names none, SENTENCE-BEFORE and LIST-BEFORE put the sentence or the list before the story, and NO-MENTION names none. A *later mention* is a token after p that repeats a candidate's word form, up to case (Appendix E.1). The *fresh stories* are 150 newly drawn cores of the same story template, with eight new mention sentences (each with its null version), a second location lexicon in half of the cores and per-core candidate orders (Appendix D.2).

Table 1. **What the architecture implies, what known circuits supply, and what we find.** "Architecture alone": what QK/OV and the causal mask imply (Elhage et al., 2021). Unmarked: preregistered and held; ^{R}: a risky prediction (recorded probability below 0.8; Appendix A).

Question	Architecture alone	Known circuits	What we find (section)
Can a mention *before* the token read its key?	No (causal mask)	Options before a context cannot attend to it (Ok & Lee, 2026)	Nor does a later token read the key through it (eight story models, passages); in the new families a list before even lowers the read below no mention^{R} (§3.1)
Mentioned after it: looked up or copied?	Either; copy route open	Duplicate-token and induction heads attend to earlier copies (Wang et al., 2022; Olsson et al., 2022)	Looked up: the key carries most of the identity in all eight story models (s_{ID} 0.612–0.784 in the new families) and three passage models; copied without a later mention. A neutral sentence opens part of the read^{R}, a null sentence none^{R}, a passage's related-articles sentence only a little (did not hold^{R}) (§3)
Does the route decide generated answers?	Not decided	Option-selecting heads (Wiegreffe et al., 2024), some by query–key alignment (Tulchinskii et al., 2024)	Stories: yes in all eight models, the sentence read in seven^{R} (not OLMo-2-7B). Passages: with lettered options^{R}; by name or in free form not shown (did not hold^{R}) (§3)
What does the matching head write?	Not decided	Binding ID vectors (Feng & Steinhardt, 2023); an option-routing feature in option rows (Sun et al., 2026)	Besides the word, a shared component naming no candidate (the flag); moved to an absent option's row it moves the answer and is read by key^{R}; not specific, not necessary as one direction (did not hold^{R}) (§4)
What sets the sign of a key read?	Not decided	S-inhibition (Wang et al., 2022); copy suppression (McDougall et al., 2023)	The question: the same reader heads carry both questions' reads, the sign applied after hop 1 (exploratory). IOI: key read negative, value read positive, applied at the listed names' key read^{R}, not by the heads ranked in lists (did not hold^{R}) (§4)
Is an edit's apparent channel fixed?	Not decided	Steering via OV (Cheng et al., 2026); keys as routers (Ma et al., 2025)	No: a released remap's channel flips with the format (discovery sample); edits do not keep the model's channel ratio, even near-natural steering vectors (did not hold) (§5)

Passages. From the SQuAD v1.1 development set (Rajpurkar et al., 2016) we take questions whose answer is a person, place or number named exactly once in the passage, with no content word of it elsewhere, and write a counterfactual entity in its place (Longpre et al., 2021): 185 items from 34 articles, with answers of mostly two to four tokens, clamped at every position of the span. Formats: options after the question (by name or letter) or before the passage, a related-articles sentence after or before it, or free form. We score the first answer token after an answer frame fixed beforehand from each model's generations (Appendix D.5).

Clamps. $\mathcal{C}_K(Z)$ sets the writing token's pre-RoPE key in every layer from ℓ_0 (0 unless stated) to that of run Z, and $\mathcal{C}_V(Z)$ its value. Later tokens see the writing token only through these, so clamping both from layer 0 reproduces run Z exactly (checked in FP32; Appendix C). A single-channel clamp creates a cache state that no prompt produces; generated answers under it show that the read reaches behaviour, not that the state is natural.

Measures. With $\Delta_Y(Z)$ the change in $\log p(Y)$ at the answer under $\mathcal{C}_K(Z)$ against a self-clamp row in the same batch,

$$\mathrm{ID}_K = \tfrac{1}{2}\big[\Delta_{\mathrm{S}}(\mathrm{S}) - \Delta_{\mathrm{S}}(\mathrm{X}) + \Delta_{\mathrm{X}}(\mathrm{X}) - \Delta_{\mathrm{X}}(\mathrm{S})\big], \quad (1)$$

so effects shared by all foreign keys, and a prior for B, cancel. ID_V is defined likewise with values, $s_{\mathrm{ID}} = \overline{\mathrm{ID}}_K/(\overline{\mathrm{ID}}_K + \overline{\mathrm{ID}}_V)$ is a ratio of means, and r is a format's $\overline{\mathrm{ID}}_K$ relative to that with the list after the story. Under the *emitted-form score*, a candidate's probability is summed over the surface forms the model itself emits on held-out stories (Appendix C.2); the discovery sample keeps its lower-case score. Under each clamp we also decode greedily: β_K (β_V) is the share of answers that move to the clamped value under the key-only (value-only) clamp, net of moves that a clamp to the third value also causes; the flip rate omits that correction.

Models, statistics and preregistration. The discovery sample has 10 models: Qwen2.5 at 1.5B–72B (Yang et al., 2024), Qwen3-8B (Yang et al., 2025), Mistral-7B (Jiang et al., 2023), Mistral-Small-24B (Mistral AI, 2025) and OLMo-2-7B (Team OLMo et al., 2024); the final round adds Llama-3.1-8B (Grattafiori et al., 2024), Gemma-2-9B and Gemma-2-2B (Gemma Team et al., 2024), Phi-4 (Abdin et al., 2024) and Falcon3-7B (Technology Innovation Institute, 2024). All are instruction-tuned and run in BF16, those of the final round from official weights (sha256-verified; Appendix J). Intervals are 95 % percentile bootstrap intervals (10,000 resamples), two-stage over clusters on fresh stories (location pair and lexicon) and passages (article), and in the mechanism tests over stories with the flag's fit fixed (so they exclude its uncertainty); the intervention law uses 90 % intervals (two one-sided tests) over base–source pairs and their stories, with the remap's two seeds as a further level that cannot estimate seed variance (Appendix F.3). A prediction over several models must hold in every evaluable model, one over the four new families in three of them (98.75 % intervals). Holm's step-down changes no verdict (Appendix A.2). Each of ten preregistrations was committed before its run, with its scoring code fixed before any output was inspected. In both conflicts between the final entry's common part and a part's own text (the title's scope, one evaluability rule) we followed the part's text; both choices favour our account (Appendix A.6 and Table 2). Unless marked, a result in Sections 3 to 5 was preregistered in this final round and held; a superscript ^{R} marks a risky prediction (probability of holding below 0.8). Where a prediction did not hold or was not evaluable we

say so in words, with what was found instead; discovery-sample, exploratory and post hoc results are labelled, and Appendix A.7 maps each sentence to its prediction and verdict.

3. Later mentions decide whether a value is looked up or copied

When the candidates are named again after the writing token, later tokens read its key and the value's identity is looked up; without such a mention it is copied from the value. The position of the mention decides, not the instruction (discovery sample), and the decision reaches generated answers. We call this the *format law*.

3.1. Fresh stories in seven families

In the discovery sample a list after the story opened a large key read, as on two further templated tasks, and a neutral sentence an unpredicted one; an instruction-matched 2×2 showed that position decides, not the instruction, and the channels carried which value was written, not its belief role (Appendix D.1). The confirmation ran on the 150 fresh stories in those four models and in Llama-3.1-8B, Gemma-2-9B, Phi-4 and Falcon3-7B (Figure 1b). In four model families never examined before, a list of the candidates after the story makes the writing token's key carry most of the value's identity ($s_{\mathrm{ID}} \geq 0.5$ in 4 of 4), as in the four discovery-sample models (s_{ID} 0.612–0.877 over all eight). Without a later mention, the identity is copied through the value in every family ($s_{\mathrm{ID}} \leq 0.1$ under NO-MENTION). The same list or sentence before the story opens no key read in any model; in the four new families a list before even lowers the key read below its no-mention level^{R}, which we do not explain (Section 7).

A neutral sentence naming the candidates after the story opens an intermediate key read (r = 0.334–0.512 of the list read) in 4 of 4 new families^{R}, as in the four discovery-sample models, with each of the eight sentences and both lexicons (exploratory). A sentence of the same length naming no candidate opens no key read, so the read needs the re-mention (^{R} in the new families); in Llama-3.1-8B it missed its threshold mainly because the model shows a small key read without any mention (r = 0.113), to which it adds almost nothing (+0.005 [0.0001, 0.011], 98.75 %; post hoc reading; Appendix D.2). At 2B (Gemma-2-2B) the test was not evaluable: the model failed the competence gate (had it passed, the prediction would not have held; post hoc).

Generated answers. Clamping only the writing token's key changes the generated answer to the clamped value in 76–100 % of stories when the candidates are listed after it and in 0 % in every model when they are not; clamping only its value does the reverse (Figure 1c). Here 'not' means no mention, a list before or a sentence before, and the reverse is a value-only change in 81–100 % of stories without a mention. The sentence read changes generated answers in seven of the eight models (flip rate 3–31 % after the sentence, 0 % before it), a small share in most^{R} (held in the new families; Phi-4 passes Holm only narrowly); at OLMo-2-7B it is visible in log-probabilities but not in generated answers (flip rate 1.0 % after, 0 % before), so in the discovery-sample models the prediction did not hold^{R} (a disclosed wording, Appendix A.6).

The measurement. Scored on the forms the models actually emit (coverage ≥ 0.8 in every cell), every re-scored verdict on the original stories is unchanged^{R}. s_{ID} moves by at most 0.05 (in the low-mass cells tested; OLMo-2-7B has none), while the sentence read relative to the list read, r, rises under emitted forms by +0.087 at Mistral-7B ([0.068, 0.105]) and by +0.048 at Qwen2.5-14B, where equivalence within 0.05 is not shown, so its predicted invariance did not hold^{R}: the lower-case score understated the sentence read. Two further validity checks held^{R} (Appendix D.3).

3.2. Reading-comprehension passages

On counterfactual SQuAD passages with multi-token answers, a later options list makes the model read the passage entity through its cached keys (s_{ID} 0.550–0.677 in Gemma-2-9B, Qwen2.5-7B and Mistral-7B), and without it through its values (s_{ID} at most 0.045 in Llama-3.1-8B, Gemma-2-9B and Qwen2.5-7B; Figure 2a). This holds for answers of three or more tokens^{R} and, with lettered options, in all four models; the derived within-model contrast held in both models where it was evaluable. Two cells these claims use failed gates (Llama-3.1-8B with options after, by one item; Mistral-7B in free form), 6 of 24 in all (Appendix D.5). Options before the passage leave a small negative key read. The prediction that a related-articles sentence after the passage opens the key read did not hold^{R}: it opens only a small key read (s_{ID} gain 0.042–0.070 in Gemma and Qwen, the evaluable models), so on passages our claim concerns a later options list.

Behaviour on passages is a boundary. In cue conflict (key and value from two counterfactual entities), with lettered options the generated answer follows the key's entity in all four models (key-source rate 0.629–0.978)^{R}, narrowly under Holm in Mistral-7B. With options answered by name the prediction that the key wins did not hold^{R}: it wins in two of the three evaluable models (Gemma, Qwen), not in Mistral (nor in Llama's cell, which is not evaluable), although a key that matches no option leaves the value to be copied^{R}. In free form the decision token follows the value while the continuation breaks (hybrids, an observa-

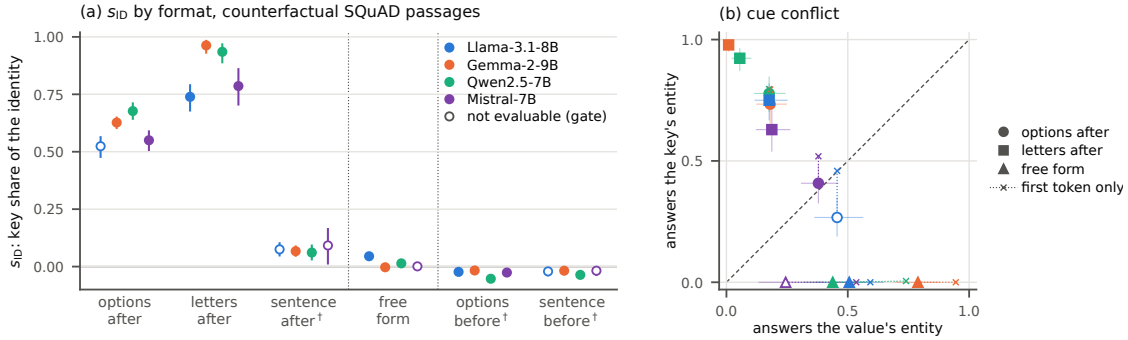


Figure 2. **On passages, a later options list makes the model read the answer entity's identity through its cached keys; answers follow the key in every evaluable model only with lettered options.** Counterfactual SQuAD passages, four models, 185 items; 95 % two-stage article-cluster intervals. (a) s_{ID} by format, at the first answer token; hollow: not evaluable (preregistered gate); †: recomputed (exploratory). (b) Cue conflict (key and value from two counterfactual entities): share of answers naming the key's or the value's entity; crosses: decision token alone; a boundary, not a confirmed claim.

tion), so too few whole answers name the value's entity and that prediction did not hold^{R} either (Figure 2b). The channel attributions are therefore not shown to decide behaviour on natural text. Two predictions on 2-bit quantization of the cached values and one on the rate of hybrids did not hold either^{R}; the head tests^{R} (with the failing gate waived, two would have held and two not) and three further passage tests (one would not have held had its gate passed) were not evaluable (post hoc; Appendix D.5).

4. A match at the later mention, read by key

In two 7B models (Qwen2.5-7B, Mistral-7B), heads enriched for duplicate-token heads match the writing token's key at each later mention and write a flag into that mention's row, which the answer reads by key. The flag suffices to move the answer but is neither specific nor the only carrier, and the question sets the sign of the read.

Two hops. In the discovery sample, 5 % of heads, the *reader heads*, read the writing token's key at the option words (hop 1): seeing the clamped key alone, they recover 0.967 and 0.942 of the option rows' read, random sets almost none. They are enriched for duplicate-token heads (Wang et al., 2022): the 20 and 16 heads carrying 80 % of the read have median duplicate-token scores of 0.370 and 0.232, against 0.028 and 0.007 over all heads (exploratory). Mean-ablating them at the option rows leaves 0.238 and 0.059 of the key read and raises the value read: a copy route stays available (Appendix E.2). The answer reads the option words, not the writing token (hop 2), mainly by key.

What the readers write. The reader heads write one vector per layer into the re-mentioned word's row. We call it the *flag* (fit on the 60 ranking stories, tested on 100 others). Injected with no key clamp into the row of an option absent from the story, a version fit without that word moves the answer there (0.997 / 0.803 of the key read's identity effect; the answer changes in 96 % / 67 % of stories)^{R}. Like the natural key clamp, the injection adds the flag at that row and removes it at the base value's row; added alone, one flag moves the answer 0.673 / 0.541 as much as the natural clamp, half a flag almost not at all (exploratory; Figure 3a). Its mean direction names no candidate: fit with or without the value's identity, it is the same direction with the same effect, although each story's write also carries its word (Appendix E.3). The answer reads it by key, as it reads the natural one (giving only the answer row the option rows' original keys removes 0.689 / 0.601 of its effect, their original values 0.047 / 0.068). But the prediction^{R} that it is specific failed against structured controls: at Qwen2.5-7B the mean reader output moves the answer 0.615 as much as the flag, and at Mistral-7B the story's own write with its flag component removed, norm-matched, 0.516 as much (the mean output contains the flag by construction; Appendix E.3). And one direction per layer does not carry the key read (ρ_K = 0.466 / 0.478): other carriers exist; the flag is sufficient but not necessary (Figure 3b). That prediction^{R} failed on its controls: removing the flag direction leaves about half of the key read in both models, and at Qwen the control directions, which overlap the flag, remove 31 % and 44 %; ablating the whole heads removes far more (above).

The sign of the read. Asked which option was *not* mentioned, both reads turn negative (ID_K −12.1 and −7.1 nats), and the same frozen reader heads carry the reads under both questions (Figure 3c); the flag fit with the belief question, injected under this one, reproduces its negative effect (1.228 and 0.899 of it; exploratory), so the sign is applied after the first hop. In indirect-object identification (IOI) with the names re-mentioned in a parenthesis, the key read is negative and the value read positive (ID_K −3.35 against ID_V 2.55 nats at Qwen2.5-7B; the same signs at Mistral-7B). The negative IOI read is applied at the an-

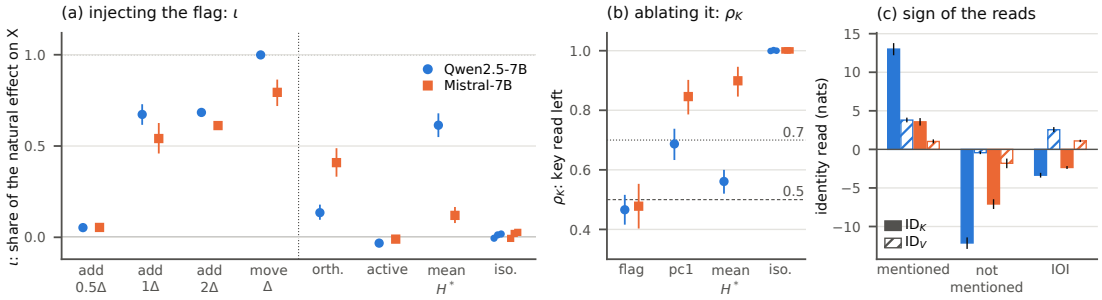


Figure 3. **What the reader heads write.** Qwen2.5-7B and Mistral-7B, BF16; flags fit on 60 discovery-sample stories, tested on 100 others; 95 % story-bootstrap intervals with the flag's fit fixed. (a) ι: effect on an absent option X as a share of the natural key clamp's, for the flag added 0.5, 1 or 2 times (exploratory) or moved from B's row (the full flag, the reference of the preregistered controls; sufficiency uses the leave-one-word-out flag, Table 38), and for controls (the text gives them relative to the flag). (b) ρ_K: key read left when one direction per layer is removed at the option rows. (c) ID_K, ID_V when asked which option is or is not mentioned, and in IOI. Scale: Figure 11.

swer's key read of the listed names (restoring their keys and values for the answer row alone removes 0.897 / 0.839 of it, their keys alone 0.431 / 0.488, their values alone 0.211 / 0.263)^{R}. The prediction^{R} that the ten answer-side heads ranked as the flag's top readers in lists apply it failed: a different (or a distributed) reader applies the negative read (the scorer's "different reader"), though even in lists these ten carry only part of hop 2. That the belief readers carry the IOI read^{R} was met at Qwen2.5-7B and is not evaluable at Mistral-7B (a gate missed). The flag is written in sentences as in lists at 1.5B–7B, but four further predictions^{R} did not hold: that it carries the size of the sentence read across scale and is read by value at 1.5B (the preregistered outcome: "Mixed."; Appendix E.5), that the initial-state flag carries binding separably from order, and that a key naming no candidate acts as removing the base value's flag.

5. Where an intervention appears to act

Even steering vectors close to the natural edit depart from the natural channel ratio; the format law describes the model's own written value and does not predict how an edit is attributed. Which cache channel an intervention appears to act through depends on the prompt (discovery sample; exploratory here) and on the edit. The motivating case is the released remap of Anonymous (2026), a rank-16 distributed alignment search subspace (Geiger et al., 2023) over the move-event span at block 4, credited under OPTIONS-AFTER partly to the writing token's keys (Appendix G): at Qwen2.5-72B its effect barely changes with the format, yet its keys carry 0.89 of it with lettered options and 0.05 without a later mention, its values the reverse (discovery sample; exploratory at this model, preregistered for a refit at 24B).

The law and its test. The preregistered law extends the format law to edits: an edit written at the writing token at depth l is read through key and value in the model's own ratio at that depth and format. With ψ_C the edit's identity effect through channel C as a share of the natural write's, from block $l{+}1$ on, and $\lambda = \log(\psi_K/\psi_V)$, the law holds in a cell when the 90 % interval of λ lies inside $\pm\log 1.25$; writing the source run's residual there is our equivalence control (Appendix F.1). We tested the edit families of Figure 4 and *non-lexical* steering vectors in up to three models, at blocks 3 to 15, in four formats (Appendix F.2).

Steering vectors. In the preregistered core test the steering vectors' departures are small but graded ($\bar{\lambda} = -0.061$ to -0.089 at Qwen2.5-7B and Mistral-7B): the format law does not transfer to edits as a channel-ratio law, and the prediction did not hold (Figure 4a). Every pooled $\bar{\lambda}$ and its interval lie inside the margin but below zero; the 90 % intervals of 7 of the 30 single statistics, all at block 7 or deeper, cross the margin, though none lies wholly outside it; at Llama-3.1-8B the vectors from unrelated sentences failed too, one statistic wholly outside. At the shallowest depth every statistic is equivalent ($|\lambda| \leq 0.053$; exploratory); pooled over depths, in the three models the keys under-read the steering vectors by 4.5–12 % ($\bar{\lambda}$ -0.046 to -0.129). The failing statistics sit in cells with a small natural key read (2.63–4.76 nats), and from block 7 on the under-read appears also where the key read is large (post hoc; Appendix F.3). The vectors' key share still follows the natural one across formats (exploratory; Figure 4b).

Edits far from the natural write. Edits that change the value through a learned remap are read mainly through the value ($\bar{\lambda} = -1.574$ / -1.707): the key channel does not read them as it reads the written word, so the prediction^{R} that the remap keeps the model's ratio did not hold. The pooled values rest on statistics near the floor of ψ at block 7; at block 3 alone $\bar{\lambda}$ is -1.03 and -0.67 (post hoc). Its key share still follows the format, below the natural one

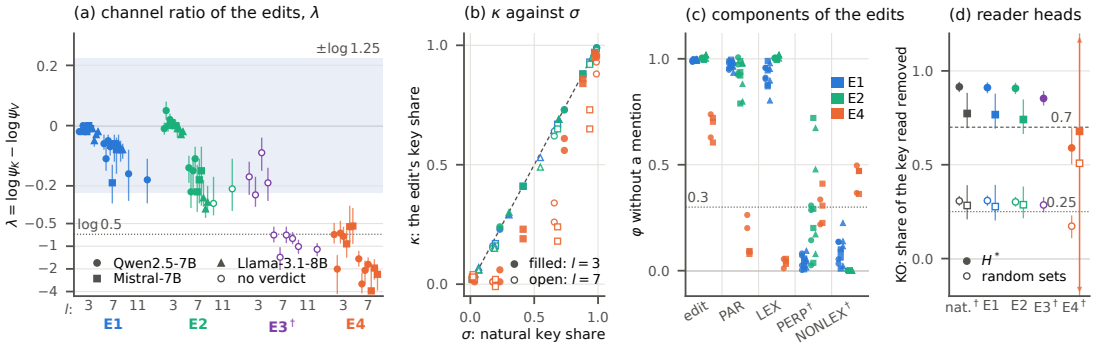


Figure 4. **The preregistered channel-ratio law failed.** Qwen2.5-7B, Mistral-7B and Llama-3.1-8B, 80 stories. (a) λ per edit family (E1, E2: steering vectors (Turner et al., 2023; Panickssery et al., 2023) from held-out stories or unrelated sentences; E3: public sparse-autoencoder features (Bussmann et al., 2024; andyrdt, 2025); E4: a remap trained at the writing token with the released code of Anonymous (2026)), depth and format; 90 % intervals; band $\pm \log 1.25$; axis linear within ± 0.25, logarithmic beyond; hollow: no verdict (gate failed; recomputed). (b) κ, the edit's key share, against σ, the natural one (exploratory). (c) Identity effect without a mention of each edit's component along the natural write (PAR), the English lexical span (LEX) and orthogonal to each. (d) Share of the key read removed by blocking the reader heads; open: random head sets, which the natural write also fails under the preregistered 0.25; carets: interval cut at the axis.

(exploratory; Appendix F.3). Unlike the released remap, it writes only the writing token and was fit without a mention (one remap, two seeds, two models). The autoencoder test was not evaluable: its alignment gate measured reconstruction at positions the dictionary's published fit excludes; under either gate tried after the data it would not have held (Appendix F.2). The predictions^{R} that identity-carrying components off the natural write or the lexical span keep the ratio, and that non-lexical vectors mark the token-form boundary, did not hold: the components are read mainly through the value (Figure 4c), and the non-lexical vectors also write far from the natural state, so the boundary was not isolated; we claim neither that the token-form account holds for edits nor that it is contradicted. The reader heads read the steering vectors like the natural value, but that prediction failed through a random-set control the natural value also fails (post hoc reading), and was not evaluable for the autoencoder and remap (Figure 4d).

Post hoc: distance from the natural write. After the data we related the departures to ν, the relative distance of an edit's cached keys and values from the natural ones ($\nu = 0$ for the natural write): of the λ statistics of identity-carrying edits, 11 of 11 with $\nu < 0.1$ are equivalent, 24 of 33 with ν between 0.1 and 0.3, and none of the 63 beyond, 52 of which have intervals wholly outside the margin (Spearman -0.75 [-0.88, -0.55]; Appendix F.4). The bands were chosen after the data, ν is collinear with the key cosine, and the pattern is largely between families. ν is a diagnostic to report, not an explanation (distances at the intervention site can miss unnatural pathways; Liu & Chen, 2026). A hypothesis for a future test ties this to Section 4: if the key route selects through a thresholded match (half a flag barely moves the answer) while the value read decays slowly with distance, an edit far from the natural write loses its key read first.

The binding swap of Prakash et al. (2026). At block 28 of Qwen2.5-14B an identity edit is copy-read like the natural value ($\kappa \leq 0.30$), while the binding swap there is key-read; key-flat reading is not a property of depth for identity edits, though this does not separate kind from depth, because the natural read there is itself copy-only and the swap changes no answer in two of the three formats (Appendix H.2). Binding forms only after the identity key route closes in every model screened, so a binding swap cannot be compared with an identity edit at a depth where the key route is open^{R}; the four earlier predictions on this swap, which failed, stand, and a window prediction was not evaluable.

6. Related Work

Options, belief stories and heads. Specific heads select among listed options (Lieberum et al., 2023; Wiegreffe et al., 2024), select-and-copy heads by query–key alignment (Tulchinskii et al., 2024); our second hop, in which the answer selects the flagged option rows by key, resembles this selection. Closest to our first hop, shallower heads write into option rows a one-dimensional option-routing feature that redirects the attention of decision heads (Sun et al., 2026); our flag plays this role, written by a duplicate-token match to the token that wrote the value, with a mean direction that is the same whichever value was written. On belief stories, belief and reality share a value slot read out under a router at the query (Steele et al., 2026a), and look-

Table 2. **How the predictions of the final preregistration fared.** Held / with a verdict (expected: sum of the recorded probabilities); risky: probability below 0.8; not evaluable (n.e.): not counted. Under the common part's evaluability rule one more risky prediction fails (10 of 28, expected 12.65). Of the 10 risky predictions that gates made not evaluable, 6 would not have held, 3 would have and one cannot be resolved, giving 13 of 36 (expected 16.70; post hoc; Appendix A.2).

Evidence (claim)	Replication	Extrapolation	Risky	n.e.
Fresh stories (1)	3/3 (2.70)	7/7 (5.80)	4/5 (2.55)	1
Passages (1)	1/1 (0.90)	3/3 (2.50)	3/9 (4.20)	7
Mechanism (2)	2/2 (1.80)	5/5 (4.20)	2/9 (3.60)	1
Interventions (3)	–	1/4 (3.30)	1/4 (1.70)	3
All	6/6 (5.40)	16/19 (15.80)	10/27 (12.05)	12
Measurement validity (fresh stories)			4/5 (2.60)	0

backs built on ordering IDs track beliefs (Prakash et al., 2026); in our terms both describe a copy route from the token that states the value, and later mentions add a second route, addressed at the mention.

Attributing interventions. Interchange interventions (Geiger et al., 2021; 2023) may act through pathways the model does not use naturally (Makelov et al., 2023; Liu & Chen, 2026; contested by Wu et al., 2024); steering vectors (Turner et al., 2023; Panickssery et al., 2023) need not reproduce natural activations (Shih et al., 2026) and are brittle to prompt changes (Tan et al., 2024). Prompt format changes what is measured (Sclar et al., 2023; Opiełka et al., 2026), and the answer encoding under which a steering gain is measured changes what it identifies (Gao et al., 2026); we find the same for the cache channel through which an edit appears to act. Cache-level accounts place routing or binding information in keys (Ma et al., 2025; Oh & Demberg, 2026) or find an operation's cache read mainly as an address (Wu & Shomali, 2026); in our stories the token that wrote a value is read mainly through its key only when a later token re-mentions the candidates.

7. Discussion

What to do differently. (i) State where the candidates are mentioned relative to the token that writes a value, and test channel claims with and without a later mention. (ii) Check localisation methods (Mueller et al., 2025) in the format of the behaviour they should explain. (iii) Do not infer an intervention's channel from the model's own read: even near-natural steering vectors were under-read by the keys except at the shallowest depth (exploratory), and the remap we trained was read mainly through its value. Measure its channels with key-only and value-only exchanges, and report its distance from the natural state (post hoc here).

How the predictions fared. The risky predictions (Table 2) held 10 of 27 against 12.05 expected, within chance under the recorded priors (probability 0.27 of so few or fewer if the predictions are independent; post hoc). Every extrapolation with a verdict on claims 1 and 2 held (probability 0.06 that all would); those on the intervention law held 1 of 4 against 3.30 expected (probability 0.018 if independent, which overstates the evidence: two of them test the same law, and one failed through its control), the one class whose priors were too high. In entries G to I, 15 of 39 predictions with a verdict failed, and in entries A to F a transfer law for interventions and a key-share threshold failed (Appendix A.5); every failure is listed in Appendix A.3.

Limitations. (i) Our passages are counterfactual SQuAD items with added options and the answer named once, scored at the first answer token. Passages in which the answer recurs later, with the counterfactual written at every mention, are the test we most lack. (ii) The confirmation uses one story template. (iii) A list before the writing token gives a small negative key read (to -2.0 nats at Qwen2.5-72B), as in IOI, where in-sentence re-mentions are read negatively too: perhaps a suppression of a word seen twice, by heads we did not identify. (iv) The read is not in the lowest layers' keys (at Qwen2.5-1.5B it lies in layers 4–15 of 28; exploratory), but with the clamp started at a third of the depth much of it disappears (on passages with options after, 0.014–0.516 remains; exploratory). (v) The readers' duplicate-token core depends on how many heads are counted; ablation and knockout disagree on whether the copy route takes over; how the answer prefers the moved-to location over the initial one, also mentioned and matched, is unresolved. (vi) The mechanism was tested in two 7B models (and at 1.5B and 3B), the interventions in three 7–8B models, all in BF16; eager attention, which Gemma-2 requires, lowers the key read by 12–17 % in one model (exploratory), though no Gemma-2-9B verdict lies within 0.03 of its threshold. (vii) Values span one to four tokens; prompts are English; all models are instruction-tuned and prompted in their chat templates (in the one base model we ran, on IOI, a later list opened no key read, a weak null; exploratory). (viii) We re-tested no third-party channel claim with and without a later mention.

Conclusion. Across seven model families on fresh stories, and for an appended options list on counterfactual reading-comprehension passages in every evaluable model, later mentions decide whether a written value is looked up through its key or copied from its value; on fresh stories the decision reaches generated answers. In two 7B models the lookup is a duplicate-token match that writes a flag the answer reads by key, sufficient but neither specific nor the only carrier. Where an intervention appears to act depends on the prompt and on the edit; a preregistered law predicting it from the model's own read failed.

Impact Statement

This work aims to make explanations of language-model internals more reliable, by showing that the cache channel that appears to carry a value or an intervention depends on the prompt under which it is measured. More reliable component-level explanations could help researchers audit and debug models, and could reduce the risk that a mechanism found under one evaluation format is trusted to explain behaviour under another. Our results also caution against over-reading interpretability findings: an attribution validated under one prompt format, or inferred from the model's own read, should not on its own certify behaviour under others. Methods for locating and editing what a model reads from its context could also be used to steer models in unwanted ways; our experiments use public models, templated stories, public reading-comprehension passages, edits built from public material, a released remap and a released state-token patch.

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Reader's guide to the appendix

The appendix is long because every preregistered line, gate and post hoc reading is reported. Table 3 gives, for each claim of the main text, the two or three places that carry its evidence; Appendix A.7 maps every main-text sentence to its prediction and verdict, and Appendix A.3 lists every prediction that did not hold or could not be evaluated.

Table 3. **Where to find the evidence for each main-text claim.**

Main-text claim	Key appendix material
Preregistration, risk classes, tallies, failures	Appendix A.2 (Tables 4 and 5), Appendix A.3, Appendix A.6
Claim 1, fresh stories and generated answers	Appendix D.2 (Table 25), Appendix D.4 (Figure 7), Appendix B.2
Claim 1, the measurement	Appendix D.3 (Tables 26 and 27), Appendix C.2
Claim 1, reading-comprehension passages	Appendix D.5 (Tables 28 and 29)
Claim 1, the discovery sample	Appendix D.1 (Tables 17 and 18)
Claim 2, reader heads and the second hop	Appendix E.2 (Tables 36 and 37), Appendix E.1
Claim 2, the flag, the sign and scale	Appendix E.3 (Table 38), Appendix E.4, Appendix E.5 (Figure 11)
Claim 3, the law and its edits	Appendix F.1, Appendix F.2, Appendix F.3 (Table 40), Appendix F.4 (Figure 12)
Claim 3, the released remap and the binding swap	Appendix G, Appendix H
Implementation, exactness, models and compute	Appendix C, Appendix J

A. Preregistrations and claim status

A.1. How the predictions were made

We made ten preregistrations, A–J. The first nine (A–I) produced the discovery sample of Appendix D.1; the tenth (J), in four parts, is the final round reported in the main text. Each was committed to version control before its run, and each scoring script was committed before its outputs were inspected. H was finalised before any output of G was inspected (item 10 of Appendix A.5), preregistration I before any stage-7 run (item 11), and every part of J before any output of its run (item 12). Tables 4 and 8 to 14 give every prediction and gate with its outcome as recorded in the preregistration log of the repository; a NOT MET is listed as plainly as a MET. The anonymised code release adds a history file that lists, without author information, every commit that changed the log or a scoring script, with its time and diff, and every commit that changed the results, experiment code or GPU scripts, so that the order of preregistration, run and scoring can be checked; each result file also records the commit of the code that produced it. In this anonymous version, commits of our repository are named by neutral labels (for example `stage1-run`) in place of their hashes; third-party commits and model revisions keep theirs. We keep the wording of the log, with two changes: format names are replaced by ours, and long entries are shortened without changing their content. In the log, the "critical token" or "state token" is the writing token, identity(K) is ID_K, and l_0 is ℓ_0; in preregistration H, "state token" keeps the meaning of Prakash et al. (2026), the token that names a state (our writing token in their stories). The log rounds independently of the tables in Appendix D.1, so a few values and intervals differ in the last digit (for example, the interval of the Qwen2.5-32B key share). The tables of G, H, I and J are generated from the score files; where a value was recomputed from the raw files for the log, we print the recomputed value and say so.

The final entry (J). Entry J has a common part and four parts: passages (A), fresh stories (B), interventions (C) and mechanism (D). The common part fixed, for every part, the scripts, the model files and their hashes, the statistics, the risk classes, the codes, the populations and the disclosure of pilots. Each part was finalised in one commit together with its scorer, the shared Holm helper, its FP32 exactness tests and its populations, before any of its outputs existed; each GPU script refused to run on a draft entry, a modified tree or code that differed from the finalising commit, and ran its part's FP32 tests before loading any model. Every pilot that ran a model ran it on the CPU in FP32 at 1.5B parameters or fewer and is listed with its numbers in the entry; no output of a final-round population was seen before its run. After each run, re-scoring the archive with the committed scorer reproduced the score file byte for byte, and four independent checks (five for part C) were made before the outcome was recorded: a recomputation from the raw files that imports no scorer code, a verdict audit from the printed numbers and the entry's rules, an anomaly hunt in the run data, and an integrity check of the manifest, the model files and the timeline (Appendix C.2). The outcome records list every discrepancy these checks found; all were cosmetic and none changed a verdict.

Risk classes and priors. Before any output of the final round, the entry recorded for every prediction ("line") a prior, the probability that it holds given that it is evaluable, with a one-line justification from data in hand, and a class that follows the prior: *replication* (L), implied by data in hand on the same models and material, prior at least 0.9; *extrapolation* (M), prior at least 0.8; *risky* (R), prior below 0.8, including every line on which no data in hand bears. Each line is an *account line*, which tests the paper's account, or a *measurement-validity line*, which tests a measurement (all five of these are in part B and all are risky); the two kinds are tallied apart, and gates are not tallied. A line is met, not met or not evaluable; evaluability is decided only by gates and by rows the tested manipulation does not change, and a quantity that the tested arm itself makes undefined counts against the line. A line over several models is an intersection-union test: not met as soon as one evaluable model fails it. For each class the scorers print the met count against the expected count (the sum of the priors of the lines with a verdict) and the Brier score; the abstract reports the risky account lines met against their expected count. For every primary line the entry pre-wrote the paper's text for both outcomes (abstract clause, title, Table 1 row, figure); the main text uses these texts, and Appendix A.6 lists every place where none fitted.

Line codes. Lines are named J-A1, J-B3-N4, J-C-BOUND, ... by part. In part B, the suffix P4f marks a line on the fresh stories in the four discovery-sample models (Qwen2.5-7B, Qwen2.5-14B, Mistral-7B, OLMo-2-7B), which must hold in every evaluable model at 95 %, and N4 a line in the four new families (Llama-3.1-8B, Gemma-2-9B, Phi-4, Falcon3-7B), which must hold in at least three of four at 98.75 % (Bonferroni over four models at one-sided 2.5 %). In part A a line must hold in every evaluable model, with at least three evaluable models including a new family; in parts C and D in every evaluable model or combination, with the minimum numbers the entry states. Gates are coded J-A-G1, ...; G-codes without a part letter (G1–G8) name rules of the common part and, in Appendix A.5, the predictions of entry G.

A.2. Entry J: every prediction

The tally. The entry has 69 counted lines (64 account lines and 5 measurement-validity lines): 36 met, 21 not met and 12 not evaluable (Tables 4 and 5). Replications held 6 of 6 (expected 5.40) and extrapolations 16 of 19 (expected 15.80): in aggregate about as often as their priors implied, but not uniformly. Those of parts A, B and D all held (15 of 15, expected 12.50; the probability that all hold under the recorded priors is 0.06), so these priors were if anything too low; those of part C, the intervention law, held 1 of 4 against 3.30 expected (probability of so few or fewer 0.018 if the four lines are independent; Brier 0.506), the one class whose priors were too high. The independence assumption overstates this evidence: J-C1 and J-C2 test the same law on overlapping steering-vector families, and J-C-READa failed through a random-set control that the natural value also fails. Risky account lines held 10 of 27 with a verdict, against 12.05 expected: 3 of 9 on passages (expected 4.20), 4 of 5 on fresh stories (2.55), 1 of 4 for the interventions (1.70) and 2 of 9 for the mechanism (3.60). Under the recorded priors, with the lines treated as independent, so few or fewer would occur with probability 0.27: the shortfall is within chance. The per-part pattern is descriptive (tail probabilities 0.32 for passages, 0.43 for the interventions and 0.22 for the mechanism, and 0.20 for so many or more on fresh stories). Measurement-validity lines held 4 of 5 (expected 2.60). The tail probabilities are post hoc: they were not part of the entry's reporting plan.

The risky lines that gates made not evaluable (post hoc). Not-evaluable lines are left out of the tallies, but the gates did not remove lines at random. Of the 10 risky account lines without a verdict, the outcome records say how 9 would have come out had the gate passed or been read otherwise: J-A6d would not have been met had Llama's options-after cell passed its BF16 gate by one more item; with Qwen's heads gate waived, J-A-HA1 and J-A-HA2 would have been met and J-A-HA3a and J-A-HA3b not; J-B-SMALL would not have been met had Gemma-2-2B passed competence; J-C3 would not have been met under either post hoc dictionary gate; J-C-READb would not have been met under the common part's rule; and J-D8-IOI would have been met had Mistral's cell passed its transfer gate. J-C-WIN cannot be resolved (no window). Resolved in this way, 6 would not have held and 3 would have, and the risky tally would read 13 of 36 against 16.70 expected (probability of so few or fewer 0.14). This reading is post hoc and does not replace the official tally.

The tally under the common part's reading of J-C-READb. The part's own text leaves J-C-READb not evaluable (the remap's key read is below the line's 2-nat floor); under the common part's evaluability rule, by which a quantity that the tested arm makes undefined counts against the line, it would be not met. The risky tally would then read 10 of 28 (expected 12.65); for part C alone 1 of 5 (expected 2.30). We use the official verdict and state the conflict (Appendix A.6).

Holm sensitivity, per part. Each scorer applied Holm's step-down at a familywise one-sided 0.025 to the interval components of its risky account lines; no verdict uses it. Passages: one decision changes (J-A5's component at Qwen2.5-7B, p = 0.0099 against 0.0031) and no verdict; J-A6b's Mistral component passes by a relative margin within the Monte Carlo error, so J-A6b should not be described as robust to Holm. Fresh stories: no decision and no verdict changes; Phi-4's component of J-B5b-N4 passes narrowly. Interventions: 4 decisions change (of 58 components) and no verdict. Mechanism: no decision and no verdict changes.

A.3. Every prediction that did not hold or could not be evaluated

Table 6 lists every line of the final round that was not met (21) or not evaluable (12), plus the derived line J-A3, sorted by what happened, with the main-text section that names it in words.

A.4. The pre-written Table 1 rows

Each part of the entry pre-wrote a "Table 1 row" for each primary line, with its text for both outcomes. The main text's Table 1 is the question-by-question table that the rewrite rule of the entry asked for, so the pre-written rows are given here, each with its numbers and verdict (Table 7); the row texts are those of the outcome that occurred.

A.5. Entries A–I: the discovery sample

Entries A–I produced the results of the discovery sample (Appendices D.1, E.1, E.2, G and H); Appendix A.7 maps each main-text claim that rests on them to its prediction. Their tables are unchanged from the earlier report. The common part of entry J promised post hoc risk labels for the predictions of entries A–I; we did not compute them, because those entries' records of what was seen before finalisation are free text that a script cannot classify without judgment; their verdicts

are in Table 8 to Table 14. Of the 41 predictions of G, H and I, 15 failed (H6 not evaluable, H12 not run; Table 15); a transfer law from clamping-estimated value completeness (C1, C2) failed at Mistral-Small-24B and held only marginally at Qwen2.5-72B, and $s_K \geq 0.80$ under OPTIONS-AFTER (C4) held only at Qwen2.5-32B of the three largest models.

Preregistration G: status of the preregistered alternatives. (a) H_{track}, context-selective heads at small scale: not declared (H_{diss} holds and G4b is met). (b) H_{replaced}: not met; neither it nor $H_{\text{redundant}}$ holds, so the preregistered outcome is a partial takeover. (c) Graded list-likeness: not supported overall (the arms that name B with two other candidates fall below NO-MENTION in both families, and lists read fully with two named candidates), but two features match it in part: in sentences at Qwen2.5-7B, r_2 is close to $2/6$, and in S2 the rows after the question carry 0.46 / 0.49 / 0.38 of the splice effect. (d) Concept-level lookup: not established; the concept pattern holds on a_v in two models and for one variant (FRMIX at the two Qwen models), not in the three the decision table needs. (e) An inhibitory key read or an additive mixture at 7B: neither at Qwen2.5-7B; Mistral-7B not evaluable. The GPT-2 small parts of G18, G21c and G22 replicate a disclosed CPU pilot on fresh cores and are not counted as confirmatory. For G16 the scorer prints the label "near-token-level"; the entry says only "graded", the label does not describe FRMIX at the Qwen models, and we do not adopt it.

Disclosures for preregistration G.

- *Re-finalisation.* The first GPU attempt stopped in a unit test before any model was loaded: one check compared batch rows with equal inputs at a tolerance of 10^{-5}, which batched matrix products on a many-core CPU can exceed. The entry was re-finalised with test-only changes (that check uses the 10^{-4} of the other batched checks, and the tests run on at most 16 CPU threads). No experiment, stimulus, scorer or threshold changed.
- *Disclosed pilots.* Before the draft, two CPU pilots of part (a) at Qwen2.5-0.5B already showed the H_{diss} pattern, so G2 is a confirmatory replication in sibling models. After the draft, a full GPT-2 small IOI run on CPU was scored while the code was built; the IOI run of the stage therefore uses fresh cores, and its GPT-2 small parts are replications.
- *Points the entry left open (no verdict depends on them).* In part (a), the emitted casing is the one with the larger clean-B candidate mass (it matters only at Qwen2.5-1.5B under SENTENCE-AFTER, where both casings pass the gate), and F is computed on the evaluation half while the key effects use all 150 cores. In part (b), the behaviour clause of H_{replaced} is applied per model in both list formats, the q_V thresholds are applied to point estimates, and G6's value criterion uses cell means, which equal the paired means.
- *Eager against sdpa attention.* The key effects of Gate a come from the eager run of part (a). At Qwen2.5-7B they are lower than the sdpa values on the same cores, both on the lowercase answer ids (ID_K 4.50 against 5.52 under SENTENCE-AFTER, 19.39 against 21.16 under OPTIONS-AFTER; under SENTENCE-AFTER the casing the model emits gives 4.73, as in Table 32); the other models agree closely. The main text quotes sdpa values; the gate is far from its bound.
- *IOI at the 7B pair rests on one model.* Mistral-7B failed the competence gate in AFTER, BEFORE and INLINE_BEFORE, and its not-evaluable cells count as not met, so the not-met verdicts of G19–G21 reflect Qwen2.5-7B-Instruct together with that gate failure.
- *GPT-2 attention probe.* The exploratory GPT-2 small name-mover and duplicate-head probe failed on the GPU (a device mismatch) and was re-run afterwards on CPU with the same arguments, model revision, cores and code, in FP32 with eager attention, under a different torch and Python version. No gate or prediction uses it.
- *Combined lines.* The scorer prints a prediction as met only when every sub-verdict is met; the counts above use these combined lines. The sub-verdicts are: G10 lists met, sentences not met; G13 THE met, MODIF not met; G14 PLURAL, SYN, FRMIX and DEMIX met, TITLE and UPPER not met.
- *Checks.* The reproduction gate checks determinism: the reproduced cells are bitwise identical to stage 3b (exploratory, recomputed). Five recomputations from the raw files, none importing the scorer, reproduce every gate value, point estimate and verdict.

Disclosures for preregistration H.

- *Timeline.* H was finalised before any output of G was inspected (item 10 below). Between the finalising commit and the commit of the results, no scorer, experiment, test or library file changed, and the text of entry H is byte-identical.

- *Gate a2 and the BF16 batch-shape offset.* Gate a2 compares rows of a batched pass with single passes. Its failure at Qwen2.5-7B under SENTENCE-AFTER ($|\text{none}-\text{clean}|$ = 0.770 against 0.500) is a batch-shape offset, not a splice error (exploratory, recomputed): rows with equal inputs agree within each batch, the reference rows of batches of different sizes differ under SENTENCE-AFTER but coincide under OPTIONS-AFTER, and the hop exactness row is 0.0000 in every story. Every scored quantity is a difference against an in-batch reference row; at Mistral-7B the gate values stay at or below 0.224. The rule still ties H5's SENTENCE-AFTER line and H6 to the gate, and both stay not evaluable.
- *Narrow pass.* Qwen2.5-7B passes Gate a2 under OPTIONS-AFTER only narrowly: $|\text{all}_T-\text{full}|$ = 0.669 against a bound of 0.719, which exceeds the 0.5-nat floor only because it scales with $\bar{d}_{\text{full}}$ (35.969 nats). Its bootstrap interval, [0.528, 0.821] (exploratory, recomputed with the scorer's resampling; the log reports an interval from an independent resampling), crosses the bound. The verdict uses the point estimate, as the entry specifies.
- *Null probabilities.* The entry quotes hypergeometric probabilities at the thresholds (overlap ≥ 3 of the top 10 for H4: 1.7×10^{-4} and 7.8×10^{-5}; overlap ≥ 10 of the top 20 for H6: 1.3×10^{-12} and 9.3×10^{-14}). The scorer prints the probability of the observed overlap or more: 1.4×10^{-10} and 6.7×10^{-9} (H4; observed 6 and 5 of the top 10), 9.2×10^{-19} and 9.9×10^{-23} (H6). Both are correct; they differ in conditioning.
- *Floors and pooling.* The H3 (d) floors of 2.7 and 2.8 nats are rounded from the stage-1 summaries, far below the observed rises. H4 (ii) pools T_{dup} over (story, word) pairs as a ratio of means, within the entry's wording; per-story averaging gives 0.468 and 0.430 against 0.459 and 0.433 (exploratory, recomputed). H5's paired condition is checked as the lower bound of the paired difference.
- *No BOS token at Qwen2.5-7B.* The duplicate-score sequences of Qwen2.5-7B have no BOS token, so position 0 is both the attention sink and the first repeated query's earlier copy. This can raise a head's D by at most $1/30$, which cannot move the median below 0.2. Mistral-7B's sequences have a BOS token.
- *Evaluability from the* NO-MENTION *sweep.* As the entry specifies, a format is evaluable in part (b) if the edit reproduces on the NO-MENTION sweep. BIND@28 and ID@28 are therefore evaluable under QNAMES and OPTIONS-AFTER, although neither flips an answer there; those cells decompose a shift of the logit margin (Appendix H.1).
- *Statistics, onsets and pairs.* κ (one-sided) and s_{ID} (symmetric) are different statistics by design; $\ell^*_{\text{ID}} = 0$ comes from the tie rule on a flat sweep; $\ell^* = 28$ rests on one pair; and the overlap of 70 of 80 pairs with the validation split of Prakash et al. (2026) is approximate, because it is computed with our filter, not theirs. Appendix H.1 gives the details and the depth confound of the identity-against-binding contrast (Gate b3).
- *FP32 re-check.* The command of the exploratory FP32 re-check passes the dtype option twice (BF16, then FP32); the last value applies, and both output files record FP32. The scorer skips these files by design, so their values are recomputed.
- *Disclosed pilots.* At Qwen2.5-0.5B (FP32), the top heads by a_3 were duplicate-token heads, consistent with H4 at 7B. The same pilot showed no head-level trade-off and a weak answer-only hop on two stories; at 7B, H3 (d) and H5 are met, unlike the pilot. The release of Prakash et al. (2026) gives per-layer IIA at Qwen2.5-14B that predicted ℓ^* = 27–28, and $\ell^* = 28$, so the replication table compares against material seen before the entry was finalised: it is a reproduction check, not an independent prediction.
- *H12 not run.* By a rule agreed before the run, for reasons of budget, the optional Llama-3-70B run was to be made only if Qwen2.5-14B failed to reproduce the edit (Gate b1). Gate b1 passed (IIA 0.993), and the run was not made. It can still be scored against the fixed 14B verdicts; with none of H7, H9 and H10 met, it would test only Gate b1, H8 and H11 Part 1.
- *Cosmetic.* For the one cell that fails the κ rule (ID@28 LETTERS-AFTER), the scorer does not print the fraction of dropped resamples, 55.0% (exploratory, recomputed); the cell enters no verdict.
- *Checks.* Four recomputations from the raw files, none importing the scorer, reproduce every gate value, point estimate and verdict, and the committed scorer reproduces the score file byte for byte.

Disclosures for preregistration I.

- *Timeline.* The entry was written after preregistrations A–H were scored and finalised at `stage7-run` with its scorer, code and GPU script. Before that only unit tests, plumbing runs at Qwen2.5-0.5B with random bases and checks on a

tiny random-weight Mistral had run; no head-level output of any model at 24B had been seen. Between `stage7-run` and the results commit only result files changed.

- *First attempt.* At `stage7-run` the script first stopped about a minute in, before the unit tests and before any model, because the release of Anonymous (2026) could not be downloaded on that host. The run, a second session on the same machine, used a copy of the same release that passed the run's checks against the hashes pinned in stage 4 before any test or model.
- *The curve point $k = 128$ was not measured.* The code built every head curve from the stored H^*, which has 64 heads, so its "top 128" rows repeat H^*. No gate or verdict uses $k = 128$ ($k^* = 64$, and 80% is reached by 64 heads or fewer), and we report no value for it.
- *Exact reproduction.* Gate I-G1 found the family run equal to stage 3b (largest difference 0.000; every candidate row equal, exploratory, recomputed), although stage 3b ran on transformers 5.9.0 over two GPUs and stage 7 on 5.18.0 on one A100 80GB PCIe (stages 1–6 used SXM4 boards).
- *The expected key route outside the option rows is absent.* The entry expected about 0.14 and 0.10 of M's key-only effect (OPTIONS-AFTER, LETTERS-AFTER) to be read outside the option rows, as under NO-MENTION; with every head at the option rows blinded, almost none remains, so ratios measured against NO-MENTION exceed 1 in places (for example under N(H^*), where M's key share falls below its NO-MENTION level; Tables 14 and 43). I4's lower bounds are unaffected.
- *I-G4 not met.* Under A^+, which also ablates the 32 heads of block 5 at the option rows, $t = 0.556$ (against 0.6), against 0.565 under A. So I5 and I6 are not read as a value takeover (the scorer prints "readable as value takeover: no"); the gate's stated rule, for I5 and I6 both met, does not arise. Block 5 alone changes nothing (Table 43).
- *The power check.* Its no-takeover scenario gave $t \approx 0.57$ with no value gain; the run gave the same t with a value gain of 2.388 nats, because the key lost more than simulated (0.90 of its key-only effect, against 0.81) and the non-additive part of M's effect, which the simulation held fixed, fell from 3.32 to 1.51 nats (exploratory, recomputed). With that part held fixed, the observed key loss and value gain give $t = 0.698$ (exploratory, recomputed), still below the preregistered 0.75.
- *Argmax rates* are over the 288 (core, fit) items for M and P and over the 96 cores for the natural rows.
- *The knockout floor is met trivially.* The null knockout's second pass reuses the unblocked mask and reproduces the unblocked rows bit for bit, so I-G1 (c) does not measure any numeric cost of the changed-mask path; it gates only the reported N lines.
- *Checks.* Four recomputations from the raw files, none importing the scorer, reproduce every gate value, point estimate and verdict, and the committed scorer reproduces the score file off the machine.

History of the claim, including post hoc steps. The account of when keys carry the value was revised twice after seeing data, and its intervention half was narrowed after a preregistered failure (item 10) and qualified at the head level after another (item 11). We list every step.

1. *Starting hypothesis.* Before the first format factorial, our hypothesis was that any re-mention of the candidates after the writing token opens the key channel.
2. *Post hoc narrowing at 1.5B.* In the first run at Qwen2.5-1.5B (seed 0, $n = 40$), SENTENCE-AFTER showed no key effect. We therefore narrowed the claim to "listed answer options after the writing token open the key channel". This narrowing was made after seeing the data, and an internal go/no-go gate written before that run was not met as written.
3. *Preregistration A* tested the narrowed claim on fresh story cores in the same model, and all three predictions were met. This confirms the narrowed claim in one model on new data, not across models.
4. *Preregistration B* tested held-out models. Both directional predictions were met in 5/5 models, but from 7B up SENTENCE-AFTER carries an intermediate amount of key identity, so the "listed options only" account does not generalise. The account given after stage 1, that from 7B up a later mention of the candidates opens the key channel, whether a list or a neutral sentence (a list most strongly), while at 1.5B and 3B only a list does, was revised again, post hoc, after stage 1. It is closer to the starting hypothesis. The residual NO-MENTION identity in the Qwen models from 7B to 32B (up to 1.25 nats) extends just beyond the ± 1-nat margin that A set at 1.5B; in the other models it stays inside. We report it as unpredicted, not as a confirmation.

5. *Preregistration C* added a transfer law derived from the fixed-value data of Anonymous (2026). It failed at Mistral-Small-24B (C1 and C2 not met) and held only marginally at Qwen2.5-72B. C3 was met in both models. The key-share threshold of C4 was met only at Qwen2.5-32B; its LIST-BEFORE control was met in all three models. The reading that the intervention is written into both channels, and that the format decides which one appears to carry it, was formed after seeing these data.

6. *Preregistration D* retested the options-list part of the account on two new tasks (D1, D2) and localised the readers at 7–14B (D3). It contains no test of whether SENTENCE-AFTER opens the key channel; SENTENCE-AFTER on the new tasks was exploratory.

7. *Preregistration E* was written in response to two internal reviews of a draft. It tested the revised account with an instruction-matched 2×2 (E1), including the first confirmatory test that a sentence after the story opens the key channel (E1c, SENTENCE-AFTER − SENTENCE-BEFORE). It also tested the value half of the post hoc reading of C (E4), the role control (E2) and the software environment (E3). All its predictions were met.

8. *Preregistration F* tested the other half of that reading: whether the format in which a remap is read, rather than the one in which it was fit, decides which channel appears to carry it, by refitting the remap without later mentions at Mistral-Small-24B. All gates and predictions were met; a smaller shift of the key share in the two mixed formats, in the direction the alternative predicts, was found post hoc.

9. *Preregistration G* was written in response to an external critique that expected the objection "duplicate-token attention in a new setting". It conceded the first hop: the re-mentions attend to the writing token with a key-matched duplicate-token pattern at every scale, including 1.5B and 3B, whose key identity under SENTENCE-AFTER is about zero (G1–G3; with option lists it is positive), so on the lower-case score the match is present where no key read was seen (the final round qualifies this: the flag is written in sentences at 1.5B and 3B, and at 1.5B the injected flag is read by key; Appendix E.5). The candidate words' attention to the writing token proved necessary for the key read, and the answer's own attention to it unnecessary (G5, G6, G8). A lead we had planned before the run, that the lookup largely replaces the value copy, was not supported (G7 a partial takeover; G11 and G20 not met) and was dropped. Three claims were narrowed: the readers to lists and to sentences naming at least three candidates (G12), the matched form to the word largely up to case (G13, G14), and the tasks to our three templated ones, since IOI did not confirm a later list's key read (G19–G21).

10. *Preregistration H* identified the reader heads and the second hop at two 7B models (H1–H5 met) and applied the exchange to an intervention from another lab. On the binding swap of Prakash et al. (2026), the law that predicts an edit's key share from the natural identity key share (H7), the shared value prediction (H8), the crossover (H9) and the rival (H10) all failed. The identity-edit control was met (H11), but at $\ell^*_{\mathrm{ID}} = 0$ it nearly restates the natural clamp, so the law for identity edits out of sample is untested; the in-sample relation for the released remaps and the refits (F4) is unchanged. H was finalised (commit `stage6-run`, 01:16 UTC) before any output of G was inspected: the stage-5 run had started at 21:25 UTC the day before, its results were first committed at 08:41 UTC, after H, and the stage-6 run started at 10:57 UTC on `stage6-run`, unchanged. The reading that at block 28 the binding swap is key-dominant while the identity edit is value-carried (Appendix H.1) was formed after these data and is post hoc.

11. *Preregistration I* was written in response to an external reading that the two halves of the paper were linked only by inference. At Mistral-Small-24B the reader heads replicated as a gate in three formats (I-G2); the released remap's key effect is read at the option words and through these heads, not through random or active sets (I1–I3); ablating them moved its key share under OPTIONS-AFTER near the NO-MENTION level (I4, a consistency check; under LETTERS-AFTER only to 0.661) and raised what its values carry (I5), but kept only part of its effect on the margin (I6, not met), the outcome the entry called a partial takeover; I-G4 was not met. The prediction that the copy keeps the remap's effect without its readers is therefore not supported; the format-level crossover stands. Qwen2.5-72B was not run. Preregistration I was finalised (`stage7-run`) before any stage-7 run.

12. *Preregistration J* was written in response to an external reading of the previous version, which raised six objections: the stories were templated, the free-form formats were scored on lower-case tokens that carry little of the probability, the novelty against QK/OV and duplicate-token heads was unclear, the intervention lesson rested on one remap, most models were from one family, and the paper relied on an anonymous submission. Its four parts, each finalised with its scorer before any stage-8 run, tested the account on fresh stories in four new families with emitted-form scoring and generated answers (B), on counterfactual reading-comprehension passages (A), on what the reader heads write and what sets the sign of the read (D), and as a channel-ratio law for identity edits obtained independently of the

natural clamp (C). The later-mention account held on fresh stories in all eight models and in generated answers, and on passages for a later options list; behaviour on passages followed the key only with lettered options; the reader heads' write is sufficient and read by key but neither specific nor necessary as one direction; the channel-ratio law failed (outcome (e)). Of the risky account lines, 10 of 27 held against 12.05 expected (Appendix A.2); every failure is in Table 6, and the decisions taken after the data are in Appendix A.6.

A.6. Pre-written texts used and decisions taken after the data

The entry pre-wrote the paper's text for each outcome of each primary line. The main text uses these texts verbatim apart from filled placeholders. Where no pre-written text fitted the outcome, or two rules of the entry conflicted, we took a decision after the data; each is listed here, with the outcome record's wording. In both conflicts between the common part and a part's own text (items 1 and 12) we followed the part's text, and both choices favour our account (the broader title is kept; a not-met line becomes not evaluable); the alternative is stated in each item and, for the tally, in Table 2.

1. The title, and a conflict between two rules of the entry. The common part (G4) says that only the natural-text lines named in Part A can change the title ("A1 and A3 in the design") and that if these are not met or not evaluable the title becomes the templates-only fallback ("… on Templated Stories"). Part A's table names the row J-A1 (with J-A2; J-A3 derived) and, for J-A1 met, says "Title kept", with conditions on J-A3 only in its not-met column. J-A1 is met; J-A3, a derived line that is not counted, is not evaluable (met in both models where it is evaluable). Read literally, the common part's "or NOT EVALUABLE" would give the fallback through J-A3, while Part A's table keeps the title. We keep the title under Part A's table and state the conflict here, as the outcome record requires.

2. The scope of the passage clause. J-A1's pre-written clause says "in four model families including two not used before". This overstates the evaluable set and is not used; the main text names the evaluable models (key read with options after in Gemma-2-9B, Qwen2.5-7B and Mistral-7B, one family new to the paper; value copy in free form in Llama-3.1-8B, Gemma-2-9B and Qwen2.5-7B). Because J-A5 was not met, the main text says "a later options list" for natural text where the pre-written clause said "later mentions".

3. Hybrids on passages. The behavioural row's clause about hybrids applies only if J-A6c alone fails. J-A6a also failed, so the hybrids are reported as an observation, not under that clause; the behavioural sentence is dropped from the abstract, and the cue-conflict panel is reported as a boundary with the decision-token rates.

4. Planned figure panels. The passage figure planned panels for the quantization drops and for the continuation decomposition and ghost markers for the templated values; these are given as Table 30 and in the text of Appendix D.5 (the quantization result is exploratory and makes no claim). The flag figure's planned inset for the hop-2 route (J-D4) is given in the text (Appendix E.3).

5. The split of the generated-sentence line. J-B5b was met in the new families and not in the discovery-sample models (OLMo-2-7B). The entry has no text for a split. The main text uses the outcome record's two sentences: the sentence read changes generated answers in seven of the eight models (flip rate 0.033–0.313 after the sentence, 0.000 before it), and at OLMo-2-7B it "is visible in log-probabilities but not in generated answers (flip rate 1.0 % after, 0 % before)". The main text adds that the flip rate is a small share in most models and that Phi-4's component passes Holm only narrowly (Appendix A.2).

6. The scoring-invariance template. J-B6b's pre-written replacement ("the key share moves by more than 0.05 under emitted forms in {cells}") names the wrong statistic: every s_{ID} difference stayed inside ± 0.05, and what moved was the sentence read relative to the list read. It is replaced, as a disclosed correction of the template, by the record's sentence given in Appendix D.3.

7. "Re-scored" for "published". J-B6c's pre-written sentence says that "every published verdict on the original stories is unchanged". The five verdicts concerned are those of this paper's discovery sample that the check re-scores, so the main text says "every re-scored verdict".

8. The null sentence at Llama-3.1-8B. J-B-NULL-N4 is met in three of the four new families. For a failing family the entry pre-wrote a replacement that withdraws, in that family, the reading that a re-mention opens the key; the record shows that the replacement does not describe Llama-3.1-8B, whose null-sentence ratio equals its no-mention baseline (Appendix D.2). That replacement is not applied, and the record's reading is reported, labelled post hoc.

9. The sign of the read in IOI. J-D-SIGN-IOI is met and J-D8-IOI is not evaluable; the entry has text for both met and both not met only. The joint abstract clause (that the frozen belief reader heads also carry an opposite-sign key read in IOI) is not used. The body reports the sign dissociation as confirmed and the frozen heads' IOI read as met at Qwen2.5-7B and not evaluable at Mistral-7B. The novelty statement (D-13) is reduced accordingly: its item (i), the flag whose mean direction names no candidate (J-D1, J-D-ADDR), is stated with J-D2's failed specificity beside it; item (ii) is reduced to the sign dissociation, which the main text presents as consistent with S-inhibition, not as a novelty; item (iii), the scale-dependent route at 1.5B, is dropped. The main text's novelty paragraph adds two points the entry's statement did not name: that the answer reads the flagged row by key (J-D4, met) and that the same reader heads carry the read under both questions (J-D8-Q, met), with the sign applied after them (D9, exploratory).

10. Mediation across scale. J-D6 and J-D6-ROUTE are not met, and the decision table gives outcome (d), "Mixed.", which the paper reports. The post hoc reading of the natural ratio (Appendix E.5) is reported as exploratory, and the paper draws no conclusion about mediation from it in either direction.

11. The lexical boundary. J-C-BOUND's pre-written sentence for not met is contradicted by the numbers (every E5 $\bar{\lambda}$ lies below E2's at the same cells) and has one slot for six sub-family values. The paper reports the line as not met with the per-sub-family values and states that the boundary was not isolated, because the effective non-lexical edits also write far from the natural state; it claims neither that the token-form account holds for edits nor that it is contradicted.

12. J-C-READb. Under the part's own text the line is not evaluable; under the common part's evaluability rule it would be not met, and the risky tally would read 10 of 28 (expected 12.65). The official verdict is used, as the part's own table was for Part A's title rule, and the conflict is stated here and in Table 2.

13. The interventions' texts. The main text opens its section with the pre-committed outcome (e) text and gives J-C1's numbers with the second clause of J-C1's pre-written sentence; the full sentence is in Table 7. The screen's pre-committed text ends "H7–H10 stand"; the main text says this in words (the four earlier predictions on the swap, which failed, stand), and Appendix H.2 quotes it with the codes. J-C4's size depends on depth, so the main text quotes ψ_K and ψ_V in one clause beside the pre-written $\bar{\lambda}$ sentence, as the record's sensitivity note asks; the per-depth values are in Appendix F.3. After the R1 sentence the main text adds that the shortfall is relative to the model's own ratio, with the remap's key shares (reported in the record, exploratory), so that "mainly through the value" is not read as an absolute share. The screen's sentence names the two screened models.

14. Table numbering. The common part's "Table 3" (the calibration summary) is the main text's Table 2. The pre-written "Table 1 rows" are in Table 7, because the main text's Table 1 is the question-by-question table that the rewrite rule of the entry asked for; each row keeps its verdict.

15. The abstract. The pre-written abstract clauses appear without their numbers, because the abstract keeps to four numbers; the numbers are in the body.

16. "In at most 0 % when they are not". J-B5's pre-written sentence is kept with this wording, and the record's definition of "not" is added: no mention, a list before or a sentence before the story, the arms the line tests (after a sentence the key moves some answers; Appendix D.4).

17. Status marks. The main text marks risky predictions with a superscript ^{R} and says in words where a prediction did not hold or could not be evaluated; earlier drafts printed "(risky; held)" and similar tags. The marks do not change any verdict; Table 16 lists them.

18. The flag's injection. J-D1's pre-written sentence says that the flag is "injected" into the row of an absent option. The line's statistic moves it: it adds the flag at that row and subtracts it at B's row, as the natural key clamp does (Appendix E.3).

The main text keeps the sentence and adds this, with the effect of adding the flag alone (the record's reported dose).

Deviations from the reporting plan. (i) The common part (G4) promised script-computed post hoc risk labels for the predictions of entries A–I; we did not compute them, so the discovery sample's failures are listed (Appendix A.5) but not classed by risk. (ii) The tail probabilities of the class tallies and the resolution of the not-evaluable risky lines (Appendix A.2) were not in the plan; they are post hoc and do not replace the official tallies.

A.7. Crosswalk: main-text claims, predictions and verdicts

Table 16 maps each claim of the main text to the predictions it rests on, their class and verdict, and the status mark and words the main text uses. Unmarked main-text claims were preregistered in the final round and held.

Table 4. **Preregistration J (the final round): every line.** Cl.: class (L replication, M extrapolation, R risky; D derived, not counted). Kind: A account line, V measurement-validity line. Prior: the recorded probability of holding, given evaluability. Met: the models (or, in part C, the combinations, sub-families, model–family pairs, formats or screened models) that meet the line, of those with a verdict. P4, N4: the discovery-sample models and the new families of part B (Appendix A.1). ‡J-A3 is derived from J-A1 and J-A2 and not counted; it is met in both models where it is evaluable. Reasons for every line that is not met or not evaluable: Table 6.

Line	Cl.	Kind	Prior	Met	Verdict
Part A: natural passages (Llama-3.1-8B, Gemma-2-9B, Qwen2.5-7B, Mistral-7B)					
J-A1	M	A	0.80	3/3	**met**
J-A1b	R	A	0.55	3/3	**met**
J-A1c	M	A	0.85	4/4	**met**
J-A2	M	A	0.85	3/3	**met**
J-A3	D	–	0.70	2/2	**not evaluable**‡
J-A4	L	A	0.90	3/3	**met**
J-A5	R	A	0.50	0/2	**not met**
J-A5B	M	A	0.80	2/2	**not evaluable**
J-A6a	R	A	0.45	2/3	**not met**
J-A6b	R	A	0.60	4/4	**met**
J-A6c	R	A	0.45	1/3	**not met**
J-A6d	R	A	0.45	2/2	**not evaluable**
J-A6e	R	A	0.40	3/3	**met**
J-A7	R	A	0.30	0/1	**not met**
J-A7b	R	A	0.35	0/1	**not met**
J-A8	M	A	0.80	2/2	**not evaluable**
J-A8d	R	A	0.60	0/1	**not met**
J-A-HA1	R	A	0.75	1/1	**not evaluable**
J-A-HA2	R	A	0.40	1/1	**not evaluable**
J-A-HA3a	R	A	0.30	1/1	**not evaluable**
J-A-HA3b	R	A	0.65	0/0	**not evaluable**
7 met, 6 not met, 7 not evaluable of 20 counted lines; risky account lines met: 3 of 9 with a verdict (expected 4.20).					
Part B: fresh samples (P4: Qwen2.5-7B, Qwen2.5-14B, Mistral-7B, OLMo-2-7B; N4: Llama-3.1-8B, Gemma-2-9B, Phi-4, Falcon3-7B)					
J-B1-P4f	L	A	0.90	4/4	**met**
J-B1-N4	M	A	0.85	4/4	**met**
J-B2-P4f	L	A	0.90	4/4	**met**
J-B2-N4	M	A	0.85	4/4	**met**
J-B3-P4f	M	A	0.80	4/4	**met**
J-B3-N4	R	A	0.50	4/4	**met**
J-B4-P4f	L	A	0.90	4/4	**met**
J-B4-N4	M	A	0.85	4/4	**met**
J-B5-P4f	M	A	0.85	4/4	**met**
J-B5-N4	M	A	0.80	4/4	**met**
J-B5b-P4f	R	A	0.45	3/4	**not met**
J-B5b-N4	R	A	0.40	4/4	**met**
J-B-NULL-P4f	M	A	0.80	4/4	**met**
J-B-NULL-N4	R	A	0.55	3/4	**met**
J-B-LB-N4	R	A	0.65	4/4	**met**
J-B-SMALL	R	A	0.40	0/0	**not evaluable**
J-B6a	R	V	0.50	4/4	**met**
J-B6b	R	V	0.45	1/3	**not met**
J-B6c	R	V	0.75	1/1	**met**
J-B7	R	V	0.35	8/8	**met**
J-B8	R	V	0.55	1/1	**met**
18 met, 2 not met, 1 not evaluable of 21 lines; risky account lines met: 4 of 5 with a verdict (expected 2.55).					

Line	Cl.	Kind	Prior	Met	Verdict
Part C: independently obtained identity edits (Qwen2.5-7B, Mistral-7B, Llama-3.1-8B; J-C6 at Qwen2.5-14B; the screen at Qwen2.5-7B and Llama-3.1-8B)					
J-C1	M	A	0.80	0/4	**not met**
J-C2	M	A	0.80	1/2	**not met**
J-C3	R	A	0.55	0/0	**not evaluable**
J-C4	R	A	0.35	0/2	**not met**
J-C5	R	A	0.35	0/1	**not met**
J-C-BOUND	R	A	0.50	2/6	**not met**
J-C-READa	M	A	0.85	0/4	**not met**
J-C-READb	R	A	0.60	0/0	**not evaluable**§
J-C6	M	A	0.85	3/3	**met**
J-C-SCREEN	R	A	0.50	2/2	**met**
J-C-WIN	R	A	0.30	0/0	**not evaluable**
2 met, 6 not met, 3 not evaluable of 11 lines; risky account lines met: 1 of 4 with a verdict (expected 1.70); headline: outcome (e). Met: combos (J-C1–J-C4), sub-families (J-C-BOUND), (model, family) pairs (J-C-READ), formats (J-C6) or screened models (J-C-SCREEN, J-C-WIN) met / with a verdict; J-C5 is one pooled test. §Not met under the common part's evaluability rule (KO(H^*) of E4, seed 101, 0.590 and 0.678 $<$ 0.7, recomputed): risky lines met 1 of 5 (expected 2.30).					
Part D: the flag (Qwen2.5-7B, Mistral-7B; J-D6, J-D6a, J-D6-ROUTE at Qwen2.5-1.5B, 3B, 7B)					
J-D1	R	A	0.55	2/2	**met**
J-D2	R	A	0.55	0/2	**not met**
J-D3	R	A	0.50	1/2	**not met**
J-D4	M	A	0.85	2/2	**met**
J-D-ADDR	M	A	0.80	2/2	**met**
J-D-KN	R	A	0.40	0/2	**not met**
J-D5	R	A	0.25	1/2	**not met**
J-D7	M	A	0.85	2/2	**met**
J-D-SIGN-Q	M	A	0.85	2/2	**met**
J-D-SIGN-IOI	L	A	0.90	2/2	**met**
J-D8-Q	M	A	0.85	2/2	**met**
J-D8-IOI	R	A	0.55	1/1	**not evaluable**
J-D-ROUTE-IOI	R	A	0.40	2/2	**met**
J-D-HOP2	R	A	0.25	0/2	**not met**
J-D6a	L	A	0.90	3/3	**met**
J-D6	R	A	0.35	0/2	**not met**
J-D6-ROUTE	R	A	0.35	0/1	**not met**
9 met, 7 not met, 1 not evaluable of 17 lines; risky account lines met: 2 of 9 with a verdict (expected 3.60).					

Parts A–D: 36 met, 21 not met, 12 not evaluable of 69 lines (64 account lines and 5 measurement-validity lines); risky account lines met: 10 of 27 with a verdict (expected 12.05; 10 of 28, expected 12.65, under the common-part reading of J-C-READb).

Table 5. **Calibration of the final round, by part and class.** Lines: account lines of the class (measurement-validity lines in their own row); met, not met and not evaluable (n.e.) as scored; expected: the sum of the recorded priors of the lines with a verdict; Brier: the mean of $(\text{prior} - 1[\text{met}])^2$ over the same lines. The totals give met of the lines with a verdict. Part C has no line of class L.

Part	Class	Lines	Met	Not met	n.e.	Expected	Brier
A, passages	L	1	1	0	0	0.90	0.010
	M	5	3	0	2	2.50	0.028
	R	14	3	6	5	4.20	0.217
B, fresh stories	L	3	3	0	0	2.70	0.010
	M	7	7	0	0	5.80	0.030
	R	6	4	1	1	2.55	0.227
	validity (R)	5	4	1	0	2.60	0.228
C, interventions	M	4	1	3	0	3.30	0.506
	R	7	1	3	3	1.70	0.186
D, mechanism	L	2	2	0	0	1.80	0.010
	M	5	5	0	0	4.20	0.026
	R	10	2	7	1	3.60	0.183
All parts	L			6 of 6		5.40	
	M			16 of 19		15.80	
	R			10 of 27		12.05	
	validity (R)			4 of 5		2.60	

Table 6. **Every line of the final round that did not hold or could not be evaluated, by kind.** Class and prior as recorded before the run (Table 4). Kinds: *contradicted*, the predicted effect is absent or reversed; *direction kept, threshold missed*; *failed in one model* of those required; *failed through a control or a definition*, where the outcome record finds that the control or the preregistered quantity, not the predicted effect, carries the failure (a post hoc reading that does not change the verdict); *not evaluable*, with the gate or rule. Where: the main-text section that names the result in words, and the appendix section with the numbers.

Line	Class	Prediction	What happened	Where
Contradicted				
J-A7	R, 0.30	2-bit KIVI quantization of the passage's values hurts free-form answers more than multiple-choice ones, beyond the keys'	one evaluable model (Qwen), low power; on all competent items 2-bit keys cost free-form answers more	Section 3.2; Appendix D.5
J-B6b	R, 0.45 (V)	under emitted forms s_{ID} and r move by less than 0.05	s_{ID} held; r(SENTENCE-AFTER) rose at Mistral-7B and Qwen2.5-14B (equivalence not shown)	Section 3.1; Appendix D.3
J-C4	R, 0.35	a remap trained at the writing token keeps the natural channel ratio	copy-only (R1) in both models	Section 5; Appendix F.3
J-C5	R, 0.35	identity-carrying components orthogonal to the natural write or outside the lexical span keep it	0 of 12 statistics equivalent; read mainly through the value	Section 5; Appendix F.3
J-D2	R, 0.55	structured controls move the answer at most 0.3 as much as the flag	mean reader output (Qwen) and the own write without the flag (Mistral) move it too; the Qwen control overlaps the flag	Section 4; Appendix E.3
J-D-KN	R, 0.40	a key naming no candidate acts as removing B's flag	removes about two (Qwen) to five (Mistral) times more support for B; per-story correlation only at Qwen	Section 4; Appendix E.3
J-D5	R, 0.25	initial-state flags carry binding in both sentence orders	below zero at Qwen, with an order effect	Section 4; Appendix E.3
J-D-HOP2	R, 0.25	the top-10 hop-2 heads of lists also carry the negative reads	they carry almost none ("different reader"); even in lists they carry only part of hop 2	Section 4; Appendix E.4
J-D6-ROUTE	R, 0.35	at 1.5B the injected flag is read by value under SENTENCE-AFTER	read by key	Section 4; Appendix E.5
Direction kept, threshold missed				
J-A5	R, 0.50	a related-articles sentence after the passage raises s_{ID} by at least 0.10	positive key read, gain below 0.10 (Gemma, Qwen)	Section 3.2; Appendix D.5
J-A6c	R, 0.45	in free form, cue conflict answers the value's entity in at least 0.6	value wins at the decision token; whole answers break into hybrids (Llama, Qwen below 0.6)	Section 3.2; Appendix D.5
J-A7b	R, 0.35	the value drop is larger in free form by at least 0.10	0.030 at the one evaluable model	Section 3.2; Appendix D.5
J-A8d	R, 0.60	hybrids under a non-matching key in at least 0.2 of free-form answers	0.105 at Gemma, the one evaluable model	Section 3.2; Appendix D.5
J-C1	M, 0.80	every statistic of the near-natural steering vectors equivalent (Qwen2.5-7B, Mistral-7B)	graded key under-reads at block 7 and deeper; every pooled $\bar{\lambda}$ inside the margin	Section 5; Appendix F.3
J-C2	M, 0.80	the same deeper and at Llama-3.1-8B	E1 held at Llama; E2 did not	Section 5; Appendix F.3
Failed in one model of those required				
J-A6a	R, 0.45	with options answered by name, cue conflict answers the key's entity in at least 0.6	key wins at Gemma and Qwen, not at Mistral	Section 3.2; Appendix D.5
J-B5b-P4f	R, 0.45	the sentence read changes generated answers in the four discovery-sample models	held in three; at OLMo-2-7B 3 of 150 answers flip	Section 3.1; Appendix D.4
Failed through a control or a definition (post hoc reading; verdict unchanged)				
J-C-READa	M, 0.85	the reader heads read the steering vectors; random sets do not	readers read them as they read the natural write; the random-set control fails for the natural write too	Section 5; Appendix F.3
J-C-BOUND	R, 0.50	non-lexical identity edits are under-read by the key where E2 is not	met in two of the 6 sub-families with a verdict; failures partly from E2's own departure; boundary not isolated	Section 5; Appendix F.3
J-D3	R, 0.50	removing the flag direction removes the key read; control directions do not	about half left in both models; Qwen's controls, which overlap the flag, remove 31 % and 44 %	Section 4; Appendix E.3
J-D6	R, 0.35	the injected sentence-row flag reproduces the natural sentence-to-list ratio	fails at 1.5B and 7B on the preregistered ratio; reproduced on contrasts specific to the absent option (post hoc)	Section 4; Appendix E.5
Not evaluable				
J-A3	derived	options after minus free form ≥ 0.4	met in both evaluable models; Llama's options-after and Mistral's free-form cells fail gates (not counted)	Appendix D.5
J-A5B	M, 0.80	the sentence after reads more than before	met in both evaluable models; three needed	Section 3.2; Appendix D.5
J-A6d	R, 0.45	the key alone moves answers with options, not in free form	met in both evaluable models; three needed (would not have been met had Llama's options-after cell passed its BF16 gate by one more item; post hoc)	Section 3.2; Appendix D.5
J-A8	M, 0.80	continuations are read through the K–V interaction	pattern in all four models; two evaluable	Section 3.2; Appendix D.5
J-A-HA1–HA3b	R	reader heads on passages (sparse set; transfer; ablation effects)	Qwen's heads BF16 floor failed; lines need both models (with Qwen's gate waived, HA1 and HA2 met, HA3a and HA3b not met; post hoc)	Section 3.2; Appendix D.5
J-B-SMALL	R, 0.40	no sentence read beyond 0.1 at Gemma-2-2B	competence gate failed (would not have been met)	Section 3.1; Appendix D.2
J-C3	R, 0.55	sparse-autoencoder features keep the channel ratio	dictionary gate failed through its specification (not met under either post hoc gate)	Section 5; Appendix F.2
J-C-READb	R, 0.60	the readers read the autoencoder and remap edits	E3 gate; E4's key read below 2 nats (not met under the common part's rule)	Section 5; Appendix F.3
J-C-WIN	R, 0.30	inside an overlap window the binding swap follows the law	no window	Section 5; Appendix H.2
J-D8-IOI	R, 0.55	the frozen belief readers carry the IOI read	met at Qwen; Mistral's transfer gate missed by 0.021 (met had it passed)	Section 4; Appendix E.4

Table 7. **The pre-written Table 1 rows of the final entry, with their numbers.** Quadruples are Llama-3.1-8B / Gemma-2-9B / Phi-4 / Falcon3-7B (the new families) or Llama-3.1-8B / Gemma-2-9B / Qwen2.5-7B / Mistral-7B (passages); pairs are Qwen2.5-7B / Mistral-7B; triples are Qwen2.5-1.5B / 3B / 7B. n.e.: not evaluable in that model. Recomputed values are marked † (post hoc).

Row (primary line)	Numbers	Verdict and row
Natural passages		
Options after: key read (J-A1, with J-A2; J-A3 derived)	s_{ID} with options after n.e. / 0.627 / 0.677 / 0.550; in free form 0.045 / -0.003 / 0.014 / n.e.	met (3 of 3 evaluable, with a new family); "supported"; title kept (Appendix A.6)
Natural mention sentence (J-A5)	s_{ID} gain over free form n.e. / 0.070 / 0.042 / n.e.	not met; "not shown on natural text"
Behaviour follows the channel (J-A6a, J-A6c; with J-A6b, d, e)	key-source rate with options by name n.e. / 0.734 / 0.778 / 0.408; value-source rate in free form 0.506 / 0.789 / 0.438 / n.e.; with letters the key wins in all four	not met (J-A6b, J-A6e met); not shown to decide behaviour on natural text; reported as a boundary
Natural readers (J-A-HA3a, with HA1, HA2)	Qwen's heads gate failed	not evaluable
Fresh stories		
Later list, new families (J-B1-N4)	s_{ID} with the list after 0.612 / 0.700 / 0.721 / 0.784	met (4 of 4)
Later sentence, new families (J-B3-N4)	r with the sentence after 0.396 / 0.512 / 0.334 / 0.449; null control (J-B-NULL-N4): r with the null sentence 0.118 / 0.003 / 0.004 / 0.028	met (4 of 4); the null control met (3 of 4; Llama's no-mention baseline, Appendix D.2)
No mention (J-B4-N4)	s_{ID} without a mention 0.074 / 0.005 / 0.005 / 0.026	met (4 of 4)
Generated answers (J-B5-N4, with J-B5-P4f)	over the eight models β_K with the list after 0.760–1.000; β_V without a mention 0.813–1.000	met (4 of 4 and 4 of 4)
Sentence, generated (J-B5b-P4f, J-B5b-N4)	flip rate after the sentence 0.033–0.313 in seven models, 0.010 at OLMo-2-7B; before it 0.000 in all	N4 met, P4f not met (OLMo-2); split worded as disclosed (Appendix A.6)
Mechanism		
What the readers write (J-D1, J-D3, J-D4)	$\mathrm{ID}_{\mathrm{inj}}^{\mathrm{LOO}}/\mathrm{ID}_K$ 0.997 / 0.803; ρ_K 0.466 / 0.478; $r_{\mathrm{ans}}^{\mathrm{inj}}(K)$ 0.689 / 0.601	J-D1 met, J-D4 met, J-D3 not met: "sufficient but not necessary"
Sign (J-D-SIGN-IOI, with J-D8-IOI)	OPTIONS-AFTER: ID_K 20.650 / 14.682 nats, $R(H^*)$ 0.967 / 0.942 (discovery sample); not mentioned: ID_K -12.138 / -7.083, ID_V -0.431 / -1.806, $R(H^*)$ 0.986 / 1.050; IOI in-sentence: ID_K -3.350 / -2.352, ID_V 2.550 / 1.096, $R(H^*)$ 0.922 / n.e.	J-D-SIGN-IOI met; J-D8-IOI not evaluable (met at Qwen); the joint clause not used (Appendix A.6)
Scale (J-D6, with J-D6-ROUTE)	ω 0.967 / 0.954 / 0.822; ρ -0.126 / -0.025† / 0.257; ρ_{nat} 0.090 / 0.266† / 0.468 (3B not evaluable; its values recomputed, post hoc)	not met; outcome (d), "Mixed."
Interventions		
Near-natural edits, E1 and E2 (J-C1)	pooled $\bar{\lambda}$ E1 -0.069 / -0.061, E2 -0.072 / -0.089	not met: "Even steering vectors close to the natural edit depart from the natural channel ratio ($\bar{\lambda}$ = -0.061 to -0.089): the format law does not transfer to edits as a channel-ratio law."
Independent edits, E3 and E4 (J-C3 with J-C4)	E4 pooled $\bar{\lambda}$ -1.574 / -1.707 (copy-only, R1); E3 not evaluable	J-C4 not met (R1), J-C3 not evaluable: "Edits that change the value through a learned remap are read mainly through the value"
Boundary, E5 (J-C-BOUND)	$\bar{\lambda}_{\mathrm{E5}}$ (E2 at the same cells): Qwen2.5-7B NL -0.705 (-0.242); Mistral-7B FR -0.242 (-0.183), NL -0.806 (-0.183); Llama-3.1-8B FR -0.558 (-0.276), SYN -1.066 (-0.129), NL -0.897 (-0.276)	not met; the boundary is not isolated; the pre-written row text is not used (Appendix A.6)
Note under "Prakash et al.'s binding swap" (J-C-SCREEN)	the swap works at blocks 15–22 (Qwen2.5-7B) and 12–16 (Llama-3.1-8B), where s_{ID} is at most -0.061 and -0.096	met: "Binding forms only after the identity key route closes in every model screened (no block with a working binding swap and $s_{\mathrm{ID}} \geq 0.4$)."

Table 8. **Preregistrations A and B.** A: Qwen2.5-1.5B-Instruct in FP32 on CPU, seed 1 (new story cores), $n = 40$ per format, all items, keys clamped from layer 0, 95% core-bootstrap CIs. Predictions committed at `prereg-AB`, run at `stageA-run`. B: GPU stage 1 on held-out models (Qwen2.5-3B/7B, Qwen3-8B, Mistral-7B-v0.3, OLMo-2-7B), $n = 150$ per format, all items, 95% core-bootstrap CIs. Predictions committed at `prereg-AB`, code at `stage1-run`, scoring script `analysis/stage1_prereg.py` committed before inspection at `stage1-score`.

	Prediction	Observed	Verdict
A1	identity(K) > 1.0 nats with a CI excluding 0, in OPTIONS-AFTER and in LETTERS-AFTER	OPTIONS-AFTER $+2.85$ [2.38, 3.33]; LETTERS-AFTER $+4.10$ [3.45, 4.77]	**met**
A2	identity(K) CI within the equivalence margin ± 1.0 in NO-MENTION, LIST-BEFORE and SENTENCE-AFTER	NO-MENTION $+0.10$ [-0.01, 0.21]; LIST-BEFORE -0.20 [-0.28, -0.13]; SENTENCE-AFTER -0.41 [-0.56, -0.26]	**met**
A3	Key share ($l_0 = 0$) in OPTIONS-AFTER within [0.25, 0.50]	0.38 [0.35, 0.41]	**met**
B1	identity(K) in OPTIONS-AFTER and LETTERS-AFTER positive, CI excluding 0, in at least 4 of 5 models	5/5 models	**met**
B2	identity(K)(OPTIONS-AFTER) $-$ identity(K)(NO-MENTION) > 0, CI excluding 0, in at least 4 of 5 models	5/5 models	**met**
B3	Exploratory, no directional claim: how the OPTIONS-AFTER key share ($l_0 = 0$ and $l_0 = 0.0625\,L$) varies with size within Qwen2.5	Rises steadily with size: 1.5B 0.37 [0.36, 0.39]; 3B 0.69 [0.68, 0.71]; 7B 0.82 [0.81, 0.83]; 14B 0.91 [0.90, 0.91]. The depth-matched l_0 gives identical values. Recorded later, with the outcome of C: non-monotonic beyond 14B.	–

Notes recorded with outcome A. LIST-BEFORE and SENTENCE-AFTER are slightly but reliably negative. They sit inside the margin but are not exactly zero. The seed-0 LETTERS-AFTER interaction ($+2.0$ [0.3, 3.7]) did not replicate: seed 1 gives $+0.6$ [-1.0, 2.2]. This confirms the post hoc option-listing claim in the same model on new data; generalisation across models is preregistration B.
Unpredicted observations recorded with outcome B (reported, not reinterpreted). SENTENCE-AFTER carries large key identity in the larger models (Qwen2.5-7B $+5.5$, Qwen2.5-14B $+12.1$, Qwen3-8B $+8.8$, Mistral-7B $+7.9$, OLMo-2-7B $+2.0$ nats) but about 0 at Qwen2.5-1.5B/3B (-0.47 and -0.04). So the post hoc "listed answer options only" refinement from 1.5B does not generalise. LIST-BEFORE is at or below 0 in every model (-0.45 to $+0.01$). NO-MENTION carries small positive identity(K) in larger models ($+1.0$ to $+1.25$ nats at 7–14B), outside the CPU-era ± 1 margin, though under 5% of OPTIONS-AFTER.

Table 9. **Preregistration C**: GPU stage 2, the released bases of Anonymous (2026) at Mistral-Small-24B and Qwen2.5-72B across formats, and the natural factorial at 24–72B. Predictions and the scoring script `analysis/stage2_score.py` committed before any stage-2 run (`prereg-C`); code at `stage2-run`. C4 is scored by `analysis/stage1_prereg.py` (committed at `stage1-score`). Primary population: native story cores with B, S and T distinct; fits averaged within core; 95% core bootstrap. κ_V was estimated before the run from the fixed-value data released with Anonymous (2026). Here ψ is our ψ_K.

Prediction	Mistral-Small-24B	Qwen2.5-72B
Gate: in OPTIONS-AFTER, the argmax of M and of P agrees with the outputs released with Anonymous (2026) on ≥ 0.95 of (core, fit) items	M 0.994, P 1.000 ($n = 360$ each): **passed**	M 1.000, P 1.000: **passed**
C1, transfer law: $\varphi_{\text{NO-MENTION}}(\text{M})$ within ± 0.15 of κ_V	0.708 [0.681, 0.734] vs 0.53 ± 0.15: **not met**	0.861 [0.837, 0.884] vs 0.73 ± 0.15: **met** (at the edge)
C2: $\varphi_{\text{OPTIONS-AFTER}}(\text{M}) - \varphi_{\text{NO-MENTION}}(\text{M}) > 0$, paired CI excluding 0	-0.003 [-0.019, $+0.013$]: **not met**	$+0.034$ [$+0.018$, $+0.049$]: **met** (small)
C3: $\psi_{\text{NO-MENTION}} < 0.5 \cdot \psi_{\text{OPTIONS-AFTER}}$	0.069 vs 0.528: **met**	0.052 vs 0.633: **met**

Prediction (natural factorial)	Outcome
C4: OPTIONS-AFTER key share ≥ 0.80 at Mistral-Small-24B, Qwen2.5-32B and Qwen2.5-72B	**not met** overall: Mistral-24B 0.78 [0.77, 0.79] no; Qwen2.5-32B 0.86 [0.84, 0.87] yes; Qwen2.5-72B 0.76 [0.75, 0.77] no.
C4: LIST-BEFORE identity(K) ≤ 0.5 nats in all three	**met** in all three: -0.53, -0.42, -1.99.

Table 10. **Preregistrations D and E.** D: GPU stage 3; predictions and the scoring script `analysis/stage3_score.py` committed before any stage-3 run (`prereg-D`); run at `stage3-run`. New tasks: $n = 150$ per format in Qwen2.5-7B, Qwen2.5-14B, Qwen3-8B, Mistral-7B-v0.3 and OLMo-2-7B; localisation: belief task, $n = 60$, in Qwen2.5-7B, Qwen2.5-14B and Mistral-7B, the order of the three D3 values. E: GPU stage 3b; predictions and the scoring script `analysis/stage3b_score.py` committed before any stage-3b run, which ran at that commit (`stage3b-run`). E1 and E2 use Qwen2.5-7B, Qwen2.5-14B, Mistral-7B and OLMo-2-7B; E4 uses Mistral-Small-24B and Qwen2.5-72B.

	Prediction	Result	Verdict
D1	Paint: per model, OPTIONS-AFTER $\mathrm{ID}_K > 0$ and OPTIONS-AFTER $-$ NO-MENTION > 0 (CIs excluding 0); met if ≥ 4 of 5 models count	5/5 models (OPTIONS-AFTER ID_K $+6.5$ to $+42.7$ nats)	**met**
D1	Schedule: same	5/5 models (OPTIONS-AFTER ID_K $+4.5$ to $+28.5$ nats)	**met**
D2	LIST-BEFORE $\mathrm{ID}_K \leq 0.5$ nats in ≥ 9 of the 10 task $\times$ model cells	10/10 task $\times$ model cells (max $+0.32$)	**met**
D3	OPTIONS-AFTER: choice-word rows recover ≥ 0.70 of the full key effect; question rows and the remaining tail rows each ≤ 0.15; in all 3 models	3/3 models: choice words 0.92 / 0.94 / 0.98; question 0.00; tail ≤ 0.01	**met**
D3	SENTENCE-AFTER, where the full key effect is positive: re-mention-word rows recover ≥ 0.50 in ≥ 2 of 3 models	3/3 models: 0.67 / 0.72 / 0.73	**met**
E1a	Instruction-matched 2×2, every arm "Answer with one word": ID_K(LIST-AFTER) > 0, CI excluding 0, in 4/4 models	$+8.7$ to $+35.8$ nats, 4/4	**met**
E1b	ID_K(LIST-BEFORE) ≤ 0.5 and ID_K(SENTENCE-BEFORE) ≤ 0.5, each in $\geq 3/4$ models	4/4 and 4/4 (max $+0.01$)	**met**
E1c	Paired LIST-AFTER $-$ LIST-BEFORE > 0 in 4/4 models; paired SENTENCE-AFTER $-$ SENTENCE-BEFORE > 0 in $\geq 3/4$; each CI excluding 0	4/4; 4/4	**met**
E2	Role control: for the direct question in OPTIONS-AFTER and NO-MENTION, $\lvert f_K \rvert \leq 0.15$ and $\lvert f_V \rvert \leq 0.15$ in $\geq 3/4$ models (Qwen2.5-72B exploratory)	4/4 and 4/4 (all fractions ≤ 0.02; also ≤ 0.01 at Qwen2.5-72B)	**met**
E3	Environment: each re-run within 0.03 of the original OPTIONS-AFTER key share (Qwen2.5-14B under transformers 5.9.0 on 2 GPUs, original 0.91; Qwen2.5-32B under transformers 5.18.0 on 1 GPU, original 0.86)	Qwen2.5-14B: 0.91 (original 0.91); Qwen2.5-32B: 0.86 (original 0.86)	**met**
E4	Value-only exchange in the frames of Anonymous (2026): ψ_V(NO-MENTION) ≥ 0.5 and ψ_V(NO-MENTION) $> \psi_K$(NO-MENTION) in both models	Mistral-24B 0.80 vs 0.07; Qwen-72B 0.90 vs 0.05	**met**

Exploratory parts of D and E, with no prediction: key shares on the new tasks; LETTERS-AFTER and SENTENCE-AFTER identity on the new tasks; layer-window and KV-group localisation at 7–14B; ψ_V and χ_V in the other formats; the role control in LETTERS-AFTER and at Qwen2.5-72B. Before preregistration E, an exploratory CPU pilot of the role control was run at Qwen2.5-1.5B ($n = 40$; recorded in the log with E).

Table 11. **Preregistration F.** GPU stage 4: the remap of Anonymous (2026) refit at Mistral-Small-24B with their released training code, under NO-MENTION (fit_none) and under their own format (fit_opt, same-code control). Predictions, gates and the scoring script `analysis/stage4_score.py` committed before the run, which ran at that commit (`stage4-run`). Thresholds refer to point estimates except F3; every statistic is per run, with the three seeds averaged within core; 96 cores, 95% core bootstrap.

	Prediction or gate	Result	Verdict
G1	Each refit P spans the released P of the same seed: mean squared principal cosine ≥ 0.99	1.000 for all six (one P for all seeds)	**met**
G2	$\varphi \geq 0.5$ under each run's fitting prompt	fit_none 0.81; fit_opt 0.72	**met**
G3	fit_opt reproduces the released crossover: $\psi_K(\text{LETTERS-AFTER}) \geq 0.5$, $\psi_V(\text{NO-MENTION}) \geq 0.5$	0.78, 0.81	**met**
F1	fit_none: $\psi_K(\text{LETTERS-AFTER}) \geq 0.5$ and $\psi_V(\text{NO-MENTION}) \geq 0.5$	0.77, 0.80	**met**
F2	fit_none: $\varphi(\text{LETTERS-AFTER}) \geq 0.5$	0.67	**met**
F3	$D = [\psi_K(\text{LETTERS-AFTER}) - \psi_K(\text{NO-MENTION})]_{\text{fit_none}} - [\cdot]_{\text{fit_opt}}$, paired over cores: CI within $[-0.25, +0.25]$	-0.008 $[-0.027, +0.009]$	**met**
F4	For each run, $\psi_K/(\psi_K + \psi_V)$ over the five formats correlates with the natural s_{ID} of Mistral-Small-24B at $r \geq 0.9$	fit_none 0.96; fit_opt 0.98	**met**

Stated alternative. If the fitting format decided which channel carries the remap, F1 and F2 would fail with $\psi_V(\text{LETTERS-AFTER}) \geq 0.5$ for fit_none and $D \leq -0.4$. This all-or-none pattern, which needs an edit that values can read and keys cannot, was not observed; the graded post hoc shift in OPTIONS-AFTER and SENTENCE-AFTER (Appendix G.3) is the more sensitive comparison. *Exploratory parts of F*, with no prediction: evaluation without the "Answer:" prefill; χ_K and χ_V; principal cosines between the remaps; loss curves; the identity-objective fits that the training code makes alongside the remap.

Table 12. **Preregistration G.** GPU stage 5, scored by `analysis/stage5_score.py`. Predictions, scoring script and code were finalised together before any stage-5 GPU run; a first attempt stopped in the unit tests before any model was loaded, and the entry was re-finalised with test-only changes at `stage5-run`, the commit the run used. Parts (a)–(d): the 150 seed-0 story cores of stages 1 and 3b (the row splice and the form probe on the first 60); part (e): 200 fresh IOI cores. Triples are Qwen2.5-7B / Qwen2.5-14B / Mistral-7B (Instruct) unless the row says otherwise; G2 and G3 concern Qwen2.5-1.5B and 3B; part (d) adds OLMo-2-7B and is met if at least three evaluable models meet it. A prediction with sub-verdicts is not met unless every sub-verdict is met (the scorer's rule); the Observed column gives the sub-verdicts. 95% core-bootstrap CIs. In all, 12 met and 10 not met; none was not evaluable. Details in Appendix E.1.

	Prediction	Observed	Verdict
G1	Anchors attend: $E \geq 0.20$ (lower > 0.10) and $F/E \geq 0.5$ under SENTENCE-AFTER at Qwen2.5-7B and 14B	E 0.869 [0.852, 0.885], F/E 1.00 (7B); E 0.915 [0.908, 0.921], F/E 0.98 (14B)	**met**
G2	Dissociation at 1.5B and 3B (H_{diss}): the same attention while ID_K(SENTENCE-AFTER) stays within ± 1 nat	E 0.789 [0.778, 0.802], F/E 1.03, ID_K (lowercase ids) -0.53 (1.5B); E 0.879 [0.866, 0.892], F/E 0.99, ID_K (lowercase ids) -0.02 (3B; 0.04 in the emitted casing)	**met**
G3	Magnitude under H_{diss}: $R_A \geq 0.5$ and $E(m) \geq 0.5\times$ the smaller anchor E	R_A 1.05 / 1.06; $E(m)$/anchor 0.91 / 1.01 (1.5B / 3B)	**met**
G4	Hop 2 (attention): (a) $G \geq 0.10$ (lower > 0.05) at the anchors; (b) $Q \leq 0.5$ (upper < 1) at 1.5B and 3B	(a) G 0.255 / 0.329; (b) Q 0.34 [0.29, 0.39] / 0.44 [0.37, 0.52]	**met**
G5	Cutting the candidate rows' attention to the writing token (M1): $r_K \leq 0.20$ (upper ≤ 0.25) in the lists, ≤ 0.40 in SENTENCE-AFTER	LIST-AFTER 0.001 / -0.026 / -0.007; OPTIONS-AFTER 0.003 / -0.028 / -0.009; SENTENCE-AFTER -0.105 / -0.036 / 0.001	**met**
G6	Matched control column (M2, LIST-AFTER): r_K within $[0.8, 1.2]$, small $\lvert\Delta\text{ID}_V\rvert$, ID_K(M1)$-$(M2) < 0	r_K(M2) 1.094 / 0.997 / 0.990	**met**
G7	$H_{\text{redundant}}$: (a) the copy takes over under M1 in $\geq 2/3$ models in both list formats; (b) the answer is kept (acc_B, $\text{on}_B \geq 0.90$) in 3/3	(a) 3/3: q_V(M1) 0.390 / 0.440 / 0.448 (LIST-AFTER) against 0.391 / 0.335 / 0.433 under NO-MENTION; (b) 1/3: acc_B 0.37 / 1.00 / 0.86 (LIST-AFTER), 0.30 / 0.99 / 0.73 (OPTIONS-AFTER); partial takeover	**not met**
G8	Routes at the answer: (a) cutting answer$\rightarrow$writing token (M3) under NO-MENTION leaves ≤ 0.6 of ID_V; (b) r_K(M3) ≥ 0.8 under LIST-AFTER; (c) M4 against M1	(a) 0.134 / 0.370 / 0.417; (b) 1.043 / 1.027 / 1.010; (c) 0.134 / 0.465 / 0.570	**met**
G9	Membership: naming the swapped candidates (S3, L3) reads the key; naming B with two others (S3out, L3out) does not	S3$-$NO-MENTION $+2.44$ / $+8.47$ / $+4.70$; L3$-$NO-MENTION $+20.66$ / $+34.78$ / $+15.11$; S3out$-$NO-MENTION -0.32 / -0.30 / -0.21	**met**
G10	Proportionality refuted, per family: lower bound of $r_2 > 1/3$ and of $r_3 > 1/2$	sentences r_2 0.28 [0.23, 0.32] / 0.58 / 0.46; lists r_2 0.96 / 0.98 / 0.98; lists met; sentences 2/3	**not met**
G11	The copy returns when S and X leave the list: ID_V(L3out)$-\text{ID}_V$(L3) > 0 and $R_V \geq 0.5$	R_V 0.11 / 0.03 / -0.08; 0/3	**not met**
G12	Reader rows (splice): $f_{\text{words}} \geq 0.5$ (lower > 0.25) in S2, S3, L2, L3	S2 0.40 / 0.38 / 0.39; S3 0.68 / 0.72 / 0.72; L2 0.91 / 0.94 / 0.92; L3 0.94 / 0.95 / 0.95; 9/12	**not met**
G13	Wrappers keep the read: $r_K \geq 0.75$ (lower > 0.5) for THE and MODIF, ≥ 3 of 4 models	THE 0.96–1.09; MODIF 0.32–0.64; THE met; MODIF not met	**not met**
G14	An exact repeat reads most: $r_K \leq 0.75$ (upper < 1) for each of six variants	TITLE, UPPER 0.83–1.14 (0/4 each); PLURAL 0.11–0.49; SYN, FRMIX, DEMIX ≤ 0.15 (4/4 each); 4/6 sub-verdicts	**not met**
G15	Token-level on the English measure: $r_K \leq 1/3$ and $\rho_s \leq 0.5$ for SYN, FRMIX, DEMIX	r_K -0.13 to 0.15; ρ_s 0.09–0.33	**met**
G16	Token pattern on r_K^{any} and a_v in ≥ 3 models, per variant	SYN Q7B t/g, Q14B t/g, M7B t/g, O7B t/t; FRMIX Q7B c/c, Q14B c/c, M7B g/g, O7B g/g; DEMIX Q7B c/g, Q14B c/g, M7B t/t, O7B t/t (t token, c concept, g graded; r^{any}/a_v); graded	**not met**
G17	Value compensation: paired $\text{ID}_V(v) - \text{ID}_V$(SENTENCE-AFTER) > 0 where the read is lost	FRMIX $+2.58$ / $+1.37$ / $+0.31$; DEMIX $+2.02$ / $+1.13$ / $+0.39$ (Qwen2.5-7B / Mistral-7B / OLMo-2-7B)	**met**
G18	No key read in plain IOI: f_K(PLAIN) within ± 0.10 (CI within ± 0.20), 3/3	-0.02 / $+0.02$ / $+0.00$ (GPT-2 small / Qwen2.5-7B / Mistral-7B)	**met**
G19	A later list opens a key read on IOI: (a) existence, (b) positive sign, 2/2	Qwen2.5-7B ID_K(AFTER) $+7.05$ [6.26, 7.81]; Mistral-7B not evaluable (Gate e); 1/2	**not met**
G20	The lookup replaces the copy on IOI: f_V(AFTER) ≤ 0.5, f_V(QUESTION) ≥ 0.75	Qwen2.5-7B f_V 0.45 (AFTER), 0.69 (QUESTION); Mistral-7B not evaluable; 0/2	**not met**
G21	Controls: (a) BEFORE $\lvert\text{ID}_K\rvert \leq 0.5$; (b) QUESTION; (c) GPT-2 small INLINE_BEFORE $\lvert\text{ID}_K\rvert \leq 0.5$	(a) -2.02 (Qwen2.5-7B); (b) met at Qwen2.5-7B, Mistral-7B not evaluable; (c) -0.25; (c) met	**not met**
G22	GPT-2 small in-sentence re-mention: (a) a key read, (b) inhibitory (replication of a disclosed pilot)	ID_K(INLINE) -2.35 [-2.47, -2.23]	**met**

Table 13. **Preregistration H.** GPU stage 6, scored by `analysis/stage6_score.py`. Predictions, gates, scoring script and code were finalised at `stage6-run`, before any stage-6 run and before any output of G was inspected; the run used that commit. Part (a), gates a1–a3 and H1–H6: Qwen2.5-7B / Mistral-7B (Instruct), OPTIONS-AFTER unless stated, BF16, eager attention, 60 ranking and 60 held-out evaluation stories, k^* = 5% of heads; a prediction over both models is met only if it is met in both. Part (b), gates b0–b3 and H7–H12: Qwen2.5-14B-Instruct on the material of Prakash et al. (2026), $n = 150$ pairs, BF16; triples are NO-MENTION / QNAMES / OPTIONS-AFTER; $\ell^* = 28$, $\ell^*_{\mathrm{ID}} = 0$. met*: met except in the cell named in the Observed column. 95% bootstrap CIs (10,000 resamples). In all, 6 met (H1–H5, H11), 4 not met (H7–H10), 1 not evaluable (H6) and 1 not run (H12). Details in Appendices E.2 and H.1.

	Gate or prediction	Observed	Verdict
a1	FP32 exactness of the head and hop splices (unit tests) before any model	7 passed, 0 failed	**met**
a2	BF16 floor per model and format: $\lvert\text{none} - \text{clean}\rvert$ and $\lvert\text{all}_T - \text{full}\rvert \le \max(0.5, 0.02\,\bar{d}_{\text{full}})$; hop exactness ≤ 0.1	Qwen2.5-7B OPTIONS-AFTER 0.606, 0.669 $\le$ 0.719; SENTENCE-AFTER 0.770 $>$ 0.500; Mistral-7B $\le$ 0.224 in both; hop 0.0000; not met: Qwen2.5-7B SENTENCE-AFTER	**met***
a3	$\bar{d}_{\text{full}} \ge 10$ (OPTIONS-AFTER), > 0 (SENTENCE-AFTER); $d_G/d_{\text{full}} \ge 0.8$	35.969 / 26.469; 10.438 / 12.254; 0.926 / 0.982	**met**
H1	Sparsity: $R(k^*) \ge 0.8$, lower bound ≥ 0.7 (top 5 % by a_3)	0.967 [0.952, 0.982] / 0.942 [0.919, 0.961]	**met**
H2	Necessity and specificity: $\mathrm{KO}(k^*) \ge 0.8$; random sets $R, \mathrm{KO} \le 0.25$	KO 0.977 / 0.971; random R 0.004 / 0.007, KO 0.011 / 0.010	**met**
H3	Ablation: $\rho_K \le 0.5$; candidate mass kept; control sets $\rho_K \ge 0.75$; ID_V rises above the floor; answer kept ≥ 0.8	ρ_K 0.238 / 0.059; $\Delta\mathrm{ID}_V$ +6.907 / +10.861; kept 0.87 / 1.00	**met**
H4	Canonical duplicate-token heads over C: median $D \ge 0.2$, median $I \le 0.1$; median $T_{\text{dup}} \ge 0.2$	D 0.370 / 0.232; I 0.011 / 0.022; T_{dup} 0.459 / 0.433	**met**
H5	Second hop: $r_{\text{ans}}(KV) \ge 0.5$; $r_{\text{ans}} > r_{\text{other}}$; $r_{\text{all}} \ge 0.8$	0.899 / 0.859; r_{other} 0.118 / 0.165; secondary (not scored) $r_{\text{ans}}(K)$ 0.797 / 0.683, $r_{\text{ans}}(V)$ 0.080 / 0.023; SENTENCE-AFTER line not evaluable	**met**
H6	The same readers for list and sentence: top-20 overlap ≥ 10	13 (Gate a2 failed) / 14	**not evaluable**
b0	Exchange exactness per cell: $\lvert m(r_4) - m(r_1)\rvert$, $\lvert m(r_0) - m(B)\rvert \le 0.3$ nats	0.099–0.194; 0.141–0.203 (13 cells)	**met**
b1	Reproduction (NO-MENTION sweep): IIA ≥ 0.7 at ℓ^* (BIND) and ℓ^*_{ID} (ID)	BIND 0.993 at $\ell^* = 28$; ID 1.000 at $\ell^*_{\mathrm{ID}} = 0$	**met**
b2	Effect size $\Phi \ge 3$ nats per cell	met in 11 of 13 cells; LETTERS-AFTER at depth 28: 0.225, 0.034; not met: LETTERS-AFTER at 28	**met***
b3	For H10 as a dissociation: identity key-read route at $\ell^* + 1$ ($s_{\mathrm{ID}} \ge 0.5$) and $\kappa_{\mathrm{ID}}(\ell^*) \ge 0.5$ under OPTIONS-AFTER	s_{ID} −0.015; κ_{ID} 0.006	**not met**
H7	The law on the binding swap: $\lvert\kappa(f) - s_{\mathrm{ID}}(f, 29)\rvert \le 0.25$ for each evaluable f and $r \ge 0.9$	gaps 0.601 / 0.934 / 0.879; $r = -0.986$; secondary, against $s_{\mathrm{ID}}(f, 0)$: gaps 0.600 / 0.181 / 0.163	**not met**
H8	Under NO-MENTION: $\psi_V \ge 0.5$, $\psi_K \le 0.25$	ψ_V 0.435, ψ_K 0.703; $\psi_V - \psi_K$ −0.268 [−0.317, −0.219]	**not met**
H9	Crossover: κ(OPTIONS-AFTER) − κ(NO-MENTION) ≥ 0.4; QNAMES between	+0.247 [0.211, 0.281]; κ(QNAMES) 0.906 not between	**not met**
H10	The rival: $\kappa(f) \le 0.25$ in every evaluable format	κ 0.618 / 0.906 / 0.864	**not met**
H11	Identity edit: Part 1 gaps to $s_{\mathrm{ID}}(f, 1) \le 0.25$ and κ_{ID} crossover ≥ 0.4; Part 2 gaps at depth 28	Part 1 gaps 0.114 / 0.048 / 0.077, crossover +0.648 [0.630, 0.666]; Part 2 gaps 0.113 / 0.036 / 0.020	**met**
H12	Optional replication at Llama-3-70B	not run (Gate b1 passed at 14B)	**not run**

Status of the preregistered alternatives. Part (a): a read distributed beyond 5% of heads (H1, H2 fail), task-tuned lookup heads instead of generic duplicate-token heads (H4 (i) fails), a three-hop route through the instruction and template rows (H5 fails with $r_{\text{all}} \ge 0.8$) and no head-level trade-off between the key and value routes (H3 (d) fails, as in the 0.5B pilot): none is supported, although the H4 margin at Mistral-7B depends on the size of the head set (Appendix E.2). Part (b): H10, the rival, is not met, and Gate b3 failed, so the reading "keys carry identity, values carry binding" is not available in either direction ("not evaluable as a dissociation at this depth"). The flat-high pattern, $\kappa \ge 0.75$ in every evaluable format, reported beside H10 without a verdict, does not hold either (NO-MENTION: 0.618). No evaluable cell is interaction-carried. H5's SENTENCE-AFTER line, reported against $r_{\text{ans}}(KV) \ge 0.35$ and $r_{\text{all}} \ge 0.5$ and not scored, holds at both models but is not evaluable at Qwen2.5-7B (Gate a2), and so not evaluable overall.

Table 14. **Preregistration I.** GPU stage 7, scored by `analysis/stage7_score.py`. Predictions, gates, scoring script and code were finalised at `stage7-run`, before any stage-7 run; the run used that commit (a first attempt stopped before the unit tests; disclosures below). Mistral-Small-24B-Instruct-2501 at the pinned revision, BF16; the released remap M and PCA control P of Anonymous (2026) on the 96 cores of Table 41, three fits averaged within each core; H^*: the top 64 of the 1120 eligible heads (5% of 1280) by a_3 on 60 separate ranking stories, per format, with the key clamped from block 6. G: the option rows. Blinding: chosen heads at G see P's key at the writing token instead of M's; g_K: the share of M's key-only effect removed by blinding every head at G; $\mathrm{KO}_x(S)$: the share of that direct read removed by blinding only the set S. A, A^+, N: mean-ablation of the heads' outputs at G, the same with the 32 heads of block 5, and a knockout of their attention from G to the writing token (Appendix G.4). $t = D^{\mathrm{A}}/D^{\emptyset}$; $c_\kappa = (\kappa^{\emptyset} - \kappa^{\mathrm{A}})/(\kappa^{\emptyset} - \kappa^{\emptyset}_{\text{NO-MENTION}})$ and $\tilde{c}_K$ the same for $\tilde{\psi}_K$ against ψ_K; $\Delta_V = V^{\mathrm{A}} - V^{\emptyset}$, with floor $F_V = 0.25\,[V^{\emptyset}(\text{NO-MENTION}) - V^{\emptyset}]$. The κ rule: κ is defined when $\psi_K + \psi_V \geq 0.3$, both are ≥ -0.1 and $D \geq 3$ nats. I1–I4 and I7 are scored under OPTIONS-AFTER and LETTERS-AFTER (met only if met in both), I5 and I6 under OPTIONS-AFTER alone; SENTENCE-AFTER is reported against the same thresholds without a verdict (Appendix G.4). 95% core-bootstrap CIs (10,000 resamples). In all, 6 met and 1 not met (I6); none was not evaluable. I-G4 only licenses reading I5 and I6 as a value takeover and is not a verdict gate.

	Gate or prediction	Observed (OPTIONS-AFTER / LETTERS-AFTER)	Verdict
I-G0	FP32 exactness of the blinding, ablation and knockout code (unit tests), before any model	17 passed, 0 failed, 0 skipped	**met**
I-G1	Reproduction of stage 3b ($\lvert\Delta\varphi\rvert$, $\lvert\Delta\psi_K\rvert$, $\lvert\Delta\psi_V\rvert \leq 0.03$, five formats); in-run rows; knockout floor; no effect at rows before the writing token	largest difference 0.000; in-run D within 0.2 %; structural check 0 nats	**met**
I-G2	The readers at 24B: $\bar{d}_{\text{full}} \geq 3$; $d_G/d_{\text{full}} \geq 0.7$; $R(64) \geq 0.7$, $\mathrm{KO}(64) \geq 0.8$; random sets ≤ 0.25; BF16 floor	$\bar{d}_{\text{full}}$ 17.020 / 18.186; d_G/d_{full} 0.853 / 0.987; R 0.832 / 0.822; KO 0.971 / 0.991; random R 0.001 / 0.001, KO 0.011 / 0.013	**met**
I-G3	The remap acts: $D \geq 3$ nats; direct read at $G \geq 1$ nat	D 13.67 / 12.87	**met**
I-G4	For reading I5/I6 as a value takeover (not a verdict gate): under A^+, $t \geq 0.6$ and $\Delta_V > 0$ (CI excluding 0)	t 0.556 [0.534, 0.578]; Δ_V +2.496 [2.250, 2.737]	**not met**
I1	The remap's key is read at the option words: $g_K \geq 0.7$, lower bound ≥ 0.6	0.976 [0.963, 0.990] / 0.999 [0.997, 1.000]	**met**
I2	Through the natural readers: $\mathrm{KO}_x(H^*) \geq 0.8$, lower bound ≥ 0.7	0.994 [0.989, 0.999] / 0.993 [0.990, 0.995]	**met**
I3	Specificity: $\mathrm{KO}_x \leq 0.25$ for three random sets and the active set	at most 0.082	**met**
I4	Under $\mathrm{A}(H^*)$ the key share falls toward NO-MENTION: c_κ (OPTIONS-AFTER), $\tilde{c}_K$ (LETTERS-AFTER) ≥ 0.5, lower bound ≥ 0.3	0.946 [0.905, 0.991] (κ 0.698 $\rightarrow$ 0.113; NO-MENTION 0.080) / 0.845 [0.818, 0.871]	**met**
I5	The value channel carries more (OPTIONS-AFTER): $\Delta_V > 0$ (CI excluding 0) and $\geq F_V$	+2.388 [2.141, 2.631] nats, F_V = 2.10; 0.366 of the lost key effect	**met**
I6	The behaviour is kept (OPTIONS-AFTER): $t \geq 0.75$, lower bound ≥ 0.6	0.565 [0.542, 0.589]; M's target is the argmax in 0.80 of (core, fit) items (0.82 unblocked)	**not met**
I7	Random and active sets change nothing: $\lvert\Delta\kappa\rvert$, $\lvert\Delta\tilde{\psi}_V\rvert$, $\lvert t - 1\rvert \leq 0.10$	at most 0.019	**met**

Status of the preregistered alternatives. Not established: H_{link} with redundant writing (I1–I7 met), because I6 failed; I-G4 was not met either. The applicable outcome is "I1–I4 and I7 met; I5 met, I6 not met", which the entry reads as a partial takeover: without its readers the value channel carries more of M's effect, but not enough to keep it. Not supported: other heads reading M's key at the option words (H_{other}; I2 met), a key read outside the option rows ($H_{\text{not_rows}}$; I1 met), backup readers (I4 met) and a non-specific disruption (I3, I7 met).

Table 15. **The 15 failed predictions of preregistrations G, H and I, by kind** (Appendix A.5). Values as in Tables 12 to 14; Qwen2.5-7B unless stated.

	Predicted	Observed	Kind
G20	On IOI, the lookup replaces the copy (2/2 of the 7B pair)	every clause met except the QUESTION baseline, f_V = 0.69 against ≥ 0.75; Mistral-7B not evaluable	direction kept, threshold missed
H9	κ(OPTIONS-AFTER) − κ(NO-MENTION) ≥ 0.4 on the binding swap, κ(QNAMES) between	$+0.247$ [0.211, 0.281]; κ(QNAMES) = 0.906 is not between	direction kept, threshold missed
G7	With the lookup cut, the copy takes over and the answer stays (3/3)	copy returns 3/3; answer kept 1/3 (Qwen2.5-7B acc_B = 0.37); preregistered "partial takeover"	met in part
G10	The dose effect is more than proportional, per family	lists 3/3; sentences 2/3 (r_2 = 0.28)	met in part
G12	The named words carry ≥ 0.5 of the read in S2, S3, L2, L3	9/12 cells; S2 fails (0.38–0.40)	met in part
G13	Wrappers keep the read (THE, MODIF)	THE 0.96–1.09; MODIF 0.32–0.64	met in part
G14	An exact repeat reads most (six variants)	four of six; case variants read nearly fully (r_K 0.83–1.14)	met in part
G21	IOI controls: list before, question wrapper, GPT-2 inline-before	question and GPT-2 met; list before gives ID_K = -2.02, not about 0; Mistral-7B not evaluable	met in part
G19	On IOI, a later list opens a positive key read (2/2)	met at Qwen2.5-7B ($+7.05$ nats); Mistral-7B failed the competence gate (two-way 0.59)	competence gate
H10	Rival: values carry the binding swap ($\kappa \leq 0.25$)	κ = 0.618, 0.906, 0.864	rival rejected
G11	Without the readers in the list, the copy carries the identity again ($R_V \geq 0.5$)	$R_V \leq 0.11$ in every model	falsified
G16	A token pattern on both measures for synonyms and translations	graded, none of the three preregistered outcomes; the French-mixed variant shows the concept pattern at Qwen2.5-7B and 14B	falsified
H7	$\kappa \approx s_{\mathrm{ID}}$ on the binding swap, depth-matched (≤ 0.25)	gaps 0.60–0.93	falsified
H8	Without a later mention values carry the swap ($\psi_V \geq 0.5$, $\psi_K \leq 0.25$)	ψ_V = 0.435, ψ_K = 0.703	falsified
I6	With the 64 reader heads mean-ablated at the option words (Mistral-Small-24B, OPTIONS-AFTER), M's effect is kept ($t \geq 0.75$)	t = 0.565 [0.542, 0.589]; its values carry $+2.388$ nats more (I5, met), the outcome the entry called a partial takeover (I-G4 not met)	falsified

Table 16. **Crosswalk.** Each main-text claim, the lines it rests on (class, verdict), the status mark (^{R}: a risky prediction) and words the main text prints, and where. Classes: L replication, M extrapolation, R risky; V measurement validity. Discovery-sample codes are those of Appendix A.5.

Main-text claim	Lines (class, verdict)	Status mark and words	Where
List after: the key carries most of the identity, eight models	J-B1-P4f (L, met), J-B1-N4 (M, met 4/4)	none	Section 3.1, Table 1, Figure 1b
No mention: copied	J-B4-P4f (L), J-B4-N4 (M), met	none	Section 3.1
Before: no key read; a list before lowers the read below no mention in the new families	J-B2-P4f (L), J-B2-N4 (M), met; J-B-LB-N4 (R, met)	^{R} for the second	Section 3.1, Table 1
Sentence after: intermediate read	J-B3-P4f (M, met), J-B3-N4 (R, met)	^{R}	Section 3.1, Table 1
Null sentence: no read	J-B-NULL-P4f (M, met), J-B-NULL-N4 (R, met 3/4; Llama reading post hoc, with the outcome record's $+0.005$ [0.0001, 0.011])	^{R} "in the new families"; "post hoc reading"	Section 3.1
Sentence read at 2B	J-B-SMALL (R, not evaluable; would not have been met, post hoc)	"not evaluable"; "post hoc"	Section 3.1
Generated answers follow the channel	J-B5-P4f, J-B5-N4 (M, met)	none	Section 3.1, Figure 1c
Sentence read in generated answers	J-B5b-N4 (R, met), J-B5b-P4f (R, not met: OLMo-2)	^{R}; "held in the new families"; "did not hold" in the discovery-sample models	Section 3.1
Scoring invariance; the sentence ratio understated	J-B6a, J-B6c (R, V, met); J-B6b (R, V, not met); J-B7, J-B8 (R, V, met; "two further validity checks")	^{R}; "did not hold" for r	Section 3.1
Passages: options after $\rightarrow$ keys; free form $\rightarrow$ values	J-A1 (M, 3/3), J-A2 (M, 3/3), J-A1b (R, met), J-A1c (M, 4/4)	^{R} for J-A1b; continuation "exploratory"	Section 3.2, Figure 2a
Passages: options before	J-A4 (L, met)	none	Section 3.2
Passages: a sentence after gives only a small read	J-A5 (R, not met); J-A5B (not evaluable)	^{R}; "did not hold"; listing sentence	Section 3.2
Passages: the derived within-model contrast; Llama-3.1-8B's cue-conflict cell	J-A3 (derived, n.e.; met in both evaluable models); J-A6a's Llama cell (n.e.)	"derived"; "not evaluable"	Section 3.2
Passages: cue conflict	J-A6b (R, met; narrowly under Holm), J-A6e (R, met), J-A6a (R, not met), J-A6c (R, not met), J-A6d (not evaluable)	^{R}; "did not hold" for J-A6a and J-A6c	Section 3.2, Figure 2b
Passages: quantization, hybrids, continuation, heads	J-A7, J-A7b, J-A8d (R, not met); J-A8 (M, n.e.); J-A-HA1–HA3b (R, n.e.); J-A6d (R, n.e.); the gate resolutions (outcome record, post hoc)	listing sentence; ^{R}; "post hoc"	Section 3.2
Scope: tasks, role, the 2×2	discovery D1, D2, E1, E2 (met)	"in the discovery sample"	Section 3.1
Two hops; a set enriched for duplicate-token heads; ablation leaves a copy route	discovery G, H1–H5 (met); the core's numbers and the hop-2 key split not scored	"in the discovery sample"; "exploratory"	Section 4
The flag is sufficient; its mean direction names no candidate (each story's write also carries its word)	J-D1 (R, met), J-D-ADDR (M, met); the dose and the per-story identity share (reported)	^{R}; dose "exploratory"	Section 4
Not specific	J-D2 (R, not met)	^{R}; "failed"	Section 4
Read by key	J-D4 (M, met)	none	Section 4
Not necessary as one direction	J-D3 (R, not met)	^{R}; "failed"	Section 4
The question sets the sign; the same readers	J-D7, J-D-SIGN-Q, J-D8-Q (M, met)	none	Section 4
The sign is applied after the first hop	D9: the list-fitted flag injected under the second question (reported, no verdict)	"exploratory"	Section 4, Table 1
IOI: key negative, value positive	J-D-SIGN-IOI (L, met)	none	Section 4
The IOI read is applied at the names' keys	J-D-ROUTE-IOI (R, met)	^{R}	Section 4
Not by the same hop-2 heads	J-D-HOP2 (R, not met)	^{R}; "failed"	Section 4
The belief readers carry the IOI read	J-D8-IOI (R, n.e.; met at Qwen)	^{R}; "not evaluable at Mistral-7B"	Section 4
The flag is written in sentences at every scale	J-D6a (L, met)	none	Section 4
Mediation across scale; the route at 1.5B; binding; a key naming no candidate	J-D6, J-D6-ROUTE (R, not met; outcome (d), "Mixed."); J-D5, J-D-KN (R, not met)	^{R}; "did not hold"	Section 4
The released remap's crossover; the lead's second sentence (the prompt and the edit)	discovery C3, E4, F (met); 72B letters exploratory; κ against σ (reported); J-C1, J-C4	"discovery sample, exploratory at this model; preregistered for a refit"; "exploratory here"	Section 5
Outcome (e), the section's lead; how J-C1 fails (none wholly outside the margin; one statistic at Llama-3.1-8B)	J-C1 (M, not met), J-C2 (M, not met); headline (e); counts over the printed statistics	"did not hold"	Section 5, Figure 4a
Equivalence at the shallowest depth; pooled under-read; small channels; the under-read at block 7	reported context; recomputed	"exploratory", "post hoc"	Section 5
A remap trained at the writing token is read through the value	J-C4 (R, not met, R1); its depth dependence (recomputed); κ against σ (reported)	^{R}; "did not hold"; "exploratory", "post hoc"	Section 5
Autoencoder features	J-C3 (R, n.e.; not met under either post hoc gate)	"not evaluable"; "post hoc"	Section 5
Orthogonal and non-lexical components; lexical boundary	J-C5, J-C-BOUND (R, not met)	^{R}; "did not hold"	Section 5
The reader heads read the steering vectors	J-C-READa (M, not met); J-C-READb (R, n.e.)	"failed"; "post hoc reading"; "not evaluable"	Section 5
Distance from the natural write; a thresholded key route (a hypothesis)	none	"post hoc"; "a hypothesis for a future test"	Section 5
An identity edit at block 28 is copy-read	J-C6 (M, met)	none	Section 5
No overlap window	J-C-SCREEN (R, met); J-C-WIN (R, n.e.)	^{R}; "not evaluable"	Section 5
Calibration; tails (lines treated as independent); the not-evaluable risky lines	G4 tallies; tail probabilities and resolutions post hoc	"post hoc"	Table 2, Section 7
Recommendation (iii): only the shallowest depth kept the ratio; the remap read through the value; report the distance	reported context (equivalence at block 3, recomputed); J-C4 (R, not met); the distance pattern	"exploratory"; "post hoc"	Section 7
Limitations: the layer window at 1.5B; the base model in IOI	discovery D (exploratory window splice); G part (e) (exploratory)	"exploratory"	Section 7

B. Prompts and items

B.1. The discovery sample: belief stories, tasks and formats

Belief task. The story generator (`ckeys/story.py`) is ported from the released generator of Anonymous (2026). Each story core draws an agent and an other agent (Alice, Bob), an object and a distractor object from twenty objects, and four locations from the six candidates *box, basket, shelf, drawer, cabinet, closet*. The initial location, the base location and the source location must differ, and the distractor's location must differ from the base and source locations. The two "initially sees" sentences are sorted by object name. The direct question is "Where does $\{A\}$ believe the $\{o\}$ is?". Base B and source S differ only at the writing token, the word after "moved to the". The third value X used for ID_K and ID_V is drawn, with a seed fixed by the core, from the candidates other than the base, source, initial and distractor locations, so it appears nowhere in the story. We keep a core only if the B, S and X prompts have the same token length and differ at exactly one token.

We show the first core generated with seed 0: agent Alice, other agent Bob, object *lamp*, distractor *coin*, initial location *basket* (the distractor is also in the basket), base location *cabinet*, source location *drawer*, third value *box*. Below are the raw user-turn prompts of the base run in all seven formats, as built by `ckeys/encoding.py`. Line breaks inside the story are added here for width; in the prompts the story is one line.

OPTIONS-AFTER (the format of the key-exchange experiments of Anonymous 2026).

```
Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Choices: box, basket, shelf, drawer, cabinet, closet
Answer with exactly one choice.
Answer:
```

LIST-AFTER. As OPTIONS-AFTER, with the last instruction line replaced by "Answer with one word.".

```
...
Question: Where does Alice believe the lamp is?
Choices: box, basket, shelf, drawer, cabinet, closet
Answer with one word.
Answer:
```

LETTERS-AFTER.

```
Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Choices: A) box, B) basket, C) shelf, D) drawer, E) cabinet, F) closet
Answer with one letter.
Answer:
```

SENTENCE-AFTER.

```
Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen. The room has a box, a
basket, a shelf, a drawer, a cabinet and a closet.
Question: Where does Alice believe the lamp is?
Answer with one word.
Answer:
```

SENTENCE-BEFORE.

```
Read the story and answer the question.

Story: The room has a box, a basket, a shelf, a drawer, a cabinet and a
closet. Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Answer with one word.
Answer:
```

NO-MENTION.

```
Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Answer with one word.
Answer:
```

LIST-BEFORE.

```
Read the story and answer the question.
Choices: box, basket, shelf, drawer, cabinet, closet

Story: Everyone initially sees that the coin is in the basket. Everyone
initially sees that the lamp is in the basket. Alice watches as the lamp is
moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Answer with one word.
Answer:
```

Chat wrapper and scoring. Each raw prompt is wrapped as in the `Answer:`-prefill interface of Anonymous (2026) and its released encoder: a system turn "You are a helpful assistant.", a user turn holding the raw prompt, the model's generation prompt, and the assistant prefill "Answer:". We also disable thinking (`enable_thinking=False`, which matters for Qwen3 and is ignored by other templates). The raw prompt itself ends in "Answer:", so this string appears twice. For Qwen2.5 the full input is

```
<|im_start|>system
You are a helpful assistant.<|im_end|>
<|im_start|>user
Read the story and answer the question.
...
Answer:<|im_end|>
<|im_start|>assistant
Answer:
```

The wrapper asserts that the raw prompt occurs exactly once in the templated text. Templates that reject a system role fall back to merging the system text into the user turn; every output file records whether this happened. No model of the discovery sample needed it; Gemma-2-9B and Gemma-2-2B, whose template rejects a system turn, do, and their files record the merge. In the discovery sample we read full-vocabulary log-probabilities at the last position and score the candidates as single tokens with a leading space (" box", " basket", . . . , or " A" to " F" under LETTERS-AFTER); the code asserts that each candidate is one token. This is the lower-case score; the final round scores the forms the models emit (Appendix C.2).

Role-swap control. The role-swapped story R exchanges the two names in the event and keeps the location word. For the core above, the base event and the role-swapped event are

```
B: Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.
R: Bob watches as the lamp is moved to the cabinet. Alice does not see this happen.
```

The question still asks for Alice's belief, so the correct answer moves from the writing token's location (*cabinet*) to the initial location (*basket*). The role effect is measured with $m = \log p(\text{initial}) - \log p(\text{LOC})$, and f_C is the effect of clamping channel C at the writing token to R's, divided by the full R−B difference in m.

Intervention frames. The stage-2, stage-3b and stage-4 runs with the bases of Anonymous (2026) use their released native story cores, rendered with the same generator, the same format builders and the same wrapper. The patched span is the move event, from the observer's name to the end of the story; the writing token is the location word inside it.

Paint and schedule tasks. The two new tasks (`ckeys/tasks.py`) follow the same structure: two sentences give the starting values of two things, sorted by name, and a final sentence writes the new value at the writing token. They use the formats OPTIONS-AFTER, LETTERS-AFTER, SENTENCE-AFTER, NO-MENTION and LIST-BEFORE, built exactly as above with the task's candidates; the neutral sentence of SENTENCE-AFTER is task-specific. The agent is drawn from six names (Alice, Bob, Carol, David, Emma, Frank) and the things from ten per task.

	paint	schedule
Candidates	red, blue, green, yellow, black, white	Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday
Things	fence, door, bench, shed, gate, chair, table, wall, boat, bike	meeting, dinner, appointment, party, class, interview, concert, match, lecture, trip
Starting value	"The {thing} was painted {value}."	"The {thing} was scheduled for {value}."
Writing sentence	"Later, {agent} repainted the {thing} {value}."	"Later, {agent} moved the {thing} to {value}."
Question	"What color is the {thing} now?"	"On which day is the {thing} now?"
SENTENCE-AFTER sentence	"The shop sells red, blue, green, yellow, black and white paint."	"The calendar shows Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday."
LETTERS-AFTER options	A) red, . . . , F) white	A) Monday, . . . , G) Sunday

The first cores generated with seed 0 give, under OPTIONS-AFTER and SENTENCE-AFTER,

```
Read the story and answer the question.

Story: The boat was painted yellow. The shed was painted yellow. Later, Emma repainted the
shed black.
Question: What color is the shed now?
Choices: red, blue, green, yellow, black, white
Answer with exactly one choice.
Answer:

Read the story and answer the question.

Story: The interview was scheduled for Friday. The match was scheduled for Monday. Later,
Frank moved the interview to Wednesday. The calendar shows Monday, Tuesday, Wednesday,
Thursday, Friday, Saturday and Sunday.
Question: On which day is the interview now?
Answer with one word.
Answer:
```

with source values *green* and *Thursday* and third values *blue* and *Tuesday*.

B.2. Fresh stories

Population. The fresh stories are the first 150 story cores of a stream with a new seed whose full tuple (agent, other agent, object, distractor, initial location, distractor's location, base, source) occurs in no earlier stage; a further 30 cores from another new seed serve only to discover each model's answer frames (calibration). The mention sentences and the second lexicon are new; the story sentences follow the template of the discovery sample. Half of the cores (those with the smallest hash of the core tuple) are rendered in a second six-word lexicon, *bin, crate, tray, jar, bucket, chest*, the others in the original one; a core is drawn in the original lexicon and rendered by index, so every rule of the generator holds in either. Each core has its own order of the six candidates, drawn with a seed fixed by its hash, used in the list and in the sentence of every arm. The bootstrap clusters are the cells (lexicon, base, source); 55 of the 60 possible cells occur. The

discovery-sample cores (the first 150 cores of seed 0) are re-measured with the original lexicon, the canonical order and the room sentence.

Arms. Every arm uses the chat wrapper and the "Answer:" prefill above and the instruction "Answer with one word.": LIST-AFTER ("Choices:" and the six candidates after the question), LIST-BEFORE (the same line before the story), NO-MENTION, SENTENCE-AFTER (a neutral sentence naming the six candidates after the story), SENTENCE-BEFORE (the same sentence before the story) and NULL-SENTENCE (the neutral sentence with six nouns that name no candidate, at the position of SENTENCE-AFTER). In the tables of Appendices D.2 and D.3 these arms are AFTER, BEFORE, NONE, POST, PRE and POST-NULL, and OPTIONS-AFTER is used on the discovery-sample cores. With the original lexicon, the canonical order and the room sentence, every builder equals the discovery-sample builder byte for byte (a unit test).

The eight sentences. Each core uses one of eight sentences (by the rank of its hash modulo 8; 18 or 19 cores each), with its six candidates in the core's order; in NULL-SENTENCE the same sentence holds the null nouns *rug, clock, mirror, poster, radio, globe* in the same order:

1. "There is a {0}, a {1}, a {2}, a {3}, a {4} and a {5} in the house."
2. "The hallway also has a {0}, a {1}, a {2}, a {3}, a {4} and a {5}."
3. "In the kitchen there are a {0}, a {1}, a {2}, a {3}, a {4} and a {5}."
4. "A {0}, a {1}, a {2}, a {3}, a {4} and a {5} stand along the wall."
5. "The house contains a {0}, a {1}, a {2}, a {3}, a {4} and a {5}."
6. "Nearby there are a {0}, a {1}, a {2}, a {3}, a {4} and a {5}."
7. "The attic holds a {0}, a {1}, a {2}, a {3}, a {4} and a {5}."
8. "Along one wall sit a {0}, a {1}, a {2}, a {3}, a {4} and a {5}."

Each of the 18 words is one token after a space and before ",", "." and " and" in every tokenizer of the round, so a null sentence has its sentence's length in every lexicon and order, and every sentence is within three tokens of the room sentence (checked by the tokenizer gate on each model's verified files). No null noun is a candidate of either lexicon or a story object. The first fresh core (sentence 2, original lexicon) gives, under LIST-AFTER, SENTENCE-AFTER and NULL-SENTENCE (the system turn, generation prompt and prefill as above):

```
Read the story and answer the question.

Story: Everyone initially sees that the hat is in the closet. Everyone
initially sees that the pen is in the closet. Bob watches as the pen is
moved to the drawer. Alice does not see this happen.
Question: Where does Bob believe the pen is?
Choices: cabinet, shelf, box, basket, drawer, closet
Answer with one word.
Answer:

Story: ... Alice does not see this happen. The hallway also has a cabinet,
a shelf, a box, a basket, a drawer and a closet.
Question: Where does Bob believe the pen is?
Answer with one word.
Answer:

Story: ... Alice does not see this happen. The hallway also has a radio,
a mirror, a rug, a clock, a poster and a globe.
```

A core of the second lexicon reads, under SENTENCE-AFTER: "Everyone initially sees that the bag is in the jar. Everyone initially sees that the cup is in the crate. Bob watches as the bag is moved to the chest. Alice does not see this happen. The hallway also has a tray, a crate, a bin, a bucket, a chest and a jar."

Generations. Greedy generations (at most 16 new tokens) stop at a newline, at an end-of-sequence or chat end-of-turn token, or once the text names a candidate; the answer is the first candidate of the core's lexicon in the text (plurals allowed), else "other". One batch per item and arm holds the clean B, S and X runs and the self-clamp, key-only and value-only rows

to S and to X; decoding reads the clamped cache, which a unit test checks against cache-free stepwise decoding under the same clamps.

B.3. Natural passages

Items. From the SQuAD v1.1 development set (Rajpurkar et al., 2016) (CC BY-SA 4.0; 48 articles, file hash asserted), a question is kept when its majority answer e_B (given by at least two of three annotators) is a person, a place (with a place head noun in the question) or a number, occurs exactly once in the passage and not in the question, and no content word of it occurs elsewhere in the passage or the question (items failing only this last rule form an exploratory stratum, LEAK). Three substitutes S, X and Z of the same type, place class and word count are drawn from the majority answers of other articles (numbers: the leading digit replaced); each shares no word with e_B, with the other substitutes or with the distractor D, and no content word of it occurs in the passage or the question. Z is only ever clamped, never shown. The fourth option D is another answer of the same paragraph and type that occurs in the passage, else an absent entity of the pool. The counterfactual passages write S, X or Z in place of e_B (Longpre et al., 2021); per tokenizer and format, the S, X and Z prompts have B's length and differ from it only inside the answer span. A type audit of the entity lists (no model output) excluded 44 entities and corrected nine place classes. Twelve articles (drawn with a fixed seed) give 71 items for the answer frames and the head ranking; the other articles give the 185 evaluation items, from 34 articles (80 persons, 77 numbers, 28 places; at most ten per article and two per paragraph). A year stratum (80 items) and the LEAK stratum (65 items) are exploratory.

Formats. Every prompt has the system turn "You are a helpful assistant.", the generation prompt with thinking disabled and the prefill "Answer:". With PRE = "Read the passage and answer the question." and a blank line, FREE = "Answer with the exact words from the passage.", MC = "Answer with exactly one of the options." and o1–o4 the four options in the item's fixed order:

- free form (NOM): PRE, "Passage: " + passage, "Question: " + question, FREE;
- options after (OPTA): as free form, with the line "Options: o1; o2; o3; o4" after the question and MC;
- options before (OPTB): PRE, "Options: o1; o2; o3; o4", a blank line, the passage and the question, MC;
- a related-articles sentence after the passage (MENA): the passage followed by " Related articles mention o1, o2, o3 and o4.", the question, FREE; before the passage (MENB): the same sentence placed before the passage;
- lettered options (LETA): the passage and question, "Options:", the lines "A. o1" to "D. o4", and "Answer with the letter of the correct option.";
- closed book (no passage): "Answer the question.", the question and the options, as MC or with letters.

Answer frame and scoring. Before any test, each model's answer frame (the text it writes between the prefill and the entity) was chosen from its greedy generations on 30 frame items, among six fixed frames; it is a single space for all four models. The decision token is the first token of the entity after the frame (the letter under LETA); the decision tokens of B, S, X, Z and D must be distinct. The clamps act at every position of the answer span in every layer from 0. Generated answers (greedy, at most 16 tokens; the clamps act in the prompt pass and decoding reads the clamped cache) are matched to the candidates after SQuAD normalisation; an answer that starts with one candidate's first word but does not complete it is "other".

B.4. Questions and IOI prompts of the mechanism tests

The mechanism tests use OPTIONS-AFTER and SENTENCE-AFTER of the discovery sample on new cores (100 evaluation stories; the flags are fit on the first 60 discovery-sample cores, the stories on which the reader heads were ranked). The question arms replace the question line of OPTIONS-AFTER by "Which of the choices is mentioned in the story?" (Q_{IN}) or "Which of the choices is not mentioned in the story?" (Q_{OUT}). The IOI prompts follow the templates of Wang et al. (2022) (Appendix E.4); the in-sentence arm (INLINE) lists the four names as a parenthetical right after the indirect object's mention, without a chat template; INLINE_CHAT puts the same sentence in the chat wrapper and instruction of the list arm. The binding test uses 100 cores whose distractor is in another location than the object, so that two initial-state sentences can be told apart.

B.5. Intervention stories

The interventions use 80 new story cores (24 ordered base–source pairs) in which the pair-swap of the released remap, π (box $\leftrightarrow$ basket, shelf $\leftrightarrow$ drawer, cabinet $\leftrightarrow$ closet), maps neither S nor X to B, under LETTERS-AFTER, OPTIONS-AFTER, SENTENCE-AFTER and NO-MENTION; 200 further cores give the steering vectors (150) and the calibration (50), and 1,000 and 100 more train and hold out the remap trained at the writing token. The prefix up to the writing token is the same in the four formats, so an edit's cached keys and values are the same tensors in all four. The steering vectors from unrelated sentences are means over 24 fixed neutral sentences, each a user turn of the chat template without prefill that contains every candidate exactly once; the boundary family uses the French, German or synonym form of each value in the same sentences.

C. Implementation, exactness and measurement

C.1. Clamps, splices and exchanges

Effects of a clamp. With $m_{\mathrm{SB}} = \log p(\mathrm{S}) - \log p(\mathrm{B})$ at the answer position, the effect of clamping channel C is measured against the self-clamp row $\mathcal{C}_{KV}(\mathrm{B})$ of the same batch:

$$d_C = m_{\mathrm{SB}}\big(\mathcal{C}_C(\mathrm{S})\big) - m_{\mathrm{SB}}\big(\mathcal{C}_{KV}(\mathrm{B})\big), \qquad C \in \{K, V, KV\}. \tag{2}$$

The key share $s_K = \bar{d}_K/(\bar{d}_K + \bar{d}_V)$ is a ratio of means over stories. A foreign key can move m_{SB} without carrying the source value, for instance by weakening B; s_K counts such effects, the identity measures ID_K and ID_V of Equation (1) remove them, and s_{ID} is their share. The discovery sample reports s_K beside s_{ID} because two of its preregistered predictions (B3, C4) were written on s_K.

Clamp hooks. Each clamp is a forward hook on the `k_proj` and/or `v_proj` module of every decoder block $\ell \geq \ell_0$. The hook overwrites the output row at the writing position p with the vector cached from the donor run (B, S or X) at the same layer. These outputs come before RoPE and before key normalisation (Qwen3, OLMo-2), and they contain all KV groups, so under grouped-query attention every group is clamped. All clamp conditions for one story run as rows of a single batch of the base prompt: the self-clamp row, $\mathcal{C}_K$, $\mathcal{C}_V$ and $\mathcal{C}_{KV}$ to S at each onset ℓ_0, and the same three to X at $\ell_0 = 0$. The onsets are $\ell_0 = 0$, $\ell_0 = \mathrm{round}(0.0625\,L)$, which matches block 6 of 80, where the exchange of Anonymous (2026) starts in Qwen2.5-72B, and $\ell_0 = \mathrm{round}(0.3\,L)$. Forward passes use `use_cache=False` and compute logits only at the last position. From preregistration A onward, every natural-factorial output file records the number of skipped cores, which is zero in every such run.

Self-clamp row. The self-clamp row $\mathcal{C}_{KV}(\mathrm{B})$ sets the writing token's keys and values to their own values, so mathematically it equals the clean base run. In BF16, however, a batched forward pass can differ slightly from an unbatched one. From preregistration A onward we therefore measure every effect against this row of the same batch (Equation (2)), not against a separately run clean base, so that batching noise cancels. Each result summary records the gap between the self-clamp row and the clean run as a noise floor. The first CPU factorials (FP32) and all row-restricted splices measure against the clean base run.

Exactness of the clamp. $\mathcal{C}_{KV}(\mathrm{S})$ from $\ell_0 = 0$ reproduces the source run at every later position. Positions before p have identical inputs in B and S and hence identical keys and values; at p both are set to S's in every layer; by induction over layers every later position receives the same inputs as in S. The clamp precedes key normalisation (Qwen3, OLMo-2) and RoPE, which both runs apply identically at p. In the FP32 CPU runs at Qwen2.5-1.5B, the recovery ratio $d_{KV}/[m_{\mathrm{SB}}(\mathrm{S}) - m_{\mathrm{SB}}(\mathrm{B})]$ equals one to the printed precision in every format (`results/format_factorial/`). The unit tests (`tests/test_interventions.py`, Qwen2.5-0.5B in FP32) check that an identity edit and a self key exchange leave the logits unchanged to within 10^{-4}, and that patching the residual stream at the writing token after one block transfers the source answer.

Row-restricted splice. The splice wraps the forward pass of each attention module. It runs the module once with the base key at p and once with the source key, which the `k_proj` hook inserts, and returns the source output in the rows of the chosen token group G and the base output elsewhere. In the layer-window and KV-group variants, the source key is inserted only in the chosen layers, or only in one group's slice of the key. A unit test checks that $G =$ all rows reproduces the full key clamp and $G = \emptyset$ reproduces the clean run, each to within 10^{-4} in the logits. Every run also evaluates $G = \emptyset$ for each story and reports its mean effect, which is zero to the printed precision.

A KV-cache pitfall. Our first splice implementation was silently wrong. Transformers 5 builds a `DynamicCache` by default, even for a single forward pass. Running the attention module twice per layer therefore appended keys to the cache, and the second (source-key) pass silently read the base keys. The fix is to run every forward pass with `use_cache=False`. The exactness test above was added in the same change, and every splice result reported here was produced after it. Anyone who patches attention by calling a module more than once per forward pass should check for this.

Exchanges in the intervention frames. The patch of Anonymous (2026) replaces the output of decoder block 4 (1-based) over every token of the move-event span by Equation (3), materialised in BF16 as in their engine, with the released rank-16

bases for the learned remap M and the PCA control P, at seeds 101–103 (and, in stage 4, the refit bases of Appendix G.3, each with the shared P). The key-only exchange replaces the writing token's `k_proj` output in blocks 6 to L (1-based) by the other patched run's, giving P+K_{M} (addition) and M+K_{P} (removal); queries, values and residual states evolve freely. The value-only exchange does the same with `v_proj`. The removal fractions are

$$\chi_K = \frac{m(\mathrm{M}{+}K_{\mathrm{P}}) - m(\mathrm{M})}{m(\mathrm{P}) - m(\mathrm{M})}, \qquad \chi_V = \frac{m(\mathrm{M}{+}V_{\mathrm{P}}) - m(\mathrm{M})}{m(\mathrm{P}) - m(\mathrm{M})},$$

with $m = m_{\mathrm{TS}}$, averaged over the three fits within each core before the ratio of means is taken. The reproduction gate compares the argmax token of M and P under OPTIONS-AFTER with the outputs released with Anonymous (2026); they agree on 0.994–1.000 of items (preregistered, ≥ 0.95; Table 9). The remap was fit without the "Answer:" prefill of our formats, which Anonymous (2026) add only at evaluation.

Attention knockout (preregistration G). A knockout sets chosen (row, column) attention weights to exactly zero in every layer and head and renormalises the row, while the writing token's key or value is clamped as in the factorial. The eight clamp rows of one story (the self-clamp row, a duplicate of it as the within-batch noise floor, and $\mathcal{C}_K$, $\mathcal{C}_V$, $\mathcal{C}_{KV}$ to S and to X) run in one batch per mask, each with its own four-dimensional attention mask, and every effect is measured against the self-clamp row under the same mask. The masks are listed in Appendix E.1.2. FP32 unit tests at Qwen2.5-0.5B check, to 10^{-4} in the logits, that the causal mask passed explicitly changes nothing, that the cut weights are exactly zero and the rest of each row is renormalised, that sdpa and eager attention agree, that per-row masks in one batch equal separate runs, and that with every edge to the writing token cut, each clamp row equals the self-clamp row. A sanity gate (Gate b) repeats the last check in every arm and model of the run: with all edges to the writing token cut, ID_K and ID_V are zero.

Head and hop splices (preregistration H). *HeadSplice* extends the row-restricted splice to heads: in the rows of the option words it returns, per head, the attention output computed with the source key at p for the chosen heads and with the base key for the others. Blinding a set of heads is the complement: all heads but these see the source key. Mean-ablation replaces a head's output at the six option rows by its position-matched mean over the ranking stories. *HopSplice* clamps K_{S} at p and gives only the chosen later rows (the answer row, every other row, or all rows) the base run's keys, values or both at the option rows, so that $r_{\mathrm{ans}} = 1 - [m(\mathrm{row}) - m(\mathrm{ID})]/[m(K_{\mathrm{S}}) - m(\mathrm{ID})]$ measures how much of the key-clamp effect reaches the answer through the option rows. Gate a1 runs seven FP32 unit tests at Qwen2.5-0.5B before any model is loaded, each to 10^{-4} in the logits: HeadSplice with all heads equals the full key clamp and with none the clean run; all heads in the option rows equal the row splice, and KV-group slices the group splice; a batched grid equals single runs; self-ablation equals the clean run; HopSplice over all rows equals a plain clamp of the option rows' keys and values, without overwrite it equals the clean and full runs, and the answer-only splice with the source run's cache equals the full clamp. All seven passed.

The exchange on Prakash et al.'s patch (preregistration H). For each pair, format and depth, one batch holds seven rows: r_0 = B+KV$_{\mathrm{B}}$ (self), r_1 = M (the patched run), r_2 = B+K_{M}, r_3 = B+V_{M}, r_4 = B+KV$_{\mathrm{M}}$, r_5 = M+K_{B} and r_6 = M+V_{B}, where the keys and values of the patched positions are exchanged in every block after the patched one, and words-only variants of r_2–r_4 exchange them at the state words alone. The residual patch is active only in r_1, r_5 and r_6. Then $\Phi = m(r_1) - m(r_0)$, $\psi_K = [m(r_2) - m(r_0)]/\Phi$, ψ_V likewise with r_3, χ_K and χ_V from r_5 and r_6, and $\kappa = \psi_K/(\psi_K + \psi_V)$, each as a ratio of means over pairs. Because the patch changes nothing before its own block and r_4 gives the clean run every patched key and value after it, r_4 equals r_1 exactly in exact arithmetic; Gate b0 requires the BF16 means of $|m(r_4) - m(r_1)|$ and $|m(r_0) - m(\mathrm{B})|$, with B run unbatched, to stay within 0.3 nats in every cell (observed 0.099–0.194 and 0.141–0.203). In FP32 at Qwen2.5-0.5B the same differences are asserted to be within 10^{-3}.

Code audit. Before the GPU runs, an independent review of the measurement code led to fixes made before any GPU result existed, among them the batched self-clamp baseline, depth-matched onsets, a seeded third value X, core-bootstrap CIs computed in the experiment scripts, the competence filter (now a secondary analysis) and provenance written into every output file.

Mistral-Small-24B tokenizer warning. When loading the Mistral-Small-24B tokenizer, transformers warns that its pre-tokenisation regex is incorrect and suggests `fix_mistral_regex=True`. We kept the default, under which the reproduction gate of preregistration C passed. In stages 4 and 7 we checked that the flag leaves the token ids unchanged, for all

3,600 frames prompts (five formats, both prefills) and for all 1,800 stage-7 ranking and evaluation prompts; we have not checked the natural-factorial prompts.

C.2. The final round: exactness, floors and scoring

Clamps over spans. On passages the writing token is the answer span: the clamps set the key, the value or both at every position of the span, in every layer from 0, to those captured in the donor run (the S, X or Z passage), whose prompt differs from B's only inside the span. Cue-conflict rows take the key from one donor and the value from another. The exactness argument above holds position by position: positions before the span are identical in all runs, and the span's keys and values are set in every layer. In the interventions an edit written at depth l reaches later tokens only through the writing token's keys and values from block $l{+}1$ on (the lemma of Appendix F.1), so every edit is applied as a clamp row whose tables come from a prefix pass with the edit written.

Exactness gates. Each part of the final round ran its own FP32 unit tests on the CPU (Qwen2.5-0.5B-Instruct and tiny random models, tolerance 10^{-4} in log-probabilities or logits) before any model was loaded, and its scorer read the test log as the part's first gate. All passed, none skipped: 67 tests in the passage part (including the clamp, quantization and head-splice tests), 91 in the fresh-story part (the emitted-form scorer against one plain forward per form, the generation under clamps against stepwise decoding, the builders against the discovery-sample prompts), 61 in the mechanism part (the flag hooks, injections, ablations and question arms) and 95 in the intervention part (the lemma, the prefix tables, the dictionary code, the remap trained at the writing token, the reader rows). Each GPU script refused to run unless the code and the entry were those of the finalising commit.

BF16 floors. In BF16 a batched row can differ from the same computation run alone. Every scored quantity is therefore a difference against a reference row of the same batch, and each part has a floor gate. On passages the gate also required the generation of the clamp-to-S row to equal the S prompt's own generation in at least 0.97 of items: it held in 23 of 24 model–format cells and failed in Llama-3.1-8B's options-after cell (0.968, one item short); in all six mismatches both answers name S and differ only in whether a full stop or the end-of-turn token follows, and two clamp-free runs of the same prompts in different batch shapes agree in the same share of items (exploratory, recomputed), so the gate sits at the model's BF16 noise in that format. The mean largest log-probability differences were 0.077–0.230 nats (bound 0.3). The passage heads gate failed at Qwen2.5-7B ($|m(\text{none}) - m(\text{clean})|$ = 0.829 and $|m(\text{all}_T) - m(\text{full})|$ = 0.759 nats, bound 0.500), which the checks traced to Qwen's eager BF16 path (Appendix D.5). The mechanism part's batch floor compares a duplicated reference row with itself in one batch and cannot fail as built; offsets between batches exist and no verdict uses them.

Attention implementation (exploratory). Gemma-2 must run with eager attention; every other model of the round runs with the fused sdpa kernel, and the head-level passes (which read attention weights) run eager. At Qwen2.5-7B, on the first 60 discovery-sample cores, FP32 sdpa agrees with BF16 sdpa, while BF16 eager lowers the sentence read: s_{ID} under SENTENCE-AFTER is 0.300 with eager attention against 0.326 in FP32, and ID_K falls by 12 % (list after) and 17 % (sentence after) (exploratory, recomputed). This exceeds the trie floor's tolerance, which compares paths under one implementation; no Gemma-2-9B verdict lies within 0.03 of its threshold.

The emitted-form score (fresh stories and interventions). A candidate w is scored as the exact chain-rule probability summed over the surface forms the models emit: 32 fixed forms (the sixteen frames `" "`, `""`, `" The "`, `" the "`, `"The "`, `"the "`, `" In the "`, `" in the "`, `" On the "`, `" on the "`, `" At the "`, `" at the "`, `" Inside the "`, `" inside the "`, `" **"` and `"**"`, each followed by w or by w capitalised) and the frames each model was found to emit. All forms of an item are scored in one forward pass over the prompt and a token trie of their proper prefixes (a four-dimensional tree mask with explicit position ids); identical token sequences count once, and no form may be a proper prefix of another. The coverage of a cell is the smallest, over its sixteen clamp and clean rows, of the mean probability mass on the six candidates; it must be at least 0.8, or the statistic falls back to its behavioural counterpart (it never had to: the lowest coverage on the fresh stories was 0.858). A trie floor compared the trie with one plain forward per form on 30 cores in three arms of every model: s_{ID} moved by at most 0.0053 (tolerance 0.02) and the mean log-probability by 0.0477–0.2876 nats (bounds 0.368–1.984). The lower-case score of the discovery sample ($\log p(\text{“ } w\text{”})$) and a 12-form score are computed in the same pass for comparison (Appendix D.3). Under LETTERS-AFTER the forms are the letter with and without a leading space and with “ **”.

Frames and generations. Before any evaluation item of a model, greedy generations of the clean B, S and X runs on calibration cores (630 per model on the fresh stories, 450 in the interventions) give its frames: the text before the first candidate, with the story's names replaced by placeholders, admitted if it occurs in at least 2 % (and at least twice) of some format's generations, at most 16 per model, and only if every item's form set still builds. On the fresh stories no model needed a frame beyond the fixed forms. Generated answers are parsed as described in Appendices B.2 and B.3; the decoder also stops at the chat template's end-of-turn token, because the generation configurations of some models list only another end-of-sequence id.

Model files. Each model's files were fetched at the revision current on the date of the final entry (equal to the revision pinned by earlier stages where one was pinned) and verified against the official sha256 (or git blob id) and size of every file before the model loaded. Llama-3.1-8B-Instruct, Gemma-2-9B-it and Gemma-2-2B-it, whose official repositories are gated, were assembled from byte-identical public copies, each file checked against the official hash; they are the official weights (sha256-verified). Mistral-7B's `tokenizer.model` is pinned by the hash of the object its repository's pointer file names. The sparse-autoencoder dictionaries and the release of Prakash et al. (2026) were checked against pinned hashes in the same way.

Checks after each run. For every part of the final round, re-running the committed scorer on the archived results reproduced its score file byte for byte, and four further checks were made before the outcome was recorded (five for the interventions): a recomputation from the raw files that imports no scorer code (every point estimate, interval decision and verdict; its own bootstrap seeds flipped no interval decision), an audit that re-derives every gate, verdict, tally, expected count, Brier score and the Holm table from the printed numbers and the entry's rules, a search for anomalies in the run data (the source of every sensitivity reported in Appendices D to F), and an integrity check of the manifest, the model files and the timeline. The discrepancies they found are cosmetic and listed in the outcome records; the release copies of the results are scrubbed of local paths.

D. Later mentions decide the route: details

D.1. The discovery sample

The discovery sample is the 10 models and the story cores of entries A–I: Qwen2.5-Instruct at 1.5B, 3B, 7B, 14B, 32B and 72B, Qwen3-8B, Mistral-7B, Mistral-Small-24B and OLMo-2-7B (three developers), with 150 stories per format and model unless stated. Its results are reported as they were scored in the earlier entries, with their labels: a preregistered result names its prediction (B1 is prediction 1 of preregistration B; Appendix A.5), and the others are marked exploratory or post hoc. These results use the lower-case score; the final round re-scored the five discovery-sample verdicts that depend on the scoring and found every one unchanged under the forms the models emit (Appendix D.3). Unless stated otherwise, intervals are 95% percentile intervals from 10,000 bootstrap resamples of stories (cores), with ratios recomputed within each resample. Each estimate is printed with its interval below it, except in Tables 20 and 21, where the intervals of the individual effects are in the result summaries released with the code.

Natural formats. Over all 10 models (Table 18), the formats fall into three regimes.

Candidates listed after the writing token. Under OPTIONS-AFTER and LETTERS-AFTER, ID_K is large and positive in every model, from 2.8 nats at Qwen2.5-1.5B to 39.0 at Qwen2.5-14B under OPTIONS-AFTER (A1 at 1.5B on new cores of seed 1 and B1 in five held-out models from 3B to 8B, both met; exploratory at 14B and 24–72B).

No later mention. Under NO-MENTION, ID_K is at most 1.25 nats, less than a tenth of its OPTIONS-AFTER value in every model (paired difference positive in 5/5 held-out models; B2, met), and the identity is mainly copied: ID_V exceeds ID_K more than tenfold (exploratory; Table 20). LIST-BEFORE names the same candidates before the writing token, which the causal mask keeps them from attending to; its ID_K is never reliably positive, and negative in nine of ten models (C4's 0.5-nat bound met; sign unexplained).

A neutral sentence after the story. SENTENCE-AFTER carries no positive key identity at Qwen2.5-1.5B and 3B, but an intermediate amount in every model from 7B up (2.0–12.1 nats), which we did not predict. The scale-free identity key share s_{ID} above 1.5B is 0.67–0.88 under OPTIONS-AFTER and at most 0.45 under SENTENCE-AFTER and 0.06 under NO-MENTION (exploratory; Table 19).

Table 17. **Position decides the key read** (discovery sample: the stories of stages 1 and 3b). The same list or neutral sentence, after or before the story, with "Answer with one word"; NO-MENTION for reference. ID_K (nats, 95% CIs), $n = 150$, keys clamped from layer 0. Preregistration E (E1).

Format	Qwen-7B	Qwen-14B	Mistral-7B	OLMo-2-7B
LIST-AFTER	+20.6 [20.2, 21.0]	+35.8 [34.9, 36.7]	+15.6 [15.1, 16.0]	+8.7 [8.4, 8.9]
SENTENCE-AFTER	+5.5 [5.2, 5.8]	+12.1 [11.5, 12.7]	+7.9 [7.3, 8.4]	+2.0 [1.8, 2.2]
LIST-BEFORE	−0.1 [−0.2, −0.1]	−0.1 [−0.2, 0.0]	0.0 [−0.1, 0.1]	−0.3 [−0.4, −0.2]
SENTENCE-BEFORE	0.0 [−0.1, 0.1]	−0.1 [−0.2, 0.0]	0.0 [−0.1, 0.1]	−0.2 [−0.3, −0.2]
NO-MENTION	+1.0 [0.9, 1.2]	+1.1 [0.9, 1.2]	+0.6 [0.5, 0.6]	+0.4 [0.3, 0.5]

An instruction-matched 2×2. Preregistration E crosses {list, sentence} with {before, after} under one instruction (Table 17); all parts of E1 were met. ID_K is 8.7–35.8 nats under LIST-AFTER and 2.0–12.1 under SENTENCE-AFTER, every CI before the story contains zero or lies below it, and both paired differences exclude zero in 4/4 models. Position, not the instruction or the list format, decides whether the key channel opens.

Across tasks, families and scale. On the paint and schedule tasks, in five 7–14B models, the paired OPTIONS-AFTER−NO-MENTION difference in ID_K is positive in 5/5 models per task (D1, met), and LIST-BEFORE stays within the preregistered 0.5-nat bound in 10/10 cells (D2, met; Table 23). On IOI (Wang et al., 2022), plain sentences carry almost no key identity (G18, met), but the preregistered test that a later list opens a key read was not met (G19; Mistral-7B failed the competence gate; Appendix E.4.1). Within Qwen2.5, the OPTIONS-AFTER key share s_K rises from 0.37 at 1.5B

to 0.91 at 14B and falls to 0.76 at 72B (B3, exploratory; Figure 5). The fall is not an artefact of the software or GPU setup (E3, met), but its cause is unknown. Other families give 0.73–0.92 (deeper clamp onsets: Table 21).

Natural factorial in all models. Table 18 gives the OPTIONS-AFTER key share and ID_K in every format for all 10 models, and Figure 6 shows ID_K relative to OPTIONS-AFTER. LIST-BEFORE is negative in every model except Mistral-7B, where it is indistinguishable from zero; at Qwen2.5-72B it reaches -2.0 nats.

Table 18. **Natural factorial, all models.** Key share s_K under OPTIONS-AFTER and value identity carried by keys, ID_K (nats), in each format, with keys and values clamped from layer 0. $n = 150$ stories per format and model; 95% bootstrap CIs over stories (ratio recomputed in each resample for s_K). Models ordered by size. Rows up to 14B are from GPU stage 1, whose predictions B1–B2 cover OPTIONS-AFTER, LETTERS-AFTER and NO-MENTION in Qwen2.5-3B/7B, Qwen3-8B, Mistral-7B and OLMo-2-7B; rows from 24B are from stage 2, whose prediction C4 covers s_K and LIST-BEFORE. The other cells, including the 1.5B and 14B rows, are exploratory; preregistration A tested 1.5B in a separate run (Table 8).

Model	s_K (OPTIONS-AFTER)	ID_K, identity carried by keys (nats)				
		LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
Qwen2.5-1.5B	0.37 [0.36, 0.39]	+3.6 [+3.3, +3.8]	+2.8 [+2.6, +3.1]	-0.5 [-0.6, -0.4]	+0.2 [+0.1, +0.2]	-0.2 [-0.3, -0.2]
Qwen2.5-3B	0.69 [0.68, 0.71]	+11.4 [+10.4, +12.4]	+10.3 [+9.6, +11.1]	-0.0 [-0.4, +0.3]	+0.1 [-0.0, +0.2]	-0.5 [-0.6, -0.3]
Mistral-7B	0.92 [0.90, 0.93]	+15.2 [+14.6, +15.7]	+14.9 [+14.5, +15.3]	+7.9 [+7.3, +8.4]	+0.6 [+0.5, +0.6]	+0.0 [-0.1, +0.1]
OLMo-2-7B	0.73 [0.71, 0.74]	+7.9 [+7.7, +8.0]	+8.2 [+8.0, +8.4]	+2.0 [+1.8, +2.2]	+0.4 [+0.3, +0.5]	-0.3 [-0.4, -0.2]
Qwen2.5-7B	0.82 [0.81, 0.83]	+20.4 [+20.0, +20.8]	+21.2 [+20.7, +21.6]	+5.5 [+5.2, +5.8]	+1.0 [+0.9, +1.2]	-0.1 [-0.2, -0.1]
Qwen3-8B	0.84 [0.83, 0.84]	+33.9 [+33.4, +34.3]	+35.4 [+34.6, +36.2]	+8.8 [+8.3, +9.3]	+1.2 [+1.0, +1.4]	-0.3 [-0.4, -0.3]
Qwen2.5-14B	0.91 [0.90, 0.91]	+34.8 [+34.5, +35.1]	+39.0 [+38.0, +40.0]	+12.1 [+11.5, +12.7]	+1.1 [+0.9, +1.2]	-0.1 [-0.2, -0.0]
Mistral-24B	0.78 [0.77, 0.79]	+9.2 [+9.1, +9.3]	+9.6 [+9.4, +9.8]	+5.6 [+5.4, +5.9]	+0.4 [+0.3, +0.5]	-0.5 [-0.6, -0.5]
Qwen2.5-32B	0.86 [0.85, 0.87]	+34.8 [+34.5, +35.1]	+32.4 [+31.6, +33.2]	+10.3 [+9.6, +10.9]	+1.1 [+0.9, +1.3]	-0.4 [-0.6, -0.3]
Qwen2.5-72B	0.76 [0.75, 0.77]	+26.4 [+26.2, +26.6]	+25.6 [+25.0, +26.2]	+8.4 [+8.0, +8.9]	+0.5 [+0.4, +0.6]	-2.0 [-2.2, -1.8]

Scale and families. Within Qwen2.5, the OPTIONS-AFTER key share rises from 0.37 at 1.5B through 0.69 (3B) and 0.82 (7B) to 0.91 at 14B, then falls to 0.86 at 32B and 0.76 at 72B (Figure 5); the 14B and 72B intervals do not overlap. The size trend was exploratory (B3), and the preregistered threshold $s_K \geq 0.80$ for the three largest models was met only at 32B (C4). Re-running Qwen2.5-14B and 32B each in the other's software and GPU setup gave 0.91 and 0.86, the originals to two decimals (E3, met). The fall beyond 14B is therefore not an artefact of the environment, and it is also seen in the identity key share (0.88 at 14B, 0.71 at 72B), although we do not know its cause. The other families give similar shares (0.73–0.92), and every model except Qwen2.5-1.5B carries more than half of the OPTIONS-AFTER effect in the key.

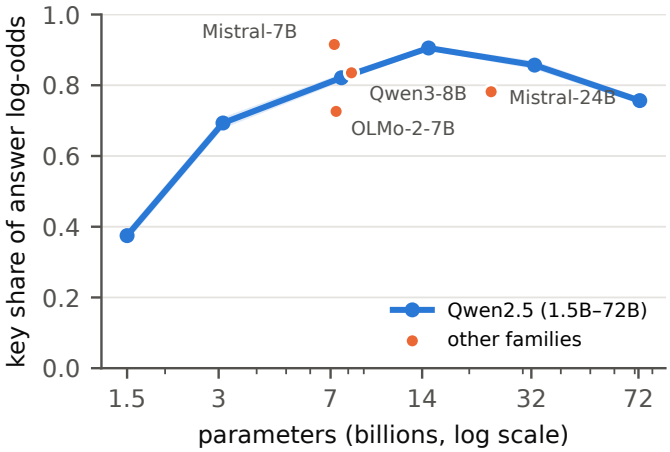


Figure 5. **Key share of the OPTIONS-AFTER answer across scale and families.** $s_K = \bar{d}_K/(\bar{d}_K + \bar{d}_V)$ with keys and values clamped from layer 0. Line and band: Qwen2.5-Instruct 1.5B–72B with 95% bootstrap CIs; points: other families. $n = 150$ stories per model; values with CIs in Table 18. s_K, the preregistered measure, includes non-specific effects; the identity key share s_{ID} shows the same rise to 14B and fall beyond (Table 19). Exploratory (B3), with the threshold of C4.

Identity key share. Table 19 normalises ID_K by the total identity effect, $s_{\mathrm{ID}} = \overline{\mathrm{ID}}_K/(\overline{\mathrm{ID}}_K + \overline{\mathrm{ID}}_V)$, so that formats and models with different confidence can be compared. It tracks s_K closely except under NO-MENTION and LIST-BEFORE, where s_K also counts the non-specific effect of a foreign key. Above 1.5B it is 0.94–1.00 under LETTERS-AFTER and at most 0.00 under LIST-BEFORE (exploratory); the table gives the other formats.

Table 19. **Identity key share s_{ID}, all models** (exploratory). Ratio of mean ID_K to mean $\mathrm{ID}_K + \mathrm{ID}_V$, keys and values clamped from layer 0; $n = 150$ stories per format and model; 95% bootstrap CIs, ratio recomputed in each resample.

Model	LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
Qwen2.5-1.5B	+0.83 [+0.81, +0.86]	+0.36 [+0.34, +0.38]	−0.08 [−0.10, −0.06]	+0.02 [+0.01, +0.02]	−0.04 [−0.05, −0.03]
Qwen2.5-3B	+0.98 [+0.97, +0.99]	+0.67 [+0.66, +0.69]	0.00 [−0.04, +0.03]	0.00 [0.00, +0.01]	−0.05 [−0.06, −0.04]
Mistral-7B	+0.94 [+0.92, +0.95]	+0.87 [+0.85, +0.88]	+0.41 [+0.39, +0.43]	+0.04 [+0.03, +0.04]	0.00 [0.00, 0.00]
OLMo-2-7B	+0.99 [+0.99, +0.99]	+0.69 [+0.68, +0.70]	+0.19 [+0.17, +0.20]	+0.03 [+0.03, +0.04]	−0.03 [−0.03, −0.02]
Qwen2.5-7B	+0.99 [+0.98, +0.99]	+0.79 [+0.77, +0.80]	+0.32 [+0.30, +0.33]	+0.06 [+0.05, +0.06]	−0.01 [−0.01, 0.00]
Qwen3-8B	+0.97 [+0.97, +0.98]	+0.81 [+0.80, +0.82]	+0.29 [+0.28, +0.30]	+0.05 [+0.04, +0.05]	−0.01 [−0.02, −0.01]
Qwen2.5-14B	+1.00 [+1.00, +1.00]	+0.88 [+0.87, +0.89]	+0.40 [+0.38, +0.41]	+0.05 [+0.05, +0.06]	0.00 [−0.01, 0.00]
Mistral-24B	+0.98 [+0.98, +0.98]	+0.73 [+0.72, +0.74]	+0.45 [+0.44, +0.46]	+0.04 [+0.03, +0.05]	−0.06 [−0.07, −0.05]
Qwen2.5-32B	+0.99 [+0.99, +0.99]	+0.82 [+0.81, +0.83]	+0.30 [+0.29, +0.32]	+0.04 [+0.04, +0.05]	−0.01 [−0.02, −0.01]
Qwen2.5-72B	+0.99 [+0.99, +0.99]	+0.71 [+0.70, +0.72]	+0.33 [+0.31, +0.34]	+0.02 [+0.02, +0.02]	−0.09 [−0.10, −0.08]

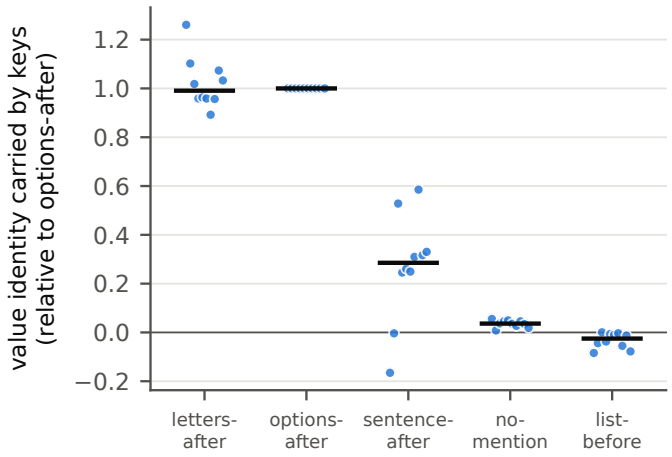


Figure 6. **Key identity by format, relative to OPTIONS-AFTER.** Each point is one of the 10 models: ID_K in that format divided by ID_K under OPTIONS-AFTER in the same model (so OPTIONS-AFTER is 1 by construction); points are jittered horizontally, and the horizontal line is the median over models. $n = 150$ stories per format and model; per-model values with 95% bootstrap CIs are in Table 18.

Decomposition. Table 20 gives, for each model and format, the effects of clamping the key, the value or both, the interaction $d_{KV} - d_K - d_V$, and ID_V. The interaction is small relative to d_{KV} in every cell, so the two channels' effects add up. Under LETTERS-AFTER the value channel carries almost no identity in any model, and under NO-MENTION and LIST-BEFORE it carries most of it: there the identity is copied. In the instruction-matched 2×2 (Table 17), ID_K is larger under LIST-AFTER than under SENTENCE-AFTER in all four models, and LIST-AFTER is close to OPTIONS-AFTER in the same model; neither comparison was preregistered.

Table 20. **Key/value decomposition.** Mean effects (nats of m_{SB}) of clamping the writing token's key (d_K), value (d_V) or both (d_{KV}) to the source run's from layer 0, measured against the self-clamp row; the interaction $d_{KV} - d_K - d_V$ with its 95% bootstrap CI; and the value identity ID_V (nats). $n = 150$ stories per format and model.

Model	Format	d_K	d_V	d_{KV}	interaction	ID_V
Qwen2.5-1.5B	LETTERS-AFTER	+9.9	+4.2	+14.5	+0.4 [-0.5, +1.3]	+0.7
	OPTIONS-AFTER	+6.3	+10.4	+16.7	-0.0 [-0.5, +0.5]	+5.0
	SENTENCE-AFTER	+0.5	+14.4	+14.8	-0.0 [-0.4, +0.3]	+6.5
	NO-MENTION	+1.8	+21.8	+23.5	-0.1 [-0.7, +0.5]	+10.2
	LIST-BEFORE	+1.2	+13.6	+14.5	-0.3 [-0.6, +0.1]	+5.9
Qwen2.5-3B	LETTERS-AFTER	+22.3	+0.8	+22.8	-0.4 [-1.1, +0.3]	+0.2
	OPTIONS-AFTER	+20.4	+9.0	+29.1	-0.4 [-1.1, +0.3]	+5.0
	SENTENCE-AFTER	+2.9	+26.8	+29.9	+0.2 [-0.6, +1.0]	+11.8
	NO-MENTION	+4.3	+41.1	+45.9	+0.5 [-0.1, +1.1]	+18.1
	LIST-BEFORE	+2.8	+23.4	+26.4	+0.2 [-0.5, +0.8]	+9.6
Mistral-7B	LETTERS-AFTER	+31.3	+1.5	+32.2	-0.5 [-1.2, +0.1]	+1.0
	OPTIONS-AFTER	+28.0	+2.6	+30.8	+0.1 [-0.6, +0.8]	+2.3
	SENTENCE-AFTER	+12.9	+19.5	+31.8	-0.6 [-2.3, +1.1]	+11.5
	NO-MENTION	+2.7	+28.8	+31.5	-0.1 [-0.4, +0.3]	+13.6
	LIST-BEFORE	+1.2	+30.9	+32.1	+0.1 [-0.3, +0.5]	+14.6
OLMo-2-7B	LETTERS-AFTER	+15.6	+0.1	+15.7	+0.0 [-0.1, +0.1]	+0.1
	OPTIONS-AFTER	+15.5	+5.8	+21.0	-0.3 [-0.9, +0.2]	+3.7
	SENTENCE-AFTER	+3.8	+16.8	+20.7	+0.0 [-0.5, +0.6]	+8.6
	NO-MENTION	+2.7	+23.5	+26.1	-0.1 [-0.5, +0.3]	+11.0
	LIST-BEFORE	+0.2	+23.6	+23.9	+0.1 [-0.1, +0.4]	+11.2
Qwen2.5-7B	LETTERS-AFTER	+40.2	+1.1	+41.2	-0.1 [-0.7, +0.4]	+0.2
	OPTIONS-AFTER	+39.6	+8.6	+47.9	-0.4 [-1.1, +0.3]	+5.8
	SENTENCE-AFTER	+11.0	+23.1	+34.4	+0.4 [-0.4, +1.1]	+11.9
	NO-MENTION	+6.0	+36.8	+42.5	-0.3 [-0.7, +0.2]	+16.6
	LIST-BEFORE	+2.4	+34.8	+36.8	-0.4 [-0.8, -0.1]	+16.0
Qwen3-8B	LETTERS-AFTER	+66.8	+1.2	+67.9	-0.1 [-0.5, +0.3]	+1.0
	OPTIONS-AFTER	+67.1	+13.2	+80.5	+0.2 [-1.4, +1.7]	+8.2
	SENTENCE-AFTER	+14.3	+40.0	+53.3	-1.1 [-2.5, +0.3]	+21.8
	NO-MENTION	+5.7	+54.4	+59.4	-0.7 [-1.6, +0.2]	+25.3
	LIST-BEFORE	+2.9	+51.8	+54.4	-0.3 [-0.8, +0.2]	+24.0
Qwen2.5-14B	LETTERS-AFTER	+70.0	+0.2	+70.2	+0.0 [-0.2, +0.3]	+0.1
	OPTIONS-AFTER	+75.0	+7.8	+82.8	-0.1 [-0.7, +0.5]	+5.4
	SENTENCE-AFTER	+21.9	+33.8	+56.9	+1.1 [-1.4, +3.6]	+18.5
	NO-MENTION	+10.8	+46.5	+57.1	-0.2 [-0.9, +0.4]	+19.1
	LIST-BEFORE	+2.4	+65.9	+68.3	+0.0 [-0.5, +0.5]	+31.1
Mistral-24B	LETTERS-AFTER	+18.7	+0.4	+19.0	-0.1 [-0.2, -0.0]	+0.2
	OPTIONS-AFTER	+17.1	+4.8	+21.9	+0.1 [-0.2, +0.4]	+3.6
	SENTENCE-AFTER	+8.1	+10.4	+18.6	+0.1 [-0.7, +0.8]	+6.8
	NO-MENTION	+2.0	+20.8	+22.6	-0.2 [-0.4, +0.0]	+9.8
	LIST-BEFORE	-0.0	+19.4	+19.4	+0.1 [-0.1, +0.3]	+9.1
Qwen2.5-32B	LETTERS-AFTER	+69.5	+0.6	+70.3	+0.2 [-0.1, +0.5]	+0.3
	OPTIONS-AFTER	+61.4	+10.2	+71.6	-0.1 [-1.8, +1.7]	+7.1
	SENTENCE-AFTER	+18.7	+44.8	+63.1	-0.4 [-2.0, +1.3]	+23.4
	NO-MENTION	+9.8	+53.8	+63.4	-0.2 [-0.9, +0.5]	+23.4
	LIST-BEFORE	+2.8	+65.1	+68.1	+0.1 [-0.5, +0.7]	+30.6
Qwen2.5-72B	LETTERS-AFTER	+52.8	+0.2	+53.0	-0.0 [-0.2, +0.2]	+0.3
	OPTIONS-AFTER	+44.6	+14.3	+59.6	+0.7 [-0.3, +1.7]	+10.5
	SENTENCE-AFTER	+14.4	+30.9	+46.4	+1.1 [-0.3, +2.6]	+17.5
	NO-MENTION	+3.1	+47.7	+50.9	+0.1 [-0.4, +0.5]	+22.9
	LIST-BEFORE	-1.7	+51.2	+49.8	+0.3 [-0.2, +0.7]	+24.1

Clamp onset. Layer-0 keys are a function of the input embedding alone, so Table 21 starts both clamps deeper (exploratory). With $\ell_0 \approx 0.06L$ the shares are essentially unchanged in eight of ten models. With $\ell_0 \approx 0.3L$ the share stays at

or above one half in seven models (up to 0.83 at Qwen2.5-14B), but falls to 0.19 at Qwen2.5-1.5B, 0.37 at Mistral-Small-24B and 0.03 at OLMo-2-7B, where the key effect depends on the lowest layers. Deeper keys therefore still carry the written word in most models. With $\ell_0 > 0$ the clamp is no longer exact, and at $\ell_0 \approx 0.3L$ the key and value effects no longer add up: the interaction is positive in every model (result summaries). The share at that onset is therefore descriptive only.

Table 21. **Key share by clamp onset.** $s_K = \bar{d}_K/(\bar{d}_K + \bar{d}_V)$ under OPTIONS-AFTER with both clamps applied from layer ℓ_0 on; the onset used in each model is given in brackets. $n = 150$ stories per model; point estimates (the $\ell_0 = 0$ intervals are in Table 18). Exploratory.

Model	layers L	s_K, onset ℓ_0=0	s_K, $\ell_0 \approx 0.06L$	s_K, $\ell_0 \approx 0.3L$
Qwen2.5-1.5B	28	0.37 (ℓ_0=0)	0.37 (ℓ_0=2)	0.19 (ℓ_0=8)
Qwen2.5-3B	36	0.69 (ℓ_0=0)	0.69 (ℓ_0=2)	0.56 (ℓ_0=11)
Mistral-7B	32	0.92 (ℓ_0=0)	0.91 (ℓ_0=2)	0.53 (ℓ_0=10)
OLMo-2-7B	32	0.73 (ℓ_0=0)	0.72 (ℓ_0=2)	0.03 (ℓ_0=10)
Qwen2.5-7B	28	0.82 (ℓ_0=0)	0.82 (ℓ_0=2)	0.66 (ℓ_0=8)
Qwen3-8B	36	0.84 (ℓ_0=0)	0.84 (ℓ_0=2)	0.73 (ℓ_0=11)
Qwen2.5-14B	48	0.91 (ℓ_0=0)	0.91 (ℓ_0=3)	0.83 (ℓ_0=14)
Mistral-24B	40	0.78 (ℓ_0=0)	0.78 (ℓ_0=2)	0.37 (ℓ_0=12)
Qwen2.5-32B	64	0.86 (ℓ_0=0)	0.86 (ℓ_0=4)	0.82 (ℓ_0=19)
Qwen2.5-72B	80	0.76 (ℓ_0=0)	0.76 (ℓ_0=5)	0.55 (ℓ_0=24)

Layer windows and KV groups. Table 22 refines the row-restricted splice at Qwen2.5-1.5B (28 layers, two KV groups). Only the option-word rows see the source key, and the swap is further limited to a window of four layers (all KV groups) or to one KV group's slice of the key (all layers). The read happens in layers 4–15, and KV group 1 carries more of it than KV group 0 in both formats. The seven windows together recover less than the option-word rows do with the swap at all layers, and the two single-group swaps together recover less than both groups at once, so the read compounds across layers and groups. We ran the same window and KV-group splices, on the option-word rows under OPTIONS-AFTER and LETTERS-AFTER, at Qwen2.5-7B, Qwen2.5-14B and Mistral-7B (exploratory; result summaries). In these models no single four-layer window and no single KV group recovers most of the option-word effect, so the dominance of one KV group seen at 1.5B does not carry over.

Table 22. **Depth and KV-group localisation of the option-word key read** (Qwen2.5-1.5B, FP32). Entries are fractions of the effect obtained when the option-word rows see the source key in every layer and both KV groups, with 95% bootstrap CIs over stories (ratio of means recomputed in each resample). $n = 40$ stories per format. Exploratory.

Rows/layers that see the swapped key	OPTIONS-AFTER	LETTERS-AFTER
layers 0–3	-0.02 [-0.02, -0.01]	-0.01 [-0.01, -0.00]
layers 4–7	+0.23 [+0.11, +0.35]	+0.36 [+0.26, +0.46]
layers 8–11	+0.14 [+0.10, +0.18]	+0.06 [+0.03, +0.09]
layers 12–15	+0.15 [+0.12, +0.19]	+0.04 [+0.02, +0.06]
layers 16–19	+0.02 [+0.01, +0.04]	+0.01 [+0.00, +0.02]
layers 20–23	+0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]
layers 24–27	+0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]
KV group 0 (all layers)	+0.22 [+0.17, +0.27]	+0.06 [+0.03, +0.09]
KV group 1 (all layers)	+0.53 [+0.40, +0.66]	+0.66 [+0.57, +0.75]

New tasks. Table 23 gives the paint and schedule results. In every model and both tasks, ID_K is large under LETTERS-AFTER and OPTIONS-AFTER, intermediate under SENTENCE-AFTER, small under NO-MENTION and near zero under LIST-BEFORE. The OPTIONS-AFTER key share ranges from 0.47 to 0.92 (paint) and from 0.35 to 0.92 (schedule), lowest at OLMo-2-7B in both.

Table 23. **Paint and schedule tasks.** Key share under OPTIONS-AFTER and ID_K (nats) in each format, clamped from layer 0. $n = 150$ stories per task, format and model; 95% bootstrap CIs. Preregistration D (D1, D2) for OPTIONS-AFTER, NO-MENTION and LIST-BEFORE; the rest exploratory.

Task	Model	s_K (OPTIONS-AFTER)	ID_K (nats)				
			LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
paint	Qwen2.5-7B	0.73 [0.72, 0.74]	+21.5 [+21.3, +21.7]	+21.4 [+21.2, +21.7]	+3.0 [+2.8, +3.1]	+1.2 [+1.1, +1.3]	+0.3 [+0.2, +0.4]
	Qwen2.5-14B	0.90 [0.90, 0.91]	+35.7 [+35.4, +36.0]	+40.3 [+39.9, +40.8]	+3.5 [+3.3, +3.7]	+0.8 [+0.7, +0.9]	+0.2 [+0.1, +0.2]
	Qwen3-8B	0.92 [0.92, 0.93]	+39.7 [+39.4, +40.1]	+42.7 [+42.3, +43.0]	+4.9 [+4.7, +5.2]	+1.1 [+1.0, +1.2]	+0.1 [+0.1, +0.2]
	Mistral-7B	0.86 [0.85, 0.87]	+16.0 [+15.7, +16.3]	+16.1 [+15.9, +16.4]	+2.2 [+1.9, +2.4]	+0.4 [+0.3, +0.5]	+0.1 [+0.1, +0.2]
	OLMo-2-7B	0.47 [0.46, 0.48]	+7.4 [+7.2, +7.5]	+6.5 [+6.3, +6.7]	+0.7 [+0.6, +0.8]	+0.1 [+0.1, +0.2]	−0.1 [−0.1, −0.1]
schedule	Qwen2.5-7B	0.58 [0.56, 0.59]	+19.6 [+19.3, +20.0]	+12.7 [+12.4, +13.0]	+4.1 [+3.8, +4.4]	+0.1 [+0.1, +0.2]	−0.1 [−0.2, −0.1]
	Qwen2.5-14B	0.87 [0.87, 0.88]	+35.6 [+35.3, +35.9]	+26.1 [+25.7, +26.5]	+10.9 [+10.3, +11.6]	+0.6 [+0.5, +0.6]	+0.2 [+0.1, +0.2]
	Qwen3-8B	0.92 [0.91, 0.92]	+39.6 [+39.2, +40.0]	+28.5 [+28.0, +29.0]	+7.3 [+6.9, +7.7]	+0.9 [+0.8, +1.0]	+0.2 [+0.1, +0.2]
	Mistral-7B	0.82 [0.81, 0.84]	+16.2 [+15.6, +16.7]	+13.2 [+12.9, +13.5]	+5.1 [+4.6, +5.6]	+0.1 [+0.1, +0.2]	0.0 [−0.1, 0.0]
	OLMo-2-7B	0.35 [0.34, 0.37]	+7.4 [+7.2, +7.6]	+4.5 [+4.3, +4.7]	+2.3 [+2.1, +2.4]	+0.1 [+0.1, +0.1]	0.0 [0.0, +0.1]

Role control. Could the writing token's channels carry more than identity, such as whether the person asked about saw the move? The role-swap control exchanges the observer's and the non-observer's names but keeps the location word (Appendix B). Table 24 gives the role-swap effect and the fractions of it that the writing token's key and value carry. The swap shifts the belief answer by 12.5 to 79.8 nats, depending on model and format, and clamping either channel alone to the role-swapped run carries at most 0.02 of that shift, under OPTIONS-AFTER, NO-MENTION and LETTERS-AFTER, in five models including Qwen2.5-72B. The prediction was preregistered for OPTIONS-AFTER and NO-MENTION in four models, with a bound of 0.15 (E2, met); LETTERS-AFTER and Qwen2.5-72B are exploratory. The writing token's key and value thus carry *which* value was written, not the observer's name; an ordering-ID address of the observer, the same in both runs, would not register (Appendix I).

Competent-only analysis. Our primary analyses pool all stories, whether or not the model answers the clean runs correctly. As a secondary analysis, every natural-factorial quantity is also computed on the stories whose base and source clean runs are both answered correctly (argmax over the candidates). From 7B up most stories are competent and the two analyses agree closely. The competent subset is smallest at Qwen2.5-1.5B and 3B, and the estimates move most at 3B. The sign of ID_K under OPTIONS-AFTER, LETTERS-AFTER, NO-MENTION and LIST-BEFORE is unchanged in every model. The near-zero SENTENCE-AFTER value at Qwen2.5-3B becomes slightly positive, with a CI that excludes zero, but stays far below the SENTENCE-AFTER values from 7B up. No conclusion in the main text changes.

D.2. Fresh stories

Design. The fresh stories (Appendix B.2) were run in eight models: the four discovery-sample models of the instruction-matched 2×2 (Qwen2.5-7B, Qwen2.5-14B, Mistral-7B, OLMo-2-7B; lines with the suffix P4f, which must hold in every evaluable model, at 95 %) and four model families never examined before (Llama-3.1-8B, Gemma-2-9B, Phi-4, Falcon3-7B; lines with the suffix N4, which need three of the four, at 98.75 %), plus Gemma-2-2B for the sentence read at small scale. Every statistic uses the emitted-form score (Appendix C.2); its coverage was at least 0.858 in every cell, so no statistic fell back to its behavioural counterpart. Intervals are two-stage bootstrap intervals over the 55 (lexicon, base, source) cells and the cores within them. Generated accuracy on the clean base run was at least 0.987 in every arm of the eight models. Table 25 gives every model and arm.

The list and no mention (J-B1, J-B4). With the list after the story the key carries most of the value's identity in all eight models: s_{ID} is 0.628–0.877 in the discovery-sample models (95 % lower bounds at least 0.608; J-B1-P4f, replication, held

Table 24. **Identity, not belief role.** Role effect: mean change in $m = \log p(\text{initial}) - \log p(\text{LOC})$ when the observer and non-observer are swapped with the same location word (nats). f_K, f_V: fraction of that change reproduced by clamping only the writing token's key or value (all layers) to the role-swapped run's, as a ratio of means. Direct question; $n = 150$ stories per format and model; 95% bootstrap CIs. Preregistration E (E2) for OPTIONS-AFTER and NO-MENTION in the four 7–14B models; LETTERS-AFTER and Qwen2.5-72B exploratory.

Model	Format	role effect (nats)	f_K	f_V
Qwen2.5-7B	OPTIONS-AFTER	+34.4	+0.006 [+0.004, +0.008]	+0.010 [+0.007, +0.013]
	NO-MENTION	+28.7	+0.014 [+0.011, +0.018]	+0.001 [0.000, +0.003]
	LETTERS-AFTER	+29.8	+0.003 [0.000, +0.005]	+0.018 [+0.013, +0.025]
Qwen2.5-14B	OPTIONS-AFTER	+79.8	+0.001 [0.000, +0.001]	0.000 [−0.001, 0.000]
	NO-MENTION	+51.4	+0.001 [0.000, +0.002]	0.000 [−0.001, +0.001]
	LETTERS-AFTER	+66.0	0.000 [−0.001, 0.000]	0.000 [−0.001, 0.000]
Mistral-7B	OPTIONS-AFTER	+28.4	+0.003 [+0.002, +0.004]	+0.004 [+0.001, +0.008]
	NO-MENTION	+26.6	+0.019 [+0.014, +0.025]	+0.001 [0.000, +0.003]
	LETTERS-AFTER	+28.9	0.000 [−0.001, +0.002]	+0.011 [+0.006, +0.017]
OLMo-2-7B	OPTIONS-AFTER	+16.9	+0.004 [+0.003, +0.006]	+0.002 [0.000, +0.003]
	NO-MENTION	+17.7	+0.014 [+0.011, +0.017]	+0.002 [0.000, +0.004]
	LETTERS-AFTER	+12.5	0.000 [−0.001, +0.001]	+0.003 [+0.001, +0.004]
Qwen2.5-72B	OPTIONS-AFTER	+50.6	+0.001 [0.000, +0.002]	+0.002 [+0.001, +0.003]
	NO-MENTION	+38.4	+0.007 [+0.006, +0.009]	−0.001 [−0.002, +0.001]
	LETTERS-AFTER	+46.8	−0.001 [−0.002, 0.000]	+0.003 [+0.002, +0.004]

in 4 of 4) and 0.612–0.784 in the new families (98.75 % lower bounds at least 0.582; J-B1-N4, extrapolation, held in 4 of 4). Without a later mention s_{ID} is at most 0.074 in every model and the value carries the identity (J-B4-P4f and J-B4-N4, held in 4 of 4 and 4 of 4).

Mentions before the story (J-B2, J-B-LB). The same list or sentence placed before the story opens no key read: the upper bounds of $r(\text{BEFORE}) - r(\text{NONE})$ and $r(\text{PRE}) - r(\text{NONE})$ are at most -0.008 in the discovery-sample models and -0.014 in the new families, and those of s_{ID} under both arms at most 0.002 and -0.006 (J-B2-P4f and J-B2-N4, held in 4 of 4 and 4 of 4). In all four new families a list before the story even lowers the key read below its no-mention level (risky line J-B-LB-N4, held in 4 of 4): the 98.75 % upper bounds of $\text{ID}_K(\text{BEFORE}) - \text{ID}_K(\text{NONE})$ lie between -0.885 and -0.205 nats. The point estimates have the same sign in the four discovery-sample models (-0.298 to -1.293 nats) and at Gemma-2-2B (-0.172; both exploratory). We cannot explain it; the discovery sample showed the same sign in nine of ten models (Table 18).

The sentence and its null control (J-B3, J-B-NULL). A neutral sentence naming the candidates after the story opens an intermediate key read in all eight models: r = 0.180–0.569 in the discovery-sample models (J-B3-P4f, extrapolation,

Table 25. **Fresh stories, eight models.** 150 fresh cores per model; emitted-form score; 95 % two-stage cluster-bootstrap intervals (the N4 lines are decided on 98.75 % bounds, given in the text). Arms: AFTER = LIST-AFTER, POST = SENTENCE-AFTER, POST-NULL = NULL-SENTENCE, PRE = SENTENCE-BEFORE, BEFORE = LIST-BEFORE, NONE = NO-MENTION. r: $\overline{\mathrm{ID}}_K$ in the arm divided by $\overline{\mathrm{ID}}_K$ under AFTER (1 by definition there); s_{ID}: the key share of the value's identity; β_K, β_V: the share of generated answers that move to the clamped value under the key-only or value-only clamp, net of moves that a clamp to the third value also causes. The first four models are the discovery-sample models (P4), the next four the new families (N4).

Model	Arm	r	s_{ID}	β_K	β_V
Qwen2.5-7B	AFTER	1	0.79 [0.77, 0.81]	0.99 [0.98, 1.00]	0.00 [0.00, 0.00]
Qwen2.5-7B	POST	0.27 [0.24, 0.30]	0.33 [0.30, 0.35]	0.11 [0.06, 0.16]	0.86 [0.79, 0.92]
Qwen2.5-7B	POST-NULL	0.03 [0.02, 0.04]	0.04 [0.03, 0.05]	0.00 [0.00, 0.00]	0.92 [0.86, 0.96]
Qwen2.5-7B	PRE	-0.01 [-0.02, -0.01]	-0.02 [-0.03, -0.01]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Qwen2.5-7B	BEFORE	-0.02 [-0.02, -0.01]	-0.02 [-0.03, -0.01]	0.00 [0.00, 0.00]	0.96 [0.93, 0.99]
Qwen2.5-7B	NONE	0.03 [0.02, 0.04]	0.04 [0.03, 0.04]	0.00 [0.00, 0.00]	0.97 [0.95, 0.99]
Qwen2.5-14B	AFTER	1	0.88 [0.86, 0.89]	1.00 [1.00, 1.00]	0.00 [0.00, 0.00]
Qwen2.5-14B	POST	0.39 [0.36, 0.43]	0.39 [0.36, 0.41]	0.23 [0.16, 0.31]	0.75 [0.66, 0.83]
Qwen2.5-14B	POST-NULL	0.03 [0.02, 0.04]	0.05 [0.04, 0.06]	0.00 [0.00, 0.00]	0.99 [0.97, 1.00]
Qwen2.5-14B	PRE	-0.00 [-0.01, 0.00]	-0.00 [-0.01, 0.00]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Qwen2.5-14B	BEFORE	-0.00 [-0.01, 0.00]	-0.00 [-0.01, 0.00]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Qwen2.5-14B	NONE	0.03 [0.02, 0.04]	0.05 [0.04, 0.06]	0.00 [0.00, 0.00]	0.99 [0.97, 1.00]
Mistral-7B	AFTER	1	0.80 [0.78, 0.82]	0.95 [0.91, 0.98]	0.05 [0.02, 0.09]
Mistral-7B	POST	0.57 [0.50, 0.63]	0.40 [0.37, 0.43]	0.31 [0.23, 0.40]	0.71 [0.62, 0.79]
Mistral-7B	POST-NULL	-0.00 [-0.01, 0.01]	-0.00 [-0.01, 0.01]	0.00 [0.00, 0.00]	0.93 [0.88, 0.97]
Mistral-7B	PRE	-0.03 [-0.05, -0.02]	-0.03 [-0.05, -0.02]	0.00 [0.00, 0.00]	0.99 [0.97, 1.00]
Mistral-7B	BEFORE	-0.02 [-0.03, -0.01]	-0.02 [-0.03, -0.01]	0.00 [0.00, 0.00]	0.98 [0.96, 1.00]
Mistral-7B	NONE	-0.00 [-0.01, 0.01]	-0.00 [-0.01, 0.01]	0.00 [0.00, 0.00]	0.97 [0.94, 0.99]
OLMo-2-7B	AFTER	1	0.63 [0.61, 0.65]	0.86 [0.80, 0.92]	0.13 [0.07, 0.19]
OLMo-2-7B	POST	0.18 [0.14, 0.22]	0.16 [0.13, 0.19]	0.01 [0.00, 0.03]	0.96 [0.93, 0.99]
OLMo-2-7B	POST-NULL	0.07 [0.05, 0.09]	0.06 [0.05, 0.07]	0.00 [0.00, 0.00]	0.74 [0.66, 0.82]
OLMo-2-7B	PRE	-0.05 [-0.07, -0.04]	-0.05 [-0.06, -0.04]	0.00 [0.00, 0.00]	1.00 [0.99, 1.00]
OLMo-2-7B	BEFORE	-0.04 [-0.06, -0.03]	-0.04 [-0.05, -0.03]	0.00 [0.00, 0.00]	1.00 [0.99, 1.00]
OLMo-2-7B	NONE	0.07 [0.05, 0.08]	0.05 [0.04, 0.06]	0.00 [0.00, 0.00]	0.81 [0.74, 0.88]
Llama-3.1-8B	AFTER	1	0.61 [0.59, 0.63]	0.76 [0.69, 0.83]	0.14 [0.08, 0.22]
Llama-3.1-8B	POST	0.40 [0.35, 0.44]	0.29 [0.26, 0.32]	0.09 [0.04, 0.13]	0.88 [0.82, 0.94]
Llama-3.1-8B	POST-NULL	0.12 [0.09, 0.15]	0.08 [0.06, 0.09]	0.00 [0.00, 0.00]	0.90 [0.84, 0.95]
Llama-3.1-8B	PRE	-0.03 [-0.05, -0.01]	-0.02 [-0.04, -0.01]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Llama-3.1-8B	BEFORE	-0.06 [-0.08, -0.05]	-0.05 [-0.07, -0.04]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Llama-3.1-8B	NONE	0.11 [0.08, 0.14]	0.07 [0.06, 0.09]	0.00 [0.00, 0.00]	0.88 [0.81, 0.93]
Gemma-2-9B	AFTER	1	0.70 [0.69, 0.71]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]
Gemma-2-9B	POST	0.51 [0.47, 0.55]	0.36 [0.34, 0.38]	0.13 [0.09, 0.19]	0.84 [0.77, 0.89]
Gemma-2-9B	POST-NULL	0.00 [-0.00, 0.01]	0.00 [-0.00, 0.01]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Gemma-2-9B	PRE	-0.02 [-0.03, -0.01]	-0.02 [-0.03, -0.02]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Gemma-2-9B	BEFORE	-0.02 [-0.02, -0.01]	-0.02 [-0.03, -0.02]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Gemma-2-9B	NONE	0.01 [-0.00, 0.01]	0.00 [-0.00, 0.01]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Phi-4	AFTER	1	0.72 [0.71, 0.74]	0.99 [0.97, 1.00]	0.02 [0.00, 0.04]
Phi-4	POST	0.33 [0.30, 0.37]	0.28 [0.26, 0.30]	0.03 [0.01, 0.06]	0.97 [0.94, 0.99]
Phi-4	POST-NULL	0.00 [-0.01, 0.01]	0.00 [-0.01, 0.01]	0.00 [0.00, 0.00]	0.97 [0.92, 1.00]
Phi-4	PRE	-0.03 [-0.03, -0.02]	-0.03 [-0.03, -0.02]	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]
Phi-4	BEFORE	-0.05 [-0.05, -0.04]	-0.06 [-0.08, -0.05]	0.00 [0.00, 0.00]	0.96 [0.92, 0.99]
Phi-4	NONE	0.01 [-0.00, 0.02]	0.01 [-0.00, 0.01]	0.00 [0.00, 0.00]	0.99 [0.97, 1.00]
Falcon3-7B	AFTER	1	0.78 [0.77, 0.79]	1.00 [1.00, 1.00]	0.00 [0.00, 0.00]
Falcon3-7B	POST	0.45 [0.42, 0.48]	0.39 [0.37, 0.41]	0.30 [0.24, 0.37]	0.72 [0.63, 0.80]
Falcon3-7B	POST-NULL	0.03 [0.02, 0.04]	0.02 [0.02, 0.03]	0.00 [0.00, 0.00]	0.99 [0.97, 1.00]
Falcon3-7B	PRE	-0.02 [-0.02, -0.01]	-0.02 [-0.02, -0.01]	0.00 [0.00, 0.00]	0.98 [0.96, 1.00]
Falcon3-7B	BEFORE	-0.02 [-0.03, -0.02]	-0.03 [-0.04, -0.02]	0.00 [0.00, 0.00]	0.99 [0.97, 1.00]
Falcon3-7B	NONE	0.03 [0.02, 0.04]	0.03 [0.02, 0.03]	0.00 [0.00, 0.00]	1.00 [0.99, 1.00]
Gemma-2-2B	AFTER	1	0.49 [0.46, 0.51]	0.38 [0.29, 0.47]	0.25 [0.17, 0.34]
Gemma-2-2B	POST	0.10 [0.07, 0.13]	0.06 [0.04, 0.08]	0.00 [0.00, 0.00]	0.64 [0.54, 0.73]
Gemma-2-2B	POST-NULL	0.01 [-0.00, 0.02]	0.00 [-0.00, 0.01]	0.00 [0.00, 0.00]	0.55 [0.45, 0.64]
Gemma-2-2B	PRE	-0.02 [-0.03, -0.01]	-0.01 [-0.02, -0.01]	0.00 [0.00, 0.00]	0.87 [0.80, 0.93]
Gemma-2-2B	BEFORE	-0.02 [-0.03, -0.01]	-0.01 [-0.02, -0.01]	0.00 [0.00, 0.00]	0.60 [0.50, 0.70]
Gemma-2-2B	NONE	0.01 [-0.00, 0.02]	0.00 [-0.00, 0.01]	0.00 [0.00, 0.00]	0.64 [0.54, 0.74]

held in 4 of 4) and 0.334–0.512 in the new families (J-B3-N4, risky, held in 4 of 4), with s_{ID} = 0.163–0.400; each line also required the read to exceed the no-mention and sentence-before reads, also on the items competent in all four arms. Split by the eight sentences and the two lexicons, r is at least 0.102 in every cell of every model (exploratory). A sentence of the same length that names no candidate opens no key read: the upper bounds of r(POST-NULL) are at most 0.093 in the discovery-sample models, and the lower bounds of the sentence's excess over its null version, δ(POST, POST-NULL), at least 0.061 (J-B-NULL-P4f, held in 4 of 4); in the new families those lower bounds are at least 0.215 (98.75 %), and the

line held in 3 of 4 (J-B-NULL-N4, risky).

Llama-3.1-8B and the null sentence (post hoc reading). The one family that missed the null line did so on its first criterion: r(POST-NULL) = 0.118 with a 98.75 % upper bound of 0.154, against the preregistered 0.1. The outcome record's reading is that this is Llama's no-mention baseline, not a read opened by the null sentence: without any mention r = 0.113 [0.085, 0.141], a key read of 0.68 nats that is large as a ratio only because Llama's list read (ID_K under AFTER, 5.99 nats) is the smallest of the eight, and r(POST-NULL) $-$ r(NONE) is $+0.005$ [0.0001, 0.011] (98.75 %, exploratory, recomputed). The null sentence adds no key read. The pre-written replacement for a failing family is therefore not applied (Appendix A.6).

The null nouns are not inert (exploratory, recomputed). Under clamps the models sometimes answer with a null noun or a room word (in 9 rows of Qwen2.5-14B, 9 of OLMo-2-7B and 4 of Llama-3.1-8B), and the coverage of the null-sentence clamp rows is lower than elsewhere (OLMo-2 0.858–0.875, Llama 0.889–0.905); every cell stays above 0.8.

The 2B model (J-B-SMALL, not evaluable). The preregistered test of the sentence read at Gemma-2-2B was not evaluable: the model failed the competence gate (generated accuracy on the clean base run 0.773 with the list after the story, 0.687 with the sentence after it, against 0.8), and its wrong answers name the object's initial location or the distractor's location, a tracking failure rather than an answer-form one (exploratory, recomputed). We report its numbers without a scale claim: r(POST) = 0.097 [0.066, 0.126] and s_{ID}(POST) = 0.060 [0.041, 0.078], a small positive key read. Had the model passed the gate, the line (which predicted r(POST) < 0.1) would not have been met, because the interval does not exclude 0.1; by sentence and lexicon its r(POST) reaches 0.177 (exploratory).

Small Qwen models (exploratory). On the discovery-sample cores, Qwen2.5-1.5B and Qwen2.5-3B give s_{ID} 0.366 and 0.709 under LIST-AFTER and -0.070 and 0.004 under SENTENCE-AFTER with emitted forms, reproducing their stage-1 values under the lower-case score; both are only partly competent under SENTENCE-AFTER (102 and 115 of 150 items).

Holm sensitivity. Over the 48 interval components of this part's risky account lines, Holm's step-down changes no decision and no verdict; Phi-4's component of the generated-sentence line J-B5b-N4 passes narrowly and could fail under another bootstrap seed, in which case that line would still hold in three of four families.

D.3. The measurement

Why the score matters. In the free-form formats the discovery sample scored each candidate by the probability of its lower-case token with a leading space, which in some cells carries a small share of the probability mass: the models write "Drawer" or "the drawer". The final round sums each candidate's probability over the forms the models emit (Appendix C.2) and re-measured the discovery-sample cores in the four discovery-sample models with both scores in the same pass. Table 26 gives the probability mass on the six candidates under the lower-case score (M^L), a 12-form score (M^Σ) and the emitted-form score (M^E), and the coverage, on the fresh and the discovery-sample cores; Table 27 gives s_{ID} under both scores and the paired differences of s_{ID} and r.

The validity lines (J-B6a–c, J-B7). All five measurement-validity lines were classed risky before the run. Coverage of the emitted forms on the discovery-sample cores was at least 0.873 in every arm (J-B6a, held in 4 of 4). The five discovery-sample verdicts that depend on the scoring (E1a, E1b and E1c of the instruction-matched 2×2, B2 of stage 1 and the bound on LIST-BEFORE in C4), recomputed under emitted forms with their original rules, are unchanged, and no per-model component changed (J-B6c, held). The record's pre-written sentence, with one word changed (Appendix A.6): "Scored on the forms the models actually emit (coverage ≥ 0.8 in every cell), every re-scored verdict on the original stories is unchanged." On the fresh stories the emitted-form argmax agrees with the generated answer in at least 0.984 of items in every cell, the clean base run never answers outside the candidates (largest "other" rate 0.000) and coverage is at least 0.858, in all eight models (J-B7, held).

The sentence read was understated (J-B6b, not held). J-B6b required the key share and the sentence read relative to the list read to stay within ± 0.05 under emitted forms in every cell whose lower-case mass was below 0.5. The key share did: every s_{ID} difference lies inside ± 0.05, the largest in size -0.019 [-0.027, -0.012] (Mistral-7B, NO-MENTION) and under SENTENCE-AFTER at most 0.014 [0.010, 0.017]. The sentence ratio did not: $r^E - r^L$ under SENTENCE-AFTER is

Table 26. **Probability mass on the candidates, by score.** M^L, M^Σ, M^E: mean over items of the clean base run's mass on the six candidates under the lower-case score, the 12-form score and the emitted-form score; coverage: the smallest mean M^E over a cell's sixteen clamp and clean rows. F: fresh cores; S0: the discovery-sample cores (arms as in Table 25, with OPTIONS-AFTER).

Model	Population	Arm	M^L	M^Σ	M^E	coverage
Qwen2.5-7B	F	AFTER	0.999	1.000	1.000	1.000
Qwen2.5-7B	F	BEFORE	1.000	1.000	1.000	1.000
Qwen2.5-7B	F	NONE	0.040	0.999	0.999	0.998
Qwen2.5-7B	F	POST	0.051	0.999	1.000	0.998
Qwen2.5-7B	F	PRE	0.021	0.999	1.000	0.999
Qwen2.5-7B	F	POST-NULL	0.067	0.998	0.999	0.995
Qwen2.5-7B	S0	OPTIONS-AFTER	1.000	1.000	1.000	1.000
Qwen2.5-7B	S0	AFTER	0.998	1.000	1.000	1.000
Qwen2.5-7B	S0	BEFORE	0.999	1.000	1.000	1.000
Qwen2.5-7B	S0	POST	0.007	1.000	1.000	0.999
Qwen2.5-7B	S0	PRE	0.004	1.000	1.000	0.999
Qwen2.5-7B	S0	NONE	0.005	0.999	1.000	0.997
Qwen2.5-14B	F	AFTER	0.890	1.000	1.000	1.000
Qwen2.5-14B	F	BEFORE	0.247	1.000	1.000	1.000
Qwen2.5-14B	F	NONE	0.000	1.000	1.000	0.997
Qwen2.5-14B	F	POST	0.000	1.000	1.000	0.999
Qwen2.5-14B	F	PRE	0.000	1.000	1.000	1.000
Qwen2.5-14B	F	POST-NULL	0.000	1.000	1.000	0.961
Qwen2.5-14B	S0	OPTIONS-AFTER	1.000	1.000	1.000	1.000
Qwen2.5-14B	S0	AFTER	0.850	1.000	1.000	1.000
Qwen2.5-14B	S0	BEFORE	0.357	1.000	1.000	1.000
Qwen2.5-14B	S0	POST	0.000	1.000	1.000	0.999
Qwen2.5-14B	S0	PRE	0.000	1.000	1.000	1.000
Qwen2.5-14B	S0	NONE	0.000	1.000	1.000	0.989
Mistral-7B	F	AFTER	0.890	0.999	0.999	0.997
Mistral-7B	F	BEFORE	0.274	0.998	0.998	0.997
Mistral-7B	F	NONE	0.053	0.947	0.948	0.906
Mistral-7B	F	POST	0.056	0.968	0.968	0.943
Mistral-7B	F	PRE	0.020	0.922	0.925	0.924
Mistral-7B	F	POST-NULL	0.038	0.943	0.943	0.923
Mistral-7B	S0	OPTIONS-AFTER	0.976	0.999	0.999	0.998
Mistral-7B	S0	AFTER	0.832	0.999	0.999	0.998
Mistral-7B	S0	BEFORE	0.149	1.000	1.000	0.998
Mistral-7B	S0	POST	0.027	1.000	1.000	0.998
Mistral-7B	S0	PRE	0.006	0.999	1.000	1.000
Mistral-7B	S0	NONE	0.029	0.999	1.000	0.982
OLMo-2-7B	F	AFTER	0.996	0.998	0.998	0.987
OLMo-2-7B	F	BEFORE	0.987	0.997	0.997	0.996
OLMo-2-7B	F	NONE	0.834	0.983	0.989	0.949
OLMo-2-7B	F	POST	0.865	0.987	0.991	0.980
OLMo-2-7B	F	PRE	0.850	0.991	0.994	0.992
OLMo-2-7B	F	POST-NULL	0.828	0.976	0.982	0.858
OLMo-2-7B	S0	OPTIONS-AFTER	0.960	0.967	0.967	0.873
OLMo-2-7B	S0	AFTER	0.996	0.997	0.998	0.985
OLMo-2-7B	S0	BEFORE	0.983	0.995	0.996	0.993
OLMo-2-7B	S0	POST	0.875	0.985	0.990	0.979
OLMo-2-7B	S0	PRE	0.853	0.991	0.993	0.991
OLMo-2-7B	S0	NONE	0.843	0.983	0.989	0.956
Llama-3.1-8B	F	AFTER	0.909	0.994	0.994	0.984
Llama-3.1-8B	F	BEFORE	0.518	0.990	0.990	0.988
Llama-3.1-8B	F	NONE	0.058	0.968	0.970	0.918
Llama-3.1-8B	F	POST	0.085	0.974	0.975	0.954
Llama-3.1-8B	F	PRE	0.077	0.979	0.981	0.975
Llama-3.1-8B	F	POST-NULL	0.081	0.969	0.972	0.889
Gemma-2-9B	F	AFTER	0.965	0.998	0.999	0.997
Gemma-2-9B	F	BEFORE	0.836	0.997	0.999	0.998
Gemma-2-9B	F	NONE	0.002	0.999	0.999	0.998
Gemma-2-9B	F	POST	0.002	0.999	0.999	0.997
Gemma-2-9B	F	PRE	0.005	0.999	0.999	0.999
Gemma-2-9B	F	POST-NULL	0.002	0.999	0.999	0.998
Phi-4	F	AFTER	0.953	1.000	1.000	0.998
Phi-4	F	BEFORE	0.919	0.999	0.999	0.999
Phi-4	F	NONE	0.010	0.999	1.000	0.996
Phi-4	F	POST	0.014	0.999	0.999	0.997
Phi-4	F	PRE	0.013	1.000	1.000	0.999
Phi-4	F	POST-NULL	0.012	0.999	0.999	0.994
Falcon3-7B	F	AFTER	0.995	1.000	1.000	0.999
Falcon3-7B	F	BEFORE	0.870	1.000	1.000	0.999
Falcon3-7B	F	NONE	0.012	0.997	0.999	0.997
Falcon3-7B	F	POST	0.021	0.999	0.999	0.995
Falcon3-7B	F	PRE	0.019	0.999	0.999	0.998
Falcon3-7B	F	POST-NULL	0.017	0.996	0.999	0.990
Gemma-2-2B	F	AFTER	0.983	0.986	0.998	0.992
Gemma-2-2B	F	BEFORE	0.549	0.976	0.993	0.992
Gemma-2-2B	F	NONE	0.004	0.948	0.988	0.987
Gemma-2-2B	F	POST	0.004	0.967	0.990	0.989
Gemma-2-2B	F	PRE	0.007	0.977	0.993	0.992
Gemma-2-2B	F	POST-NULL	0.005	0.959	0.988	0.986

+0.087 [0.068, 0.105] at Mistral-7B and +0.048 [0.039, 0.058] at Qwen2.5-14B, where the point lies inside the margin but equivalence is not shown. The line held only at Qwen2.5-7B (1 of 3 evaluable models; OLMo-2-7B had no cell with

Table 27. **Scoring invariance on the discovery-sample cores.** M^L as in Table 26; s^L_{ID}, s^E_{ID}: the key share under the lower-case and the emitted-form score; $s^E - s^L$ and $r^E - r^L$: paired differences (the sentence read relative to the list read, r, is 1 by definition under AFTER). 95 % paired cluster-bootstrap intervals.

Model	Arm	M^L	s^L_{ID}	s^E_{ID}	$s^E - s^L$	$r^E - r^L$
Qwen2.5-7B	OPTIONS-AFTER	1.000	0.79 [0.77, 0.80]	0.79 [0.77, 0.80]	0.000 [0.000, 0.000]	0.003 [0.002, 0.004]
Qwen2.5-7B	AFTER	0.998	0.78 [0.76, 0.79]	0.78 [0.76, 0.79]	0.000 [-0.000, 0.001]	–
Qwen2.5-7B	BEFORE	0.999	-0.01 [-0.02, -0.00]	-0.01 [-0.02, -0.00]	0.000 [-0.000, 0.000]	0.000 [-0.000, 0.000]
Qwen2.5-7B	POST	0.007	0.32 [0.29, 0.34]	0.32 [0.29, 0.34]	0.000 [-0.003, 0.004]	0.009 [0.003, 0.014]
Qwen2.5-7B	PRE	0.004	-0.00 [-0.01, 0.01]	-0.01 [-0.02, 0.00]	-0.005 [-0.009, -0.000]	-0.004 [-0.007, -0.001]
Qwen2.5-7B	NONE	0.005	0.06 [0.05, 0.07]	0.04 [0.04, 0.05]	-0.015 [-0.020, -0.009]	-0.012 [-0.017, -0.007]
Qwen2.5-14B	OPTIONS-AFTER	1.000	0.88 [0.86, 0.89]	0.88 [0.86, 0.89]	0.000 [0.000, 0.000]	0.024 [0.018, 0.031]
Qwen2.5-14B	AFTER	0.850	0.86 [0.84, 0.88]	0.86 [0.84, 0.88]	-0.001 [-0.002, 0.000]	–
Qwen2.5-14B	BEFORE	0.357	-0.00 [-0.01, -0.00]	-0.00 [-0.01, 0.00]	0.001 [-0.001, 0.003]	0.001 [-0.001, 0.003]
Qwen2.5-14B	POST	0.000	0.39 [0.37, 0.42]	0.39 [0.37, 0.41]	-0.003 [-0.007, 0.000]	0.048 [0.039, 0.058]
Qwen2.5-14B	PRE	0.000	-0.00 [-0.01, 0.00]	-0.00 [-0.01, 0.00]	-0.000 [-0.002, 0.002]	-0.000 [-0.002, 0.001]
Qwen2.5-14B	NONE	0.000	0.05 [0.04, 0.06]	0.04 [0.03, 0.05]	-0.009 [-0.014, -0.003]	0.000 [-0.004, 0.004]
Mistral-7B	OPTIONS-AFTER	0.976	0.87 [0.84, 0.89]	0.86 [0.84, 0.89]	-0.003 [-0.008, 0.000]	-0.002 [-0.005, 0.001]
Mistral-7B	AFTER	0.832	0.82 [0.79, 0.84]	0.82 [0.79, 0.84]	-0.001 [-0.002, 0.001]	–
Mistral-7B	BEFORE	0.149	0.00 [-0.01, 0.01]	0.00 [-0.01, 0.01]	0.000 [-0.003, 0.003]	0.000 [-0.003, 0.003]
Mistral-7B	POST	0.027	0.41 [0.38, 0.44]	0.42 [0.39, 0.45]	0.014 [0.010, 0.017]	0.087 [0.068, 0.105]
Mistral-7B	PRE	0.006	0.00 [-0.01, 0.01]	-0.01 [-0.02, -0.00]	-0.010 [-0.015, -0.006]	-0.011 [-0.016, -0.006]
Mistral-7B	NONE	0.029	0.04 [0.03, 0.05]	0.02 [0.00, 0.03]	-0.019 [-0.027, -0.012]	-0.016 [-0.023, -0.009]
OLMo-2-7B	OPTIONS-AFTER	0.960	0.69 [0.67, 0.71]	0.68 [0.66, 0.70]	-0.008 [-0.011, -0.005]	-0.006 [-0.009, -0.003]
OLMo-2-7B	AFTER	0.996	0.59 [0.57, 0.62]	0.59 [0.57, 0.62]	0.001 [-0.000, 0.002]	–
OLMo-2-7B	BEFORE	0.983	-0.03 [-0.04, -0.02]	-0.03 [-0.04, -0.02]	-0.001 [-0.002, 0.001]	-0.001 [-0.002, 0.001]
OLMo-2-7B	POST	0.875	0.19 [0.17, 0.21]	0.19 [0.16, 0.21]	-0.003 [-0.007, 0.001]	-0.008 [-0.014, -0.002]
OLMo-2-7B	PRE	0.853	-0.02 [-0.03, -0.01]	-0.02 [-0.04, -0.01]	-0.002 [-0.004, 0.000]	-0.002 [-0.004, 0.000]
OLMo-2-7B	NONE	0.843	0.03 [0.02, 0.05]	0.03 [0.02, 0.05]	0.001 [-0.000, 0.003]	0.001 [-0.001, 0.003]

lower-case mass below 0.5 and was not evaluable). The corrected sentence the paper uses (Appendix A.6): "s_{ID} moves by at most 0.05, while the sentence read relative to the list read, r, rises under emitted forms by $+0.087$ at Mistral-7B and by $+0.048$ at Qwen2.5-14B, where equivalence within 0.05 is not shown." The reason, from the record (exploratory, recomputed): in the sentence arms of these two models most of the mass sits on capitalised forms (" Drawer", "Draw" + "er"), and the lower-case form's share falls by 1–2 nats when a clamp promotes a candidate, so the lower-case score understates ID_K under SENTENCE-AFTER by 11–15 % (the emitted-form value is 12–17 % above it), while the list anchor barely changes. The discovery sample's lower-case values of r under SENTENCE-AFTER (0.337 and 0.506) therefore understated the sentence read.

The intervention frames. The same comparison on the frames of the released remap at Mistral-Small-24B (J-B8, risky, held) is in Appendix G.5: all twelve differences lie within [-0.013, 0.018].

Relative changes (reported). On the discovery-sample cores the relative changes of ID_K and ID_V under emitted forms are within 15 % in every OPTIONS-AFTER and list-after cell. Larger ones occur where ID_K is itself near zero (at most about one nat: the list before the story at Qwen2.5-14B, the sentence before it and no mention) and in three cells with little lower-case mass: ID_K under SENTENCE-AFTER at Mistral-7B ($+17$ %), ID_V under NO-MENTION at Qwen2.5-14B ($+20$ %) and ID_V under SENTENCE-BEFORE at Mistral-7B ($+17$ %).

D.4. Generated answers

The lines (J-B5, J-B5b). With the list after the story, the key-only clamp moves 0.860–1.000 of generated answers to the clamped value in the discovery-sample models and 0.760–1.000 in the new families; without a mention, with the list before or with the sentence before, it moves none (0.000, 0.000 and 0.000 at most), and the value-only clamp moves 0.813–1.000 of answers without a mention (J-B5-P4f and J-B5-N4, held in 4 of 4 and 4 of 4). The sentence read changes generated answers in seven of the eight models: the key-only clamp's flip rate is 0.033–0.313 after the sentence and 0.000 before it in those seven (J-B5b-N4, risky, held in 4 of 4); at OLMo-2-7B the flip rate is 0.010 after the sentence and 0.000 before it, with a lower bound of 0.000 for the difference, so the discovery-sample line J-B5b-P4f was not met (risky; 3 of 4). The record's reading (exploratory, recomputed): 3 of 150 items flip, in 3 of 55 cells, and 11.9 % of the resamples are exactly zero, so the failure does not depend on the seed; the key clamp under SENTENCE-AFTER closes about 29 % of a 10-nat

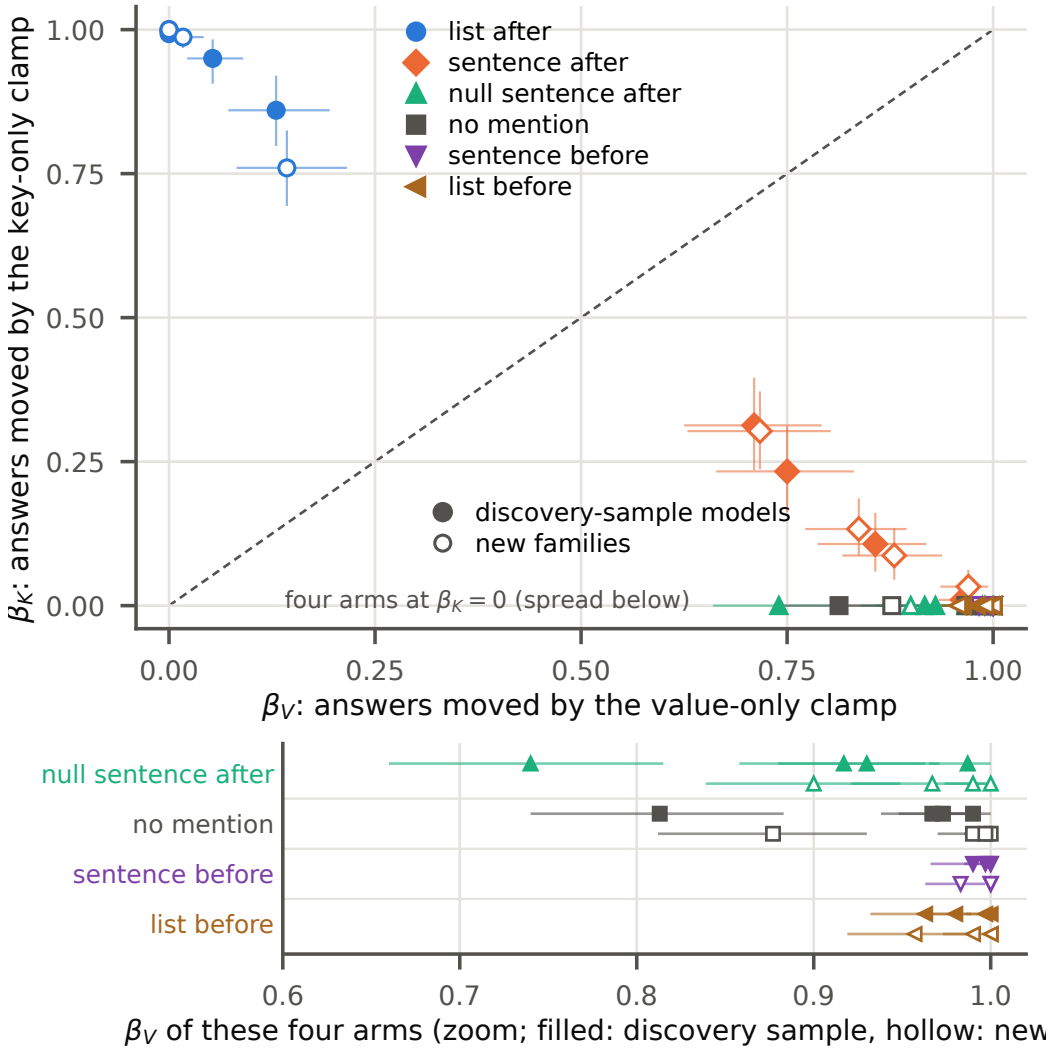


Figure 7. **Generated answers follow the channel the later mentions open** (fresh stories, eight models; the full version of Figure 1c). Each point is one model and arm: β_K, the share of generated answers that move to the clamped value under the key-only clamp, against β_V, the same under the value-only clamp, net of moves to the third value; 95 % cluster intervals. The strip below spreads out the four arms whose β_K is zero in every model (no mention, list before, sentence before, null sentence).

base–source margin at OLMo-2-7B, and the generations agree with the emitted-form argmax: the read is present in log-probabilities but too small to change OLMo-2's answers. The pre-written texts had no wording for this split; the wording used is a disclosed decision (Appendix A.6).

The crossover (reported). Over the eight models, β_K under LIST-AFTER is 0.760–1.000 against β_V = 0.000–0.143; without a mention β_V is 0.813–1.000 against β_K = 0.000; under SENTENCE-AFTER β_K is 0.010–0.313 and β_V 0.710–0.970; under NULL-SENTENCE β_K is 0.000 in every model and β_V 0.740–1.000. A single-channel clamp creates a cache state that no prompt produces; the generated answers show that the channel the later mentions open is the one that changes what the model says.

D.5. Natural passages

Design and gates. Four models (Llama-3.1-8B and Gemma-2-9B, two families never examined before, and Qwen2.5-7B and Mistral-7B) answered the 185 counterfactual passages in six formats (Appendix B.3). Each line is decided on the items competent in its formats (the clean run answers B, the clamp to S answers S and the clamp to X answers X) and must hold in every evaluable model, with at least three evaluable models including a new family. Intervals are 95 % two-stage

bootstrap intervals over articles and the items within them. Three gates decide evaluability (Appendix C.2). The BF16 floor failed in one cell, Llama-3.1-8B's options-after cell, by one item (generation agreement 0.968 against 0.97); every line that uses that cell is not evaluable in Llama. The competence floor of 80 items failed in five cells: Llama-3.1-8B with the related-articles sentence after and before the passage, and Mistral-7B in free form and with the sentence after and before. Mistral's free-form answers are mostly sentences ("The passage states that ..."), which the answer rule does not read as the entity, while its clamped answers never name B (exploratory, recomputed): a failure of answer form, not of compliance with the counterfactual. The emitted-form gate held in every cell (median probability of the decision token 0.855–1.000; the first generated token was the decision token in at least 0.977 of competent items). Table 28 gives every model and format.

Identity (J-A1–J-A5). With options after the passage the key carries most of the answer entity's identity in the three evaluable models (J-A1, extrapolation, held: s_{ID} 0.627, 0.677 and 0.550 at Gemma-2-9B, Qwen2.5-7B and Mistral-7B, lower bounds above the preregistered 0.35; Llama-3.1-8B's cell, 0.524, not evaluable), also for answers of three or more tokens (J-A1b, risky, held: 0.661 [0.623, 0.699], 0.726 [0.680, 0.771] and 0.587 [0.533, 0.635]), and with lettered options in all four (J-A1c, held: s_{ID} 0.739–0.963). In free form the value carries it (J-A2, held: s_{ID} 0.045, -0.003 and 0.014 at Llama, Gemma and Qwen; Mistral not evaluable). The derived contrast J-A3 (options after minus free form at least 0.4; not counted) holds in both models where it is evaluable (0.634 and 0.670) and is not evaluable, because Llama's options-after cell and Mistral's free-form cell fail their gates. Options before the passage give a small negative key read in every model (J-A4, replication, held: ID_K -0.900 to -0.180 nats, ratios to the options-after read -0.058 to -0.018); the line's criteria are one-sided. A related-articles sentence after the passage opens only a small key read: its gain in s_{ID} over free form is 0.070 at Gemma and 0.042 at Qwen, below the preregistered 0.10 (J-A5, risky, not held; Llama and Mistral not evaluable), although its key read is positive (Table 28). Its contrast with the same sentence before the passage, J-A5B, holds in both evaluable models (ID_K difference 1.334 and 2.215 nats) and is not evaluable (two of the three required models).

Behaviour (J-A6a–e). Table 29 gives the cue-conflict rates. With lettered options the generated answer follows the key's entity in all four models (J-A6b, risky, held), and when the key matches no option the value is copied (J-A6e, risky, held in the three evaluable models). With options answered by name the key wins at Gemma and Qwen but not at Mistral (key-source rate 0.408 against the preregistered 0.6; J-A6a, risky, not held). In free form the value-source rate misses 0.6 at Llama and Qwen (J-A6c, risky, not held; Mistral not evaluable): many of the answers that name no candidate are hybrids that start with the value source's first word and do not complete the entity (34 % of the "other" answers at Llama to 76 % at Gemma; recomputed), and at the decision token the value-source rate is 0.535–0.945 while the key-source rate is at most 0.005. Scored on the answer's first word, which is not the preregistered criterion, the value-source rates would be 0.674 / 0.950 / 0.740 / 0.547 (Llama / Gemma / Qwen / Mistral; Llama's lower bound 0.55, exploratory, recomputed) and J-A6c would be met; as preregistered (the whole answer) it is not. The key-only flag (J-A6d) moves answers to the key's entity with options after the passage and never in free form in every model, but the line is not evaluable (two evaluable models). Llama's options-after cue conflict, whose cell is not evaluable, favours the value (key 0.267, value 0.456), driven by numbers, 67 of its 70 single-token items, and by person names whose tokens are split between the key and the value source (exploratory, recomputed). Because J-A6a and J-A6c both failed, the hybrids are reported as an observation, not under the pre-written clause for a failure of J-A6c alone (Appendix A.6), and the channel attributions are not shown to decide behaviour on natural text.

Continuations (J-A8, not evaluable; J-A8d, not held). The continuation of a multi-token answer is read through an interaction of the two channels in all four models: in free form, the key and value effects on the continuation tokens add up to far less than the joint effect (I_{cont} = 1.192–1.515), while at the decision token they add up (I_{dec} = -0.173 to 0.039), and with options listed the joint clamp hardly moves the continuation (0.002–0.017 of its free-form effect). The key alone moves the continuation away from S (d_K^{cont} = -14.38 to -12.23 nats), the pattern the line predicted: the key addresses the next piece and the value supplies it. This pattern is seen in all four models, but its line (J-A8, extrapolation) is not evaluable (two evaluable models: Llama's options-after cell fails the BF16 gate and Mistral's free-form cell the competence gate); it is a channel-level replication of induction-head K-composition (Elhage et al., 2021; Olsson et al., 2022), not a novelty claim. The hybrid rate under a key that matches no candidate (J-A8d, risky) rests on Gemma alone, the one model with enough items (0.105 [0.030, 0.217], 6 of 57 items, against 0.2), many of whose B entities are famous names that the model completes from memory; not held.

Table 28. **Natural passages: identity by format.** Counterfactual SQuAD passages, 185 items per model; n_{comp}: items competent in the format; s_{ID}, ID_K and ID_V on those items, with 95 % two-stage article-cluster intervals. Gate: the preregistered gate that fails in the cell (J-A-G1: BF16 floor; J-A-G2: fewer than 80 competent items); such a cell enters no line. †Computed from the raw results with the scorer's functions, not printed by the scorer (exploratory).

Model	Format	n_{comp}	s_{ID}	ID_K (nats)	ID_V (nats)	Gate
Llama-3.1-8B	options after	148	0.524 [0.473, 0.568]	6.859 [5.903, 7.829]	6.237 [5.702, 6.805]	J-A-G1
	letters after	156	0.739 [0.675, 0.794]	6.726 [5.873, 7.572]	2.371 [1.805, 3.080]	–
	sentence after†	70	0.075 [0.044, 0.106]	0.916 [0.514, 1.339]	11.260 [10.555, 11.969]	J-A-G2
	free form	86	0.045 [0.026, 0.063]	0.552 [0.315, 0.794]	11.786 [11.122, 12.434]	–
	options before†	164	−0.023 [−0.035, −0.011]	−0.252 [−0.383, −0.121]	11.176 [10.440, 11.972]	–
	sentence before†	69	−0.021 [−0.042, −0.002]	−0.227 [−0.429, −0.020]	10.784 [10.084, 11.531]	J-A-G2
Gemma-2-9B	options after	158	0.627 [0.599, 0.653]	9.868 [9.073, 10.596]	5.869 [5.379, 6.389]	–
	letters after	158	0.963 [0.927, 0.987]	9.676 [9.121, 10.128]	0.375 [0.132, 0.750]	–
	sentence after†	107	0.067 [0.043, 0.092]	1.047 [0.653, 1.496]	14.685 [14.010, 15.349]	–
	free form	109	−0.003 [−0.010, 0.002]	−0.057 [−0.164, 0.037]	16.900 [16.204, 17.582]	–
	options before†	167	−0.017 [−0.024, −0.011]	−0.183 [−0.262, −0.117]	11.234 [10.786, 11.692]	–
	sentence before†	117	−0.018 [−0.024, −0.012]	−0.247 [−0.332, −0.164]	14.176 [13.500, 14.855]	–
Qwen2.5-7B	options after	162	0.677 [0.639, 0.715]	15.491 [13.983, 16.966]	7.385 [6.287, 8.550]	–
	letters after	163	0.935 [0.885, 0.972]	19.782 [18.416, 20.972]	1.366 [0.591, 2.438]	–
	sentence after†	106	0.061 [0.027, 0.096]	1.335 [0.564, 2.144]	20.573 [19.205, 21.983]	–
	free form	104	0.014 [0.004, 0.025]	0.311 [0.081, 0.548]	21.430 [20.006, 22.911]	–
	options before†	166	−0.053 [−0.067, −0.041]	−0.878 [−1.087, −0.671]	17.310 [16.163, 18.440]	–
	sentence before†	99	−0.036 [−0.051, −0.022]	−0.725 [−0.993, −0.454]	20.770 [19.338, 22.218]	–
Mistral-7B	options after	157	0.550 [0.503, 0.593]	9.817 [8.568, 11.071]	8.029 [7.187, 8.907]	–
	letters after	155	0.786 [0.701, 0.864]	7.424 [6.211, 8.638]	2.016 [1.219, 2.987]	–
	sentence after†	43	0.092 [0.009, 0.168]	1.545 [0.139, 3.045]	15.330 [14.205, 16.444]	J-A-G2
	free form	43	0.001 [−0.011, 0.015]	0.019 [−0.184, 0.257]	16.426 [15.215, 17.516]	J-A-G2
	options before†	150	−0.026 [−0.037, −0.016]	−0.387 [−0.540, −0.240]	15.195 [14.444, 15.971]	–
	sentence before†	44	−0.018 [−0.040, 0.000]	−0.278 [−0.586, 0.004]	15.313 [14.206, 16.307]	J-A-G2

Quantization of the cached values (J-A7, J-A7b; exploratory reading). The passage part also tested whether quantizing the passage's cached values to 2 bits with KIVI (Liu et al., 2024) hurts free-form answers more than multiple-choice ones, beyond the same quantization of the keys. Only Qwen2.5-7B had enough prior-free items (33): the difference in differences was -0.091 [-0.243, 0.038] (J-A7, risky, not held) and the value drop's difference between formats 0.030 [0.000, 0.107] (J-A7b, risky, not held). The test had low power (the drops come from four and one items) and points the other way: on all competent items 2-bit keys cost free-form answers more than 2-bit values do (Table 30), and neither costs

Table 29. **Cue conflict and the copy fallback on passages.** The cache gets two sources: the key from one counterfactual entity and the value from another (S and X, both directions). Key / value: the share of generated answers naming the key's or the value's entity; "other": answers naming no candidate; dec.: the value-source rate at the decision token alone. Flag only: the key from S and the value from a third entity Z, share answering S; copy fallback: the key from Z and the value from S, share answering S. Whole answers; competent items; cells of Llama's options-after and Mistral's free-form formats fail a gate (not evaluable).

	options after		lettered options		free form			flag only	copy fallback
Model	key	value	key	value	value	other	value (dec.)	(options after)	(options after)
Llama-3.1-8B	0.267	0.456	0.750	0.176	0.506	0.494	0.593	0.541	0.845
Gemma-2-9B	0.734	0.180	0.978	0.009	0.789	0.211	0.945	0.842	0.880
Qwen2.5-7B	0.778	0.176	0.923	0.055	0.438	0.553	0.740	0.883	0.840
Mistral-7B	0.408	0.379	0.629	0.187	0.244	0.744	0.535	0.637	0.834

multiple-choice answers (accuracy at least 0.974). The relative reconstruction errors are 0.172–0.239 for the keys (per channel) and 0.414–0.421 for the values (per token), without strength matching. We make no claim about quantization.

Table 30. **KIVI 2-bit quantization of the passage's cache** (exploratory). Faithful accuracy (the generation names S on the S passage) on competent items with no quantization, with 2-bit keys and with 2-bit values (every passage token, every layer; keys per channel, values per token, groups of 32), and the relative reconstruction errors $\|x' - x\|/\|x\|$.

Model	Format	none	keys 2-bit	values 2-bit	error (keys)	error (values)
Llama-3.1-8B	free form	1.000	0.849	0.965	0.188	0.420
	options after	1.000	0.993	0.986	0.189	0.420
	lettered options	1.000	0.987	0.981	0.189	0.420
Gemma-2-9B	free form	1.000	0.954	0.982	0.239	0.421
	options after	1.000	0.994	0.994	0.239	0.421
	lettered options	1.000	0.994	1.000	0.239	0.421
Qwen2.5-7B	free form	1.000	0.865	0.971	0.172	0.420
	options after	1.000	0.988	0.988	0.173	0.421
	lettered options	1.000	0.988	0.988	0.173	0.421
Mistral-7B	free form	1.000	0.860	0.953	0.193	0.414
	options after	0.994	0.987	0.987	0.193	0.414
	lettered options	1.000	0.974	0.994	0.193	0.414

Reader heads on passages (J-A-HA1–HA3b, not evaluable). The head lines needed Qwen2.5-7B and Mistral-7B. At Qwen2.5-7B the heads' BF16 floor failed ($|m(\text{none}) - m(\text{clean})| = 0.829$ nats against 0.500): plain eager passes in different batch shapes differ by 0.7–0.9 nats at Qwen (0.1 at Mistral), and Qwen's eager teacher-forced chain fails in most items whose continuation contains the digit "0" (exploratory, recomputed), so every head line is not evaluable. The option rows are the readers in both models ($d_G/d_{\text{full}} = 0.914$ and 0.945). At Mistral-7B, whose gates passed (exploratory): the top 52 heads ranked on the passages recover $R = 0.938$ of the option rows' read and blinding them removes 1.003, against -0.001 and 0.047 for random sets; the reader heads found on the templated stories recover 0.555 and remove 0.987 (23 heads shared with the natural set); and mean-ablating the natural set at every row after the question lowers faithful multiple-choice accuracy on prior-free items by 0.449. Ablating the heads with the largest direct effect on the answer in free form (a copy set) breaks both formats (faithful accuracy with options after $0.816 \to 0.281$ at Qwen and $0.892 \to 0.481$ at Mistral; in free form $0.643 \to 0.097$ and $0.281 \to 0.022$), and ablating the natural readers also lowers free-form accuracy (to 0.330 and 0.086; exploratory): the readers found at the option rows are needed for free-form answers as well. Because ablating the readers and cutting their attention disagree on whether the copy route takes over in the discovery sample (Appendix E.2), these passage tests would have bridged the two; they are not evaluable.

Sensitivity: Llama's options-after cell (disclosed). Had the BF16 gate passed in that cell (180 of 185 items instead of 179), J-A3 would be met (three evaluable models including a new family) and J-A6d not met (Llama's flag-only rate 0.541 against 0.6); J-A1, J-A4, J-A6a and J-A6e would not change, and the risky tally of this part would read 3 of 10 (expected 4.65).

Holm sensitivity. Over the 43 interval components of this part's risky lines, Holm's step-down changes one decision, J-A5's component at Qwen2.5-7B ($p = 0.0099$ against a threshold of 0.0031), and no verdict. J-A6b's Mistral component

passes Holm by a relative margin of 0.2 % (p 0.002772 against 0.002778), within the Monte Carlo error of the bootstrap standard error, so J-A6b should not be described as robust to Holm.

Reported and exploratory. Without the passage, the question and options alone favour the original answer: the closed-book argmax is e_B in 0.573–0.665 of items (persons 0.738–0.850, numbers 0.325–0.481); the identity measures are contrasts between S and X, from which this prior cancels to first order. On all 185 valid items (not only competent ones) the pattern is the same: s_{ID} with the sentence after the passage is 0.064–0.093, and with options or the sentence before it -0.051 to -0.013; the option mass at the decision token is 0.915–1.000 with the options listed and 0.579–0.864 without. In the LEAK stratum (a surname re-mentioned in the passage after the full name; 65 items) s_{ID} is 0.002–0.024 in free form and 0.536–0.780 with options after. This says nothing about whether a natural in-passage re-mention opens a key read: the S, X and Z passages differ from B only inside the answer span, so the re-mention still names the original entity, and its match to a foreign key is lost alike under S and X and cancels in the S-versus-X contrast of Equation (1). A test needs the counterfactual written at every mention (Section 7, (i)). On years (80 items), at the first digit that differs, the free-form read is value-only (d_K -0.44 to 1.12 nats against d_V 11.27–30.34), and with options listed the key reads (d_K 7.64–21.97), against the written expectation, which was value-only for both formats. Scored on the full answer string from the decision token on, s_{ID} in free form is negative (-1.218 to -0.344), the continuation pattern above. With the clamp started at 30 % of the depth the free-form value copy keeps 0.987–1.000 of its full effect, while the options-after key read keeps 0.014–0.516; the first span piece carries 0.752–0.973 of the value copy, while the later pieces carry 0.402–0.594 of the key read (the first 0.102–0.454). All of these are exploratory.

E. A match at the later mention, read by key: details

This appendix gives the discovery-sample results on which later tokens read the writing token's key and through which heads (Appendices E.1 and E.2; preregistrations D, G and H), and the final round's tests of what those heads write (Appendix E.3), what sets the sign of the read (Appendix E.4) and how the flag behaves across scale (Appendix E.5). The final round's mechanism tests use Qwen2.5-7B and Mistral-7B (and Qwen2.5-1.5B and 3B for scale), the reader heads frozen from the discovery sample, flags fit on its 60 ranking stories and 100 new evaluation stories; intervals are 95 % story-bootstrap intervals, and a line over the two 7B models holds only if it holds in both.

E.1. Which later tokens read, and what a re-mention must be

Letting only the option words see the source key recovers 0.92–0.98 of the full key effect at 7–14B under OPTIONS-AFTER, and every other group at most 0.01; under SENTENCE-AFTER the sentence's candidate words recover 0.67–0.73 (D3, met; Figure 9a). Conversely, forbidding the six candidate words to attend to the writing token removes the key identity (r_K, the fraction left, is -0.105 to 0.003 in 9/9 format–model cells; G5, met), and cutting a matched control column does not (G6, met). The words must name the candidates at stake: listing B with two other candidates instead of S and X gives no read (G9, met). In lists two named candidates suffice (G10, met for lists); in sentences the read grows with the number named (G10, not met for sentences), and with only S and X named the words carry 0.38–0.40 of the splice effect, most of it not identity-specific (G12, not met; Appendix E.1.3). The candidate words are thus the main readers in lists and in sentences naming at least three candidates.

What a re-mention must be. Varying the re-mentions of SENTENCE-AFTER in four 7–14B models shows a read of the word form, largely up to case (r_K: ID_K relative to an exact repeat). Capitalised and upper-case forms read nearly fully (0.83–1.14; G14, which predicted at most 0.75, not met), but a modified noun phrase such as "wooden shelf" reads only partly (0.32–0.64; G13, not met). Synonyms and mixed-language translations carry little on our English-scored measure (at most 0.15; G15, met). No preregistered token or concept outcome holds (G16, not met). Where the read is lost, the value identity rises (G17, met), but no more than after a sentence naming other words (exploratory; Appendix E.1.4).

Localisation. Table 31 gives the row-restricted splice for each token group. The later-mention words are the listed options under OPTIONS-AFTER and LETTERS-AFTER, and the candidate words of the neutral sentence under SENTENCE-AFTER. Under OPTIONS-AFTER, the question, the story after the writing token and the remaining tokens, including the answer position, each recover at most 0.01 at 7–14B. Under SENTENCE-AFTER the "other later tokens" recover 0.05–0.11, and 0.19–0.22 of the effect is recovered by no single group, so the groups do not add up. At Mistral-Small-24B, with the key clamped from block 6, the option rows carry 0.853, 0.987 and 0.682 of the key effect under OPTIONS-AFTER, LETTERS-AFTER and SENTENCE-AFTER (Gate I-G2, a head splice over all heads; Appendix G.4); we did not localise the read at 72B. An earlier exploratory run at Qwen2.5-1.5B gives the same picture: the option words recover 0.87 of the key effect under OPTIONS-AFTER and 0.98 under LETTERS-AFTER. Restricting the splice further, the read is concentrated in layers 4–15 of 28, and one of the two KV groups carries most of it (exploratory; Table 22).

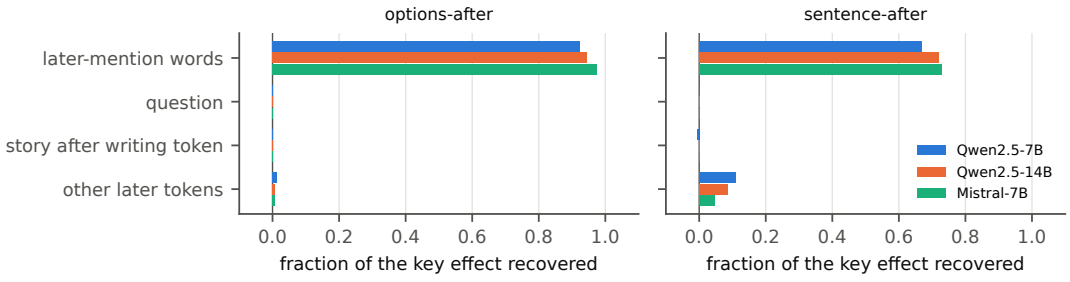


Figure 8. **Which later tokens read the writing token's key** (discovery sample; preregistration D, D3). Fraction of the full key effect recovered when only one group of later tokens sees the source key, at Qwen2.5-7B, Qwen2.5-14B and Mistral-7B, under OPTIONS-AFTER and SENTENCE-AFTER; $n = 60$ stories; values with intervals in Table 31.

Attention probe. An early exploratory probe at Qwen2.5-1.5B ($n = 25$ cores, no CIs, a different prompt) found that the option word naming the source location attends to the writing token as in the source run when the writing token's key is

Table 31. **Which later tokens read the writing token's key.** Fraction of the full key effect (all tokens see the source key, in every layer) recovered when only one group of later tokens sees it; ratio of mean log-odds changes. "Other later tokens" covers the instruction, chat template, prefill and answer position. Qwen2.5-7B, Qwen2.5-14B and Mistral-7B: BF16, $n = 60$ stories per format; preregistration D (D3) for OPTIONS-AFTER and SENTENCE-AFTER, LETTERS-AFTER exploratory. Qwen2.5-1.5B: FP32, $n = 40$, exploratory. 95% bootstrap CIs. The groups need not add up to one.

Model	Format	later-mention words	question	story after writing token	other later tokens
Qwen2.5-1.5B	OPTIONS-AFTER	+0.87 [+0.80, +0.95]	0.00 [−0.01, +0.01]	−0.01 [−0.01, 0.00]	0.00 [−0.04, +0.03]
Qwen2.5-1.5B	LETTERS-AFTER	+0.98 [+0.96, +1.00]	+0.01 [0.00, +0.01]	0.00 [−0.01, 0.00]	0.00 [0.00, 0.00]
Qwen2.5-7B	OPTIONS-AFTER	+0.92 [+0.90, +0.94]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.01 [+0.01, +0.02]
Qwen2.5-7B	LETTERS-AFTER	+1.00 [+0.99, +1.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Qwen2.5-7B	SENTENCE-AFTER	+0.67 [+0.60, +0.73]	0.00 [0.00, +0.01]	−0.01 [−0.01, 0.00]	+0.11 [+0.09, +0.13]
Qwen2.5-14B	OPTIONS-AFTER	+0.94 [+0.93, +0.96]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.01 [+0.01, +0.01]
Qwen2.5-14B	LETTERS-AFTER	+1.00 [+1.00, +1.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Qwen2.5-14B	SENTENCE-AFTER	+0.72 [+0.64, +0.80]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.09 [+0.07, +0.10]
Mistral-7B	OPTIONS-AFTER	+0.98 [+0.96, +0.99]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.01 [0.00, +0.01]
Mistral-7B	LETTERS-AFTER	+1.00 [+1.00, +1.01]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Mistral-7B	SENTENCE-AFTER	+0.73 [+0.67, +0.78]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	+0.05 [+0.03, +0.06]

swapped. Part (a) of preregistration G supersedes it (Appendix E.1.1).

Preregistration G. Preregistration G (GPU stage 5) asked what the duplicate-token account predicts and where it stops. Parts (a)–(d) use the 150 seed-0 story cores of stages 1 and 3b (the row splice and the form probe the first 60 of them), and part (e) 200 fresh IOI cores. Every contrast is paired by core, with 95% percentile intervals from 10,000 core-bootstrap resamples and ratios of means recomputed within each resample. Table 12 gives every verdict; here we define the measures, give the per-cell values and report the observations that the entry did not predict, without reinterpreting them. A shared gate required the stage-5 factorial to reproduce the stage-3b ID_K of SENTENCE-AFTER, NO-MENTION and LIST-AFTER in the four models of parts (c) and (d); every value agrees at the printed precision. Triples are Qwen2.5-7B / Qwen2.5-14B / Mistral-7B unless stated.

E.1.1. (A) RE-MENTION ATTENTION AT EVERY SCALE

Under SENTENCE-AFTER the neutral sentence mentions the written word B again. With eager attention, we take for each head the attention from that mention to the writing token, minus the mean attention to the writing token from the candidates not mentioned before it. The three heads with the largest mean excess on the odd-indexed cores form H^*. On the even-indexed cores, E is the mean excess at H^*, and F is the gain in the attention from the mention of S to the writing token under the key clamp $\mathcal{C}_K(\mathrm{S})$, so $F/E = 1$ means that the attention follows the clamped key one for one. G is the same excess for the answer position's attention to the mention of B (hop 2), at its own three heads, and Q is G at a small model divided by the smaller G of the two anchors, Qwen2.5-7B and 14B. Table 32 gives every cell.

At 1.5B and 3B the re-mention attends to the writing token about as strongly as at 7B and 14B, and the attention follows the clamped key, while ID_K under SENTENCE-AFTER stays within the preregistered ± 1-nat margin (G2, H_{diss}; G3). By the entry's criterion the heads are not embedding-level, and the same heads serve both formats at 1.5B and 7B (overlap of the two H^*, as a Jaccard index, 1.00 and 1.00; 0.50 at 3B). Hop-2 attention is weaker at the small models (Q = 0.34 and 0.44; G4), but at 3B it reaches the presence bar that G4a sets for the anchors without any key read, so it does not alone account for the scale difference. G measures attention; it does not intervene on hop 2.

Table 32. **Re-mention attention** (preregistration G, part (a)). E: excess attention of the later mention of B to the writing token at the three selected heads; F/E: the share of it that follows the key clamp; G: excess attention of the answer position to the mention of B (hop 2) at its own three heads; ID_K and the non-specific key effect d_K (nats) from the same eager run, scored in the casing the model emits. Heads selected on odd-indexed and evaluated on even-indexed cores of 150; 95% bootstrap CIs. Under SENTENCE-AFTER: G1 at 7B and 14B, G2 and G3 at 1.5B and 3B, G4 in all four; OPTIONS-AFTER is the gate and the reference of G3.

Model	Format	E	F/E	G (hop 2)	ID_K (nats)	d_K (nats)
Qwen2.5-1.5B	SENTENCE-AFTER	0.789 [0.778, 0.802]	1.03	0.086 [0.075, 0.098]	−0.53 [−0.64, −0.41]	+0.31
	OPTIONS-AFTER	0.750 [0.734, 0.767]	1.04	0.453 [0.423, 0.484]	+2.63 [2.41, 2.86]	+5.96
Qwen2.5-3B	SENTENCE-AFTER	0.879 [0.866, 0.892]	0.99	0.113 [0.095, 0.132]	+0.04 [−0.35, 0.44]	+3.03
	OPTIONS-AFTER	0.832 [0.821, 0.844]	1.03	0.395 [0.355, 0.435]	+10.37 [9.62, 11.13]	+20.57
Qwen2.5-7B	SENTENCE-AFTER	0.869 [0.852, 0.885]	1.00	0.255 [0.222, 0.288]	+4.73 [4.39, 5.07]	+9.21
	OPTIONS-AFTER	0.887 [0.874, 0.900]	1.00	0.770 [0.739, 0.798]	+19.39 [19.00, 19.78]	+36.90
Qwen2.5-14B	SENTENCE-AFTER	0.915 [0.908, 0.921]	0.98	0.329 [0.304, 0.353]	+13.48 [12.76, 14.20]	+24.09
	OPTIONS-AFTER	0.866 [0.856, 0.876]	0.99	0.865 [0.859, 0.872]	+38.95 [37.88, 39.97]	+74.79

Unpredicted observations. (i) At 3B the key clamp has a non-specific effect without identity (d_K = 3.03 nats against ID_K = 0.04). (ii) Under SENTENCE-AFTER only 0.66 and 0.75 of the cores at the small models are answered correctly in both clean runs, and almost every wrong answer is the initial location; on the competent cores the 3B value is slightly positive, as in the stage-1 competent-only analysis (Appendix D.1; recomputed, not preregistered). (iii) Under the key clamp, the answer's hop-2 attention does not follow the clamped key at 1.5B (change towards the mention of S 0.015, towards B 0.028), partly follows it at 3B (0.127 and -0.105) and follows it at 7B and 14B (0.343 and -0.243; 0.321 and -0.213). (iv) Under SENTENCE-AFTER at 7B and 14B, the answer row at the hop-2 heads attends mostly to the writing token itself (0.534 and 0.468), not to the re-mentions; under OPTIONS-AFTER its five largest columns are all re-mentioned words. (v) The two formats' H^* overlap little at 14B (0.20), but each format's heads still work on the other (cross-format E = 0.768 and 0.734).

E.1.2. (B) ATTENTION KNOCKOUT

A knockout sets the attention from a set of rows to a set of columns to zero in every layer and head (Appendix C). The preregistered masks are M0, none; M1, the six candidate words after the writing token to the writing token; M2, the same rows to the earlier occurrences of the initial-location word, a matched control column; M3, the answer position to the writing token; M4, the union of M1 and M3; and M8, every later row to the writing token, a sanity gate. The exploratory masks are M2b, the candidate rows to the token before the writing token; M5, only the rows of the three clamped candidates B, S and X; M6, the other three candidate rows; M7, every later row that is not a candidate word; Mq, the question rows; and Mpost, the rows between the candidates and the answer position. r_K is ID_K under the mask divided by ID_K without it; $q_V = \mathrm{ID}_V/d_{KV}$ is the value identity as a share of the joint clamp's effect; acc_B is the fraction of cores whose answer, restricted to the candidates, is still B. Table 33 gives every cell.

Cutting the candidate words' attention to the writing token removes the key read in every format and model, the control column does not, and cutting the answer's own attention leaves the read intact (G5, G6, G8b). The copy returns to its NO-MENTION share on the identity measure (G7a), but the answer is kept only at Qwen2.5-14B (G7b): the preregistered outcome is a partial takeover. Under NO-MENTION, M3 leaves 0.134 / 0.370 / 0.417 of ID_V (G8a), but it also removes most of the writing token's whole effect and of the models' ability to answer (Table 33).

Unpredicted observations. (i) At Qwen2.5-7B, the answer that replaces B under M1 is mostly the initial location (exploratory, recomputed, descriptive); G7b's location-mass criterion cannot see these answers, because they are location words too. (ii) Under SENTENCE-AFTER, cutting only the answer's attention to the writing token (M3) raises the key read (r_K = 1.660 / 1.800 / 1.394), and so does M7. (iii) Cutting only the rows of the three clamped candidates (M5) removes the

Table 33. **Attention knockout** (preregistration G, part (b); masks in the text). r_K: ID_K under the mask relative to no mask; q_V: ID_V as a share of the joint clamp's effect d_{KV}; acc_B: fraction of cores whose candidate-restricted answer is B. $n = 150$ cores per cell. M1, M2, M3, M4 and M8 enter G5–G8 and Gate b; the other masks, and every mask under SENTENCE-AFTER, are exploratory. Under NO-MENTION only the masks that do not need candidate rows are defined.

Format	Mask	Qwen2.5-7B			Qwen2.5-14B			Mistral-7B		
		r_K	q_V	acc_B	r_K	q_V	acc_B	r_K	q_V	acc_B
LIST-AFTER	M0	1.000	0.125	1.00	1.000	0.075	1.00	1.000	0.105	1.00
	M1	0.001	0.390	0.37	−0.026	0.440	1.00	−0.007	0.448	0.86
	M2	1.094	0.104	1.00	0.997	0.065	1.00	0.990	0.113	1.00
	M2b	1.001	0.125	1.00	1.000	0.075	1.00	1.000	0.107	1.00
	M3	1.043	0.022	1.00	1.027	0.017	1.00	1.010	0.053	1.00
	M4	0.001	0.512	0.00	0.001	0.470	0.66	−0.001	0.457	0.39
	M5	−0.008	0.418	0.73	−0.025	0.446	1.00	−0.007	0.457	0.89
	M6	0.961	0.117	1.00	1.003	0.071	1.00	1.003	0.106	1.00
	M7	0.947	0.003	0.97	1.022	0.004	1.00	0.948	0.020	0.97
	M8	0.000	–	0.00	0.000	–	0.13	0.000	–	0.17
	Mq	0.981	0.123	1.00	1.002	0.076	1.00	0.988	0.112	1.00
	Mpost	1.001	0.119	1.00	1.002	0.068	1.00	0.999	0.096	1.00
OPTIONS-AFTER	M0	1.000	0.120	1.00	1.000	0.066	1.00	1.000	0.074	1.00
	M1	0.003	0.377	0.30	−0.028	0.427	0.99	−0.009	0.422	0.73
	M2	1.071	0.099	1.00	0.998	0.057	1.00	0.972	0.083	1.00
	M2b	1.000	0.120	1.00	0.999	0.066	1.00	1.001	0.077	1.00
	M3	1.041	0.019	1.00	1.017	0.013	1.00	1.006	0.040	1.00
	M4	−0.000	0.523	0.00	0.002	0.464	0.58	0.001	0.446	0.29
	M5	−0.004	0.404	0.58	−0.027	0.435	1.00	−0.008	0.429	0.77
	M6	0.971	0.113	1.00	1.005	0.059	1.00	1.007	0.077	1.00
	M7	0.944	0.001	0.98	0.983	0.004	1.00	0.926	0.015	0.99
	M8	0.000	–	0.00	0.000	–	0.10	0.000	–	0.19
	Mq	0.980	0.119	1.00	0.997	0.068	1.00	0.993	0.079	1.00
	Mpost	1.001	0.118	1.00	0.998	0.065	1.00	1.000	0.070	1.00
SENTENCE-AFTER	M0	1.000	0.346	1.00	1.000	0.325	1.00	1.000	0.361	1.00
	M1	−0.105	0.423	1.00	−0.036	0.443	1.00	0.001	0.460	0.99
	M2	1.027	0.346	1.00	0.922	0.333	1.00	0.963	0.368	1.00
	M2b	1.007	0.346	1.00	0.994	0.327	1.00	0.991	0.362	1.00
	M3	1.660	0.099	0.77	1.800	0.117	1.00	1.394	0.188	0.99
	M4	0.007	0.496	0.07	0.014	0.480	0.91	0.002	0.446	0.29
	M5	−0.113	0.423	1.00	−0.035	0.443	1.00	−0.003	0.464	1.00
	M6	0.974	0.348	1.00	1.053	0.319	1.00	0.995	0.359	1.00
	M7	1.184	−0.036	0.35	1.939	−0.016	1.00	1.456	0.012	0.86
	M8	0.000	–	0.00	0.000	–	0.14	0.000	–	0.05
	Mq	0.929	0.349	0.99	1.005	0.323	1.00	1.000	0.359	1.00
	Mpost	1.028	0.329	0.99	1.327	0.255	1.00	1.209	0.290	1.00
NO-MENTION	M0	1.000	0.391	1.00	1.000	0.335	1.00	1.000	0.433	1.00
	M3	0.023	0.566	0.00	0.041	0.459	0.20	0.117	0.445	0.15
	M8	0.000	–	0.00	0.000	–	0.00	0.000	–	0.12
	Mq	0.967	0.388	1.00	1.005	0.331	1.00	0.988	0.432	1.00
	Mpost	1.040	0.378	1.00	1.004	0.318	1.00	0.995	0.402	1.00

read as completely as M1 (under LIST-AFTER, r_K = −0.008 / −0.025 / −0.007), while cutting the other three (M6) leaves it intact (0.961 / 1.003 / 1.003). (iv) Under SENTENCE-AFTER, M1 raises q_V to 0.423 / 0.443 / 0.460. (v) At Qwen2.5-7B, cutting the control column (M2) slightly strengthens the key read (r_K = 1.094).

E.1.3. (C) MEMBERSHIP AND DOSE

The arms replace the neutral sentence (family S) or the option list (family L) by one that names a subset of the candidates, in the canonical order and with "Answer with one word": S2 and L2 name S and X; S3 and L3 name B, S and X; S3out and L3out name B and two candidates other than S and X; S3half names B, S and one other; S4 and L4 name B, S, X and one other; S4out and L4out name S, X and two others, but not B; and S6 and L6 are SENTENCE-AFTER and LIST-AFTER.

r_k is ID_K in the arm naming k candidates divided by ID_K in the arm naming all six. The row splice (first 60 cores) lets only the named words' rows see the source key, giving f_{words}, their share of the total splice effect d. Table 34 gives every arm.

Table 34. **Membership and dose** (preregistration G, part (c)). Top: ID_K and ID_V (nats) per arm, seed-0 cores, $n = 150$. Bottom: row splice ($n = 60$), the share of the total key-swap effect d read by the named words' rows (f_{words}) and by the rows after the question. Qwen2.5-7B, Qwen2.5-14B and Mistral-7B enter G9–G12; OLMo-2-7B and the arms S3half, S4, S4out, L4, L4out, S6, L6 (as splice arms) and S3out, L3out (as splice arms) are exploratory.

Arm	Qwen2.5-7B		Qwen2.5-14B		Mistral-7B		OLMo-2-7B	
	ID_K	ID_V	ID_K	ID_V	ID_K	ID_V	ID_K	ID_V
NO-MENTION	+1.04	+16.60	+1.05	+19.12	+0.55	+13.65	+0.38	+10.97
S2	+1.52	+13.97	+6.99	+23.51	+3.66	+13.22	+0.86	+8.55
S3	+3.49	+11.94	+9.52	+19.67	+5.25	+11.83	+1.51	+8.75
S3half	+2.11	+12.13	+5.41	+16.16	+2.66	+11.79	+0.67	+8.37
S3out	+0.73	+12.61	+0.75	+13.40	+0.34	+12.18	+0.29	+8.84
S4	+4.59	+12.26	+10.88	+18.88	+6.15	+11.87	+1.64	+8.69
S4out	+4.18	+14.08	+10.82	+24.36	+6.17	+13.58	+1.59	+8.96
S6	+5.52	+11.91	+12.08	+18.46	+7.87	+11.46	+2.01	+8.63
L2	+19.76	+18.63	+35.19	+29.48	+15.27	+16.24	+5.87	+10.03
L3	+21.70	+4.06	+35.83	+2.97	+15.66	+3.29	+7.89	+5.14
L3out	+0.69	+5.49	+0.92	+3.49	+0.31	+2.47	+0.37	+4.30
L4	+21.50	+4.71	+35.87	+4.19	+15.38	+3.62	+8.35	+5.33
L4out	+20.29	+14.90	+35.45	+30.76	+15.45	+13.68	+8.00	+9.07
L6	+20.60	+5.93	+35.81	+5.69	+15.55	+3.44	+8.65	+5.93
Reader rows (splice, $n = 60$): f_{words} / share read by the rows after the question								
S2	0.40 / 0.46		0.38 / 0.49		0.39 / 0.38		0.26 / 0.62	
S3	0.68 / 0.13		0.72 / 0.11		0.72 / 0.10		0.90 / −0.01	
S6	0.67 / 0.11		0.72 / 0.09		0.73 / 0.05		0.76 / 0.06	
S3out	0.04 / 0.90		0.11 / 0.74		0.11 / 0.68		−0.16 / 1.40	
L2	0.91 / 0.14		0.94 / 0.17		0.92 / 0.08		0.72 / 0.13	
L3	0.94 / 0.01		0.95 / 0.00		0.95 / 0.01		0.91 / −0.01	
L6	0.91 / 0.01		0.93 / 0.01		0.95 / 0.01		0.88 / 0.00	
L3out	0.58 / 0.13		0.39 / 0.09		0.41 / 0.18		0.58 / 0.31	

The key read needs the swapped candidates themselves: naming B with two others gives none (G9, met in both families and replicated at seed 1). In lists two named candidates give the full read; in sentences the read grows with the number named, and at Qwen2.5-7B r_2 = 0.28 (0.32 at seed 1) is not above proportional (G10, met for lists, not for sentences). Removing S and X from a list does not bring the copy back (R_V = 0.11 / 0.03 / −0.08; G11, not met). In a sentence naming only S and X (S2), the named words carry 0.40 / 0.38 / 0.39 of the splice effect and the rows after the question 0.46 / 0.49 / 0.38 (G12, not met, 9/12). G12 splits the total key-swap effect d, not ID_K, and in S2 most of d is not identity-specific: ID_K/d_K is 0.14 / 0.29 / 0.38 in S2 against 0.44 / 0.56 / 0.61 in S3 (arithmetic on printed values). The main text's narrowing to "lists and sentences naming at least three candidates" rests on this.

Unpredicted observations. (i) Naming B with two other candidates pushes ID_K below NO-MENTION, although the written word is re-mentioned (S3out−NO-MENTION −0.32, L3out−NO-MENTION −0.35 at Qwen2.5-7B; Table 12). (ii) In sentences every added candidate raises ID_K: at Qwen2.5-7B the paired steps are S3−S2 = 1.96 (adds B), S4−S3 = 1.11 (adds one non-matching candidate) and S6−S4 = 0.93 (adds two), each with a CI excluding zero; a pure duplicate-token account does not predict the last two. Naming one of the two swapped candidates (S3half) gives about half of S3 (0.61 at Qwen2.5-7B). (iii) In lists that do not name the written word B, the key read is at full strength while the value identity stays near or above NO-MENTION: ID_V(L4out) is 14.90 / 30.76 / 13.68 against 16.60 / 19.12 / 13.65 under NO-MENTION and 4.06 / 2.97 / 3.29 in L3. This suggests, post hoc, that the copy is suppressed when the written word itself is re-mentioned; it is a hypothesis for a future test, not a claim. (iv) Under NO-MENTION and every sentence arm, the clean answer puts almost no probability on the six lowercase candidates at the three primary models, so the sentence-family contrasts are scored on tokens the model is not about to emit, as in stage 3b; in lists the candidates carry most of it.

E.1.4. (D) NON-IDENTICAL RE-MENTIONS

Each variant replaces the re-mentions of the SENTENCE-AFTER sentence: THE adds "the"; MODIF uses a modified noun phrase ("wooden shelf"); TITLE and UPPER change the case; PLURAL uses the plural; SYN a synonym; FRMIX and DEMIX put the French or German word in the English frame; FR and DE translate the whole sentence (exploratory); and three structure-matched floor sentences name other words. With the floor and the exact repeat (SENTENCE-AFTER) scored on the English candidates, $r_K(v) = [\mathrm{ID}_K(v) - \mathrm{ID}_K(\text{floor})]/[\mathrm{ID}_K(\text{SENTENCE-AFTER}) - \mathrm{ID}_K(\text{floor})]$; r_K^{any} also scores the answer in the variant's own form; ρ_s is the ratio of the identity key shares; and a_v is the variant's attention to the writing token, summed over its tokens, relative to the exact repeat's ($n = 60$). Gates required clean accuracy of at least 0.95 in the variant, its floor and its anchor, and that at least five of the six words map correctly in a six-way prompt (the passing words are the population of the cell). A prediction is met if at least three evaluable models meet it. Table 35 gives every cell.

Table 35. **Non-identical re-mentions** (preregistration G, part (d)). r_K: key identity relative to an exact repeat, above a structure-matched floor, on the English candidates; r_K^{any}: the same with the answer also scored in the variant's own form; a_v: attention of the re-mention to the writing token relative to the exact repeat, summed over the re-mention's tokens ($n = 60$). $n = 150$ cores per cell before the per-word competence gate; DEMIX and DE cover fewer cores. THE, MODIF, TITLE, UPPER, PLURAL, SYN, FRMIX and DEMIX enter G13–G16; FR, DE and the list variant are exploratory.

Variant	Qwen2.5-7B			Qwen2.5-14B			Mistral-7B			OLMo-2-7B		
	r_K	r_K^{any}	a_v	r_K	r_K^{any}	a_v	r_K	r_K^{any}	a_v	r_K	r_K^{any}	a_v
THE	1.09	1.09	1.09	0.99	0.99	1.02	1.06	1.06	1.11	0.96	0.96	1.02
MODIF	0.32	0.32	1.22	0.64	0.64	1.24	0.64	0.64	1.08	0.55	0.55	0.94
TITLE	0.94	1.19	0.92	1.09	1.37	0.97	1.07	1.53	1.79	0.87	0.89	0.90
UPPER	1.08	1.51	1.74	1.14	1.57	1.93	1.01	1.26	2.58	0.83	0.90	1.84
PLURAL	0.33	0.33	0.87	0.37	0.37	0.88	0.49	0.50	1.40	0.11	0.11	0.89
SYN	−0.03	0.09	0.41	0.00	0.09	0.41	0.06	0.26	0.53	−0.13	−0.13	0.21
FRMIX	0.15	0.95	0.86	0.10	1.06	0.90	0.07	0.56	0.55	0.09	0.44	0.39
DEMIX	−0.01	0.69	0.66	0.02	1.32	0.65	0.02	0.31	0.33	0.00	−0.13	0.09
FR	0.18	0.55	1.16	0.18	0.47	1.21	0.14	0.28	0.83	0.16	0.41	0.58
DE	0.00	0.59	0.92	0.06	1.06	1.01	0.05	0.55	0.58	−0.05	0.43	0.30
list SYN	0.05	0.33	–	0.05	0.23	–	0.07	0.47	–	−0.00	0.05	–

On our English-scored measure the read is bound to the word form (G15): synonyms and mixed-language words carry at most 0.15 of an exact repeat's key identity. It is largely insensitive to case (TITLE and UPPER read 0.83–1.14; G14 not met for them), sensitive to number (PLURAL 0.11–0.49), and a modified noun phrase reads only partly (MODIF 0.32–0.64; G13 not met for it, THE met). No preregistered token-or-concept outcome holds for any variant (G16, not met): the profile is graded and model-dependent. Per model, and exploratory: at the two Qwen models FRMIX shows the concept pattern on both measures (r_K^{any} = 0.95 and 1.06; a_v = 0.86 and 0.90), and scored in its own form it reads fully (r_K = 0.99 and 1.28); DEMIX is concept-like on r_K^{any} but graded on a_v there; at Mistral-7B and OLMo-2-7B the read is graded or token-level. Where the read is lost, the value identity rises (G17, met), but by no more than after the floor sentence that names other words (exploratory: the floor sentence raises ID_V over SENTENCE-AFTER by 2.71, 2.51 and 0.54 nats at Qwen2.5-7B, Mistral-7B and OLMo-2-7B, recomputed, against 2.58, 1.37 and 0.31 for FRMIX).

Unpredicted observations. (i) A capitalised or upper-case re-mention reads nearly as fully as the exact repeat, although its token ids differ; the CI lies below 1 in three of the eight cells. (ii) MODIF attends to the writing token about as much as the exact repeat (a_v = 1.22, 1.24, 1.08 and 0.94 in the four models) but reads the key only partly: attention without the read. Values of a_v above 1 come from summing attention over multi-token spans; with the last token only, the FRMIX attention leg falls below the concept bar (exploratory, recomputed in the log). (iii) The raw rise of $\log p(\mathrm{S})$ under the key clamp does not separate the variants, so only the identity contrasts are reported.

E.2. Reader heads and the second hop

Preregistration H (GPU stage 6) has two parts: the heads that read the writing token's key and the route from them to the answer, at two 7B models (this subsection), and the key-only and value-only exchange applied to the binding intervention of Prakash et al. (2026) at Qwen2.5-14B (Appendix H.1). Table 13 gives every gate and verdict, and Appendix A.5 the disclosures.

Summary (discovery sample). Preregistration H looked inside the option rows at Qwen2.5-7B and Mistral-7B under OPTIONS-AFTER (Table 36); H1–H5 were met in both. When only the top 5% of heads by a_3 (40 and 52) see the source key, they recover $R(k^*) = 0.967$ and 0.942 of the option rows' read (H1), and blinding them removes 0.977 and 0.971 (H2); random sets of equal size recover 0.004 and 0.007 and remove 0.011 and 0.010 (three-draw means; Figure 9b). Ranking by single-head effect does not find the set, nor does leave-one-out at Qwen2.5-7B, and no layer alone recovers more than 0.05 (exploratory; below): the read is joint across layers.

Mean-ablating them at the option rows cuts ID_K to $\rho_K = 0.238$ and 0.059 of its value (control sets ≥ 0.953), raises ID_V by 6.907 and 10.861 nats, and leaves the base answer in 0.87 and 1.00 of stories (H3). The lookup thus suppresses a copy route that stays available. Cutting the candidate words' edges instead (G5) also restores the value identity in 3/3 models but keeps the answer only at Qwen2.5-14B (G7, not met: a preregistered partial takeover); we do not reconcile the two. We do not claim the lookup replaces the copy: its preregistered tests failed (G11; on IOI, G20).

The second hop. The answer need not attend to the writing token (forbidding it leaves r_K at 1.010–1.043; G8b, met at 7–14B); it reads the option words. With the source key clamped, giving only the answer row the option rows' base keys and values removes $r_{\mathrm{ans}} = 0.899$ and 0.859 of the key-clamp effect, against 0.118 and 0.165 for all other later rows (H5, met; Figure 9c). Unscored (H5), the answer reads them by key: $r_{\mathrm{ans}}(K) = 0.797$ and 0.683, against $r_{\mathrm{ans}}(V) = 0.080$ and 0.023. The SENTENCE-AFTER line is not evaluable (below). Without a later mention, the same cut leaves 0.134–0.417 of ID_V (G8a, met).

What the duplicate-token account explains. The first hop, a re-mention attending to the writing token, is the pattern of duplicate-token heads (Wang et al., 2022). Under SENTENCE-AFTER, the re-mentioned word attends to the writing token far more than the other candidates do, and this attention follows the clamped key one for one, at 7B and 14B (E, the excess at the top three heads, = 0.869, 0.915; F/E, its share following the clamp, = 1.00, 0.98; G1) and equally at 1.5B and 3B (E = 0.789, 0.879; F/E = 1.03, 0.99; G2, G3), whose SENTENCE-AFTER key identity was about zero on the lower-case score (within the preregistered ± 1-nat margin). All three were met. Whether the flag that this match writes at 1.5B carries the size of the sentence read was tested in the final round (Appendix E.5): the flag is written in sentences as in lists at every scale, the preregistered mediation test came out mixed, and at 1.5B the injected flag is read by key. The core of the causally selected readers is of this kind (H4, met): over the top 20 and 16 heads by a_3, which carry 80% of the read, the median duplicate-token score D (on random repeated sequences) is 0.370 and 0.232 (whole-model medians 0.028 and 0.007; exploratory, recomputed), the median induction score (Olsson et al., 2022) 0.011 and 0.022. Only this core is canonical: over the full 5% set the median D is 0.179 and 0.065 (exploratory, recomputed).

We therefore claim no new head type. What the discovery sample added is that the answer then reads the re-mentioned words, mainly by key (H5, secondary); the final round tested what these heads write (Appendix E.3). The channels carry which value was written, not its belief role (E2, met; Table 24).

Design. Qwen2.5-7B (784 heads) and Mistral-7B (1024 heads), BF16 with eager attention; OPTIONS-AFTER confirmatory, SENTENCE-AFTER reported. Heads are ranked on 60 ranking stories by a_3, the attention change of the option rows of S and B towards p under the key clamp ($a_3 = \frac{1}{2}[(A^{K_{\mathrm{S}}} - A^{\mathrm{B}})_{\text{row S}\to p} + (A^{\mathrm{B}} - A^{K_{\mathrm{S}}})_{\text{row B}\to p}]$), and every curve is evaluated on 60 other stories. $R(k)$ is the fraction of the option rows' key effect d_G recovered when only the top k heads see the source key in the option rows (HeadSplice; Appendix C), and $\mathrm{KO}(k)$ the fraction lost when only they are blinded; $k^* = \lceil 0.05 \times \text{heads} \rceil$ = 40 and 52, and k_{80} is the smallest k on the grid with $R(k) \geq 0.8$. The other rankings are the single-head effect f^+ (each head alone sees the source key) and the leave-one-out effect d^- (each head alone is blinded); the controls are three random sets (the first k^* heads of fixed random permutations) and the k^* heads most active at the option rows outside the top $2k^*$ by a_3. The duplicate-token score D and the induction score I come from 100 random repeated-token sequences, and the in-task score T_{dup} is the attention from each option row to the earlier occurrences of the same word other than p. The causal set C is the top $\min(k_{80}, k^*)$ heads by a_3. Table 36 gives H1–H5, Figure 10 the curves

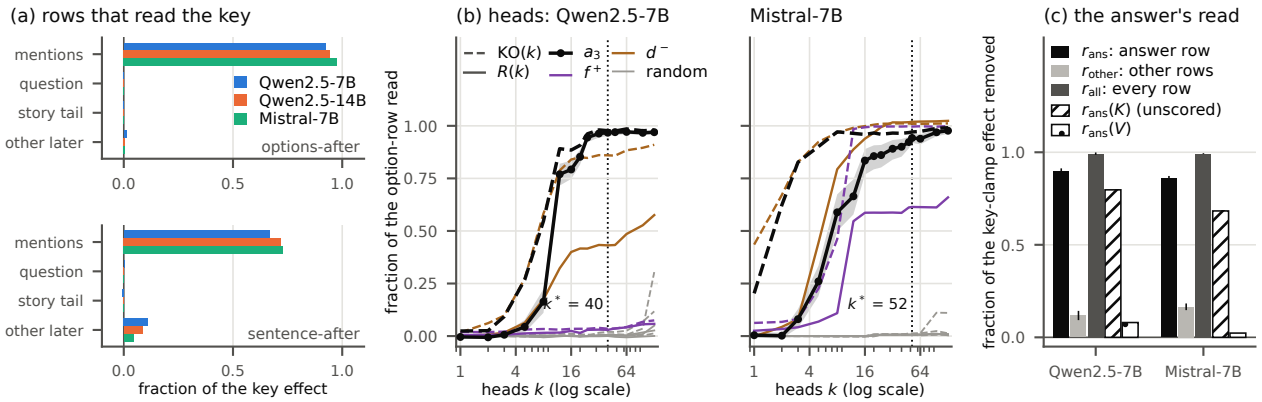


Figure 9. **The later mentions read the writing token's key through a small set of heads, and the answer reads the later mentions** (discovery sample). (a) Row-restricted splice ($n = 60$; D3): the fraction of the full key effect recovered when only one group of later tokens sees the source key (top OPTIONS-AFTER, bottom SENTENCE-AFTER). (b) OPTIONS-AFTER, 60 held-out stories: $R(k)$ (solid) and $\mathrm{KO}(k)$ (dashed) for heads ranked by a_3 (band: 95% CI), by single-head effect f^+ and leave-one-out effect d^-, and three random sets (grey); dotted: k^* (H1, H2). (c) Key-clamp effect removed when only the answer row, all other later rows or all rows get the option rows' base keys and values (H5; hatched, open: keys or values only, not scored). Values in Tables 31 and 36.

and Table 37 every ablation condition.

Table 36. **Reader heads and the second hop** (preregistration H, part (a); OPTIONS-AFTER, 60 held-out stories, BF16). $R(k^*)$, $\mathrm{KO}(k^*)$: fraction of the option rows' key effect recovered when only the top k^* heads by a_3 see the source key, or lost when only they are blinded. ρ_K: ID_K under mean-ablation of these heads at the option rows, relative to no ablation. C: the top heads by a_3 that carry 80% of the read. r_{ans}: fraction of the key-clamp effect removed when only the answer row gets the option rows' base keys and values; r_{other}, r_{all}: the same for every other later row and for all rows. 95% bootstrap CIs over stories. The last row is H5's secondary contrast, reported, not scored.

	Quantity (OPTIONS-AFTER)	Qwen2.5-7B	Mistral-7B
	heads k^* (of all heads)	40 (784)	52 (1024)
H1	$R(k^*)$, top-k^* by a_3 see K_S	0.967 [0.952, 0.982]	0.942 [0.919, 0.961]
	k_{80} by a_3 / by f^+ / by d^-	20 / none / none	16 / none / 12
H2	$\mathrm{KO}(k^*)$, top-k^* blinded	0.977 [0.966, 0.987]	0.971 [0.954, 0.984]
	random sets: R / KO (mean of 3)	0.004 / 0.011	0.007 / 0.010
H3	ρ_K under mean-ablation of the top-k^*	0.238 [0.205, 0.270]	0.059 [0.040, 0.079]
	ρ_K, control sets (min–max)	0.953–1.018	1.002–1.009
	rise of ID_V (nats)	+6.907 [6.264, 7.561]	+10.861 [9.995, 11.677]
	answer kept (base argmax)	0.87	1.00
H4	$\lvert C\rvert$ (top heads by a_3 carrying 80 %)	20	16
	median duplicate score D over C	0.370 [0.361, 0.378]	0.232 [0.227, 0.238]
	median induction score I over C	0.011	0.022
	median in-task T_{dup} over C	0.459	0.433
	top-10 by a_3 ∩ top-10 by D	6	5
H5	r_{ans}: only the answer row restored	0.899 [0.886, 0.913]	0.859 [0.845, 0.872]
	r_{other} / r_{all}	0.118 / 0.992	0.165 / 0.992
	$r_{\mathrm{ans}}(K)$ / $r_{\mathrm{ans}}(V)$ (secondary, not scored)	0.797 / 0.080	0.683 / 0.023

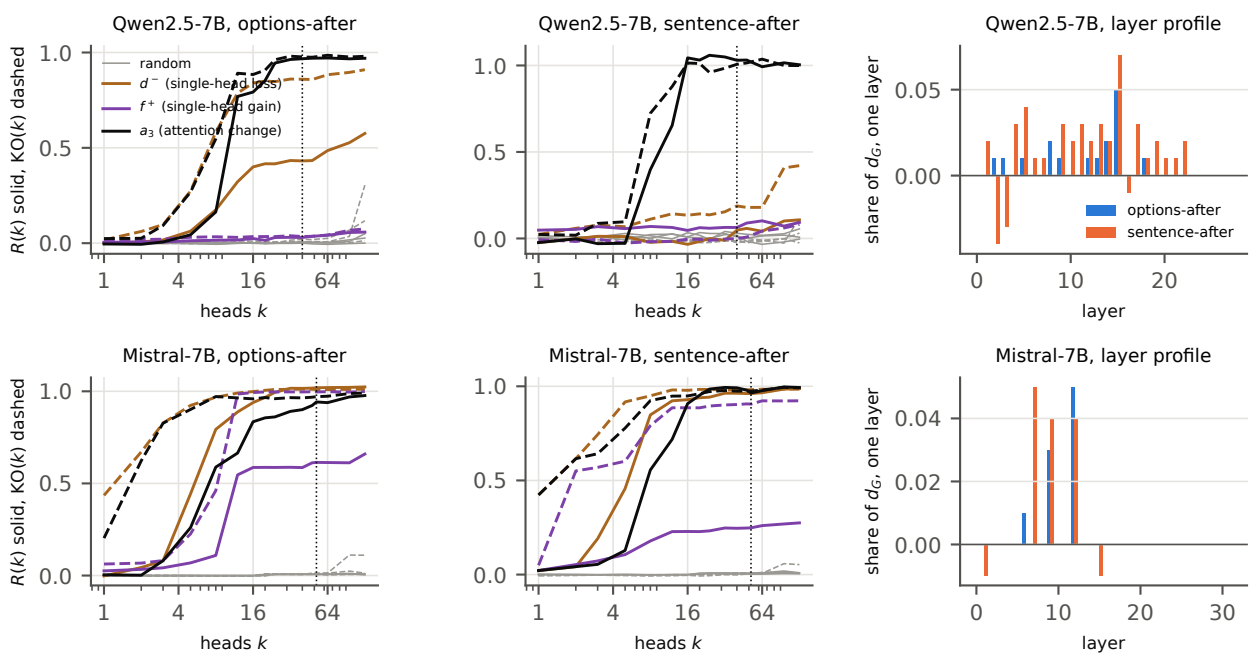


Figure 10. **Head curves and layer profiles** (preregistration H, part (a); exploratory except $R(k^*)$ and $\mathrm{KO}(k^*)$ of the a_3 ranking under OPTIONS-AFTER). Left and middle: $R(k)$ (solid) and $\mathrm{KO}(k)$ (dashed) against the number of heads k for the rankings by a_3, single-head effect f^+ and leave-one-out effect d^-, and for three random sets (grey), under OPTIONS-AFTER and SENTENCE-AFTER; dotted: k^*. Right: the share of the option rows' key effect recovered when all heads of one layer see the source key. 60 held-out stories per model and format.

Sufficiency and necessity (H1, H2). The top 5% of heads recover almost all of the option rows' read and blinding them removes almost all of it, while random sets of equal size do neither; no single random draw exceeds R = 0.006 / 0.007 or KO = 0.020 / 0.013 (exploratory, recomputed). The ranking is stable: the split-half Spearman correlation of a_3 between ranking and evaluation stories is 0.973 / 0.956 under OPTIONS-AFTER and 0.985 / 0.981 under SENTENCE-AFTER.

The read is a joint property of many heads (exploratory). Ranking by single-head effect does not find the set, and ranking by leave-one-out effect finds it at one of the two models: at Qwen2.5-7B, f^+ reaches only $R(128)$ = 0.06 and d^- 0.58; at Mistral-7B f^+ reaches 0.66, while d^- reaches k_{80} = 12, fewer heads than a_3 needs (16). The a_3 curve rises steeply (Qwen2.5-7B: $R(8)$ = 0.16, $R(12)$ = 0.77), and at Mistral-7B blinding the top three heads removes 0.83 of the read while the same three alone recover 0.08. Letting all heads of a single layer read the source key recovers at most 0.05 (layer 15) and 0.05 (layer 12) of d_G, and the single-layer shares sum to only 0.163 and 0.094 (exploratory, recomputed); the heads of C span 12 and 10 layers (exploratory, recomputed). No single head or layer is necessary or sufficient; the top 20 and 16 heads by a_3, over many layers, are, jointly.

Ablation (H3). Mean-ablating the top-k^* heads at the option rows removes most of the key identity, raises the value identity and keeps the answer; the random and most-active sets of equal size leave ρ_K near 1 (Table 37). Zero-ablation (exploratory) goes the same way (ρ_K = 0.081 and 0.057). Without ablation the answer is B in 0.97 of Qwen2.5-7B stories, against 0.87 under ablation. The rise of the value identity is partial at Qwen2.5-7B (ID_V under ablation 11.55 nats against 16.60 under NO-MENTION in stage 1), and at Mistral-7B it reaches about the NO-MENTION level (13.45 against 13.65); both comparisons are across stages and stories (arithmetic on printed values). The stage-5 edge knockout keeps the answer only at Qwen2.5-14B (G7b); the two interventions differ (mean-ablation at six rows leaves ρ_K = 0.238 at Qwen2.5-7B, the knockout leaves $r_K \approx 0$), and we do not reconcile them.

Duplicate-token heads (H4) and the size of C. Over C the median previous-token score is 0.017 / 0.004 (largest 0.083 / 0.049, exploratory, recomputed), and the in-task score of control words that are not candidates is 0.027 / 0.065, against T_{dup} = 0.459 / 0.433. T_{dup} is measured in the task the heads were selected on, so we treat it as supporting rather than

Table 37. **Every ablation condition** (preregistration H, part (a); 60 held-out stories). ρ_K: ID_K under ablation at the six option rows, relative to none; $\Delta\mathrm{ID}_V$: rise of the value identity (nats); kept: fraction of stories whose candidate-restricted clean answer is still B. "mean": mean-ablation with the head's position-matched mean over the ranking stories; "zero": zero-ablation. H3 uses the top-k^* mean row, the three random sets and the most-active set under OPTIONS-AFTER; every other cell is exploratory.

Condition	Qwen2.5-7B, OPTIONS-AFTER			Qwen2.5-7B, SENTENCE-AFTER			Mistral-7B, OPTIONS-AFTER			Mistral-7B, SENTENCE-AFTER		
	ρ_K	$\Delta\mathrm{ID}_V$	kept	ρ_K	$\Delta\mathrm{ID}_V$	kept	ρ_K	$\Delta\mathrm{ID}_V$	kept	ρ_K	$\Delta\mathrm{ID}_V$	kept
none	1.000	+0.000	0.97	1.000	+0.000	1.00	1.000	+0.000	1.00	1.000	+0.000	1.00
top-k^* (mean)	0.238	+6.907	0.87	−0.080	+2.295	1.00	0.059	+10.861	1.00	0.012	+2.640	1.00
top-10 (mean)	0.859	+1.978	0.98	0.616	+1.436	1.00	0.139	+12.326	1.00	0.233	+2.302	0.98
top-20 (mean)	0.654	+4.285	0.95	0.126	+2.282	1.00	0.104	+12.728	1.00	0.056	+2.648	1.00
top-k^* (zero)	0.081	+11.315	1.00	−0.060	+2.756	1.00	0.057	+9.030	0.95	0.006	+2.609	1.00
random set 1	1.002	+0.091	0.97	0.969	+0.081	1.00	1.002	−0.096	1.00	1.003	+0.048	1.00
random set 2	0.978	+0.851	1.00	0.655	+0.820	1.00	1.009	−0.292	1.00	1.289	−1.563	1.00
random set 3	0.953	+0.241	0.98	0.975	+0.114	1.00	1.006	−0.326	1.00	1.100	−0.336	1.00
most active at G	1.018	−0.177	0.97	0.934	+0.012	1.00	1.009	+0.049	1.00	0.949	+0.521	1.00
next k^* by a_3	1.019	−0.062	1.00	1.140	−0.028	1.00	0.996	+0.173	1.00	0.823	+0.682	1.00

independent evidence; D and I, on random sequences, are the test. H4 depends on the size of the head set (exploratory, recomputed): only 13 of the 20 heads of C at Qwen2.5-7B and 9 of the 16 at Mistral-7B have $D \geq 0.2$, against 26 and 11 in the whole model, whose median D is 0.028 / 0.007. Over the full top-k^* set (exploratory) the median D is 0.179 / 0.065; at Mistral-7B it is 0.156 over the top 20 and 0.294 over the top 12. At Qwen2.5-7B (exploratory), $R(16) = 0.792$, just under 0.8, so C has 20 heads; over the top 16 the median D would also be 0.370. Over all heads, a_3 correlates weakly with D (Spearman 0.185 / 0.230) and with T_{dup} (0.146 / 0.251). We therefore say that the core of the readers, the top heads by a_3 that carry 80% of the read, is canonical duplicate-token heads, not that every reader is one.

The second hop in detail (H5). The routes add up: $r_{\mathrm{ans}} + r_{\mathrm{other}}$ = 1.017 / 1.024 against r_{all} = 0.992 / 0.992. Restricted to single option rows, the source word's row gives r_{ans} = 0.605 / 0.557, the base word's row 0.414 / 0.391 and the separators 0.049 / 0.016 (exploratory). As H5's secondary contrast, reported and not scored, the answer reads the option rows mainly through their keys ($r_{\mathrm{ans}}(K) - r_{\mathrm{ans}}(V)$ = 0.717 / 0.660, CIs excluding zero), and the strong version $r_{\mathrm{ans}}(KV) \geq 0.7$ holds in both models.

SENTENCE-AFTER (reported, not scored). Gate a2 failed at Qwen2.5-7B under SENTENCE-AFTER (Appendix A), so H6 and H5's SENTENCE-AFTER line are not evaluable; the values below are exploratory. The option-row ceiling covers less of the full effect than under OPTIONS-AFTER (d_G/d_{full} = 0.572 / 0.728, exploratory, recomputed). At Qwen2.5-7B, $R(k)$ and $\mathrm{KO}(k)$ exceed 1 (Figure 10), and mean-ablation of the top k^* drives ρ_K to -0.080; one random set changes ρ_K substantially in each model (Table 37). The answer row's read gives r_{ans} = 0.788 / 0.780, mainly through keys ($r_{\mathrm{ans}}(K)$ = 0.875 / 0.732, $r_{\mathrm{ans}}(V) = -0.046$ / 0.052). The top-20 heads by a_3 of the two formats share 13 and 14 heads (H6's threshold is 10; met at Mistral-7B, not evaluable at Qwen2.5-7B), the top 10 share 7 and 9, and the full a_3 grids correlate at Spearman 0.520 / 0.573 (exploratory, recomputed). In this free-form format the candidates carry little of the clean answer's probability, so H3's candidate-mass criterion fails in every condition, as expected.

Ablation against knockout (not reconciled). Mean-ablating the reader heads at the six option rows (H3) keeps the base answer, while cutting the candidate words' attention to the writing token (G5) keeps it only at Qwen2.5-14B (G7b): the two interventions differ, and the discovery sample does not reconcile them. The passage tests of the final round that would have bridged them were not evaluable (Appendix D.5).

E.3. What the reader heads write: the flag

Definitions. Let H^* be the reader heads of the discovery sample (the top 5 % by a_3 under OPTIONS-AFTER, frozen: 40 heads at Qwen2.5-7B, 52 at Mistral-7B). Under the natural key clamp K_{S}, the heads' summed write at layer l into the rows of S and B changes; the *flag* Δ_l is the mean over the 60 ranking stories of half the sum of the change at S's row and minus the change at B's row, projected through each head's output matrix, one vector per layer that holds an H^* head. Leave-one-word-out flags Δ^{-w} are fit on the stories in which w is neither B nor S. An *injection* adds a vector to the attention output at chosen rows, in the clean base run, with no key clamp; "move" adds the flag at an option's row and subtracts it at B's row. With ℓ_w the six-way log-probability of candidate w and N_X the natural key clamp's effect on X, ι =

the injection's mean effect on ℓ_X divided by N_X; π_X = the share of stories whose answer becomes X; $\mathrm{ID}_{\mathrm{inj}}$ = the identity contrast of moving the flag to S against X; $\rho_K(u)$ = ID_K with the direction u removed at the six option rows, relative to none; $r^{\mathrm{inj}}_{\mathrm{ans}}(C)$ = the share of the injection's effect removed when only the answer row reads the option rows' clean channel C. Scale (Appendix E.5): ω, how far the flag written in a sentence matches the list's along the list flag; c_{fl}, the cosine of the sentence and list flags; ρ and ρ_{nat}, the injected and natural sentence-to-list ratios. Binding: Ψ_{bind}, the double difference of initial-state flags under two binding questions. The runs are BF16 with sdpa attention, except the passes that fix the head sets (eager), and every ratio is taken within one run; the intervals are over stories, conditional on the fitted flag and the frozen heads. Table 38 gives every line. The intervals of this subsection resample stories with the flag's fit held fixed: they exclude the uncertainty of the fit itself, which matters because several criteria were decided near their thresholds (J-D2's differences, Mistral-7B's ρ_K); a refit inside each resample was not run.

Sufficiency and identity (J-D1, risky, held; J-D-ADDR, held). Moved into the row of an option absent from the story (added there and subtracted at B's row, as the natural key clamp moves it), the flag fit without that option's word moves the answer there: $\mathrm{ID}^{\mathrm{LOO}}_{\mathrm{inj}}/\mathrm{ID}_K$ = 0.997 [0.980, 1.014] and 0.803 [0.746, 0.856], and the answer changes in π_X = 0.96 and 0.67 of stories. The mean flag names no candidate: fit from the natural source run instead of the key clamp it is the same direction (norm-weighted cosine 0.993 and 0.962) with the same effect (ratio 1.005 [1.003, 1.008] and 0.981 [0.949, 1.013]). Added alone, without the subtraction at B's row, the flag has less effect, with a threshold and saturation (adding half, one and two flags at X's row gives ι = 0.053, 0.673, 0.684 and 0.054, 0.541, 0.612); the full flag gives $\mathrm{ID}_{\mathrm{inj}}/\mathrm{ID}_K$ = 0.994 and 0.814; isotropic, head-span, layer-permuted and wrongly placed flags (at the "Choices" and "Question" rows) give $|\iota| \leq 0.048$ (reported).

Specificity (J-D2, risky, not held). The line required three structured controls to move the answer at most 0.3 as much as the flag. The active-set flag (the same construction over the heads most active at the option rows) does not move it (ι = -0.032 and -0.010), but at Qwen2.5-7B the mean reader output at the option rows, norm-matched, moves it 0.615 as much as the flag, and at Mistral-7B the story's own write with its flag component removed, norm-matched, 0.516 as much. The checks found (exploratory, recomputed) that the Qwen control overlaps the flag by construction: the clean reader output at the option rows contains the flag written at the mentioned words' rows (norm-weighted cosine 0.636), 54 % of the flag's energy lies along the mean-output directions and 29 % along the top principal direction, and ablating the mean output reproduces the flag's value-channel signature. The overlap explains much but not all of the failure (a linear energy estimate gives ρ_K about 0.71 and 0.85 for the two Qwen controls), and Mistral's controls overlap the flag too (cosine 0.58) yet pass. The Mistral failure on the orthogonal own write points to a story-specific write (rank-one capture 0.851, against 1.008 at Qwen).

Necessity (J-D3, risky, not held). Removing the flag direction at the six option rows, one direction per layer, leaves ρ_K = 0.466 [0.416, 0.516] and 0.478 [0.403, 0.553] of the key read: about half in both models. The line also required two control directions, the top principal direction of the heads' output and the mean reader output, to leave at least 0.7. At Mistral they do (0.846 and 0.899) and the line holds there, with the flag's point estimate close to its 0.5 floor; at Qwen they remove 31 % and 44 % (they leave 0.687 and 0.561) because they overlap the flag, and the line fails. Isotropic directions leave 0.999–1.002 (reported). The pre-written replacement used in the main text: "One direction per layer does not carry the key read (ρ_K = 0.466 / 0.478): other carriers exist; the flag is sufficient but not necessary." Whole-head ablation of the reader heads removes far more of the read (discovery sample: ρ_K = 0.238 and 0.059; Table 37), so the heads write more than this direction.

Read by key (J-D4, held). Giving only the answer row the option rows' clean keys and values removes $r^{\mathrm{inj}}_{\mathrm{ans}}(KV)$ = 0.869 and 0.762 of the injected flag's effect; their clean keys alone 0.689 and 0.601, their clean values alone 0.047 and 0.068 ($K - V$ = 0.642 [0.589, 0.692] and 0.533 [0.478, 0.588]); every other later row carries 0.018 and 0.078 (reported). The answer reads the injected flag by key, as it reads the natural one (discovery sample, H5).

What the flag is (reported). Its largest cosine with the unembeddings of the locations is 0.071 at Qwen and 0.121 at Mistral (both at layer 21), against at most 0.053 and 0.017 for random token rows: it is not a readable location word. The share of the per-story flag's variance explained by the identity of B is 0.24–0.94 across layers, against a null of 0.08: the mean flag is identity-free, while each story's write also carries its own word.

Table 38. **The flag: every line of the mechanism tests.** Top: Qwen2.5-7B and Mistral-7B, flags fit on 60 discovery-sample stories and tested on 100 new ones; bottom: the scale lines at Qwen2.5-1.5B, 3B and 7B (Appendix E.5). 95 % story-bootstrap intervals. Q_{IN}, Q_{OUT}: OPTIONS-AFTER asked which option is (is not) mentioned; IOI in-sentence: the four names listed right after the indirect object's mention (Appendix E.4). n.e.: not evaluable (gate in brackets). †Recomputed: at 3B the line is not evaluable (too few competent cores), and the post hoc rows are not printed by the scorer.

Line	Statistic	Qwen2.5-7B	Mistral-7B	Verdict
J-D1	$\mathrm{ID}_{\mathrm{inj}}^{\mathrm{LOO}}/\mathrm{ID}_K$	0.997 [0.980, 1.014]	0.803 [0.746, 0.856]	**met**
	π_X	0.96	0.67	
J-D2	ι(move Δ)	0.999 [0.992, 1.006]	0.794 [0.719, 0.864]	**not met**
	ι(own write $\perp$ flag)	0.135 [0.096, 0.179]	0.409 [0.332, 0.488]	
	ι(mean H^* output)	0.614 [0.549, 0.679]	0.120 [0.078, 0.166]	
	ι(active-set flag)	−0.032 [−0.037, −0.027]	−0.010 [−0.015, −0.006]	
J-D3	ρ_K(flag)	0.466 [0.416, 0.516]	0.478 [0.403, 0.553]	**not met**
	ρ_K(pc1)	0.687 [0.633, 0.738]	0.846 [0.786, 0.902]	
	ρ_K(mean H^* output)	0.561 [0.520, 0.600]	0.899 [0.846, 0.946]	
J-D4	$r_{\mathrm{ans}}^{\mathrm{inj}}(KV)$	0.869 [0.855, 0.884]	0.762 [0.737, 0.786]	**met**
	$r_{\mathrm{ans}}^{\mathrm{inj}}(K)$	0.689 [0.642, 0.735]	0.601 [0.540, 0.659]	
	$r_{\mathrm{ans}}^{\mathrm{inj}}(V)$	0.047 [0.030, 0.067]	0.068 [0.035, 0.104]	
	$r_{\mathrm{ans}}^{\mathrm{inj}}(K) - r_{\mathrm{ans}}^{\mathrm{inj}}(V)$	0.642 [0.589, 0.692]	0.533 [0.478, 0.588]	
J-D-ADDR	weighted $\cos(\Delta^K, \Delta^{KV})$	0.993	0.962	**met**
	$\mathrm{ID}_{\mathrm{inj}}(KV)/\mathrm{ID}_{\mathrm{inj}}(K)$	1.005 [1.003, 1.008]	0.981 [0.949, 1.013]	
J-D-KN	Pearson (per story)	0.894 [0.868, 0.917]	0.147 [0.041, 0.252]	**not met**
	B-loss ratio β	5.167 [3.878, 7.704]	9.095 [2.862, 58.110]	
	mean($\Delta\ell_S - \Delta\ell_X$) (nats)	−0.073 [−0.763, 0.612]	0.290 [−0.457, 1.024]	
J-D5	Ψ_{bind} pooled (nats)	−0.578 [−0.863, −0.308]	0.269 [0.096, 0.456]	**not met**
	object-first	0.376 [0.156, 0.619]	0.459 [0.120, 0.848]	
	distractor-first	−1.299 [−1.678, −0.955]	0.126 [−0.034, 0.283]	
	stratum difference	1.675 [1.272, 2.115]	0.333 [−0.049, 0.751]	
J-D7	$\mathrm{ID}_K(Q_{\mathrm{OUT}})$ (nats)	−12.138 [−12.905, −11.413]	−7.083 [−7.726, −6.463]	**met**
	$\mathrm{ID}_K(Q_{\mathrm{IN}})$ (nats)	13.005 [12.206, 13.777]	3.575 [3.085, 4.059]	
	paired difference (nats)	25.143 [24.121, 26.159]	10.658 [9.897, 11.413]	
J-D-SIGN-Q	$\mathrm{ID}_V(Q_{\mathrm{OUT}})$ (nats)	−0.431 [−0.616, −0.254]	−1.806 [−2.442, −1.192]	**met**
J-D-SIGN-IOI	ID_K, IOI in-sentence (nats)	−3.350 [−3.655, −3.042]	−2.352 [−2.542, −2.162]	**met**
	ID_V, IOI in-sentence (nats)	2.550 [2.244, 2.889]	1.096 [0.945, 1.247]	
J-D8-Q	$R(H^*)$, Q_{IN}	1.027 [1.016, 1.038]	1.058 [1.032, 1.087]	**met**
	$\mathrm{KO}(H^*)$, Q_{IN}	0.974 [0.968, 0.980]	0.931 [0.892, 0.964]	
	$R(H^*)$, Q_{OUT}	0.986 [0.971, 1.002]	1.050 [1.011, 1.093]	
	$\mathrm{KO}(H^*)$, Q_{OUT}	1.021 [1.001, 1.041]	0.958 [0.918, 0.995]	
J-D8-IOI	$R(H^*)$, IOI in-sentence	0.922 [0.882, 0.960]	n.e. (J-D-G4 INLINE)	**not evaluable**
	$\mathrm{KO}(H^*)$, IOI in-sentence	0.998 [0.980, 1.016]	n.e. (J-D-G4 INLINE)	
J-D-ROUTE-IOI	$r_{\mathrm{ans}}(KV)$	0.897 [0.821, 0.976]	0.839 [0.769, 0.910]	**met**
	$r_{\mathrm{ans}}(K)$	0.431 [0.318, 0.536]	0.488 [0.406, 0.567]	
	$r_{\mathrm{ans}}(V)$	0.211 [0.163, 0.267]	0.263 [0.231, 0.298]	
	$r_{\mathrm{ans}}(K) - r_{\mathrm{ans}}(V)$	0.220 [0.081, 0.343]	0.225 [0.131, 0.314]	
	r_{other}	0.103 [0.064, 0.146]	0.190 [0.157, 0.223]	
J-D-HOP2	carry, top-10 hop-2 heads, Q_{OUT}	0.077 [0.062, 0.092]	0.161 [0.117, 0.208]	**not met**
	carry, top-10 hop-2 heads, IOI	−0.085 [−0.141, −0.036]	−0.029 [−0.093, 0.025]	

Line	Statistic	Qwen2.5-1.5B	Qwen2.5-3B	Qwen2.5-7B	Verdict
J-D6a	ω	0.967 [0.952, 0.982]	0.954 [0.937, 0.971]	0.822 [0.810, 0.834]	**met**
	c_{fl}	0.966	0.895	0.872	
J-D6	ρ	−0.126 [−0.179, −0.077]	−0.025† (n.e.) [−0.104, 0.050]	0.257 [0.222, 0.292]	**not met**
	ρ_{nat}	0.090 [0.042, 0.137]	0.266† (n.e.) [0.196, 0.341]	0.468 [0.431, 0.505]	
	$\rho - \rho_{\mathrm{nat}}$	−0.216 [−0.273, −0.163]	−0.290† (n.e.) [−0.376, −0.213]	−0.211 [−0.245, −0.178]	
J-D6-ROUTE	$r_{\mathrm{ans}}^{\mathrm{inj}}(V)$	−0.140 [−0.276, −0.054]	–	–	**not met**
	$r_{\mathrm{ans}}^{\mathrm{inj}}(K)$	1.114 [1.006, 1.285]	–	–	
	$r_{\mathrm{ans}}^{\mathrm{inj}}(V) - r_{\mathrm{ans}}^{\mathrm{inj}}(K)$	−1.254 [−1.543, −1.075]	–	–	
J-D6, post hoc	natural ratio, X-specific contrast†	−0.113 [−0.213, −0.030]	–	0.271 [0.234, 0.308]	–
	ρ minus it†	−0.013 [−0.099, 0.083]	–	−0.014 [−0.049, 0.022]	
	natural ratio on ID_K†	−0.115 [−0.182, −0.062]	–	0.290 [0.270, 0.311]	
	ρ minus it†	−0.011 [−0.079, 0.061]	–	−0.033 [−0.066, −0.002]	
D6b	$\Delta^{\mathrm{sentence}}/\Delta^{\mathrm{list}}$ at the list row	0.926 [0.908, 0.942]	0.926 [0.892, 0.961]	0.943 [0.924, 0.961]	reported

A key that names no candidate (J-D-KN, risky, not held). Clamping the writing token's key to a word that is not a candidate ("garage", "kitchen", ...) was predicted to act like removing B's flag. Per story, its effects on the six candidates correlate with those of subtracting the flag at B's row at Qwen (Pearson 0.894) but not at Mistral (0.147), and the loss of support for B is far larger than the flag's (β = 5.167 and 9.095 against the preregistered 0.7–1.3); the third criterion held (-0.073 and 0.290 nats against tolerances of 2.065 and 1.468). The printed sizes are inflated, because at Qwen $p(\mathrm{B}) > 0.99$ in every clean story and at Mistral the denominator is nearly inert; on the six-way logit of B, β is 1.93 [1.78, 2.10] and 5.20 [3.38, 10.49] (exploratory, recomputed): a non-candidate key removes about two (Qwen) to five (Mistral) times more support for B than subtracting the mean flag at B's row.

Binding (J-D5, risky, not held). The flags of the two initial-state sentences were predicted to carry which object they bind, in both sentence orders. Pooled, Ψ_{bind} = -0.578 [-0.863, -0.308] at Qwen (below zero) and 0.269 [0.096, 0.456] at Mistral; the two orders differ at Qwen (object first 0.376, distractor first -1.299 nats; difference 1.675 [1.272, 2.115]). The pre-written replacement: "The initial-state flag does not carry binding in a way separable from order." The order effect is a property of the model, not of the fit (the fit set is balanced, 23 against 22 stories), and the two initial-state flags are nearly collinear (weighted cosine 0.97 in both models), so Ψ_{bind} is a 0.2–1.1-nat contrast on injection effects of 12–19 nats; Mistral's pass is fragile (distractor first 0.126 [-0.034, 0.283]).

Other tasks (exploratory). The belief flag injected into the paint and schedule tasks gives 0.434 of the tasks' own flag at Qwen, and 0.178 and 0.569 at Mistral.

Holm sensitivity. Over the 27 interval components of this part's risky lines with a verdict, Holm's step-down changes no decision and no verdict.

E.4. The sign of the read, and IOI

The question sets the sign (J-D7, J-D-SIGN-Q, J-D8-Q, held). Asked which option is *not* mentioned in the story (Q_{OUT}), the key read turns negative: ID_K = -12.138 [-12.905, -11.413] and -7.083 [-7.726, -6.463] nats, against 13.005 and 3.575 when asked which option is mentioned (Q_{IN}; paired differences 25.143 and 10.658). The value read turns negative too (ID_V = -0.431 [-0.616, -0.254] and -1.806 [-2.442, -1.192]), so the sign follows the task, not a polarity of the readers. The same frozen reader heads carry the read under both questions: when only they see the clamped key they recover R = 1.027 and 1.058 of it under Q_{IN} and 0.986 and 1.050 under Q_{OUT}, blinding them removes 0.974–1.021 at Qwen and 0.931–0.958 at Mistral, and random sets of the same size do neither (their R and KO lie between -0.006 and $+0.015$). Behaviourally, under Q_{OUT} the key clamp makes the story's own location the answer in 31 % and 33 % of stories, and the list flag injected at X's row moves the answer 1.228 and 0.899 as much as the natural clamp (reported; under Q_{IN} 0.999 and 0.665): the readers' write need not change with the question, and the sign is applied after the first hop (exploratory). A negative use of duplicate detection is known (Wang et al., 2022; McDougall et al., 2023); we do not present the polarity as a discovery.

IOI separates the channels (J-D-SIGN-IOI, held). In the in-sentence re-mentions of indirect-object identification (the four names listed right after the indirect object's mention), the key read is negative while the value read stays positive: ID_K = -3.350 [-3.655, -3.042] and -2.352 [-2.542, -2.162] nats, ID_V = 2.550 and 1.096 (fresh IOI cores; the discovery sample had shown the same, Appendix E.4.1).

Where the negative read is applied (J-D-ROUTE-IOI, risky, held). With the key clamp at the indirect object, restoring the listed names' clean keys and values for the answer row alone removes $r(KV)$ = 0.897 and 0.839 of the read, their keys alone 0.431 and 0.488, their values alone 0.211 and 0.263 ($K - V$ = 0.220 [0.081, 0.343] and 0.225 [0.131, 0.314]); every other row carries 0.103 and 0.190. The line tests $K > V$; the key alone carries less than half of the read.

Which heads apply it (J-D-HOP2, risky, not held). The ten answer-side heads ranked as the flag's top readers in lists were predicted to carry the negative reads too. They carry 0.077 and 0.161 of the injected flag's effect under Q_{OUT} and -0.085 and -0.029 in IOI, while every head together carries 0.696 and 0.851 (0.855 and 0.555 in IOI); the scorer's reading is "different reader". The reading is confounded: in lists (OPTIONS-AFTER), the format in which they were ranked, the same ten heads carry only 0.339 and 0.292 of hop 2 (every head 0.846 and 0.810), so ten heads do not carry hop 2 even there. The pre-written replacement: "a different (or a distributed) reader applies the negative read".

The belief readers in IOI (J-D8-IOI, risky, not evaluable). At Qwen2.5-7B the frozen belief reader heads carry the IOI read (R = 0.922 [0.882, 0.960], blinding them removes 0.998, random sets 0.002 and 0.031). At Mistral-7B the transfer gate (the option rows must carry at least half of the in-sentence key read) missed by 0.021 (0.479), so the line, which needs both models, is not evaluable; with the gate passed, Mistral's values (0.938 and 1.030) would meet the criteria and the line would be met. At Qwen the same heads also carry the IOI list read (R = 0.972, KO = 0.997; reported). The line is therefore reported as met at Qwen2.5-7B and not evaluable at Mistral-7B, not as a result about both models.

The IOI factorial of the discovery sample. The in-sentence IOI arm was first run in preregistration G (part (e), below), whose later-list predictions failed; the final round used the in-sentence arm because it dissociates the channels.

E.4.1. The IOI factorial of the discovery sample (preregistration G, part (e))

The IOI cores follow the templates of Wang et al. (2022), alternating ABBA and BABA, with four names: the indirect object of the base and source runs and of the third-value run, and the subject. The clamps act at the indirect object's mention. Arms: PLAIN (the sentence); AFTER (a "Choices:" list of the four names after the sentence, in the chat wrapper on instruct models); BEFORE (the list before it); QUESTION ("Which name completes the sentence?", without candidates); and INLINE and INLINE_BEFORE (the list as a parenthetical right after the indirect object's mention, or before the sentence; preregistered only at GPT-2 small). $f_K = \mathrm{ID}_K/\mathrm{ID}_{KV}$ and f_V likewise. A cell is evaluable only if the clean model passes Gate e: two-way accuracy at least 0.75, four-way at least 0.50, and a positive mean logit difference. GPT-2 small and GPT-2 XL ran in FP32 with exactness asserted, the 7B and 14B models in BF16. Table 39 gives every cell.

Plain IOI carries almost no key identity in the three preregistered models (f_K within -0.02 to 0.02; G18). At Qwen2.5-7B-Instruct a later list gives a large positive key read (ID_K(AFTER) = 7.05 [6.26, 7.81]), but G19 required it in both 7B models, and Mistral-7B failed the competence gate in AFTER. G20 fails at Qwen2.5-7B-Instruct on its baseline clause (f_V(QUESTION) = 0.69 against the predicted 0.75), and G21 because the list before the sentence gives a negative read (-2.02 nats).

Unpredicted observations. (i) BEFORE gives a negative key read in every model where it passes the gate (Qwen2.5-7B-Instruct -2.02, Qwen2.5-14B-Instruct -3.83, Qwen2.5-7B base -0.95 nats) and in the cells that fail it. (ii) The sign of the AFTER read varies by model: positive at Qwen2.5-7B- and 14B-Instruct (7.05 and 10.24; 14B exploratory, where G20 fails on its baseline clause as at 7B), absent at Qwen2.5-7B base (-0.03, a weak null: the joint effect there is only 1.72 nats, against 11.96 at the instruct model) and negative in the gate-failed cells (Mistral-7B -1.98, GPT-2 small -1.77, GPT-2 XL -1.87). (iii) An in-sentence re-mention (INLINE) gives a negative key read in all six models (-3.29 to -2.33 nats), with the gate passed in every INLINE cell. (iv) In a row splice under AFTER ($n = 60$), the listed names carry the whole key effect (1.05 at Qwen2.5-7B-Instruct, 1.07 at Mistral-7B, whose AFTER cell failed the gate), the rows of the two clamped names alone 0.27 and 0.84. (v) In the GPT-2 small attention probe (re-run on CPU; Appendix A), the canonical name-mover heads attend from the end to the indirect object, and under INLINE the duplicate-token heads attend from the listed copy of the name to its earlier mention; under the key clamp this attention moves to the row of the listed swapped name.

E.5. Scale

The flag is written in sentences (J-D6a, replication, held). At Qwen2.5-1.5B, 3B and 7B the flag written by the reader heads at the sentence's re-mentions matches the list's: ω = 0.967, 0.954 and 0.822, cosine c_{fl} = 0.966, 0.895 and 0.872 (Table 38, bottom). The sentence flag injected at the list row moves the answer 0.926, 0.926 and 0.943 as much as the list flag (reported). The reader heads at 1.5B and 3B were ranked in the final round with the discovery sample's procedure (17 and 29 heads).

Mediation (J-D6, risky, not held; J-D6-ROUTE, risky, not held). J-D6 predicted that the list flag injected at the sentence's row of the absent option reproduces the natural ratio of the sentence read to the list read, in sign and within $\pm\max(0.1, 0.5|\rho_{\mathrm{nat}}|)$. At 1.5B the signs differ (ρ = -0.126 [-0.179, -0.077] against ρ_{nat} = 0.090 [0.042, 0.137]); at 7B ρ = 0.257 against ρ_{nat} = 0.468, a difference of -0.211 [-0.245, -0.178] against a tolerance of $\pm$0.234; 3B is not evaluable (35 of 60 competent cores, against 40). J-D6-ROUTE predicted that at 1.5B the injected flag is read by value under SENTENCE-AFTER; it is read by key ($r^{\mathrm{inj}}(V)$ = -0.140, $r^{\mathrm{inj}}(K)$ = 1.114, difference -1.254 [-1.543, -1.075]). The preregistered decision table gives outcome (d), whose text is "Mixed.": J-D6 is not met, and ρ_{nat} is positive in both evaluable models, so the

Table 39. **IOI** (preregistration G, part (e); 200 fresh cores). Gate e: two-way / four-way accuracy of the clean run (failed cells are not evaluable). ID_K with 95% bootstrap CI, ID_V and ID_{KV} in nats; f_K, f_V: shares of ID_{KV}, unstable where ID_{KV} is near zero. GPT-2 small, Qwen2.5-7B-Instruct and Mistral-7B enter G18–G22 (the GPT-2 small parts replicate a disclosed pilot); Qwen2.5-7B base, Qwen2.5-14B-Instruct and GPT-2 XL are exploratory.

Model	Arm	Gate e (2-way / 4-way)	ID_K (nats)	ID_V	ID_{KV}	f_K	f_V
GPT-2	PLAIN	0.98 / 0.98	−0.15 [−0.18, −0.11]	+14.08	+8.61	−0.02	+1.64
	AFTER	0.48 / 0.36 (failed)	−1.77 [−1.84, −1.71]	+2.37	+0.93	−1.92	+2.57
	BEFORE	0.55 / 0.35 (failed)	−0.05 [−0.07, −0.04]	+1.00	+0.89	−0.06	+1.13
	QUESTION	0.49 / 0.49 (failed)	−0.04 [−0.06, −0.02]	+5.35	+4.37	−0.01	+1.23
	INLINE	0.94 / 0.91	−2.35 [−2.47, −2.23]	+6.71	+3.74	−0.63	+1.79
	INLINE_BEFORE	0.91 / 0.82	−0.25 [−0.29, −0.21]	+4.80	+3.27	−0.08	+1.47
Qwen2.5-7B	PLAIN	1.00 / 1.00	+0.20 [0.10, 0.31]	+6.72	+10.33	+0.02	+0.65
	AFTER	0.98 / 0.89	+7.05 [6.26, 7.81]	+5.34	+11.96	+0.59	+0.45
	BEFORE	0.97 / 0.82	−2.02 [−2.18, −1.86]	+6.37	+7.67	−0.26	+0.83
	QUESTION	1.00 / 1.00	+0.61 [0.43, 0.78]	+17.77	+25.90	+0.02	+0.69
	INLINE	1.00 / 0.67	−3.26 [−3.47, −3.05]	+2.63	+2.02	−1.61	+1.30
	INLINE_BEFORE	1.00 / 0.62	−0.74 [−0.80, −0.67]	+1.17	+1.43	−0.52	+0.82
Mistral-7B	PLAIN	0.99 / 0.99	+0.02 [−0.02, 0.07]	+6.06	+7.94	+0.00	+0.76
	AFTER	0.59 / 0.42 (failed)	−1.98 [−2.38, −1.58]	+3.35	+3.16	−0.63	+1.06
	BEFORE	0.81 / 0.27 (failed)	−0.96 [−1.12, −0.80]	−2.32	−1.01	+0.95	+2.30
	QUESTION	0.96 / 0.96	+0.11 [0.03, 0.18]	+11.33	+12.98	+0.01	+0.87
	INLINE	1.00 / 0.71	−2.33 [−2.46, −2.19]	+1.27	+1.01	−2.29	+1.25
	INLINE_BEFORE	0.99 / 0.32 (failed)	−0.42 [−0.48, −0.37]	−0.26	−0.01	+43.18	+26.44
Qwen2.5-7B base (expl.)	PLAIN	1.00 / 1.00	+0.18 [0.10, 0.26]	+5.45	+8.53	+0.02	+0.64
	AFTER	0.85 / 0.77	−0.03 [−0.15, 0.08]	+1.02	+1.72	−0.02	+0.59
	BEFORE	0.83 / 0.66	−0.95 [−1.01, −0.89]	+0.76	+0.92	−1.03	+0.82
	QUESTION	0.55 / 0.59 (failed)	+0.27 [0.19, 0.35]	+4.46	+8.25	+0.03	+0.54
	INLINE	1.00 / 0.74	−2.83 [−2.96, −2.70]	+1.72	+1.21	−2.34	+1.42
	INLINE_BEFORE	1.00 / 0.82	−0.76 [−0.82, −0.70]	+1.53	+1.68	−0.45	+0.91
Qwen2.5-14B (expl.)	PLAIN	1.00 / 0.99	+0.29 [0.18, 0.39]	+6.36	+9.66	+0.03	+0.66
	AFTER	0.99 / 0.92	+10.24 [8.60, 11.89]	+8.52	+21.47	+0.48	+0.40
	BEFORE	0.97 / 0.88	−3.83 [−4.07, −3.59]	+12.60	+14.31	−0.27	+0.88
	QUESTION	0.99 / 0.99	+0.88 [0.66, 1.10]	+20.86	+30.79	+0.03	+0.68
	INLINE	1.00 / 0.68	−2.85 [−3.01, −2.70]	+1.52	+1.14	−2.51	+1.33
	INLINE_BEFORE	1.00 / 0.46 (failed)	−1.06 [−1.16, −0.97]	−0.31	+0.33	−3.27	−0.94
GPT-2 XL (expl.)	PLAIN	0.99 / 0.99	−0.08 [−0.13, −0.03]	+8.82	+7.89	−0.01	+1.12
	AFTER	0.52 / 0.26 (failed)	−1.87 [−1.95, −1.79]	+0.72	+0.06	−30.82	+11.88
	BEFORE	0.52 / 0.32 (failed)	−0.57 [−0.62, −0.53]	+0.41	+0.31	−1.84	+1.30
	QUESTION	0.47 / 0.47 (failed)	−0.01 [−0.04, 0.03]	+4.00	+5.02	−0.00	+0.80
	INLINE	0.99 / 0.67	−3.29 [−3.41, −3.17]	+2.28	+1.09	−3.03	+2.09
	INLINE_BEFORE	0.99 / 0.55	−1.09 [−1.18, −1.01]	+1.41	+0.87	−1.25	+1.62

outcome "not mediated" does not apply.

A post hoc reading of the natural ratio. The preregistered ρ_{nat} divides the natural clamp's effect on X under SENTENCE-AFTER by that under OPTIONS-AFTER; under SENTENCE-AFTER it includes a gain that every candidate other than B gets when B loses its key match (at 1.5B the clamp to S raises X more than the clamp to X does). On a contrast specific to the absent option (the clamp to X minus the clamp to S, on X), the natural ratio is -0.113 [-0.213, -0.030] at 1.5B and 0.271 [0.234, 0.308] at 7B, and ρ minus it is -0.013 [-0.099, 0.083] and -0.014 [-0.049, 0.022]; on the ratio of ID_K, from which the entry's prior was taken, it is -0.115 and 0.290, and ρ minus it -0.011 [-0.079, 0.061] and -0.033 [-0.066, -0.002] (post hoc, recomputed). On these contrasts the injected sentence-row flag reproduces the natural ratio in sign and size, and the decision table would read (b). This reading came after the data and cannot be scored: the verdict stays not met and the outcome (d), and we draw no conclusion about mediation from it in either direction. At 3B, on its 35 competent cores, the line would have failed: ρ = -0.025 against ρ_{nat} = 0.266 (post hoc, recomputed).

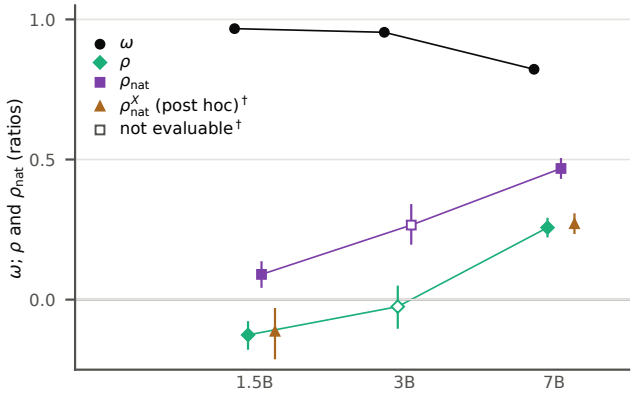


Figure 11. **The flag across scale** (Qwen2.5-1.5B, 3B and 7B; 95 % story-bootstrap intervals). ω: how far the flag written in a sentence matches the list's; ρ and ρ_{nat}: the injected and the natural sentence-to-list ratio (J-D6, not met: outcome "Mixed."); ρ^{X}_{nat}: the natural ratio on the contrast specific to the absent option (post hoc). Hollow and †: not evaluable at 3B, values recomputed.

F. Where an intervention appears to act: details

This appendix gives the design and every result of the final round's test of the channel-ratio law (preregistration J, part C): five families of identity edits obtained independently of the natural clamp, in Qwen2.5-7B, Mistral-7B and Llama-3.1-8B, on 80 new stories (Appendix B.5). The motivating case, the released remap, is in Appendix G; the binding swap at block 28 and the overlap screen are in Appendix H.2.

F.1. The law, the lemma and the statistics

The law. For an edit Z that writes a value at the writing token at depth l (the output of 0-based block l), channel $C \in \{K, V, KV\}$ and format f, let $\psi_C^Z(f,l) = \overline{\mathrm{ID}}_C^Z / \overline{\mathrm{ID}}_C^{\mathrm{nat}}$ be the edit's identity effect through channel C as a share of the natural write's, from block $l{+}1$ on. The channel-ratio law states that one scaling per edit holds for both channels and in every format: $\psi_K^Z = \psi_V^Z$. Its statistic is $\lambda = \log(\psi_K/\psi_V)$ (Section 5; each ψ floored at 0.01, so that a dead channel is a departure), computed in three ways at each depth: $W(f,l)$ within one format, for OPTIONS-AFTER and SENTENCE-AFTER where both channels are used; $A_L(l)$, the key read under LETTERS-AFTER against the value read under NO-MENTION; and $A_P(l)$, the key read under OPTIONS-AFTER against the value read under NO-MENTION. A channel is used in a cell when its natural identity effect is at least 2 nats and at least a tenth of the joint effect. A statistic is equivalent when its 90 % interval lies inside $\pm \log 1.25$ (two one-sided tests at 5 %); it counts as a departure when the edit's key and value effects interact (its interaction $\iota_{KV} > 0.5$ in one of its cells; not the ι of Appendix E.3). The law holds in a combination of model, edit and depths (a combo) when it has at least three statistics and every one is equivalent; otherwise the combo's pattern is copy-only (rival R1: pooled $\bar{\lambda} \le \log 0.5$), key-flat (R3: $\bar{\lambda} \ge \log 2$) or a graded departure. A line over several combos holds only if every evaluable combo holds (an intersection-union test).

The lemma. Because the base and source prompts differ only at the writing token p, an edit confined to the residual at (p,l) reaches every other position only through p's keys and values in blocks $\ge l{+}1$: under the causal mask p's residual reaches other tokens only through attention, and p's keys and values in blocks $\le l$ are computed from inputs the edit does not change. So the base run with the edit's keys and values written at p from block $l{+}1$ is exactly the edited run, and its key-only and value-only rows are exact. Writing the source run's residual $h_{\mathrm{S},l}(p)$ gives the source's keys and values at p from $l{+}1$: this edit is the natural clamp itself. Every edit is therefore applied as a clamp row whose tables come from a prefix pass with the edit written; the prefix through p is the same in all four formats, so an edit's tables are the same tensors in every format and only the readers after p differ. The unit tests check the lemma in FP32 to 10^{-4}.

Controls of the design. The natural write written as an edit (T) is the equivalence control: its tables and rows equal the natural ones bitwise in every evaluable cell of all three models (14 cells each), so every statistic of T is equivalent. A sensitivity gate checked that the statistics can see a departure: synthetic rows whose key table is interpolated halfway toward the base run (SK50) were classified as copy-only ($\bar{\lambda}$ = -2.319, -1.563 and -2.373), and rows with the value table halved (SV50) as graded departures (0.603, 0.586, 0.619). The design's earlier key-share rule ($\kappa = K/(K+V)$ within a margin of the natural share) passes SV50, a two-fold channel departure: this compression is why the law is stated on log ratios. The natural format effect on the key share is large in all three models (σ(LETTERS-AFTER, 3) $-$ σ(NO-MENTION, 3) = 0.956, 0.957 and 0.904). Intervals of the law are 90 % hierarchical bootstrap intervals over the 24 ordered base–source pairs and the stories within them (with the remap's two seeds as a further level); others are 95 %.

F.2. The edit families and their gates

Families. Each edit writes, at (p,l) for $l \in \{3, 7, 11, 15\}$, a target value t (S or X) in place of B:

- **E1**, in-task steering vectors: $h_{\mathrm{B}} + \mu_1(t) - \mu_1(\mathrm{B})$, with $\mu_1(x)$ the mean residual at (p,l) over 150 held-out stories with x written (Turner et al., 2023; Panickssery et al., 2023);
- **E2**, steering vectors from unrelated sentences: the same with $\mu_2(x)$, the mean block-l output at the token of x over 24 neutral sentences;
- **E3**, features of a public sparse autoencoder (Qwen2.5-7B only): the BatchTopK dictionaries of andyrdt (2025) (Bussmann et al., 2024) at layers 3, 7, 11 and 15, at a pinned revision; the edit sets the latents most selective for t to their mean on stories with t written and removes those selective for B, leaving every other latent and the reconstruction error at B's; the number of latents (4 at every layer) was set on calibration stories;

- **E4**, a remap trained at the writing token: a rank-16 distributed alignment search basis fitted at (p, l) with the released training recipe of Anonymous (2026) (pair-swap objective, six-way cross-entropy, one epoch over 1,000 training stories, two seeds, one from the PCA basis and one random), without a later mention; its losses fell from 5.05–8.12 to 0.00–0.63 and its held-out flip rates are 0.84–1.00;
- **E5**, the boundary family: steering vectors that write t as its French (FR) or German (DE) word or a synonym (SYN), and the part of E1 outside the English lexical span (NL); scaled by 1 or 2 on calibration stories.

Each edit's displacement is split into its component along the story's natural displacement (PAR) and the rest (PERP), and into its projection on the five-dimensional span of the English candidate words' mean residuals (LEX) and the rest (NONLEX). A component carries identity when its effect without a later mention is at least 0.3 of the natural one with a lower bound above zero. The distance ν of an edit's cached keys and values from the natural ones, relative to the natural displacement, is reported per cell (post hoc use in Appendix F.4).

Gates. The FP32 tests passed (95 tests). T reproduced the natural tables in every evaluable cell; the natural format effect held; the sensitivity gate held in all three models. The cell gate (natural joint effect at least 2 nats, interaction at most 0.5, coverage at least 0.8) failed in 6 of 48 cells: LETTERS-AFTER at blocks 11 and 15 in every model, where the natural effect is small. The efficacy gate (flip rate at least 0.8 of the natural and φ above 0.5 without a mention) held for E1 and E2 at every depth in all three models (φ 0.985–1.019) and for E3 at all four layers (0.871–0.928); E4 is effective at blocks 3 and 7 at Qwen2.5-7B and 3, 7 and 11 at Mistral-7B (φ 0.605–0.738) and not deeper; the E5 sub-families are mostly ineffective. Random directions of the natural norm move nothing beyond the specificity bound at any depth.

The dictionary gate failed through its specification. The gate required each dictionary's fraction of variance explained (FVE) on its block's outputs, over every prompt position after the first, to be within 0.05 of its published value. It failed at all four layers (FVE $-5{,}761$, $-8{,}416$, $-9{,}151$ and $-11{,}222$ against published values of 0.931, 0.862, 0.827 and 0.807), so E3 is not evaluable in every line that uses it. The published FVE is computed after the dictionary's training buffer drops the first tokens of each sample and activations with norms above ten times the batch median; the gate dropped only the first position and kept the chat template's tokens, identical in all prompts, where large-norm activations appear from block 3 on (inferred: the layer-3 dictionary explains 0.811 of the variance on block 2 and fails on block 3; the per-position activations were not saved). At the writing position the layer-3 dictionary reproduces its published value (FVE 0.928 against 0.931), and E3 flips every answer at every layer. The gate measured reconstruction over positions that the dictionary's published fit excludes; a corrected gate needs a rerun. After the data (post hoc, recomputed): admitting only layer 3, whose FVE at the writing position is within 0.05 of the published value, the E3 law would not hold (1 of 4 statistics equivalent, $\bar{\lambda}$ = -0.170); admitting all four layers, 1 of 10; no component of E3 carries identity. These statistics appear in Table 40 and Figure 12 marked post hoc.

F.3. Results per line

Steering vectors (J-C1, not held; J-C2, not held). J-C1 (extrapolation) required every statistic of E1 at every depth and of E2 at blocks 3 and 7 to be equivalent at Qwen2.5-7B and Mistral-7B. Equivalent were 8 of 10 statistics of E1 at Qwen ($\bar{\lambda}$ = -0.069), 5 of 6 at Mistral (-0.061), 7 of 8 of E2 at Qwen (-0.072) and 3 of 6 at Mistral (-0.089): graded departures, every pooled $\bar{\lambda}$ within the margin but not every statistic. The line's pre-written sentence (the main text opens instead with the outcome text below; Appendix A.6): "Even steering vectors close to the natural edit depart from the natural channel ratio ($\bar{\lambda}$ = -0.061 to -0.089): the format law does not transfer to edits as a channel-ratio law." J-C2 (extrapolation) covered E2 at blocks 11 and 15 (not evaluable at Qwen and Mistral: 2 and 0 statistics) and Llama-3.1-8B: E1 held there (7 of 7 equivalent, $\bar{\lambda}$ = -0.046), E2 did not (4 of 7, $\bar{\lambda}$ = -0.129). Every failing statistic is a key under-read at block 7 or deeper (Table 40).

Where the steering vectors depart (post hoc). In the core test (J-C1), 7 of its 30 λ statistics are not equivalent, all at block 7 or deeper, and the number of them with a 90 % interval wholly outside the margin is 0: each failure is an inconclusive test rather than a shown departure, and all but one point estimate lie inside the margin (recomputed). A rule that requires every one of six to ten statistics to be equivalent has limited power against a small, consistent under-read. At Llama-3.1-8B, 3 statistics of the vectors from unrelated sentences are not equivalent and 1 lies wholly outside the margin (J-C2). Every non-equivalent E1 or E2 statistic is a key-channel under-read (ψ_K 0.754–0.867 with ψ_V 0.961–1.092) in a cell whose natural key identity effect is small, 2.63–4.76 nats (natural key shares 0.152–0.196 in the cells of the sentence after

Table 40. **Every λ statistic of the channel-ratio law.** $\lambda = \log \psi_K - \log \psi_V$ with its 90 % hierarchical bootstrap interval, per model, edit and depth l; $W(\cdot, l)$ within OPTIONS-AFTER (first column) or SENTENCE-AFTER; $A_L(l)$ and $A_P(l)$: the key read under LETTERS-AFTER or OPTIONS-AFTER against the value read under NO-MENTION. *Not equivalent (the interval is not inside $\pm \log 1.25 = \pm 0.223$); $^{\iota}$a departure by the interaction rule ($\iota_{KV} > 0.5$); –: a cell not used or not evaluable. Line: the line the statistic belongs to (n.e.: in a combo with fewer than three statistics). †Recomputed through the scorer's code, not printed by it (post hoc): E3, whose layers fail the dictionary gate; the E5 sub-families, whose line prints only pooled values; Qwen2.5-7B E2 at $l = 11$.

Model	Edit	l	$W(l)$, OPTIONS-AFTER	$W(l)$, SENTENCE-AFTER	$A_L(l)$	$A_P(l)$	Line
Qwen2.5-7B	E1	3	−0.02 [−0.03, −0.02]	−0.02 [−0.03, 0.00]	0.00 [−0.01, 0.00]	−0.02 [−0.03, −0.02]	J-C1
		7	−0.06 [−0.08, −0.03]	−0.11 [−0.15, −0.07]	−0.05 [−0.07, −0.03]	−0.07 [−0.10, −0.05]	J-C1
		11	−0.16 [−0.25, −0.08]$^{*\iota}$	–	–	−0.18 [−0.27, −0.11]$^{*\iota}$	J-C1
	E2	3	−0.01 [−0.03, 0.01]	0.05 [0.02, 0.08]	0.00 [−0.02, 0.01]	0.02 [0.00, 0.04]	J-C1
		7	−0.14 [−0.19, −0.10]	−0.22 [−0.31, −0.16]*	−0.15 [−0.21, −0.11]	−0.11 [−0.15, −0.07]	J-C1
		11†	−0.27 [−0.39, −0.17]*	–	–	−0.21 [−0.34, −0.11]*	J-C2 (n.e.)
	E3†	3	−0.17 [−0.23, −0.12]*	−0.23 [−0.30, −0.17]*	−0.09 [−0.15, −0.04]	−0.19 [−0.25, −0.14]*	post hoc
		7	−0.71 [−0.89, −0.55]*	−1.40 [−2.07, −1.03]*	−0.71 [−0.91, −0.55]*	−0.79 [−0.94, −0.65]*	post hoc
		11	−1.01 [−1.26, −0.82]*	–	–	−1.10 [−1.36, −0.91]*	post hoc
	E4	3	−0.70 [−0.92, −0.49]*	−2.01 [−4.31, −1.31]*	−0.67 [−0.92, −0.46]*	−0.74 [−0.95, −0.55]*	J-C4
		7	−1.47 [−1.87, −1.15]$^{*\iota}$	−3.17 [−4.23, −2.25]*	−2.09 [−3.17, −1.55]$^{*\iota}$	−1.74 [−2.16, −1.41]$^{*\iota}$	J-C4
	E4, seed 101	3	−0.80 [−1.00, −0.63]*	−3.10 [−4.35, −2.07]*	−0.73 [−0.98, −0.53]*	−0.78 [−0.99, −0.60]*	J-C4 (seed)
	E4, seed 102	3	−0.59 [−0.80, −0.42]*	−1.50 [−1.96, −1.18]*	−0.61 [−0.87, −0.39]*	−0.70 [−0.93, −0.51]*	J-C4 (seed)
		7	−1.30 [−1.63, −1.05]$^{*\iota}$	−3.70 [−4.28, −2.41]*	−1.77 [−2.30, −1.40]$^{*\iota}$	−1.59 [−1.93, −1.31]$^{*\iota}$	J-C4 (seed)
		11	−1.30 [−1.73, −1.02]*	–	–	−1.59 [−2.04, −1.29]*	J-C4 (seed)
	PERP of E4	3	−1.70 [−2.09, −1.39]*	–	–	−1.77 [−2.22, −1.41]*	J-C5
	NONLEX of E4	3	−1.77 [−2.11, −1.47]*	–	–	−1.88 [−2.23, −1.58]*	J-C5
		7	−1.97 [−2.63, −1.52]*	–	–	−2.37 [−3.03, −1.92]*	J-C5
	E5-NL†	11	−0.58 [−1.08, −0.27]*	–	–	−0.83 [−1.34, −0.51]*	J-C-BOUND
Mistral-7B	E1	3	–	−0.02 [−0.03, −0.01]	0.00 [−0.01, 0.01]	−0.01 [−0.02, 0.00]	J-C1
		7	–	−0.19 [−0.27, −0.13]*	−0.06 [−0.12, −0.03]	−0.08 [−0.16, −0.04]	J-C1
	E2	3	–	0.01 [0.00, 0.02]	0.00 [0.00, 0.01]	0.00 [−0.01, 0.02]	J-C1
		7	–	−0.22 [−0.35, −0.13]*	−0.18 [−0.27, −0.11]*	−0.15 [−0.29, −0.07]*	J-C1
	E4	3	–	−0.93 [−1.42, −0.62]*	−0.55 [−0.90, −0.29]*	−0.54 [−0.93, −0.31]*	J-C4
		7	–	−3.92 [−4.24, −1.96]*	−1.95 [−2.78, −1.49]*	−2.36 [−3.87, −1.82]*	J-C4
	E4, seed 101	7	–	−2.63 [−4.02, −1.69]*	−1.90 [−2.50, −1.48]*	−2.53 [−3.92, −1.96]*	J-C4 (seed)
	E4, seed 102	3	–	−0.83 [−1.25, −0.54]*	−0.44 [−0.71, −0.22]*	−0.44 [−0.67, −0.27]*	J-C4 (seed)
		7	–	−4.22 [−4.27, −2.58]*	−1.99 [−3.16, −1.47]*	−2.23 [−3.52, −1.70]*	J-C4 (seed)
	PERP of E4	3	–	–	–	−1.91 [−3.28, −1.31]*	J-C5
		7	–	–	–	−3.30 [−3.46, −3.10]$^{*\iota}$	J-C5
	NONLEX of E4	3	–	–	–	−1.19 [−2.08, −0.72]*	J-C5
		7	–	–	–	−2.02 [−3.17, −1.57]*	J-C5
	E5-FR†	7	–	−0.35 [−0.88, 0.06]*	−0.33 [−0.67, 0.01]*	−0.05 [−0.52, 0.32]*	J-C-BOUND
	E5-NL†	7	–	−1.82 [−3.45, −1.11]*	−0.33 [−0.59, −0.12]*	−0.27 [−0.78, 0.01]*	J-C-BOUND
Llama-3.1-8B	E1	3	−0.01 [−0.02, 0.00]	−0.05 [−0.07, −0.04]	−0.03 [−0.04, −0.02]	−0.02 [−0.03, −0.01]	J-C2
		7	−0.06 [−0.09, −0.03]	–	−0.08 [−0.12, −0.05]	−0.08 [−0.11, −0.06]	J-C2
	E2	3	−0.01 [−0.04, 0.01]	−0.01 [−0.03, 0.01]	−0.03 [−0.04, −0.02]	−0.02 [−0.04, −0.01]	J-C2
		7	−0.24 [−0.30, −0.18]*	–	−0.32 [−0.42, −0.24]*	−0.26 [−0.34, −0.20]*	J-C2
	PERP of E2	7	−1.47 [−2.12, −1.06]*	–	–	−1.75 [−2.41, −1.34]*	J-C5
	E5-FR†	7	−0.50 [−0.83, −0.24]*	–	−0.55 [−0.85, −0.27]*	−0.62 [−0.90, −0.40]*	J-C-BOUND
	E5-SYN†	3	−0.78 [−1.05, −0.57]*	−0.01 [−0.37, 0.30]*	−1.92 [−2.35, −1.58]*	−1.02 [−1.35, −0.75]*	J-C-BOUND
		7	−0.87 [−1.11, −0.68]*	–	−1.82 [−2.27, −1.48]*	−1.04 [−1.35, −0.76]*	J-C-BOUND
	E5-NL†	7	−1.09 [−1.46, −0.78]*	–	−0.50 [−0.78, −0.29]*	−1.10 [−1.45, −0.81]*	J-C-BOUND

the story and of Qwen's OPTIONS-AFTER cell at block 11), against natural value effects of up to 17.12 nats (exploratory, recomputed). At block 3 every statistic of the three models is equivalent ($|\lambda| \leq 0.053$; exploratory). At block 7 the key under-read is also present where the natural key effect is large, but inside the margin (Qwen E2 $A_L(7)$: λ = -0.154, ψ_K = 0.875 at 14.2 nats; post hoc). Pooled over depths, the keys under-read the steering vectors by 4.5–12 % ($\bar{\lambda}$ -0.046 to -0.129). After the data, a channel floor of 4.8 nats in place of the preregistered 2 would make J-C1 met (all 4 combos) and the headline outcome (d); floors of 3 and 4 nats leave it not met, and under the lower-case score J-C1 stays not met (post hoc, recomputed). This floor is not used.

Sparse-autoencoder features (J-C3, risky, not evaluable). Not evaluable through the dictionary gate (Appendix F.2); under either gate considered after the data the line would not have been met (post hoc, recomputed; Appendix F.2).

The remap trained at the writing token (J-C4, risky, not held). No statistic of E4 is equivalent; the pooled $\bar{\lambda}$ is -1.574 [-2.083, -1.267] at Qwen2.5-7B and -1.707 [-2.072, -1.326] at Mistral-7B, copy-only (R1) in both models, for both seeds (per seed -1.355 and -1.465 at Qwen, -2.355 and -1.691 at Mistral) and under the lower-case score. The seed level of the

hierarchical bootstrap rests on two seeds and cannot estimate seed variance; the per-seed values are given for that reason. The pre-written R1 sentence, used for the remap only: "Edits that change the value through a learned remap are read mainly through the value ($\bar{\lambda}$ = -1.574 / -1.707): the key channel does not read them as it reads the written word." Its size depends on depth (post hoc, recomputed): at block 3 alone $\bar{\lambda}$ is -1.03 at Qwen and -0.67 at Mistral (a graded departure), and at block 7 -2.12 and -2.74, where the statistics under SENTENCE-AFTER sit near the floor of ψ (ψ_K = 0.026 and 0.012 against ψ_V = 0.626 and 0.608); we therefore quote ψ_K and ψ_V beside the pooled magnitudes, which the pre-written sentence fixes, and give the block-3 values in the main text. The remap does not act through the natural lexical code: its components along the natural displacement and along the lexical span carry at most 0.42 and 0.09 of its effect, while E1 and E2 carry at least 0.94 of theirs through one of them (reported).

Identity-carrying components (J-C5, risky, not held). 12 PERP or NONLEX components of the effective edits carry identity without a mention (φ 0.307–0.721); 8 of them have 12 statistics in all, between -3.30 and -1.19, and 0 are equivalent: identity written outside the natural direction or outside the lexical span is read mainly through the value.

The lexical boundary (J-C-BOUND, risky, not held). The line predicted that a non-lexical identity edit is under-read by the key channel ($\bar{\lambda}_{\mathrm{E5}} \leq \log 0.5$ with an upper bound below $\log 0.8$) where E2 at the same cells is not ($\bar{\lambda} \geq \log 0.8$). Of the 6 sub-families with a verdict, two met it, Mistral-7B NL ($\bar{\lambda}$ = -0.806; E2 -0.183) and Llama-3.1-8B SYN (-1.066; E2 -0.129), and four did not: Qwen2.5-7B NL (-0.705; E2 -0.242), Mistral-7B FR (-0.242 [-0.657, 0.109]), Llama-3.1-8B FR (-0.558; E2 -0.276) and Llama-3.1-8B NL (-0.897; E2 -0.276). The boundary was not isolated (post hoc, recomputed): every E5 $\bar{\lambda}$ lies below E2's at the same cells, but the two NL failures come only from E2's own departure at those cells, the line stays not met without the E2 control (both FR sub-families fail by themselves), and Mistral NL's pass rests on one near-floor statistic. The effective E5 edits write far from the natural state (post hoc; ν 0.65–1.29) and follow the same distance pattern as the remap and the components (Appendix F.4); FR, DE and SYN are mostly ineffective because (post hoc) they write displacements 2.15–3.39 times the natural norm, nearly orthogonal to it (cosine at most 0.32). The pre-written sentence for this outcome is contradicted by the numbers and has one slot for six values; we claim neither that the token-form account holds for edits nor that it is contradicted (Appendix A.6).

Reader heads (J-C-READa, not held; J-C-READb, risky, not evaluable). At OPTIONS-AFTER and block 7, J-C-READa required the discovery sample's reader heads in blocks after 7 (32 heads at Qwen, 43 at Mistral) to remove at least 0.7 of each steering vector's key read when blinded, and three size-matched random sets at most 0.25 on average. The readers removed 0.741–0.911, about what they remove of the natural write (0.915 and 0.773); the random sets removed 0.277–0.309, and the line was not met. The random-set control fails for the natural value as well (post hoc, recomputed): the natural donor's random-set mean is 0.307 at Qwen and 0.283 at Mistral, and the edits match the natural donor within 0.007 on the readers and on the random-set mean (Mistral E2 on the readers -0.032, post hoc). The sets are size-matched among the heads of blocks after 7 (560 and 768 heads) but not drawn disjoint from the readers; at Qwen one of them contains the top-ranked reader and removes 0.664 of the natural read (post hoc). The bound of 0.25 was carried over from stage 6, whose sets were drawn from all blocks with the key clamped from block 0; the failure is of the control in this design, not evidence that other heads read the edits, and a criterion referred to the natural donor would make the line met (post hoc reading). J-C-READb (the same for E3 and E4) is not evaluable under the line's text: E3 through its gate, and E4 because its own key read at block 7 is below 2 nats (0.852 and 0.182). Under the common part's evaluability rule, by which a quantity the tested arm makes undefined counts against the line, it would be not met (E4's readers remove 0.590 and 0.678, below 0.7, post hoc and recomputed), and the risky tally would read 10 of 28 (expected 12.65); the official verdict is used (Appendix A.6).

The headline. The scorer checks the pre-committed outcomes for Section 5 in the order (a), (a′), (c), (b), (e), (d), (f) and prints the first whose condition holds: outcome (e), "FAILED". Outcomes (a) and (a′) need J-C3 and J-C4, or J-C5, met (the law for the autoencoder features and the trained remap, or for the identity-carrying components); (c) needs J-C-BOUND met; (b) needs every effective edit to act through the natural lexical code and no component to carry identity, but the trained remap does not act through it (its parallel and lexical components carry at most 0.42 and 0.09 of its effect) and 12 components carry identity; (e) applies because J-C1 is not met. Its pre-committed text, which the main text uses verbatim: "Even steering vectors close to the natural edit depart from the natural channel ratio; the format law describes the model's own written value and does not predict how an edit is attributed." The binding-swap lines (J-C6 met, J-C-SCREEN met, J-C-WIN not evaluable) are in Appendix H.2.

Holm sensitivity. Over the 58 interval components of this part's risky lines with a verdict, Holm's step-down changes 4 decisions (one statistic of the remap at block 3 and three upper bounds of the boundary line) and no verdict.

F.4. Post hoc and exploratory

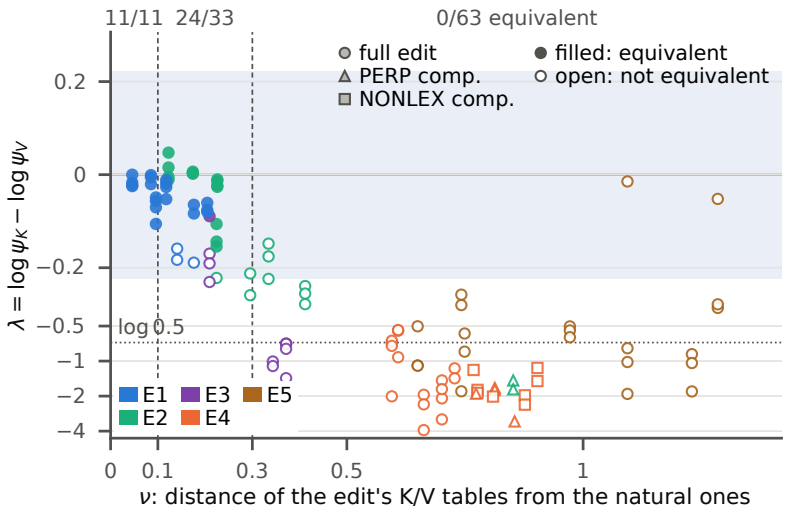


Figure 12. **Post hoc: departures from the law follow how far an edit writes from the natural write.** Each point is one λ statistic of an instance that carries identity (φ without a mention at least 0.3), in the three models: full edits (circles; colour by family, E3 included although its gate failed) and PERP and NONLEX components (triangles, squares); filled: equivalent. ν: the distance of the edit's cached keys and values from the natural ones, relative to the natural displacement. Band: $\pm \log 1.25$; dotted: $\log 0.5$, the copy-only bound; above the panel: equivalent of all statistics in each band of ν (cut-points chosen after the data). Linear ν axis; symmetric-log λ axis, linear within ± 0.25. Recomputed; post hoc.

Distance from the natural write (post hoc). Over the 107 λ statistics of identity-carrying instances, λ has Spearman correlation -0.75 with ν (95 % interval -0.88 to -0.55 from a bootstrap over the 38 units of model, instance and depth, since each unit contributes several statistics) and 0.80 with the cosine of the edit's key displacement to the natural one; 11 of 11 statistics with $\nu < 0.1$ are equivalent, 24 of 33 with ν between 0.1 and 0.3, and 0 of 63 beyond, of which 52 have a 90 % interval wholly outside the margin (a shown departure) and the rest are inconclusive (Figure 12). Four caveats limit the reading. (i) The band cut-points 0.1 and 0.3 were chosen after the data. (ii) At $\nu = 0$ the edit is the natural write and equivalence holds by construction, so a monotone association is partly guaranteed; only the rate of departure beyond small ν is informative. (iii) ν is largely confounded with the edit family: of the 44 statistics with $\nu < 0.3$, 40 are steering vectors (E1, E2) and 4 E3, while of the 63 beyond, 6 are steering vectors, 6 E3 and 51 the remap, the non-lexical vectors and the components. Within the steering vectors (n = 46) the correlation is -0.59; within the remap, non-lexical vectors and components (n = 51) it is 0.14: beyond $\nu \approx 0.3$ every statistic comes from edits that are copy-read whatever their distance. (iv) The table statistics are collinear (pairwise |Spearman| 0.84–0.98 among ν, the key, value and residual cosines and the first-order key projection), so ν cannot be told apart from the key cosine. The first-order prediction from the projections of the key and value displacements on the natural ones ranks the statistics only weakly (Spearman 0.20) and misses by -0.68 on average: beyond $\nu \approx 0.3$ the key read collapses non-linearly while the value read decays more slowly (exploratory, recomputed). We therefore offer ν as a diagnostic to report alongside an intervention, not as an explanation of the departures. The pattern was found after the data and was not preregistered. A hypothesis for a future preregistration links it to the flag (Appendix E.3): the flag's dose response is threshold-like (half a flag barely moves the answer, one and two flags about equally), so the key route may select through a thresholded match while the value read degrades gradually; an edit that writes far from the natural state would then lose its key read first. It predicts that shrinking the key displacement of a near-natural steering vector past a threshold collapses ψ_K while ψ_V falls smoothly.

The key share against the natural share (reported). At block 3 the steering vectors' key share is within 0.02 of the natural share in every format of all three models, while the remap's is lower. From block 3 to block 15 the change in

the steering vectors' key share tracks the change in the natural share in 14 of 14 qualifying pairs (largest gap 0.032), a consistency check that never enters the headline.

Generated answers under the edits (exploratory). At block 7, on the first 40 stories, E1 and E2 flip every generated answer without a mention, as the natural value does (1.00, 1.00 and 1.00 at Qwen2.5-7B), and almost as many with the options listed (0.96, 0.95, 0.94); the trained remap flips 0.69 and 0.99 of answers without a mention (two seeds) but only 0.14 and 0.25 with the options listed: its key part, which the options' readers would need, is weak.

G. The released remap

This appendix describes, in our words, the intervention released with Anonymous (2026) and everything we measured on it: the reproduction, the crossover across formats that motivates Section 5, the refit without later mentions (preregistration F), the reader heads at 24B under the remap (preregistration I) and the same frames under the emitted-form score (final round). All of it is discovery sample except the last part. We describe what the released analysis does, not what its authors concluded from it.

G.1. What the released analysis does

The released analysis of Anonymous (2026) patches Mistral-Small-24B and Qwen2.5-72B at the output of (1-based) decoder block 4 over every token of the event span (from "{A} watches as" to the end of the story). For a rank-16 basis U with orthonormal rows, each span position becomes

$$h \leftarrow h_{\mathrm{B}} + U^{\top} U\,(h_{\mathrm{S}} - h_{\mathrm{B}}). \tag{3}$$

In the learned remap M, U is a distributed alignment search basis (Geiger et al., 2023) fit with a six-choice cross-entropy objective so that the patched model ranks $\mathrm{T} = \pi(\mathrm{S})$ first, where π is a fixed pair-swap of the six locations. In the control P, U is an unfitted PCA basis of source-minus-base differences, which transfers S. We reuse the released bases (three seeds each; the Qwen2.5-72B PCA bases are reconstructions, labelled so by Anonymous (2026)). Following the released design, a *key-only exchange* replaces the writing token's key in every block from (1-based) block 6 on by the key of the other patched run, while queries, values and residual states evolve freely; we add the symmetric *value-only exchange*. Unlike our clamp, these exchanges are not exact, because the patch also changes the other span tokens. With $m_{\mathrm{TS}} = \log p(\mathrm{T}) - \log p(\mathrm{S})$, writing $\mathrm{P}{+}K_{\mathrm{M}}$ for P with M's keys and $m(\cdot)$ for m_{TS} in a run,

$$\varphi = \frac{m(\mathrm{M}) - m(\mathrm{P})}{m(\mathrm{T}) - m(\mathrm{S})}, \tag{4}$$

$$\psi_K = \frac{m(\mathrm{P}{+}K_{\mathrm{M}}) - m(\mathrm{P})}{m(\mathrm{M}) - m(\mathrm{P})}, \tag{5}$$

$$\psi_V = \frac{m(\mathrm{P}{+}V_{\mathrm{M}}) - m(\mathrm{P})}{m(\mathrm{M}) - m(\mathrm{P})}. \tag{6}$$

φ is the shift of M beyond P towards T, as a fraction of the natural S→T shift; ψ_K and ψ_V are the fractions of the learned-minus-PCA difference that M's keys or values add to P. All are ratios of means over the 96 of their 120 native story cores in which B, S and T are distinct, after averaging the three fits within each core.

The exchanges are not exact, because the patch also changes the other span tokens; the removal fractions χ_K and χ_V and the implementation are in Appendix C.1. The remap was fit without the "Answer:" prefill of our formats, which the released analysis adds only at evaluation.

G.2. Reproduction and the crossover across formats

Reproduction. Under OPTIONS-AFTER, the format of the released analysis, our runs of the released bases reproduce the released answers: the argmax of M and of P agrees with the released outputs on 0.994–1.000 of (core, fit) items (a preregistered gate, Table 9). As in the released analysis, M's keys alone reproduce a substantial part of the M−P difference in both models (ψ_K = 0.53 and 0.63 under OPTIONS-AFTER), and, in a comparison the release does not make, more of it than M's values do. We did not re-test the release's colour-lookup questions, and we score log-odds where the release scores argmax agreement. The addition fractions sum to less than one and the removal fractions mostly to more than one (Table 41), so the channels do not combine additively, and part of the edit may travel through the other patched span tokens.

Behaviour barely changes across formats; the carrying channel flips. We then changed only the format and added a value-only exchange. Two comparisons are confirmatory and met in both models: without a later mention M's keys carry less than half of what they carry under OPTIONS-AFTER (C3), and M's values carry most of the effect (E4). φ lies between 0.70 and 0.75 at Mistral-Small-24B and between 0.86 and 0.91 at Qwen2.5-72B, and the PCA control transfers the source location in every format. Yet at Qwen2.5-72B, M's keys reproduce 0.89 of the M−P difference under LETTERS-AFTER but 0.05 under NO-MENTION, while M's values do the reverse (Figure 13b). Mistral-Small-24B and the removal fractions

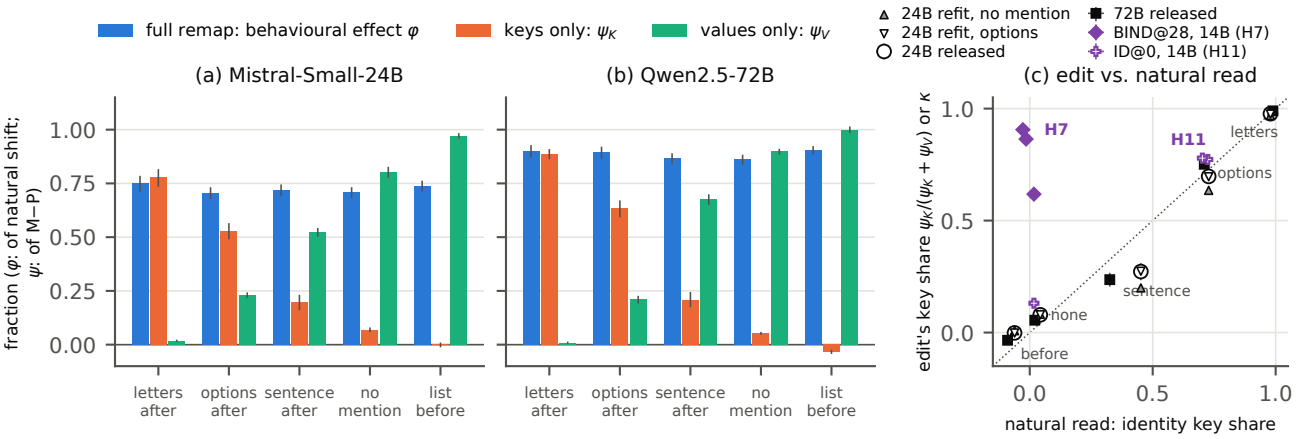


Figure 13. **The same learned intervention has nearly the same behavioural effect in every format, but whether the writing token's keys or values carry it depends on the later mentions** (discovery sample). The released remap M of Anonymous (2026) against the PCA patch P at Mistral-Small-24B (a) and Qwen2.5-72B (b): φ, ψ_K and ψ_V (Equations (4) to (6)), exchanged from block 6 on; $n = 96$ cores $\times$ 3 fits, 95% bootstrap intervals. (c) The remap's key share $\psi_K/(\psi_K+\psi_V)$ against the unpatched model's s_{ID} in the same format: post hoc for the released remaps, preregistered (F4) for the refits (triangles; Appendix G.3). Diamonds: the binding swap of Prakash et al. (2026) at Qwen2.5-14B, κ at block ℓ^* against s_{ID} from block ℓ^*+1 (H7, failed); crosses: the identity edit at block 0 against s_{ID} from block 1 (H11, met). Values in Tables 41, 42 and 45.

show the same crossover. Across the ten format–model cells, the remap's key share tracks the identity key share s_{ID} of the unpatched model in the same format (Figure 13c; $r = 0.98$): post hoc for the released remaps, preregistered for the refits (Appendix G.3), and failed out of sample for a binding edit (Appendix H).

Interpretation (post hoc, discovery sample). In LETTERS-AFTER, OPTIONS-AFTER, SENTENCE-AFTER and NO-MENTION the prompt is identical up to the end of the story, so under the causal mask M writes the same key and the same value at the writing token in all four; only their readers differ, and which channel appears to carry the edit follows them. Its value half (E4) and the refit test (F) were then preregistered and confirmed. Whether this reading extends to edits obtained independently of the released remap is the channel-ratio law of the final round, which failed (Section 5 and Appendix F).

Intervention frames. Table 41 gives every frame quantity with its interval. At Qwen2.5-72B, M's keys reproduce 0.89 of the M−P difference under LETTERS-AFTER and 0.63 under OPTIONS-AFTER, but only 0.21 under SENTENCE-AFTER, 0.05 under NO-MENTION and -0.03 under LIST-BEFORE, while M's values do the reverse (ψ_V = 0.01, 0.21, 0.67, 0.90, 1.00; exploratory except C3 and E4). The removal fractions follow the addition fractions: χ_K is large where ψ_K is, and χ_V where ψ_V is. Among the six candidates, M's top answer is the target T in 0.76–0.87 of (core, fit) items at Mistral-Small-24B and 0.92–0.95 at Qwen2.5-72B, depending on the format.

G.3. A remap fit without later mentions (preregistration F)

The remap edits block 4, upstream of every exchanged key and value. Preregistration F asked whether the format it was fit in, rather than its readers, decides its channel: at Mistral-Small-24B it refit the remap with the released code under NO-MENTION (fit_none) and, as a same-code control, under OPTIONS-AFTER (fit_opt). All gates and all four predictions were met (Tables 11 and 42): fit_none is carried by keys under LETTERS-AFTER ($\psi_K = 0.77$) and by values under NO-MENTION ($\psi_V = 0.80$), transfers to lettered options ($\varphi = 0.67$), shows the control's crossover (F3), and its key share tracks the natural read ($r = 0.96$; Figure 13c). The reading format decides which channel dominates.

Recipe. We ran the released training code of Anonymous (2026) (`gpu/train.py` and `gpu/behavior_engine.py`, unmodified) through `experiments/refit_remap.py`. Its recipe: a rank-16 basis at the output of (1-based) block 4 over every event-span token; the pair-swap objective with a six-way conditional cross-entropy at the start of the reply (no prefill); AdamW, learning rate 10^{-3}, batch 1, one shuffled epoch of their 1000 training pairs (2000 stories); seeds 101–103, which change only the order of the training pairs: every fit starts from the

Table 41. **The intervention of Anonymous (2026) across formats.** φ: behavioural shift of the learned remap M relative to the PCA patch P, as a fraction of the natural source-to-target shift (Equation (4)). ψ_K, ψ_V: fraction of the M−P difference that P reproduces when given M's keys or values at the writing token (Equations (5) and (6)). χ_K, χ_V: fraction of that difference that M loses when given P's keys or values (Appendix C). Exchanges from block 6 on. $n = 96$ story cores with B, S and T distinct, three fits averaged within each core; 95% bootstrap CIs over cores. Preregistrations C (C1–C3) and E (E4) for the cells they name; the rest exploratory.

Model	Format	φ	ψ_K	χ_K	ψ_V	χ_V
Mistral-Small-24B	LETTERS-AFTER	0.75 [0.71, 0.79]	0.78 [0.73, 0.82]	0.99 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.70 [0.68, 0.73]	0.53 [0.49, 0.57]	0.76 [0.75, 0.78]	0.23 [0.22, 0.24]	0.44 [0.41, 0.48]
	SENTENCE-AFTER	0.72 [0.69, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.71 [0.68, 0.73]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.74 [0.71, 0.76]	-0.00 [-0.01, 0.01]	0.01 [-0.00, 0.02]	0.97 [0.96, 0.98]	0.96 [0.95, 0.98]
Qwen2.5-72B	LETTERS-AFTER	0.90 [0.87, 0.93]	0.89 [0.86, 0.91]	0.99 [0.99, 1.00]	0.01 [0.00, 0.01]	0.17 [0.13, 0.20]
	OPTIONS-AFTER	0.89 [0.86, 0.92]	0.63 [0.59, 0.67]	0.79 [0.78, 0.81]	0.21 [0.19, 0.23]	0.44 [0.39, 0.48]
	SENTENCE-AFTER	0.87 [0.84, 0.89]	0.21 [0.17, 0.25]	0.34 [0.31, 0.36]	0.67 [0.65, 0.70]	0.80 [0.76, 0.84]
	NO-MENTION	0.86 [0.84, 0.88]	0.05 [0.05, 0.06]	0.11 [0.09, 0.12]	0.90 [0.88, 0.91]	0.95 [0.94, 0.96]
	LIST-BEFORE	0.91 [0.88, 0.92]	-0.03 [-0.04, -0.02]	-0.02 [-0.04, -0.01]	1.00 [0.98, 1.01]	1.01 [0.99, 1.02]

same PCA basis of source-minus-base differences (the randomised sketch differs by seed but returns the same subspace, mean squared principal cosine 1.000 across seeds), and that basis is the control P of every fit. The code also fits an identity-objective basis next to each remap (exploratory): its shift of the answer towards T, relative to P and as a fraction of the natural S→T shift, is at most 0.18 for fit_none (under LETTERS-AFTER) and 0.04 for fit_opt. The training stories are disjoint from the 120 evaluation cores, but most evaluation event sentences also occur in some training story, as for the released remap.

What the wrapper replaces. At import time, and recorded with hashes in each run's `FRAME.json`: the prompt function, for fit_none only, by our NO-MENTION prompt, which agrees token for token with the original prompt up to the end of the story in all 1000 training pairs; the release-integrity check, by one that tolerates only the release's mismatched `README.md`; and the model loader, by one with the same load arguments (BF16, sdpa, eval mode, no gradients, no cache, pad = eos, `fix_mistral_regex`) at the pinned revision, waiving the release's exact Python, package, device and memory pins and its model-asset receipt (the weights are identified instead by the pinned commit, which the loader checks). Under the causal mask the activations up to the end of the story are the same in both runs, so both start from the same P; the PCA bases came out bitwise identical across runs.

Results. Table 42 gives every frame quantity for the two refits and for the released bases re-run in the same job; Table 11 gives the gates and predictions. The rest of this paragraph is exploratory. The control matches the released remap to within 0.03 in every quantity and format. Every fitted remap keeps 0.34–0.36 of its subspace (mean squared principal cosine) in the shared PCA start, against 0.003 for random subspaces, so overlaps between fits lie above that floor: fit_opt overlaps the released remap at 0.71–0.79, while fit_none overlaps the released remap at 0.37–0.39 and fit_opt at 0.37–0.39, less than two seeds of one fit overlap each other (0.41–0.44). Without the "Answer:" prefill, which matches the interface the remaps were fit at, the difference in differences is $+0.004$ $[-0.014, +0.021]$. The training loss of the remap fits fell from 2.9–4.8 nats over the first 100 updates to 0.11–0.48 over the last 100; it starts above chance ($\ln 6 \approx 1.79$) because the PCA start transfers the source, which the pair-swap objective scores as wrong.

Where the fitting format still matters (post hoc). The fitting format shifts the balance where both channels read, in the direction the stated alternative predicts. fit_none's ψ_K is lower than the control's by 0.09 [0.07, 0.12] under OPTIONS-AFTER and by 0.07 [0.05, 0.10] under SENTENCE-AFTER. It transfers better under NO-MENTION ($\varphi = 0.81$ against 0.72) and worse under LETTERS-AFTER (0.67 against 0.77), where its values carry almost nothing ($\psi_V = 0.02$). These comparisons were made after seeing the data; the preregistered all-or-none alternative (Table 11) was not observed.

G.4. The reader heads at 24B under the remap (preregistration I)

Preregistration I (GPU stage 7; Table 14) was written after an external reading noted that the two halves of the paper were linked only by inference: the reader heads were shown at two 7B models (Appendix E.2), the format dependence of the remap at 24B and 72B (Appendix G.2). It tests the link at Mistral-Small-24B on the 96 cores and released bases of the frames (Table 41), with M and its PCA control P patched at block 4 and the key or value of the writing token exchanged

Table 42. **Refit remaps in the frames of Anonymous (2026)** (Mistral-Small-24B, "Answer:" prefill). fit_none: refit under NO-MENTION; fit_opt: refit under OPTIONS-AFTER with the same code; released: the released bases, re-run in the same job. All fits share one PCA control P. $n = 96$ cores, three fits averaged within core; 95% core bootstrap.

Remap	Format	φ	ψ_K	χ_K	ψ_V	χ_V
fit_none	LETTERS-AFTER	0.67 [0.64, 0.70]	0.77 [0.73, 0.80]	0.98 [0.97, 0.98]	0.02 [0.02, 0.03]	0.15 [0.13, 0.19]
	OPTIONS-AFTER	0.72 [0.70, 0.74]	0.43 [0.39, 0.47]	0.74 [0.73, 0.76]	0.25 [0.23, 0.26]	0.54 [0.50, 0.58]
	SENTENCE-AFTER	0.81 [0.79, 0.83]	0.12 [0.10, 0.15]	0.47 [0.44, 0.49]	0.50 [0.47, 0.52]	0.85 [0.81, 0.88]
	NO-MENTION	0.81 [0.79, 0.84]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.77, 0.82]	0.93 [0.91, 0.94]
	LIST-BEFORE	0.76 [0.74, 0.78]	0.01 [-0.00, 0.02]	-0.00 [-0.01, 0.01]	0.97 [0.96, 0.99]	0.95 [0.94, 0.96]
fit_opt	LETTERS-AFTER	0.77 [0.74, 0.80]	0.78 [0.73, 0.82]	0.98 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.72 [0.69, 0.74]	0.52 [0.48, 0.56]	0.76 [0.74, 0.77]	0.23 [0.22, 0.24]	0.44 [0.40, 0.48]
	SENTENCE-AFTER	0.73 [0.71, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.72 [0.69, 0.74]	0.07 [0.06, 0.08]	0.17 [0.15, 0.19]	0.81 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.77 [0.74, 0.79]	-0.00 [-0.01, 0.01]	0.00 [-0.01, 0.02]	0.98 [0.96, 0.99]	0.96 [0.95, 0.98]
released	LETTERS-AFTER	0.75 [0.71, 0.79]	0.78 [0.74, 0.82]	0.99 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.70 [0.68, 0.73]	0.53 [0.49, 0.57]	0.76 [0.75, 0.78]	0.23 [0.22, 0.24]	0.44 [0.41, 0.48]
	SENTENCE-AFTER	0.72 [0.69, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.71 [0.68, 0.73]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.74 [0.71, 0.76]	-0.00 [-0.01, 0.01]	0.01 [-0.00, 0.02]	0.97 [0.96, 0.98]	0.96 [0.95, 0.98]

from block 6, as there.

The readers at 24B (Gate I-G2). Heads are ranked by a_3 (Appendix E.2) on 60 ranking stories that share no story with the 96 cores, separately for each format, with the natural key clamped at the writing token from block 6, where the exchange starts; heads in blocks 1–5 cannot see the exchanged key and are not eligible (1120 of the 1280 heads remain). H^* is the top 64, 5% of all heads. On the 96 cores the option rows carry 0.853, 0.987 and 0.682 of the natural key effect under OPTIONS-AFTER, LETTERS-AFTER and SENTENCE-AFTER. H^* alone recovers $R(64)$ = 0.832, 0.822 and 0.912 of the option rows' read, and blinding H^* removes $\mathrm{KO}(64)$ = 0.971, 0.991 and 1.012; random sets of 64 recover at most 0.002 and remove at most 0.037 (means of three). $R(64)$ is lower than at 7B only on a story type the ranking stories lack, the 36 of the 96 cores whose distractor location equals B or S: on the other cores $R(64)$ = 0.944, 0.942 and 0.974, on those 0.612, 0.602 and 0.749 (exploratory, recomputed, exploratory). Also exploratory: the sets of OPTIONS-AFTER and LETTERS-AFTER share 59 of 64 heads, those of OPTIONS-AFTER and SENTENCE-AFTER 43; ranking heads on the remap itself (P against P with M's key, in-sample on the 96 cores) gives sets that share 57, 55 and 59 heads with H^*.

The remap's key read (I1–I3). In the exchange-blinding operation, chosen heads at the option rows see P's own key at the writing token instead of M's, pinned from block 6, while every other head and row sees M's key. Blinding every head at the option rows removes g_K = 0.976 and 0.999 of M's key-only effect (0.901 under SENTENCE-AFTER, reported). Blinding only H^* removes 0.994 and 0.993 of that read (1.023), and random or active-at-G sets at most 0.082. Little of M's key effect is read outside the option rows: the key route of about one nat seen without a later mention does not persist when the candidates are mentioned, which is why some ratios against NO-MENTION exceed 1 (Appendix A).

Ablation and knockout (I4–I7). Table 43 gives every condition. The ablation A replaces each head's output at the option rows by its mean over the ranking stories, in every row of the frames batch, as in H3; A^+ adds the 32 heads of block 5; the knockout N stops the heads from attending from the option rows to the writing token. Under OPTIONS-AFTER, $\mathrm{A}(H^*)$ removes nearly all of M's key-only effect, moves its key share near its NO-MENTION level (I4) and raises its value-only effect by 2.388 nats, 0.366 of the lost key effect (I5). The non-additive part of M's effect ($D - K - V$, which also holds reads of the other patched span tokens and, in block 5, of the writing token) falls from 3.32 to 1.51 nats (exploratory, recomputed), and t = 0.565. Post hoc, most of the fall in $D = m(\mathrm{M}) - m(\mathrm{P})$ is in P's rows: ablating the readers compresses every margin, so $m(\mathrm{P})$ rises from -9.66 to -5.68 nats while $m(\mathrm{M})$ falls only from 4.02 to 2.05, and the unedited model's margin for its own answer over S falls from 11.51 to 7.12 nats while that answer stays the top one in every core. Under LETTERS-AFTER (reported two-sided) the remap and the natural task both lose: t = 0.385, M's target is the top answer in 0.46 of cases (0.76 unblocked) and the natural answer in 0.83 of cores, consistent with the entry's caution that a lettered answer may need the readers. Under SENTENCE-AFTER every threshold is met (t = 0.944; reported). The knockout $\mathrm{N}(H^*)$ keeps more of M's effect than the ablation under OPTIONS-AFTER (t = 0.722), and less under LETTERS-AFTER (t = 0.296),

where answers move to the initial location in about a fifth of cases, as with the all-heads knockout at Qwen2.5-7B (G7).

Table 43. **The remap under each blocking at Mistral-Small-24B** (preregistration I; values as printed by the scorer). $t = D^O/D^\emptyset$; superscripts O and $\emptyset$: with and without the blocking; $\kappa = K/(K+V)$ with the κ rule (–: κ undefined in 99% of resamples; Δ_V not applicable or M→T not printed); $\tilde{\psi}_K = K^O/D^\emptyset$ and $\tilde{\psi}_V = V^O/D^\emptyset$; $\Delta_V = V^O - V^\emptyset$, paired by core; D, K, V: M's effect, its key-only and its value-only part, in nats; M→T: share of (core, fit) items whose top answer is M's target. A: mean-ablation of the heads' outputs at the option rows; A$^+$: with block 5; N: no attention from the option rows to the writing token. †Exploratory: N(all at G), every head at the option rows G; A(block 5), the 32 heads of block 5 alone; N(rand0), the first random set of 64; A($H^*_{\text{OPTIONS-AFTER}}$), the OPTIONS-AFTER set ablated under LETTERS-AFTER. 95% core-bootstrap CIs.

Format	Condition	t	κ	$\tilde{\psi}_K$	$\tilde{\psi}_V$	Δ_V (nats)	D / K / V (nats)	M→T
OPTIONS-AFTER	unblocked	1	0.698	0.528	0.229	–	13.67 / 7.22 / 3.13	0.82
	A(H^*)	0.565 [0.542, 0.589]	0.113 [0.080, 0.143]	0.052	0.404	+2.388 [2.141, 2.631]	7.73 / 0.70 / 5.52	0.80
	A$^+$(H^*)	0.556 [0.534, 0.578]	0.100 [0.067, 0.129]	0.046	0.411	+2.496 [2.250, 2.737]	7.60 / 0.62 / 5.63	0.79
	N(H^*)	0.722 [0.697, 0.748]	0.070 [0.042, 0.097]	0.046	0.605	+5.140 [4.835, 5.452]	9.87 / 0.62 / 8.27	0.83
	N(all at G)†	0.739 [0.713, 0.767]	−0.009	−0.006	0.684	+6.220 [5.899, 6.541]	10.11 / −0.08 / 9.35	–
	A(block 5)†	1.006 [1.002, 1.010]	0.709	0.561	0.230	+0.013 [−0.020, 0.048]	13.76 / 7.67 / 3.14	–
	N(rand0)†	0.985 [0.977, 0.992]	0.684	0.497	0.230	+0.011 [−0.006, 0.029]	13.47 / 6.79 / 3.14	–
LETTERS-AFTER	unblocked	1	0.977	0.777	0.018	–	12.87 / 9.99 / 0.23	0.76
	A(H^*)	0.385 [0.358, 0.410]	0.661 [0.615, 0.708]	0.179	0.092	+0.945 [0.791, 1.102]	4.95 / 2.30 / 1.18	0.46
	A$^+$(H^*)	0.373 [0.347, 0.398]	0.666 [0.627, 0.706]	0.205	0.102	+1.086 [0.927, 1.244]	4.80 / 2.63 / 1.32	0.47
	N(H^*)	0.296 [0.252, 0.342]	0.466 [0.427, 0.507]	0.134	0.154	+1.746 [1.408, 2.122]	3.81 / 1.73 / 1.98	0.39
	N(all at G)†	0.190 [0.156, 0.224]	–	0.040	0.137	+1.524 [1.180, 1.896]	2.44 / 0.51 / 1.76	–
	A(block 5)†	1.017 [1.011, 1.023]	0.976	0.849	0.021	+0.038 [0.020, 0.058]	13.09 / 10.92 / 0.27	–
	N(rand0)†	0.978 [0.967, 0.988]	0.977	0.748	0.018	−0.007 [−0.027, 0.011]	12.59 / 9.62 / 0.23	–
	A($H^*_{\text{OPTIONS-AFTER}}$)†	0.373 [0.347, 0.398]	0.646	0.170	0.093	+0.961 [0.797, 1.129]	4.80 / 2.18 / 1.19	–
SENTENCE-AFTER	unblocked	1	0.274	0.196	0.522	–	12.16 / 2.39 / 6.35	0.87
	A(H^*)	0.944 [0.928, 0.960]	0.012 [−0.000, 0.025]	0.011	0.869	+4.220 [3.816, 4.622]	11.48 / 0.13 / 10.57	0.88
	A$^+$(H^*)	0.943 [0.927, 0.959]	0.011 [−0.001, 0.024]	0.010	0.871	+4.242 [3.833, 4.650]	11.47 / 0.12 / 10.59	0.88
	N(H^*)	0.968 [0.956, 0.982]	0.019 [0.007, 0.031]	0.017	0.910	+4.722 [4.430, 5.020]	11.78 / 0.21 / 11.07	0.88
	N(all at G)†	0.949 [0.934, 0.964]	0.024	0.022	0.888	+4.459 [4.114, 4.807]	11.54 / 0.27 / 10.81	–
	A(block 5)†	1.005 [1.003, 1.007]	0.294	0.214	0.516	−0.075 [−0.113, −0.033]	12.23 / 2.61 / 6.27	–
	N(rand0)†	0.998 [0.996, 1.000]	0.250	0.180	0.540	+0.220 [0.144, 0.306]	12.14 / 2.19 / 6.57	–

Summary. The remap's key-only effect at Mistral-Small-24B is read at the option words and through the natural reader heads, not through random or active sets (I1–I3); ablating these heads moves its key share under OPTIONS-AFTER near the NO-MENTION level (I4) and raises what its values carry (I5), but keeps only part of its effect on the margin (I6, not met; the entry's partial takeover), and Gate I-G4, which reading the kept part as a value takeover requires, failed. Qwen2.5-72B was not run.

G.5. The frames under the emitted-form score (final round)

The discovery sample scored the frames on the lower-case candidate tokens. In the final round the frames at Mistral-Small-24B were re-measured on the 96 native cores of the release with the released bases, scored both ways in the same pass, the emitted-form score using the model's own frames (risky measurement-validity line J-B8, met). All twelve differences between the two scores (of φ, ψ_K and ψ_V, under OPTIONS-AFTER, NO-MENTION, LIST-BEFORE and SENTENCE-AFTER) have 95 % paired core-bootstrap intervals within [-0.013, 0.018], inside the preregistered ± 0.05, and the emitted forms carry 0.961–0.990 of the natural run's probability mass. For example, without a later mention ψ_V is 0.802 under the lower-case score and 0.793 under emitted forms, and under OPTIONS-AFTER ψ_K is 0.529 and 0.528. The crossover is therefore not an artefact of the lower-case score.

H. The binding swap of Prakash et al.

Preregistration H applied the key-only and value-only exchange to an edit we did not design: the state-token patch BIND of Prakash et al. (2026), at Qwen2.5-14B on their material, with an identity edit ID as a control. Its four predictions on the swap (H7–H10) failed. The final round asked two narrower questions on the same material: whether an identity edit is copy-read at the swap's depth (J-C6), and whether any model has a depth at which the binding swap works while the identity key route is still open, the only depth at which the two kinds of edit could be compared (the overlap screen). Table 13 gives the verdicts of H, and Appendix A.2 those of the final round.

H.1. The exchange at Qwen2.5-14B (preregistration H, part (b))

BIND replicates (IIA 0.993 at block 28 under NO-MENTION; Gate b1).

The binding swap is key-dominant; four predictions failed. Without a later mention, keys carry most of BIND's effect and values a substantial share ($\psi_K = 0.703$, $\psi_V = 0.435$, $\kappa = 0.618$). Under QNAMES and OPTIONS-AFTER, $\kappa = 0.906$ and 0.864, but BIND flips 0/150 answers there (exploratory count), so these decompose a shift of the logit margin; LETTERS-AFTER failed the effect-size gate. Not met: values carrying BIND without a later mention (H8); a crossover of at least 0.4 (later mentions raised κ by 0.247 [0.211, 0.281], and QNAMES is not in between; H9); the rival $\kappa \leq 0.25$ (H10; nor is $\kappa \geq 0.75$ throughout); and the law $\kappa \approx s_{\mathrm{ID}}$, which missed by 0.60–0.93 (H7; Figure 13c).

The identity edit, and depth. ID at block 0 matched the law (κ_{ID} = 0.131, 0.773, 0.779 against s_{ID} = 0.017, 0.724, 0.702), and later mentions raised κ_{ID} by 0.648 [0.630, 0.666] (H11, met). This is consistency on their material, not out-of-sample support: at block 0 ID nearly equals the natural clamp, and H11's block-28 part compares values near zero. Post hoc, at block 28 BIND is key-dominant while ID is value-carried ($\psi_V \geq 0.811$ in the evaluable formats), which is consistent with the H7 failure but does not explain it. Depth is an alternative explanation (below): no depth has both the binding swap and a key-read identity route, so whether the kind of edit or its depth sets how far the format moves the attribution is untested.

Material and population. We use the template-2 stories of Prakash et al. (2026), their raw prompt (an instruction, the story, "Question:" and "Answer:", without a chat template) and their pool of 320 reversed-sentence pairs drawn with their seed. In each story two characters each fill an opaque bottle with a drink, in two sentences, and the question asks what one character believes their bottle contains. The release is fetched at run time and checked by hash, and our port reproduces their pool byte for byte. Their language-model filter under NO-MENTION passed 315 pairs at Qwen2.5-14B-Instruct; the population is the first 150 passing pairs in pool order, of which about 70 are in their 80-pair validation split. The formats differ only after the story: NO-MENTION is their question; QNAMES names four candidate drinks in the question (the two story drinks and two drinks S and X absent from the story); OPTIONS-AFTER lists the same four as choices with "Answer with exactly one choice"; LETTERS-AFTER letters them (exploratory); and QNAMES2 (exploratory) names only the two story drinks, as the questions of their preliminary second dataset do.

The two edits. BIND is their state-token patch: at the output of block ℓ^*, the residuals at the two drink words and the punctuation after each are set to those of the counterfactual in which the two sentences are exchanged, each position taking the other sentence's. The drink words' identity does not change; their ordering information does. These are the ordering IDs on which the binding, answer and visibility lookbacks of Prakash et al. (2026) are built, so the swap is not an identity edit. ID substitutes, at block ℓ^*_{ID}, the queried drink's span from a donor story in which that drink is replaced by S. With $m = \log p(\text{target}) - \log p(\text{queried drink})$, where the target is the other story drink for BIND and S for ID, IIA is the fraction of pairs whose patched answer is the target, and ℓ^* and ℓ^*_{ID} are the earliest blocks whose IIA under NO-MENTION is within 0.01 of the maximum.

Replication and sweeps. The swap reproduces their released per-layer IIA (Table 44). Its IIA is 1.00 at blocks 29–34 and 0.993 at 28, where one pair fails; the tie rule selects $\ell^* = 28$, and one more failure would have selected 29. Below block 24 it flips no answer under NO-MENTION (IIA 0.00 at block 23, 0.72 at 24). Figure 14 shows every sweep. The swap reproduces only under NO-MENTION: under QNAMES, OPTIONS-AFTER and LETTERS-AFTER its IIA is at most 0.01 at every swept block from 18 to 40, and its effect on the margin peaks at 6.5 nats (OPTIONS-AFTER, block 28), 6.1 (QNAMES, block 31) and 2.8 (LETTERS-AFTER, block 23). The identity edit flips every answer under NO-MENTION at every block through 35, so its curve is flat and the tie rule picks $\ell^*_{\mathrm{ID}} = 0$ by default rather than by localisation. When the candidates

Table 44. **Replication of the binding swap** (Qwen2.5-14B-Instruct, NO-MENTION, IIA of BIND by patched block). Released: the per-layer values published with Prakash et al. (2026) on their 80 validation pairs, transcribed from their release (not part of our results). Ours: 150 pairs, and the 70 pairs shared with their validation split (exploratory). Recomputed from our sweep files; a reproduction check, since the released values were seen before preregistration H was finalised.

Block	Prakash et al. (80 pairs)	ours (150 pairs)	ours, 70 shared pairs
24	0.69	0.720	0.700
26	0.72	0.747	0.729
28	1.00	0.993	1.000
30–34	1.00	1.000	1.000
36	0.06	0.080	0.057

are named after the question it stops flipping answers from about block 20: under OPTIONS-AFTER its IIA is 1.00 through block 11, 0.89 at 14, 0.66 at 16, 0.17 at 19 and 0.00 from 20; under QNAMES it is 0.90 at block 0 and 0.00 from 20.

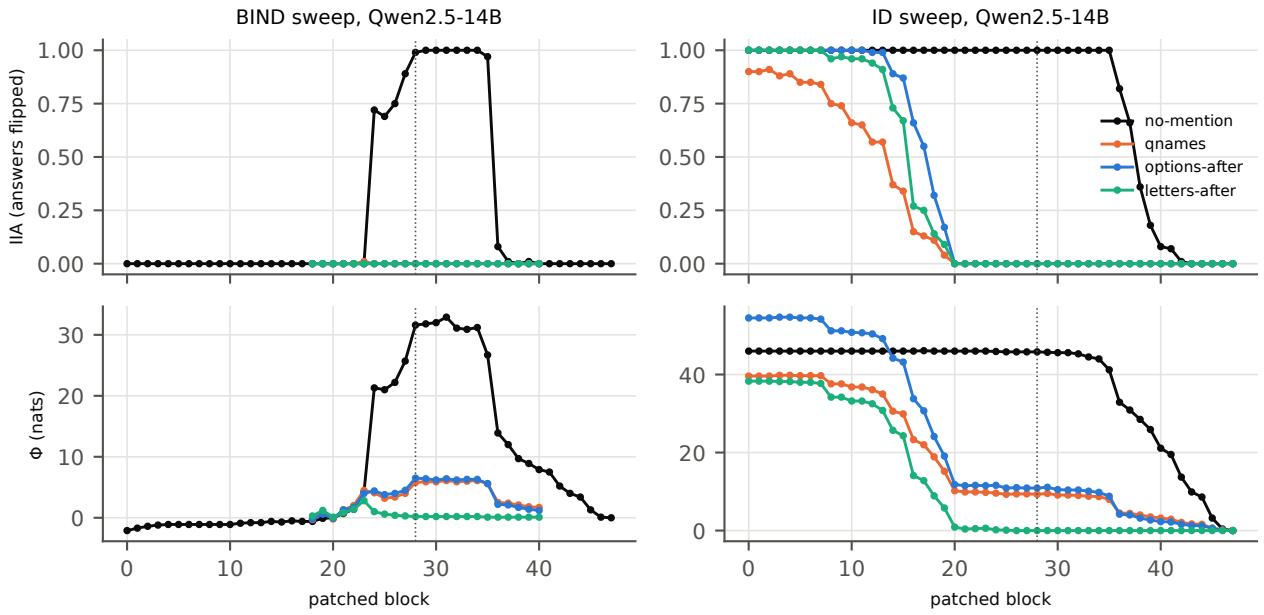


Figure 14. **Sweeps of the two edits over the patched block** (Qwen2.5-14B-Instruct, 150 pairs, BF16). Top: IIA, the fraction of answers flipped to the target; bottom: Φ, the mean shift of $m = \log p(\text{target}) - \log p(\text{queried drink})$. Left: the binding swap BIND (swept over blocks 18–40 in the formats that name candidates); right: the identity edit ID. Dotted: block $\ell^* = 28$. ℓ^* and ℓ^*_{ID} are chosen on the NO-MENTION sweeps (preregistered); the other formats are reported.

The exchange. The exchange (Appendix C) exchanges the patched positions' keys or values in every block after the patched one. A cell is evaluable if Gates b0 and b2 pass in it ($\Phi \geq 3$ nats), $\psi_K + \psi_V \geq 0.5$ with both at least -0.1, and the edit reproduces at that depth on the NO-MENTION sweep. Table 45 summarises the cells and Table 46 gives all thirteen. LETTERS-AFTER fails Gate b2 at block 28 for both edits (Φ = 0.225 and 0.034 nats), and ID@28 under LETTERS-AFTER also fails the κ rule; LETTERS-AFTER is exploratory throughout. BIND reproduces fully under NO-MENTION (it flips 149 of 150 answers; exploratory count), and that cell carries H8 by itself. Under QNAMES and OPTIONS-AFTER it flips 0 and 0 answers, so those cells decompose a shift of the logit margin (Φ = 5.814 and 6.523 nats against base margins of -20.06 and -27.69, exploratory, recomputed), not a reproduced swap.

Further exchange results (exploratory). Part of BIND's value share under NO-MENTION sits at the punctuation positions: with the drink words alone exchanged, ψ_V falls from 0.435 to 0.238 and κ_w = 0.779 (exploratory, recomputed); under QNAMES and OPTIONS-AFTER, κ_w is close to κ (Table 46). Split by the queried character, BIND under NO-MENTION gives ψ_K = 0.648 and ψ_V = 0.464 for the first ($n = 71$) and 0.772 and 0.400 for the second ($n = 79$). Under QNAMES2, which names only the two story drinks, the margin shift is carried by keys alone (κ = 1.014, $\psi_V = -0.014$), but no an-

Table 45. **The exchange on the interventions of Prakash et al. (2026)** (Qwen2.5-14B-Instruct, 150 pairs). Each cell: $\kappa = \psi_K/(\psi_K + \psi_V)$ with (ψ_K, ψ_V), the number of answers the full patch flips, and its effect Φ on the margin (nats). BIND@28: the binding swap at block $\ell^* = 28$; ID@0, ID@28: the identity edit at blocks 0 and 28. $s_{\mathrm{ID}}(f, \ell_0)$: identity key share of the natural clamp at the queried drink from block ℓ_0, with 95% CI. †Exploratory formats. ‡Gate b2 failed ($\Phi < 3$ nats). n/e: the κ rule fails. H7 compares BIND@28 with $s_{\mathrm{ID}}(f, 29)$, H11 ID@0 with $s_{\mathrm{ID}}(f, 1)$ (Part 1) and ID@28 with $s_{\mathrm{ID}}(f, 29)$ (Part 2).

	NO-MENTION	QNAMES	OPTIONS-AFTER	LETTERS-AFTER†	QNAMES2†
BIND@28	κ 0.618 (0.703, 0.435) flips 149/150, Φ 31.616	κ 0.906 (1.020, 0.106) flips 0/150, Φ 5.814	κ 0.864 (1.025, 0.161) flips 0/150, Φ 6.523	κ 0.658 (1.652, 0.859) flips 0/150, Φ 0.225‡	κ 1.014 (1.016, −0.014) flips 0/150, Φ 7.285
ID@0	κ 0.131 (0.131, 0.871) flips 150/150, Φ 46.005	κ 0.773 (0.777, 0.229) flips 135/150, Φ 39.606	κ 0.779 (0.789, 0.224) flips 150/150, Φ 54.550	κ 0.969 (0.977, 0.031) flips 150/150, Φ 38.255	–
ID@28	κ 0.129 (0.129, 0.872) flips 150/150, Φ 45.797	κ 0.007 (0.006, 0.827) flips 0/150, Φ 9.321	κ 0.006 (0.005, 0.811) flips 0/150, Φ 10.885	n/e (−0.171, 3.024) flips 0/150, Φ 0.034‡	–
$s_{\mathrm{ID}}(f, 29)$	0.017 [0.007, 0.026]	−0.028 [−0.039, −0.019]	−0.015 [−0.020, −0.010]	0.098 [−2.155, 1.750]	–
$s_{\mathrm{ID}}(f, 1)$	0.017 [0.007, 0.026]	0.724 [0.710, 0.739]	0.702 [0.689, 0.716]	0.984 [0.976, 0.992]	–

Table 46. **All thirteen exchange cells** (Qwen2.5-14B-Instruct, 150 pairs). Φ: effect of the full patch on m (nats) with 95% CI; flips: answers moved to the target; ψ_K, ψ_V: fractions of Φ reproduced in the clean run by the patched run's keys or values at the patched positions; interaction $1 - \psi_K - \psi_V$; κ with 95% CI; κ_w: κ with the drink words alone exchanged (punctuation positions excluded); χ_K, χ_V: fractions of Φ lost when the patched run gets the clean keys or values. LETTERS-AFTER and QNAMES2 are exploratory; κ_w, χ_K and χ_V are exploratory everywhere.

Edit	Format	Φ (nats)	flips	ψ_K	ψ_V	interaction	κ	κ_w	χ_K	χ_V
BIND@28	NO-MENTION	31.616 [30.193, 32.985]	149/150	0.703	0.435	−0.138	0.618 [0.595, 0.640]	0.779	0.565	0.297
	QNAMES	5.814 [5.187, 6.456]	0/150	1.020	0.106	−0.127	0.906 [0.879, 0.931]	0.919	0.894	−0.020
	OPTIONS-AFTER	6.523 [5.905, 7.171]	0/150	1.025	0.161	−0.186	0.864 [0.839, 0.889]	0.875	0.839	−0.025
	LETTERS-AFTER	0.225 [0.170, 0.282]	0/150	1.652	0.859	−1.511	0.658 [0.559, 0.854]	0.278	0.141	−0.652
	QNAMES2	7.285 [6.734, 7.861]	0/150	1.016	−0.014	−0.002	1.014 [0.992, 1.034]	1.019	1.014	−0.016
ID@0	NO-MENTION	46.005 [45.276, 46.725]	150/150	0.131	0.871	−0.002	0.131 [0.119, 0.142]	0.132	0.129	0.869
	QNAMES	39.606 [38.747, 40.488]	135/150	0.777	0.229	−0.006	0.773 [0.759, 0.786]	0.749	0.771	0.223
	OPTIONS-AFTER	54.550 [53.652, 55.462]	150/150	0.789	0.224	−0.012	0.779 [0.764, 0.793]	0.737	0.776	0.211
	LETTERS-AFTER	38.255 [37.690, 38.788]	150/150	0.977	0.031	−0.008	0.969 [0.960, 0.977]	0.957	0.969	0.023
ID@28	NO-MENTION	45.797 [45.069, 46.514]	150/150	0.129	0.872	−0.001	0.129 [0.118, 0.140]	0.131	0.128	0.871
	QNAMES	9.321 [8.653, 9.995]	0/150	0.006	0.827	0.166	0.007 [−0.004, 0.018]	0.009	0.173	0.994
	OPTIONS-AFTER	10.885 [10.314, 11.513]	0/150	0.005	0.811	0.185	0.006 [−0.003, 0.013]	0.003	0.189	0.995
	LETTERS-AFTER	0.034 [−0.012, 0.078]	0/150	−0.171	3.024	−1.854	n/e	−0.263	−2.024	1.171

swer flips. At block 28 the identity edit is value-carried in every evaluable format (ψ_V = 0.872 / 0.827 / 0.811), while BIND is key-dominant (Table 45); this is the post hoc reading of Appendix H, with its caveats there: under QNAMES and OPTIONS-AFTER neither edit flips an answer, BIND also moves a payload, and Gate b3, built for the reverse dissociation, failed.

Table 47. **Identity key share of the natural clamp on the stories of Prakash et al. (2026)** (Qwen2.5-14B-Instruct, 150 pairs). $s_{\mathrm{ID}} = \overline{\mathrm{ID}}_K/(\overline{\mathrm{ID}}_K + \overline{\mathrm{ID}}_V)$ for keys and values clamped at the queried drink word from block ℓ_0 on. The onsets 0, 1, 3, 14 and 29 are preregistered in the four formats; the others form an exploratory sweep in NO-MENTION and OPTIONS-AFTER. Under LETTERS-AFTER at $\ell_0 = 29$ the denominator is near zero, so s_{ID} is undetermined (CI in Table 45).

Format	ℓ_0=0	ℓ_0=1	ℓ_0=3	ℓ_0=6	ℓ_0=12	ℓ_0=14	ℓ_0=18	ℓ_0=24	ℓ_0=27	ℓ_0=29	ℓ_0=30	ℓ_0=33	ℓ_0=36	ℓ_0=42
NO-MENTION	0.018	0.017	0.017	0.017	0.018	0.017	0.017	0.017	0.017	0.017	0.017	0.018	0.021	0.036
QNAMES	0.724	0.724	0.726	–	–	0.488	–	–	–	−0.028	–	–	–	–
OPTIONS-AFTER	0.702	0.702	0.702	0.688	0.634	0.587	0.218	0.041	−0.014	−0.015	−0.016	−0.015	−0.018	−0.070
LETTERS-AFTER	0.984	0.984	0.984	–	–	0.975	–	–	–	0.098	–	–	–	–

Depth. The identity key route at the drink word closes with depth when candidates are named after the question (Table 47): under OPTIONS-AFTER, s_{ID} is 0.702 for $\ell_0 \leq 3$, 0.587 at 14, 0.218 at 18, 0.041 at 24 and −0.014 at 27, while under NO-MENTION it stays near zero at every onset. Gate b3, which H10's reading as a dissociation needed, failed: at $\ell_0 = 29$ under OPTIONS-AFTER, ID_K = −0.092 nats and s_{ID} = −0.015, and κ_{ID} at block 28 is 0.006. The binding swap flips no answer below block 24 under NO-MENTION and at most 0.01 of answers at any swept block (18–40) when can-

didates are named, while the identity edit stops flipping answers from about block 20 when they are named, and the key route of identity has closed by block 24. No depth has both the binding swap and a key-read identity route, so the contrast between the identity edit at block 0 (a format shift of $+0.648$) and the binding swap at block 28 ($+0.247$) is confounded with depth: whether the kind of edit or its depth decides how far the format moves the attribution is untested.

κ and s_{ID} are different statistics. κ is one-sided (m compares the target with the queried drink) and exchanges the drink word and the punctuation after it; s_{ID} is the symmetric S-against-X identity contrast at the word alone. With $\ell^*_{\mathrm{ID}} = 0$, the ID@0 exchange and the natural clamp from block 1 are nearly the same intervention: applied to both, the one-sided κ agree within 0.001 (NO-MENTION), 0.028 (QNAMES), 0.044 (OPTIONS-AFTER) and 0.012 (LETTERS-AFTER) (exploratory, recomputed). The H11 Part 1 gaps therefore split as follows: under NO-MENTION almost the whole gap is the difference between the statistics (0.115); under QNAMES, 0.028 is exchange against clamp and 0.020 the difference between the statistics; under OPTIONS-AFTER, 0.044 and 0.032 (exploratory, recomputed; the log's split, from three-decimal printed values, differs in the last digit). H11 Part 2, at block 28, compares values within a few hundredths of zero under QNAMES and OPTIONS-AFTER (κ_{ID} = 0.007 and 0.006 against $s_{\mathrm{ID}} = -0.028$ and -0.015) and repeats Part 1's NO-MENTION cell (0.129 against 0.017). H11 is thus consistency of the exchange with the clamp on their material, not an independent test of the law. For H7, the one-sided clamp analogue from block 29 is 0.131 / 0.012 / 0.005 (exploratory, recomputed), which would leave every gap far above 0.25; no verdict depends on the difference between the statistics. H7's correlation, $r = -0.986$, is computed over three values of s_{ID} within a few hundredths of zero and carries no information. Against s_{ID} from block 0 (H7's secondary comparison, not scored), the gaps are 0.600, 0.181 and 0.163.

FP32 re-check (exploratory, recomputed). Re-running the NO-MENTION sweeps in FP32 at blocks 26–30 gives BIND IIA of 0.747, 0.880, 0.993, 1.000 and 1.000, against 0.747, 0.887, 0.993, 1.000 and 1.000 in BF16, so the tie rule again selects block 28. At block 28 the flips agree in 0.987 of pairs (a different single pair fails in FP32), and Φ differs by 0.12 nats. The identity edit's IIA is 1.000 at blocks 0–2 in both precisions.

H.2. The final round: an identity edit at block 28 and the overlap screen

An identity edit at block 28 (J-C6, met). The identity edit of H was the residual of a donor story, which at block 28 nearly restates the natural clamp. J-C6 replaced it by a steering vector (Panickssery et al., 2023): the mean difference of the block-28 states at the drink word and the punctuation after it between stories with the drink S and with the queried drink, taken over the pool pairs outside the population, written at block 28 of Qwen2.5-14B. Its anchor gate reproduced H's binding swap (IIA 0.993; κ_{BIND} = 0.618, 0.906 and 0.864 under NO-MENTION, QNAMES and OPTIONS-AFTER, equal to stage 6). The steering vector flips every answer without a later mention (IIA 1.000), and its keys and values lie at a relative distance ν = 0.214 from those of H's identity edit. The pre-written text, used verbatim: "At block 28 of Qwen2.5-14B an identity edit is copy-read like the natural value ($\kappa \leq 0.30$), while the binding swap there is key-read; key-flat reading is not a property of depth for identity edits." Its key share is κ_{CAA} = 0.129 [0.117, 0.140], 0.005 and 0.002 in the three formats, and $\kappa_{\mathrm{BIND}} - \kappa_{\mathrm{CAA}}$ = 0.489 [0.462, 0.515], 0.901 [0.871, 0.929] and 0.863 [0.837, 0.888] (95 % pair-bootstrap intervals). This does not separate kind from depth, because the natural read there is itself copy-only, and the swap changes no answer in two of the three formats: under QNAMES and OPTIONS-AFTER the binding swap's IIA is 0.00 and 0.00 (0.99 under NO-MENTION), so its κ there describes margin shifts of 5.81 and 6.52 nats that change no answer.

The overlap screen (J-C-SCREEN, met; J-C-WIN, not evaluable). At Qwen2.5-7B and Llama-3.1-8B the binding swap was swept over every block under NO-MENTION (150 pairs passing the language-model filter in each model), and the natural identity key share s_{ID} under OPTIONS-AFTER was measured from the block after each working block and on a grid. A window would be a block where the swap works ($\Phi \geq 3$ nats and IIA ≥ 0.5) and $s_{\mathrm{ID}}(l{+}1) \geq 0.4$. The swap works at blocks 15–22 of Qwen2.5-7B (IIA up to 0.947) and 12–16 of Llama-3.1-8B (up to 0.773), where $s_{\mathrm{ID}}(l{+}1)$ is at most -0.061 and -0.096; the identity key route is open early (s_{ID} from block 0: 0.686 and 0.596) and has closed before the swap begins to work. The pre-written text, with its codes as written: "Binding forms only after the identity key route closes in every model screened, so a binding swap cannot be compared with an identity edit at a depth where the key route is open; H7–H10 stand." Without a window, J-C-WIN, which would have tested the law inside it, is not evaluable.

I. Competing accounts and further related work

What four accounts predict (as of the earlier entries). Table 48 sets four accounts against the observations of the discovery sample. Each, ours included, is contradicted somewhere. Two rows matter most: the binding swap, where our account's preregistered extension failed (H7, H9) and the lookback account is partly compatible, post hoc (one edit, one model); and I6, where at 24B less of the remap's effect was kept than predicted once its readers were ablated.

Table 48. **What four accounts predict, as of the entries of the discovery sample.** DT: duplicate-token attention alone; LB: lookback addresses and payloads (Prakash et al., 2026; Wu & Shomali, 2026); FF: an edit's channel is set by the format it was fit in; LM: later-mention readers (ours). ✓: predicted and observed; ×: predicted otherwise (refuted); –: silent; (ph): compatible, post hoc; ✓/×: the attention is predicted, its presence without a read is not; ✓/(ph): the attention is predicted, its presence without a read is explained post hoc (the read is decided downstream). Binding-swap row: Appendix H; last two rows: Appendix G.4. The final round's results are added in the text.

Observation (evidence)	DT	LB	FF	LM
Hop 1: a sparse set of heads, whose core is duplicate-token heads, attends from the re-mentions to the writing token, key-matched, at every scale, also at 1.5B and 3B where $\mathrm{ID}_K \approx 0$ (G1–G3, H1–H4)	✓/×	–	–	✓/(ph)
The same words placed before the writing token give no key read (E1)	✓	×	–	✓
The answer reads the option words, not the writing token (H5, G8b)	–	×	–	✓
Without a later mention the identity is copied through the answer's edge to the writing token (G8a, E4)	–	✓	–	✓
Removing the readers raises the value copy (G7a, H3, I5)	–	(ph)	–	✓
A remap refit without later mentions shows the same crossover (F1–F4)	–	–	×	✓
Prakash et al.'s binding swap is key-dominant in every evaluable format (κ 0.618, ψ_V 0.435 without a later mention); later mentions shift κ by +0.247 (H7–H10)	–	(ph)	–	×
At 24B the remap's key read runs through the natural reader heads; ablating them removes most of its key-only effect (I1–I4)	–	–	–	✓
Without its readers, the remap keeps its effect on the margin (I6)	–	–	–	×

What the final round adds. Four results of the final entry bear on the accounts. (i) The first hop's output: moved into the row of an option absent from the story (and out of B's row), the reader heads' write moves the answer there and is read by key; added alone it moves the answer 0.673 and 0.541 as much as the natural clamp (Appendix E.3), so a duplicate-token match writes something the answer uses (Appendix E.3); but structured controls also move the answer, and removing one direction per layer leaves about half of the key read, so the write is sufficient, not specific, and not necessary as one direction. Duplicate-token attention alone (DT) predicts the attention, not this write; it does not decide between LB and LM. (ii) The sign of the key read follows the task: asked which option was not mentioned, the key read and the value read both turn negative, and the same frozen reader heads carry both questions' reads; in IOI's in-sentence re-mentions the key read is negative while the value read stays positive, applied at the answer's key read of the listed names, not by the answer-side heads ranked in lists (Appendix E.4). A negative use of duplicate detection is known (Wang et al., 2022; McDougall et al., 2023); we do not present the polarity as a discovery. (iii) The channel-ratio law, which would have made an intervention's apparent channel a property of the model's own read in each format, failed, even for steering vectors close to the natural edit (Appendix F); so the LM column's "✓" for the refit row describes the remap of the release and its refit, not edits in general. (iv) On counterfactual reading-comprehension passages a later options list opens the key read, but behaviour follows the key only with lettered options (Appendix D.5).

Relation to lookback, address and payload accounts. Prakash et al. (2026) find, with causal mediation and causal abstraction, lookback mechanisms built on ordering IDs (binding, answer and visibility lookbacks) that track beliefs; Wu & Shomali (2026) find that transplanting an operation's span redirects which state is read (a routing record), and that the writing model reads mainly the address. In our terms, a lookback whose payload is the state's identity would read identity by copy, through the value. The exchange of Appendix H.1 shows that their state-token patch, which swaps ordering information rather than identity, reaches the answer mainly through the state tokens' keys, with a substantial value share without a later mention (ψ_K = 0.703, ψ_V = 0.435 under NO-MENTION); because one patch moves both, the decomposition does not separate an address from a payload, and it is one edit at one model. What is copied through the value without a later mention is the state's identity: an identity edit at the same positions is value-carried under NO-MENTION (ψ_V = 0.871), as the released remap of Anonymous (2026) is (E4), and in the final round so is a steering vector at block 28

(J-C6, Appendix H.2). Later mentions add a second route for identity, a lookup in which a re-mentioned word finds the writing token through its key, as duplicate-token heads do (Wang et al., 2022); at two 7B models it runs through a sparse set of heads enriched for duplicate-token heads, and the answer then reads the re-mentioned words (Appendix E.2). On their material this route is visible for an identity edit at block 0 (κ_{ID} rises from 0.131 under NO-MENTION to 0.779 under OPTIONS-AFTER) but closes with depth, and the final round's screen found no depth with both it and a working binding swap in two further models (Appendix H.2), so whether it runs alongside a lookback at matched depth is untested. Which of the positional, lexical and reflexive mechanisms of Gur-Arieh et al. (2025) our lookup corresponds to, if any, we do not test. In our QNAMES2 format, which names both candidate states, the binding swap shifts the margin through keys alone ($\kappa = 1.014$) but flips no answer (exploratory). As for the role-swap control: the observer is the first character named in both runs, so the control cannot detect an ordering ID of the observer at the writing token.

Further related work. On heads and circuits: attention weights alone do not determine a head's output (Kobayashi et al., 2020), induction can copy tokens or whole concepts (Feucht et al., 2025), and circuit components are reused across tasks (Merullo et al., 2023). Further work on options and retrieval: specific heads select among options (Lieberum et al., 2023; Wiegreffe et al., 2024), select-and-copy heads do so by query–key alignment (Tulchinskii et al., 2024), and a winner is chosen before it is bound to a symbol (Wong et al., 2026); shallower heads that read persuasive keywords write a one-dimensional option-routing feature into the option tokens, which redirects the attention of decision heads that copy the option they attend to (Sun et al., 2026), the closest analogue of our two hops; belief and reality share a value slot read out under a router at the query (Steele et al., 2026a); answering with option symbols requires binding the options to the symbols (Robinson et al., 2022); questions address facts by their order of mention (Zhou, 2026), compatible with (not tested by) the key-dominant binding swap; value-slot routers (Steele et al., 2026b) and cached-KV edits (Li, 2026) concern storage. On the validity of interventions: steering vectors are not identifiable from input–output behaviour (Venkatesh & Kurapath, 2026); diagnostics include control tasks (Hewitt & Liang, 2019), benchmarks (Huang et al., 2024; Arora et al., 2024) and input partitions (Li et al., 2026); the format can change what a steering gain identifies (Gao et al., 2026); models' detection and localisation of injected concept vectors relies on gate and router heads, and better localisation is associated with better alignment of the key and value changes the injection induces in their QK and OV computations (Zou et al., 2026). Activation patching (Vig et al., 2020) depends on the metric and the corruption (Zhang & Nanda, 2023; Heimersheim & Nanda, 2024), and precomputed KV caches substitute exactly for a joint pass (Pustovit, 2026).

J. Models, compute and licences

Table 49. **Models of the final round and the tests that use them.** Instruction-tuned models, BF16; revisions are the official repositories' commits on the date of the final entry, each file verified against the official hash before the model loaded (Appendix C.2); the three gated models were assembled from byte-identical public copies. Attention: sdpa, except eager for Gemma-2 and for the passes that read attention weights. Fresh: the fresh stories (Appendix D.2); passages: Appendix D.5; mechanism: Appendix E; interventions: Appendix F; the frames: the released remap at 24B under the emitted-form score (Appendix D.3). The discovery sample's models are listed in Appendix J.1.

Model	Revision	Attention	Tests
Qwen2.5-7B-Instruct	`a09a35458c702b33eeacc393d103063234e8bc28`	sdpa	fresh, passages, mechanism, interventions
Qwen2.5-14B-Instruct	`cf98f3b3bbb457ad9e2bb7baf9a0125b6b88caa8`	sdpa	fresh; binding swap at block 28
Qwen2.5-1.5B-Instruct	`989aa7980e4cf806f80c7fef2b1adb7bc71aa306`	sdpa	mechanism (scale); fresh (exploratory)
Qwen2.5-3B-Instruct	`aa8e72537993ba99e69dfaafa59ed015b17504d1`	sdpa	mechanism (scale); fresh (exploratory)
Mistral-7B-Instruct-v0.3	`c170c708c41dac9275d15a8fff4eca08d52bab71`	sdpa	fresh, passages, mechanism, interventions
OLMo-2-1124-7B-Instruct	`470b1fba1ae01581f270116362ee4aa1b97f4c84`	sdpa	fresh
Llama-3.1-8B-Instruct	`0e9e39f249a16976918f6564b8830bc894c89659`	sdpa	fresh, passages, interventions
Gemma-2-9B-it	`11c9b309abf73637e4b6f9a3fa1e92e615547819`	eager	fresh, passages
Gemma-2-2B-it	`299a8560bedf22ed1c72a8a11e7dce4a7f9f51f8`	eager	fresh (the 2B sentence test)
Phi-4	`2db69c1c3e91a05d2c64a3185acfbaf36f744e25`	sdpa	fresh
Falcon3-7B-Instruct	`1e57a0ecd176c7c139f289c60a74e57f887c3dfb`	sdpa	fresh
Mistral-Small-24B-Instruct-2501	`9527884be6e5616bdd54de542f9ae13384489724`	sdpa	the frames under emitted forms

Compute of the final round. Every part ran on one NVIDIA A100 80GB (PCIe for the passages, SXM4 for the other three parts) in BF16, with Python 3.12.14, torch 2.11.0 (CUDA 12.8), transformers 5.18.0 and numpy 2.5.3, from a clean checkout of the finalising commit, in one session each: the passages from 13:13 to 15:44 UTC, the mechanism from 09:17 to 10:33 UTC, the fresh stories from 11:25 to 13:51 UTC and the interventions from 14:40 to 17:33 UTC, on two consecutive days, each well before its preregistered deadline. No step failed, was skipped or was cut by a deadline in any part. The only training is the rank-16 remap fit at the writing token (losses and held-out flip rates in Appendix F.2); the steering vectors are means and the dictionaries are used as released. Pilots ran only on the CPU in FP32, with models of at most 1.5B parameters, and are listed with their numbers in the entry; no output of a final-round population was seen before its run.

Third-party material and licences. SQuAD v1.1 (Rajpurkar et al., 2016) is used under CC BY-SA 4.0, and the counterfactual passages are derived from it. The sparse-autoencoder dictionaries are those of andyrdt (2025) for Qwen2.5-7B-Instruct (BatchTopK, $k = 64$, 131,072 latents; Bussmann et al. 2024) at layers 3, 7, 11 and 15, at a pinned revision, each file checked against its sha256 at download. The release of Prakash et al. (2026) is fetched at run time at its pinned commit and checked against stored file hashes; none of it is copied into our code. The released bases and training code of Anonymous (2026) are checked against the hashes pinned in stage 4 before any model is loaded. Each model is used under the licence on its model card.

J.1. The discovery sample

Compute of the discovery sample. All experiments except the stage-4 refits are forward passes; the refits train only rank-16 bases (Appendix G.3), and no model weights are ever trained. Pilots with models of at most 1.7B parameters, the first format factorials, preregistration A, the 1.5B row-restricted splices, the early attention probe and the FP32 unit tests ran on CPU in FP32; the preregistration-A output records transformers 5.18.0 and Python 3.11.15. All GPU runs used NVIDIA A100 80GB GPUs (SXM4 in stages 1–6, PCIe in stage 7) in BF16, torch 2.11.0 with CUDA 12.8 and Python 3.12.14, as recorded in each stage's `ENV.txt` and output files:

- stage 1 (preregistration B; Qwen2.5-1.5B to 14B, Qwen3-8B, Mistral-7B, OLMo-2-7B): one GPU, transformers 5.18.0, commit `stage1-run`;
- stage 2 (preregistration C; frames at 24B and 72B, natural factorial at 24–72B): two GPUs with `device_map="auto"`, transformers 5.9.0, pinned by the stage-2 script for the reproduction gate, commit `stage2-run`;

- stage 3 (preregistration D; new tasks and localisation at 7–14B): one GPU, transformers 5.18.0, commit `stage3-run`;
- stage 3b (preregistration E): two GPUs, commit `stage3b-run`; the 2×2, the role control at 7–14B and the Qwen2.5-32B environment re-run under transformers 5.18.0 (the re-run on one GPU), and the Qwen2.5-14B environment re-run, the value-only frames and the role control at 72B under transformers 5.9.0;
- stage 4 (preregistration F): two GPUs, one per refit (55 minutes each), then the frames on both, transformers 5.9.0, commit `stage4-run`;
- stage 5 (preregistration G): one GPU, transformers 5.18.0, commit `stage5-run`; eager attention for the re-mention attention and form probes, sdpa elsewhere; GPT-2 small and GPT-2 XL in FP32 with exactness asserted; the GPT-2 attention probe re-run on CPU (Appendix A);
- stage 6 (preregistration H): one GPU, transformers 5.18.0, commit `stage6-run`; eager attention for part (a), sdpa for part (b), and the exploratory re-check sweeps of part (b) in FP32. The release of Prakash et al. (2026) is fetched at run time at its commit `0579347` and checked against stored file hashes; none of it is copied into our repository;
- stage 7 (preregistration I; Mistral-Small-24B at the pinned revision): one GPU, transformers 5.18.0, commit `stage7-run`; eager attention for the head ranking and the exploratory remap ranking, sdpa elsewhere; about 2.4 hours from the unit tests to the score. The release of Anonymous (2026) was checked against the hashes recorded in stage 4 before any model was loaded.

Model revisions of the discovery sample. For the two models of Anonymous (2026) we pin the revisions used for the reproduction: Mistral-Small-24B-Instruct-2501 at `9527884be6e5616bdd54de542f9ae13384489724` and Qwen2.5-72B-Instruct at `495f39366efef23836d0cfae4fbe635880d2be31`. The pins apply to the frames, the stage-4 refits, every stage-7 run and the stage-2 natural factorial at those sizes. All other runs, including Qwen2.5-32B and the role control at Qwen2.5-72B, load the default revision available at run time.

Released code. The code release contains the story generator, encoder, tasks and hook primitives (`ckeys/`); one script per experiment (`experiments/`); the preregistered scoring scripts (`analysis/stage1_prereg.py`, `stage2_score.py`, `stage3_score.py`, `stage3b_score.py`, `stage4_score.py`, `stage5_score.py`, `stage6_score.py`, `stage7_score.py`, and for the final round `stage8a_score.py` to `stage8d_score.py` with their shared Holm helper); the refit wrapper (`experiments/refit_remap.py`); a port of the story generator and counterfactuals of Prakash et al. (2026) that reads their release at run time (`ckeys/causaltom.py`); one shell script per GPU stage, which pins the library version and records the commit and environment (`scripts/`); the exactness tests (`tests/`); and `paper/make_figures.py`, which generates every result figure, every table in Appendices D.1, E.1, E.2, G.3 and G.4, the tables of preregistrations G, H, I and J, and every result macro in the text from the saved results and score files (the schematic in Figure 1 is drawn by `paper/make_schematic.py`).